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# Crazy 3

**Papers 1–4 combined · Sections 10–13**

All four answer books merged section by section; overlapping answers are printed once with every fact kept, tagged P1–P4 by where they were asked.


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# Section 10 · Testis, Penis & Urethra

## Long answer questions (LAQ)

### LAQ 1 Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma

P1 · P2 **merged from 2**

*Asked as:* P1 LAQ "Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications" (asked as "Lymphatic drainage of testis and principles for designing templates for RPLND" and "Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies") · P2 LAQ "Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis."

*The P1 second part (Summer 2021) is taken as the management implications of lymphatic drainage in penile cancer, i.e. management of the inguinal and pelvic nodes.*

#### Embryological basis

- **Why retroperitoneal:** the testis develops at L1–L2 beside the mesonephros (near the kidney) and descends, carrying its vessels and lymphatics; its artery, vein and lymphatics stay retroperitoneal. So its lymph returns to where it began: the **retroperitoneal (lumbar) nodes near the renal hilum**, not the groin. The scrotum is skin and drains to the **groin**.

#### Lymphatic drainage of the testis

- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run up the **spermatic cord** with the testicular (gonadal) vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum (**para-aortic and interaortocaval nodes**). Lymph capillaries do not enter seminiferous tubules.
- **Onward:** lumbar trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta, behind the right crus; hence retrocrural nodes), then the **thoracic duct** to the left innominate vein. This explains **left supraclavicular** node metastases.
- Lymph flows **cephalad and right to left**.
- **Epididymis:** caput and corpus drain with the testis; **cauda epididymis** drains with the vas to the **external iliac** nodes.

##### Retroperitoneal nodal regions

| Region | Extent |
|---|---|
| Right paracaval (with precaval, retrocaval) | IVC (inferior vena cava) midline to right ureter |
| Interaortocaval | IVC midline to aortic midline (between IVC and aorta) |
| Left para-aortic (with pre-aortic, retroaortic) | Aortic midline to left ureter |

- **Also know:** ten mapped regions include right and left common iliac, interiliac and right and left suprahilar.

### Primary landing zones

| | Right testis | Left testis |
|---|---|---|
| Primary (landing zone) | **Interaortocaval** (below renal vessels) | **Left para-aortic** (and left hilar) |
| Secondary / next echelon | Right paracaval, precaval, then para-aortic | Interaortocaval; pre-aortic |
| Crossover / contralateral | Small but real to left para-aortic (another source: common; right to left occurs) | None significant; rare to the right, bulky disease only (another source: rare; implies bulk) |

- **Crossover is only right to left**, matching the overall lymph flow.
- **Donohue mapping (1982):** contralateral and suprahilar spread is rare but rises with bulk; interiliac spread is very rare.
- **Pelvic, iliac and inguinal nodes** are involved only when drainage is altered (or with retrograde spread from bulky disease): previous scrotal or inguinal surgery (orchidopexy, hernia repair), scrotal violation, or tumour invading the tunica vaginalis or scrotum.
- The retroperitoneum is the first metastatic site in 70–80% (choriocarcinoma spreads by blood). CT (computed tomography) understages 25–35% of stage I NSGCT (non-seminomatous germ cell tumour); 5–9 mm landing-zone nodes are suspicious.

### Lymphatic drainage of the scrotum

- Scrotal skin drains to the **ipsilateral superficial inguinal nodes**. Lymphatics, like vessels and nerves, **do not cross the median raphe**.

### Clinical implications: testis and scrotum

- **Trans-scrotal orchidectomy or biopsy is contraindicated.** It leaves cord behind and opens scrotal skin lymphatic routes, so tumour can reach inguinal and pelvic nodes, causing local and inguinal or pelvic recurrence. Use **radical inguinal orchidectomy** with high cord clamping.
- Inguinal or pelvic spread from the testis means altered drainage: scrotal violation, previous orchidopexy or hernia repair, scrotal invasion.
- RPLND (retroperitoneal lymph node dissection) templates and para-aortic or "dog-leg" radiotherapy fields follow the landing zones.
- Scrotal skin cancer is staged at the ipsilateral groin.

### Principles of designing RPLND templates

**1. Map-based design: follow the side-specific landing zones** to remove likely disease and spare the rest.

- Historical dissection: bilateral infrahilar + suprahilar + interiliac: most men anejaculatory.
- **Ray (1974)** first templated landing zones; **Donohue (1982)** mapping confirmed side-specific spread; **Weissbach and Boedefeld (1987)** reduced left template with frozen section.
- So a template must include the primary landing zone, the next echelon and the known crossover route.

**2. Fixed boundaries**

- **Superior:** renal hila (renal vessels). **Lateral:** ureters. **Inferior:** ureter crossing the common iliac artery. Always remove the **gonadal vein and cord stump** to the internal ring.

**3. Omit suprahilar and interiliac regions** in non-bulky disease, as spread there is rare. This reduces chylous ascites, renal, pancreatic and duodenal injury, and protects the hypogastric plexus.

**4. Unilateral modified templates** leave contralateral sympathetic fibres untouched. Because of right-to-left crossover, both templates take the para-aortic nodes **above the IMA (inferior mesenteric artery)**.

| Template | Content |
|---|---|
| **Right modified** | Paracaval, precaval, retrocaval, interaortocaval, right common iliac; para-aortic **above IMA**; omit para-aortic below IMA |
| **Left modified** | Left para-aortic, preaortic, retroaortic, left common iliac; interaortocaval above IMA; omit paracaval/precaval/retrocaval |
| **Full bilateral infrahilar** | All three regions both sides (paracaval, precaval, retrocaval, interaortocaval, pre-aortic, para-aortic, retroaortic, both common iliacs), between renal vessels, ureters and ureteric crossing of iliacs |
| Historical extended | Plus suprahilar and interiliac (thoracoabdominal); most anejaculatory |

- Omitting para-aortic above the IMA (right) or interaortocaval (left) is controversial.

**5. Nerve sparing to preserve ejaculation:** antegrade ejaculation (emission and bladder-neck closure) depends on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the **superior hypogastric plexus at or below the IMA** and running over the sacral promontory. Identify them prospectively before dividing tissue.

- **Unilateral templates** leave contralateral fibres untouched (antegrade ejaculation 75–97%). **Prospective nerve sparing** (Jewett 1988) identifies fibres: 90–100%.

**6. Completeness ("duplicate open RPLND"): split-and-roll** over the great vessels, **divide lumbar vessels** to clear retrocaval and retroaortic tissue, clip lymphatics at the cephalad limit (left renal vein) to prevent chylous ascites. Incomplete RPLND or mass excision gives worse outcomes (GETUG T93BP; 33% local relapse after post-chemotherapy mass-only resection).

**7. Extend if needed:** enlarged nodes at operation call for the full bilateral template (frozen section, Weissbach 1987). Convert to bilateral if nodes are suspicious at surgery or the primary has somatic-type malignancy.

## Template by clinical setting

| Setting | Template |
|---|---|
| **Primary RPLND** (high-risk stage I or low-volume stage II NSGCT, normal markers) | Default: modified unilateral nerve-sparing **or** full bilateral infrahilar nerve-sparing; no randomised trial. Modified allowed: low-volume, clinically negative nodes. Main risk: out-of-template nodes 3–23% (right > left) |
| **PC-RPLND (post-chemotherapy)**, residual mass >1 cm, normal markers | **Full bilateral infrahilar + residual mass** (standard). Main risk: crossover with bulk; viable tissue unrecognisable |
| Modified PC template allowed only if | Mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good or intermediate risk |

- Against unilateral templates: 3–23% of node-positive men have nodes outside the modified template, more with right tumours.
- *Sources differ:* Campbell testis-neoplasm chapter calls full bilateral nerve-sparing the standard; the surgery chapter accepts either.

### RPLND outcomes and complications

- Primary RPLND: pN+ about 20–30%; cancer-specific survival (CSS) about 98.8–100%; retroperitoneal recurrence <2% after full template.
- **Antegrade ejaculation >90%** with templates plus nerve sparing (modified template alone: early 75–87%, refined 97%).
- Complications: ileus (commonest), wound infection, chylous ascites (0.2–2.1% primary; 2–12% post-chemotherapy), pulmonary (bleomycin lung toxicity; low $FiO_2$), ejaculatory loss.
- **Also know:** nephrectomy needed at PC-RPLND (left-sided mass >10 cm); GETUG T93BP showed incomplete RPLND predicts relapse.

### Stage I seminoma: definition, staging and work-up

- Disease confined to the testis: any pT, N0, M0 on imaging. Stage IA = pT1; IB = pT2–4; IS = markers still raised after orchidectomy.
- **Work-up:** AFP (alpha-fetoprotein), β-hCG (human chorionic gonadotropin) and LDH (lactate dehydrogenase) before and after orchidectomy. CT abdomen and pelvis; chest CT or radiograph. Baseline testosterone; **offer sperm banking**. A raised AFP means non-seminoma.
- **Radical inguinal orchidectomy:** early cord control at the internal ring. It cures 80–85% of stage I seminoma.
- **Risk factors for relapse: tumour >4 cm** and **rete testis invasion**. Five-year relapse is 12%, 16% and 32% with 0, 1 and 2 factors. LVI (lymphovascular invasion) predicts relapse in non-seminoma, not seminoma.

### Stage I seminoma: options after orchidectomy (cancer-specific survival about 100% with each)

| | Surveillance | Carboplatin | Radiotherapy |
|---|---|---|---|
| Method | Clinical review, markers, CT | **One cycle, AUC (area under the curve) 7** | **20 Gy** in 10 fractions to the para-aortic strip |
| Relapse | 14–20% | About 5% | 3–5% |
| Relapse site | Retroperitoneum (84–100%) | Retroperitoneum | Pelvis, chest, left supraclavicular |
| Drawbacks | Late relapse, many scans, compliance | Underdosing; late effects unknown | Second cancers, GI (gastrointestinal) toxicity, oligospermia |
| Status | **Preferred** | If unwilling or unsuitable for surveillance; risk factors | If unwilling or unsuitable for surveillance |

- **Carboplatin (Calvert):** 7 × (GFR [glomerular filtration rate] + 25) mg, using isotope GFR.
- **Trials: MRC TE18** (Medical Research Council): 20 Gy equals 30 Gy, with less toxicity. **MRC TE19/EORTC (European Organisation for Research and Treatment of Cancer) 30982:** one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE10: para-aortic field equals dog-leg, with 2% pelvic relapse.

- **Risk-adapted (EAU [European Association of Urology]):** surveillance for low risk; with risk factors discuss surveillance or carboplatin (treating all would overtreat two-thirds).

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Inguinal orchidectomy + markers + CT │ Stage I seminoma │ Surveillance (preferred) ── unwilling / unreliable ──► Carboplatin AUC7 ×1(–2) │ (or radiotherapy 20 Gy) Relapse (mostly retroperitoneal) ├─ small nodes ► dog-leg radiotherapy with boost, or chemotherapy └─ bulky/distant ► BEP ×3 or EP ×4
```

- BEP: bleomycin, etoposide, cisplatin; EP: etoposide, cisplatin.

## Stage I seminoma: relapse, IGCCCG and follow-up

- Relapses are almost always cured; median 14 months, but 10–20% occur after 4 years.
- **IGCCCG:** seminoma is good risk unless there are non-pulmonary visceral metastases (intermediate). **There is no poor-risk seminoma**, and markers do not change the risk group.
- **Surveillance follow-up:** examination and markers every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly; CT every 4–6 months (years 1–3), then 6-monthly to year 7, then yearly. CT finds >85% of relapses.

## Lymphatic drainage of the penis

| Superficial system | Deep system |
|---|---|
| Prepuce and shaft skin | Glans, frenulum, corpora |
| Dorsal network divides at the base and drains to both superficial inguinal groups | Trunks circle the corona and run under Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes |

- **Two rules:** drainage is **bilateral and unpredictable in side** (presymphyseal crossover), but **orderly in sequence**: superficial inguinal, then deep inguinal, then external iliac and obturator. **No skip metastases.**
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**; sentinel nodes are mostly **superomedial**, then central. **Deep nodes** (1–3) lie medial to the femoral vein; **Cloquet's node** is the highest.

## Management implications in penile cancer

### 1. Prognosis and timing

- Nodal status is the **strongest prognostic factor**. Penile SCC (squamous cell carcinoma) has a long **locoregional phase**, so the groin is a **curable station**.
- **Kroon (2005):** early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about **84% vs 35%** when delayed until palpable.
- Manage both groins **within 3 months**; remove **at least 8 nodes per side**.

### 2. Bilateral management

- Crossover means **both groins** are staged or treated; with one palpable groin, the other holds micrometastasis in about 30%.

### 3. Impalpable groin (cN0): risk-based

| Risk | EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023 | Groin |
|---|---|---|
| Low | pTis, pTa, pT1a G1 | Surveillance |

| Risk | EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023 | Groin |
|---|---|---|
| Intermediate | pT1a G2 | Surveillance if compliant, or surgical staging |
| High | **pT1b or higher**, G3 | **DSNB (dynamic sentinel node biopsy)**; ILND if DSNB unavailable |

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk and is staged surgically (ILND first).
- **DSNB:** ultrasound ± FNAC (fine-needle aspiration cytology) first, because a tumour-replaced node diverts flow (false negative). Then technetium-99m nanocolloid, blue dye and gamma probe.
- **No-skip spread and superomedial sentinels** justify **modified ILND (Catalona)**: smaller field, saphenous vein preserved; a positive frozen section leads to complete radical ILND.

**4. Palpable nodes (cN+)**

- About half are reactive, but do not delay with antibiotics. Ultrasound-guided **FNAC**; if positive, **radical ILND** on that side and stage the other.
- **Bulky (≥4 cm), fixed or bilateral nodes:** neoadjuvant **TIP** (paclitaxel, ifosfamide, cisplatin) × 4, then surgery in responders.

**5. Pelvic nodes**

- Stepwise spread means a negative groin needs no pelvic dissection.
- **PLND (pelvic lymph node dissection)**, ipsilateral, if **more than 2 inguinal nodes or ENE (extranodal extension)**. ENE in any node is pN3.

**6. Complications of ILND**

- Wound infection, **flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis). Reduced by modified templates, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.
- **Also know:** adjuvant therapy for pN2–3; Cabanas sentinel biopsy abandoned.

## Recent advances

- **Robotic RPLND** for low-volume disease (same templates, same rules).
- RPLND for small-volume stage II seminoma (**SEMS, PRIMETEST**).
- Serum **miR-371a-3p** microRNA for selection, and it detects relapse better than standard markers.
- MRI (magnetic resonance imaging) instead of CT for seminoma surveillance (TRISST trial) to reduce radiation.

## Diagrams to draw

- Retroperitoneum showing great vessels with the three nodal regions, right (interaortocaval) and left (para-aortic) landing zones and the right-to-left crossover arrow; outline the right and left modified templates, IMA level and hypogastric plexus.
- Beside it, the femoral triangle with Daseler's zones and Cloquet's node.

**Key points to remember:**

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila, ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND: full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum (ipsilateral superficial inguinal) and penis drain to inguinal nodes; never operate trans-scrotally.
- Seminoma relapse risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

## LAQ 2 Grafts in urology and buccal mucosa

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "Enlist various grafts used in urology and role of BMG in urology" · P3 SAQ "Uses of buccal mucosa in urology"

### Graft versus flap

| | Graft | Flap |
|---|---|---|
| Blood supply | Separated; gets a new supply from the bed | Moved with its own pedicle |
| Depends on | A well-vascularised host bed | Its pedicle |
| Example | BMG (buccal mucosal graft) | Penile fasciocutaneous island flap |

### Graft take (about 96 hours)

| Phase | Duration | Mechanism |
|---|---|---|
| **Imbibition** | about 48 h | Graft "drinks" nutrients from the bed; graft is cooler than core |
| **Inosculation** | about 48 h | Bed and graft vessels connect; true microcirculation returns |

- A graft lives on its bed, so it does best on thick vascular beds. Take fails with poor beds (scar, radiation), haematoma, seroma, infection or shear. Hence **quilting or pie-crusting** and **immobilisation for 5–10 days** (bolster or vacuum dressing).

### Grafts used in urology

| Group | Graft | Main uses |
|---|---|---|
| **Skin** | **STSG** (split-thickness skin graft), meshed | Staged urethroplasty only; penile and scrotal cover (Fournier gangrene, buried penis) |
| | **FTSG** (full-thickness skin graft) | Fossa navicularis, penile urethra (avoid in lichen sclerosus) |
| | Posterior auricular (Wolfe) FTSG | Staged hypospadias repair |
| **Mucosa** | **Buccal, lingual, labial (oral mucosa)** | Urethra, hypospadias, ureter |
| | Bladder epithelium (mucosa) | Urethra, hypospadias; meatal prolapse |
| | Rectal (colonic) mucosa | Panurethral stricture when oral mucosa is unavailable |
| **Other autologous** | Dermis | Tunica albuginea in Peyronie disease |
| | Tunica vaginalis | Peyronie disease, urethral cover; easy harvest |
| | Saphenous vein | Peyronie defects; vascular repair |
| | Rectus fascia, fascia lata | Pubovaginal sling |
| **Allograft / xenograft** | Pericardium, porcine SIS, porcine dermis, cadaveric fascia | Peyronie grafting ("off the shelf"); slings |
| **Synthetic** | Polypropylene (macroporous type I mesh) | Mid-urethral sling, prolapse |
| | Dacron, PTFE (polytetrafluoroethylene) | Vascular grafts; not advised for Peyronie (infection, fibrosis) |

| Group | Graft | Main uses |
|---|---|---|
| **Newer** | Collagen fleece (TachoSil); acellular matrix, tissue-engineered oral mucosa | Peyronie; experimental urethra |

- Vein is not a true graft: it survives by direct endothelial perfusion and its vasa vasorum.
- **Peyronie grafts:** pericardium (thin, strong, does not contract) and SIS (cut about 25% oversize; contracts) are commonest; AUA gives no preference.
- **STSG versus FTSG:** STSG carries little dermis, takes easily, can be meshed (up to sixfold) but contracts. FTSG carries full dermis, contracts less, looks better but is more "fastidious" in take.

### Buccal mucosa: history and why it works

- **BMG** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).
- Thick, **non-keratinised**, elastin-rich, hairless epithelium used to a **wet environment**; resists infection.
- Thin, **highly vascular lamina propria** with a panlaminar plexus, so imbibition and inosculation are rapid and take is excellent.
- Little contraction; handles and holds sutures well; low donor morbidity.

### Harvest

1. Nasotracheal intubation; mouth gag; mark the graft (about 4–5 × 2 cm per cheek; P3: 4–6 × 2–2.5 cm).
2. Stay **1 cm from the commissure** and avoid **Stensen's (parotid) duct** opposite the upper second molar.
3. Infiltrate, incise, raise the mucosa off buccinator; haemostasis.
4. Donor site: leave open or close; both acceptable.
5. Back table: **defat** the graft, cut about 10% oversize, fenestrate with a needle; quilt to the bed. - **Lingual** (under the tongue) and **labial (lip)** grafts are alternatives. Buccal and lingual give equal urethroplasty success; lip mucosa is thinner, suits the glans.

### Uses in urethroplasty

| Technique | Principle | Best for |
|---|---|---|
| **Dorsal onlay** (Barbagli, 1995) | Urethra rotated, dorsal urethrotomy; graft spread-fixed on corpora | Bulbar; reliable bed |
| **Ventral onlay** | Ventral urethrotomy; graft covered by spongioplasty | Bulbar with healthy spongiosum |
| **Dorsal inlay** (Asopa) | Ventral approach (urethrotomy); dorsal plate incised; graft inlaid | Penile and bulbar, no mobilisation |
| **Augmented anastomosis** | Excise narrowest part, floor anastomosis, graft roof | Long, dense bulbar stricture |
| **Kulkarni** (2012) | One-sided dissection keeps ventral blood supply; penile invagination; dorsal graft (onlay) | Panurethral, one stage |
| **Staged** (Johanson, Bracka) | Stage 1 lay open and graft; stage 2 tubularise after 6 months or more | **Lichen sclerosus**, failed hypospadias |

| Technique | Principle | Best for |
|---|---|---|
| **BMG perineal urethrostomy** | Graft completes dorsal wall to skin | Panurethral or failed repairs |

- Dorsal and ventral results are similar (dorsal equals ventral); graft equals flap; onlay beats tubes.
- **Skin is avoided in lichen sclerosus** because the disease recurs in skin; use staged BMG.

### Other uses of BMG

- **Ureter:** BMG onlay ureteroplasty (first human report 1999), now robotic with an **omental wrap**; for long proximal or mid strictures in dense scar. CORRUS (Collaborative of Reconstructive Robotic Ureteral Surgery) failure about 15–25% by site.
- Redo and staged hypospadias; urethral closure in rectourethral fistula; bladder neck contracture; female urethral stricture; Peyronie grafting (small defects only).

### Complications

- **Donor:** pain, swelling (first month), numbness (up to 6 months), mouth tightness, salivary duct injury.
- **Recipient:** graft failure, restricture, sacculation, fistula, ED.

### Outcomes

- Bulbar BMG urethroplasty about 85–90% success; most failures within 12 months.
- Results fall with time and in penile, redo and lichen sclerosus cases.

### Recent advances

- Tissue-engineered oral mucosa; acellular matrix; robotic BMG ureteroplasty.

**Diagram to draw:** bulbar cross-section showing dorsal onlay, ventral onlay and dorsal inlay (Asopa) positions.

> **Key points to remember:**
> - Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
> - BMG: non-keratinised, thin vascular lamina propria (excellent take), wet-adapted.
> - Avoid Stensen's duct; buccal equals lingual in success.
> - Staged BMG, not skin, in lichen sclerosus.
> - Bulbar success about 85–90%.

## LAQ 3 Urethral stricture: etiology, classification, DVIU, management and failed VIU

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Urethral stricture disease: etiology, presentation and management (including recent advances)" (asked as "Describe etiology, clinical presentation and management of urethral strictures" and "Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease") · P3 LAQ "Urethral stricture: classification, DVIU and surgery after failed VIU" (asked as "Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU" and SAQ "DVIU (Direct Visual Internal Urethrotomy)")

*"Structure" is taken to mean urethral stricture disease.*

### Definition

- A **urethral stricture** is a fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar, often with variable **spongiofibrosis**.
- Posterior narrowing is named differently (WHO [World Health Organization] 2004, ICUD [International Consultation on Urological Diseases] 2010): after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery or radiation, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses about three-quarters of the lumen area, so symptoms appear late.

### Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries plus retrograde flow from the glans) lets the bulb be divided and mobilised.

### Aetiology

| Category | Examples and mechanism |
|---|---|
| **Idiopathic** (commonest, about equal to iatrogenic) | No cause found |
| **Iatrogenic** | Catheters (very common in LMICs [low- and middle-income countries]), TURP (transurethral resection of prostate), TURBT (transurethral resection of bladder tumour), cystoscopy, hypospadias repair, radiotherapy. Mechanism: mucosal injury, pressure ischaemia, infection |
| **Traumatic** | **Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI) gives a posterior obliterative defect |
| **Inflammatory** | Gonorrhoea (historically commonest, now rare); **lichen sclerosus (LS)**. Chlamydia: no proven link |
| Congenital | Rarest; infants only (short anterior stricture) |

- **LS**: chronic skin disease starting at the glans and meatus. High-pressure voiding forces urine into Littre glands, causing inflammation and deep spongiofibrosis. It **recurs in any genital skin** used to repair it.

## Classification

### 1. By site (WHO 2004 / ICUD 2010 terminology)

| Segment | Term |
|---|---|
| Anterior: meatus, fossa navicularis, penile (pendulous), bulbar | **Stricture** |
| Membranous urethra after pelvic fracture | **PFUI** / distraction defect |
| Posterior urethra after prostatectomy or radiotherapy; bladder neck | **Stenosis or contracture** |

**2. By aetiology:** idiopathic, iatrogenic, traumatic, inflammatory, congenital (table above).

**3. By length:** short (up to about 2 cm), long, multifocal, **panurethral** (whole anterior urethra).

**4. By depth (Jordan's spectrum):** (A) mucosal fold, (B) iris constriction, (C) full-thickness with minimal spongiofibrosis, (D) full-thickness (deep) spongiofibrosis, (E) fibrosis beyond the spongiosum (periurethral fibrosis), (F) complex stricture with fistula or abscess.

**5. By complexity:** simple (short, bulbar, first treatment) versus complex (long, multiple, LS, radiation, failed previous treatment, fistula).

- **PFUI** (Ch 125): major distraction, minor distraction, partial injury.
- **Also know: LSE system** (Lumen calibre, Segment involved, Etiology) for standard reporting.

## Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- **Retention** after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

## Evaluation (define the 4 Ls: location, length, depth, density)

| Test | Role |
|---|---|
| **Uroflowmetry and PVR (post-void residual)** | Low, flat plateau-shaped flow curve; screen and follow-up |
| **Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)** | Site and length; steep oblique views; done by the surgeon |
| **Flexible urethroscopy** | Calibre, appearance; paediatric scope sees beyond |
| **Sonourethrography** | Length and depth of spongiofibrosis |
| MRI (magnetic resonance imaging) | PFUI defect alignment only; not routine |

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation and instrumentation before surgery. In retention, place a **suprapubic catheter** rather than force a dilation.
- Before urethroplasty: treat infection; look for LS at the meatus; examine the mouth; record erectile function.

## Management principles

- First agree the aim: **palliation or cure**.

- The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned (archaic), because repeated endoscopy makes later urethroplasty harder.

## 1. Dilation

- Soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Cures only epithelial strictures without spongiofibrosis.

## 2. DVIU (direct vision internal urethrotomy, VIU)

### Principle

- Incision of the stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged by **secondary intention**: a **race between epithelialisation and wound contraction**. The stricture recurs when contraction wins, which is usual when spongiofibrosis is deep. Deep spongiofibrosis cannot be cured endoscopically.

### Indications

| Favourable | Unfavourable |
|---|---|
| **Bulbar** | Penile, fossa navicularis, any non-bulbar |
| **Short (<1.5–2 cm)** | Longer |
| Superficial, no dense spongiofibrosis | Dense scar (straddle, LS) |
| **First procedure** | Repeat; luminal narrowing >74% |

### Technique

1. RUG and VCUG define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED [erectile dysfunction], fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit. - **Laser** (holmium, Nd:YAG [neodymium:yttrium–aluminium–garnet], others): **no advantage over cold knife** (laser = cold knife).

### Adjuncts

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C injection (reduces recurrence), steroid injection (limited data); removable stents; permanent UroLume stents withdrawn.

### Complications

- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI, sepsis; ED from deep cuts; spongio-cavernous fistula; incontinence if sphincter incised.

## Why repeat VIU is futile after failure

- VIU succeeds only in **bulbar strictures <1.5 cm without dense spongiofibrosis, at the first attempt**. Recurrence shows deep spongiofibrosis, which an incision cannot remove.
- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none. **A second urethrotomy adds virtually nothing.**

- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Each repeat dilation or urethrotomy lengthens the scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure.
- **AUA (American Urological Association) guideline (2016)** and EAU (European Association of Urology): DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **early urethroplasty, not repeated endoscopy**, after one failed dilation or DVIU; no permanent stents (no stent if urethroplasty is possible).

### 3. Urethroplasty (including surgery after failed VIU)

| Stricture | Operation | Principle and key points |
|---|---|---|
| **Bulbar, short (1–2 cm, up to 3–5 cm proximally)** | **EPA (excision and primary anastomosis)**: gold standard, about 90–95% success | Excise all scar, spatulate both ends widely, tension-free anastomosis. **Non-transecting** or **vessel-sparing** variants keep bulbar arteries |
| **Bulbar, longer** | **BMG (buccal mucosal graft) onlay**: dorsal (**Barbagli**), ventral, dorsal inlay (**Asopa**) | Graft spread-fixed on a well-vascular bed (corpora or spongiosum) |
| Bulbar with an obliterated segment; long dense | **Augmented anastomosis** | Excise the narrowest part, anastomose the floor, graft the roof |
| **Penile** | One-stage BMG inlay or onlay; **penile skin flap** (Orandi, Quartey, circular) | Skin island carried on dartos; not in LS; avoid EPA here (chordee) |
| **LS, failed hypospadias, infection** | **Staged repair** (Johanson): stage 1 open and graft; tubularise after about 6 months or more | Oral mucosa, not genital skin, in LS |
| **Panurethral** | Kulkarni one-sided, one-stage dorsal graft, or staged | — |
| **Elderly, unfit, complex, multiple failures** | **Perineal urethrostomy** (BMG-augmented) | Simple, durable bypass |
| Meatal stenosis; fossa navicularis | Meatotomy; flap or graft | Topical clobetasol for early LS meatal disease |
| **PFUI** (including after failed VIU) | Suprapubic catheter, then delayed **perineal anastomotic urethroplasty** after 3 months or more | Progressive steps: bulbar mobilisation, intracrural separation, inferior pubectomy, rerouting of the urethra |

- **Onlay is better than tube** repairs. Graft and flap results are equal; grafts are simpler.
- **Avoid genital skin in LS** (use oral mucosa).
- **Not fit or declines surgery: Optilume** paclitaxel-coated balloon; intermittent self-dilation; long-term suprapubic catheter.
- **Urethral stents** (UroLume) are withdrawn.

### Recent advances in management

1. **Ladder abandoned**: repeated endoscopy lowers later urethroplasty success, so **early urethroplasty** is offered after one failed DVIU (AUA and EAU guidelines).
2. **Optilume paclitaxel-coated balloon** (FDA [Food and Drug Administration] 2023): paclitaxel inhibits fibroblasts. **ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after (repeated) failed endoscopy; IPSS (International Prostate Symptom Score) 9 vs 19.9.
3. **Anti-scarring adjuncts after DVIU**: mitomycin C injection reduces recurrence; intermittent self-dilation; removable stents. Permanent UroLume stents withdrawn.
4. **Vessel-sparing and non-transecting EPA**: preserve bulbar arteries, theoretically less ED.
5. **Augmented anastomosis**: excise the narrowest segment, graft the rest.
6. **Kulkarni one-sided dorsal BMG** through a perineal incision for panurethral strictures.
7. **Staged oral mucosa repair for LS**; avoid skin. **BMG perineal urethrostomy** reduces neomeatal stenosis.
8. **Robotic reconstruction** of bladder neck contracture and posterior stenosis (Y-V plasty, BMG).
9. **Tissue-engineered oral mucosa** (investigational). **Prevention**: silicone and hydrophilic catheters, smaller endoscopes.
10. **PROMs (patient-reported outcome measures)**: USS-PROM (Urethral Stricture Surgery PROM) is increasingly used to judge success. - **Also know:** dartos-island fossa repair; lingual graft; clobetasol for early LS.

### Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

### Follow-up

- Voiding cystourethrogram at catheter removal (about 3 weeks); then uroflow, PVR and symptom score; RUG, cystoscopy or urethroscopy if flow falls.
- Most EPA recurrences occur in the first 6 months; graft results decline with time.

### Diagrams to draw

- RUG of a bulbar stricture.
- Cross-section of dorsal onlay versus ventral onlay.
- EPA with spatulated ends.
- Urethrotome cutting at 12 o'clock over a guidewire.

> **Key points to remember:**
>
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
> - DVIU only once, for bulbar strictures <1.5–2 cm at first attempt; do not repeat it.
> - Santucci: about 8% long-term success; repeat DVIU futile.
> - AUA: urethroplasty after failure; laser = cold knife; ladder abandoned.
> - EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
> - Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
> - ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

## LAQ 4 Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy

P2 · P3 · P4 **merged from 3**

*Asked as:* P2 LAQ "Importance and management of Inguinal lymph nodes in CA penis." · P3 SAQ "Inguinal Block Dissection?" · P4 SAQ "Videoendoscopic inguinal lymphadenectomy"

*"Inguinal block dissection" is taken as ILND (inguinal lymph node dissection), mainly for penile cancer.*

### Importance

- Presence and extent of inguinal metastasis is the **strongest prognostic factor** in penile SCC (squamous cell carcinoma).
- A long **locoregional phase** precedes distant spread, so the groin is a **curable station**; lymphadenectomy alone can cure.
- **Kroon (2005)**: early ILND for impalpable nodes gave 3-year disease-specific survival about 84% vs about 35% when delayed until nodes were palpable (ENE [extranodal extension] about 20% vs 95%).
- Manage both groins **within 3 months** of diagnosis; aim for **at least 8 nodes per side**.
- About 80% 5-year survival with 2 or fewer unilateral nodes, no ENE, no pelvic nodes.

### Lymphatic drainage and surgical anatomy

- **Superficial system** (prepuce, shaft skin) drains to superficial inguinal nodes. **Deep system** (glans, frenulum) drains to the corona, then within Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes.
- Presymphyseal crossover makes drainage **bilateral** and unpredictable in side; with one palpable groin, the other has micrometastasis in about 30%.
- **Stepwise, no skip**: superficial inguinal, then deep inguinal, then external iliac; so superficial status gates deep dissection and inguinal status gates pelvic dissection.
- **Femoral triangle:** superior **inguinal ligament**, lateral **sartorius**, medial **adductor longus**; floor fascia lata; apex where the muscles cross.
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**:

| Zone | Related vessel |
|---|---|
| Central | Saphenofemoral junction |
| Superolateral | Superficial circumflex iliac vein |
| Inferolateral | Lateral femoral cutaneous and superficial circumflex veins |
| **Superomedial** (sentinel nodes mostly here) | Superficial external pudendal and superficial epigastric veins |
| Inferomedial | Great saphenous vein |

- **Deep nodes** (1–3) medial to the femoral vein; **Cloquet's node** is the highest.

### Staging (TNM [tumour–node–metastasis] 8th edition)

| N | Clinical | Pathological |
|---|---|---|
| 1 | One mobile unilateral node | 2 or fewer unilateral, no ENE |

| N | Clinical | Pathological |
|---|---|---|
| 2 | Mobile, 2 or more unilateral or bilateral | 3 or more (unilateral or bilateral) |
| 3 | Fixed mass or pelvic nodes | ENE or pelvic nodes |

- **T1a**: no LVI (lymphovascular invasion), no PNI (perineural invasion), not high grade. **T1b**: LVI or PNI or high grade.

### Evaluation

- **Careful bilateral palpation**: cN0 or cN+ (number, side, mobile or fixed). Imaging (US [ultrasound], CT [computed tomography], MRI) only if examination is unreliable (obesity, previous groin surgery).
- **cN+ or high risk**: FDG-PET/CT (fluorodeoxyglucose positron emission tomography–CT) or CT chest–abdomen–pelvis.
- About half of palpable nodes are reactive, yet an antibiotic trial is **no longer recommended** (delays treatment). US-guided **FNAC (fine-needle aspiration cytology)** is specific; a negative result never ends the pathway.

### Management of the impalpable groin (cN0): risk-based

| Risk (EAU-ASCO [European Association of Urology–American Society of Clinical Oncology] 2023) | Primary | Groin |
|---|---|---|
| Low | pTis, pTa, pT1a G1 | Surveillance |
| Intermediate | pT1a G2 | Surveillance (compliant) or surgical staging |
| High | **pT1b or more**, G3 | **DSNB (dynamic sentinel node biopsy)**; ILND if unavailable or DSNB positive |

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk, staged surgically (ILND first).
- **DSNB**: US ± FNAC first (a tumour-replaced node diverts flow and causes a false negative); then 99mTc (technetium-99m) nanocolloid, blue dye and gamma probe; serial sectioning. False-negative about 7% at expert centres.
- **Also know:** Cabanas anatomical sentinel biopsy abandoned.

### Management of palpable nodes (cN+)

`Palpable node → US-guided FNAC positive → radical ILND same side + stage other side (superficial ILND, FS-ILND or DSNB) negative → excisional biopsy → negative: observe Bulky (≥4 cm), fixed, bilateral or pelvic nodes → neoadjuvant TIP × 4 → responders: ilioinguinal resection → cisplatin-unfit: chemoradiation`

- **PLND (pelvic lymph node dissection)**: if **more than 2 inguinal nodes or ENE** (pN2–3); ipsilateral; external iliac commonest, then obturator.
- **Neoadjuvant TIP** (paclitaxel, ifosfamide, cisplatin), **Pagliaro (2010)**: objective response 50%; survival mainly in responders. InPACT (International Penile Advanced Cancer Trial) is testing the sequence.

- **Adjuvant**: chemotherapy for pN2–3 (weak evidence); radiotherapy for multiple nodes or ENE (at least 50 Gy; 54 Gy for ENE).

### Inguinal block dissection (ILND): indications

- Penile SCC: **proven cN+**; **cN0 high risk** (pT1b or more, G3) if DSNB is positive or unavailable. Also vulval cancer, melanoma.
- Within 3 months; at least 8 nodes per side. Bulky or fixed nodes: chemotherapy first.

### ILND templates

| Template | Extent |
|---|---|
| **Radical** (cN2 or more) | Superficial + deep; saphenous often ligated; sartorius transposed |
| **Fascia-sparing ILND (FS-ILND)** | May be used for cN1 |
| **Modified (Catalona)** | Smaller field: nothing lateral to femoral artery or below fossa ovalis; **saphenous spared**; no sartorius transposition; positive frozen section: convert to complete (radical) ILND |
| Superficial | Above fascia lata only |

### Technique: open radical ILND

1. Frog-leg position; incision **2 cm below and parallel** to the inguinal ligament.
2. Flaps **deep to Camper's fascia** (preserves the skin capillaries).
3. Boundaries: inguinal ligament, sartorius, adductor longus. Divide saphenous at apex; open fascia lata; clear along the femoral vessels; **never lateral to the femoral artery** (femoral nerve).
4. **Sartorius transposition** over the vessels; suction drains until <30 mL/day.
5. Pelvic dissection if more than 2 inguinal nodes or ENE.

### Videoendoscopic inguinal lymphadenectomy (VEIL)

#### Definition and aim

- **VEIL:** ILND through thigh ports, beneath Scarpa's fascia; laparoscopic (Tobias-Machado) or robotic (**RVEIL**).
- **Aim:** open-surgery template without the groin incision that causes flap necrosis.

#### Indications

- Penile SCC, **impalpable nodes**, intermediate/high risk (pT1b+, G3, LVI).
- **Mobile, non-bulky palpable nodes.**
- **Not for** bulky, fixed (cN3) or skin-involving nodes: these need chemotherapy, then open surgery.
- Also vulval cancer, melanoma.
- DSNB still comes first for impalpable nodes (EAU-ASCO).

#### Technique

- Mark an inverted triangle on the skin **before insufflation**, because gas distorts landmarks.
  `Position: supine frog-leg / low lithotomy → 2 cm camera incision below the femoral triangle apex → finger dissection beneath Scarpa's fascia → two lateral and medial working ports (+ assistant port) → CO2 at 15 mmHg (higher`

`briefly to create space) → develop space up to the inguinal ligament → remove nodal packet off fascia lata, apex to ligament → preserve saphenous vein if possible; clip tributaries → frozen section → deep dissection medial to femoral vein if positive → bag retrieval, suction drain in dependent part`

- **Template:** same as open: superficial ± deep; Daseler's five zones; Cloquet's node is the highest deep node. **Never dissect lateral to the femoral artery** (femoral nerve).

**Outcomes vs open**

| | Open ILND | VEIL/RVEIL |
|---|---|---|
| Wound infection, necrosis, dehiscence | Higher | **Lower** |
| Hospital stay, drain time | Longer | Shorter |
| Operative time | Shorter | About 2 h (slightly longer per side) |
| Nodal yield, recurrence | Reference | **Similar** |
| Evidence | Retrospective | Retrospective, selection-biased |

- Commonest complication of VEIL: **lymphocele**, then lymphoedema and cellulitis.

**Robotic VEIL**

- 3D vision, wristed instruments, ICG (indocyanine green) for lymphatics; adequate dissection on open re-exploration (phase I). No VEIL vs RVEIL comparison.

## Complications of ILND

- Wound infection, **flap necrosis**, dehiscence, seroma, **lymphocele**, **lymphoedema**, DVT/PE (deep vein thrombosis/pulmonary embolism).
- Reduced by transverse incision, correct flap plane, modified template, saphenous sparing, drain removal by volume (<30 mL/day), thromboprophylaxis and minimally invasive access (VEIL); sartorius transposition increases them.

## Follow-up

- Examination 3-monthly for 2 years, then 6-monthly to 5 years (recurrence mostly within 2 years); then self-examination.

## Diagrams to draw

- Femoral triangle with Daseler's zones and Cloquet's node.
- Femoral triangle with VEIL port sites below the apex.

> **Key points to remember:**
>
> - Nodal status is the strongest prognostic factor; ILND can cure.
> - pT1b or higher: stage groins surgically (DSNB preferred) within 3 months; at least 8 nodes per side.
> - Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
> - PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT testing the sequence).
> - Femoral triangle: ligament, sartorius, adductor longus; flaps deep to Camper's; stay medial to femoral artery.
> - VEIL: same template, fewer wound complications, equal nodal yield; not for bulky, fixed or skin-involving nodes.

## LAQ 5 Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias

P1

**Asked as:**

- Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures. — *Summer 2019 Paper I Q1 (LAQ)*
- Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias. — *Practice Set 5 Paper I Q1 (LAQ)* (male urethral anatomy also answered in 2019-Summer Paper 1, Q1; isolated epispadias in 2022-Summer Paper 3, Q3f)

**Type:** LAQ (25 marks)

### Embryology

- **Cloaca** is divided by the **urorectal septum** (weeks 4–7) into the anterior **UGS (urogenital sinus)** and the posterior anorectal canal.
- The UGS has three parts: **vesical** (bladder), **pelvic** (prostatic and membranous urethra) and **phallic** (urethral plate of the penis).
- **Posterior urethra:** from the pelvic UGS. Prostatic buds appear at 10 weeks; the **utricle** (Müllerian remnant) and ejaculatory ducts (Wolffian) open on the verumontanum.
- **Anterior urethra:** Leydig cells make testosterone from about 8 weeks; type 2 5α-reductase converts it to DHT (dihydrotestosterone), which masculinises the genital tubercle. Germ layers: ectoderm gives skin and prepuce, mesoderm the corpora, endoderm the **urethral plate**.
- **Penile urethra: "double zipper"** 1. **Opening zipper:** the urethral plate canalises proximal to distal into a urethral groove between the urethral folds. 2. **Closing zipper:** the folds fuse in the ventral midline, forming the tube and the median raphe. Folds fused by 16 weeks; urethra complete by about 18 weeks.

| Glanular urethra | Ectodermal ingrowth theory (Glenister) | Endodermal theory (current) |
|---|---|---|
| Formed by | Surface ectoderm grows into the glans to meet the endodermal urethra | Direct canalisation of the urethral plate inside the glans |
| Epithelium | Mixed ectoderm and endoderm | Endoderm (UGS) along the whole urethra |

- **Clinical link:** the shaft–glans junction (change of mechanism) is the commonest hypospadias site. **Also know:** urethral duplication; PUV (posterior urethral valves); utricular cyst.

### Anatomy of the male urethra

- About 18–20 cm long in adults; normal diameter 8–9 mm (wide). **Posterior urethra** runs from the bladder neck to the **perineal membrane**; **anterior urethra** from there (perineal membrane) to the meatus, inside the corpus spongiosum.

| Segment | Length / features | Lining |
|---|---|---|
| Prostatic | Length of prostate; urethral crest, **verumontanum** with utricle (Müllerian remnant) and ejaculatory ducts; angle divides preprostatic and prostatic parts | Urothelium |
| Membranous | 2–2.5 cm; shortest and narrowest; striated sphincter | Pseudostratified columnar |
| Bulbar | Widens as the bulb; **Cowper ducts at 3 and 9 o'clock**; lies **eccentric, close to (nearer) the dorsum** of spongiosum | Pseudostratified columnar |
| Penile | About 15 cm; central, uniform calibre; glands of Littré | Pseudostratified columnar |
| Fossa navicularis | Glanular; lacuna magna on the roof | Stratified squamous |

- **Sphincters:** (1) **internal (preprostatic) smooth muscle sphincter** at the bladder neck, autonomic, gives continence at rest; (2) **external striated rhabdosphincter**, horseshoe-shaped around the membranous urethra, supplied by the pudendal nerve (mainly S3, some S2; Onuf nucleus), plus its intrinsic smooth muscle.
- **Blood supply (arteries):** internal pudendal → common penile artery → artery to the bulb (proximal; bulbourethral branches) and **retrograde flow via the dorsal arteries through the glans**. This **dual supply** allows bulbar transection in anastomotic urethroplasty.
- **Venous** (veins) to pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.
- **Female urethra:** about 4 cm; smooth muscle plus striated sphincter over the distal two-thirds; **Skene glands** (prostate homologue) near the meatus; oestrogen-dependent mucosal seal.

## Endoscopic management of stricture

**Principle:** healing after incision or dilation is by **secondary intention**, a race between epithelialisation and wound contraction. No endoscopic method removes **spongiofibrosis**, so cure is limited to short, superficial bulbar strictures.

| Favourable | Unfavourable |
|---|---|
| Bulbar, **<1.5–2 cm** | Penile, fossa, longer |
| No dense spongiofibrosis | Straddle injury, LS (lichen sclerosus) |
| First procedure | Repeat; luminal narrowing >74% |

### 1. Dilation

- Stretches the scar without tearing it (bleeding means tearing).
- **Safest: balloon dilation over a guidewire** under vision; blind filiforms condemned. Sounds or self-dilation for palliation.

### 2. DVIU (direct vision internal urethrotomy)

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define site and length.
2. Guidewire across; **normal saline** irrigation.
3. **Sachse cold knife** at 12 o'clock, through scar to healthy tissue; avoid deep cuts (corpora: ED (erectile dysfunction), fistula).

4. Catheter **3–7 days**; longer adds nothing. - **Laser urethrotomy** (holmium, others): **no advantage over cold knife**. - Results: **Steenkamp and Heyns** (randomised): dilation equals urethrotomy; a second attempt rarely and a third never succeeds. **Santucci (2010):** about 8% stricture-free long term.

**3. Adjuncts against recurrence**

- **CISC (clean intermittent self-catheterisation)** tapering over 3–6 months: delays recurrence, which returns on stopping.
- **Mitomycin C** injection reduces recurrence; steroid injection, colchicine: weak data.

**4. Urethral stents**

- **UroLume** (permanent): **withdrawn** (ingrowth). **Memokath**, removable stents: small series only.

**5. Optilume (paclitaxel-coated balloon)**

- FDA (Food and Drug Administration) 2023. **ROBUST III:** 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy.

**Complications:** recurrence (commonest), bleeding with erections, irrigant extravasation, UTI (urinary tract infection) and sepsis, ED, spongio-cavernous fistula, incontinence.

**Guideline (AUA (American Urological Association) 2016):** urethrotomy or dilation for short (<2 cm) bulbar strictures; **urethroplasty, not repeat endoscopy**, after failure. The reconstructive ladder is abandoned.

## Classification of congenital urethral anomalies

| Group | Anomalies |
|---|---|
| Abnormal meatal position | **Hypospadias** (ventral); **epispadias** (dorsal) |
| Obstructive | **PUV (posterior urethral valves)**, Young types I–III; **anterior urethral valves** (with diverticulum); urethral atresia or hypoplasia (congenital stricture); congenital meatal stenosis |
| Dilatation | **Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); anterior urethral diverticulum; Cowper duct cyst (syringocele); enlarged prostatic utricle |
| Duplication | **Effmann classification**: type I blind incomplete; type II complete (IIA2 "Y-type"); type III with caudal duplication. The ventral channel is the functional one |
| Accessory openings | Congenital urethrocutaneous fistula; parameatal urethral cyst |
| Female | Epispadias; urethral prolapse; persistent urogenital sinus |

## Epispadias: definition and classification

- A **dorsal urethral wall defect**, replaced by a broad dorsal mucosal strip (urethral plate); part of the **exstrophy–epispadias complex**, mostly seen with exstrophy.
- Embryology: an overdeveloped cloacal membrane blocks mesenchymal ingrowth; genital tubercles fuse caudal to it.

| | Male | Female |
|---|---|---|
| Incidence | About 1 in 117,000 | About 1 in 484,000 |

| | Male | Female |
|---|---|---|
| Types | Glanular, penile, **penopubic** (about 70%, incontinent) | **Davis:** vesicular (patulous meatus), subsymphyseal, retrosymphyseal (incontinent) |
| Genitalia | Short, upturned penis with **dorsal chordee** | **Bifid clitoris**, flat glabrous mons |
| VUR (vesicoureteric reflux) | 30–40% | 30–75% |

- The more proximal the meatus, the worse the continence (glanular never incontinent). Pubic diastasis is less than in exstrophy.

## Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram) for reflux; cystoscopy and **bladder capacity** under anaesthesia.
- **Key decision:** use the exstrophy pathway (osteotomy and closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter tissue or tiny bladder.

## Management

**Aims:** continence with safe upper tracts, and a straight penis of adequate length that works for intercourse.

- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

**Stage 1: urethroplasty and penile reconstruction (6–12 months)**

1. Release dorsal chordee: divide suspensory ligaments; free corpora from the inferior pubic rami.
2. **Modified Cantwell–Ransley:** an 18-mm plate strip is outlined; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; the plate is tubularised over an 8-Fr stent and moved **ventral to the corpora**, which are rotated medially over it (diamond corporotomies if curvature persists).
3. Glans wings closed; skin flaps resurface the dorsum; base Z-plasty; stent 10–12 days. - **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias. - Urethroplasty adds outlet resistance, so the bladder grows before stage 2.

**Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)**

- **Young–Dees–Leadbetter** repair with Marshall–Marchetti–Krantz suspension and **ureteric reimplantation**. Osteotomy is rarely needed.
- **Good capacity at presentation:** single-stage repair is possible.
- **Female:** urethroplasty and genitoplasty (clitoral halves and labia joined, mons reconstructed) at 6–12 months; bladder neck repair at 4–5 years.

## Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula about 1 in 6 (penile base); stricture rare; neourethra catheterisable; erections and paternity preserved.
- Better than exstrophy: the bladder was never exposed or scarred. Failures: injectables, augmentation.

**Diagram to draw:** sagittal male urethra with segments, sphincters and a 12 o'clock urethrotomy; Cantwell–Ransley with plate transposed under the corpora.

> **Key points to remember:**
>
> - Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
> - Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
> - DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
> - Optilume ROBUST III 74.6% vs 26.8%.
> - Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
> - About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

## LAQ 6 Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.

P1

**Type:** LAQ (25 marks)
**Source:** 2024-Winter Paper 1, Q1 (related: 2019-Summer Paper 1, Q2; 2020-Summer Paper 1, Q3b)

### Anatomy: definition and boundaries

- The space behind the parietal peritoneum. **Anterior:** posterior peritoneal reflection; **posterior:** posterior abdominal wall; **cranial:** diaphragm; **caudal:** extraperitoneal pelvis.

| Group | Contents |
|---|---|
| Primarily retroperitoneal (formed there) | Kidneys, ureters, adrenals, aorta, IVC (inferior vena cava) |
| Secondarily retroperitoneal (mesentery fused to the wall) | Ascending and descending colon, duodenum D2–D3, pancreas (not the tail) |
| Vessels, lymphatics, nerves | Lumbar nodes, cisterna chyli, sympathetic trunks, autonomic plexuses, lumbosacral plexus |

### Posterior abdominal wall

- **Flank layers:** external oblique, internal oblique, transversus abdominis, transversalis fascia. **Posterior:** psoas, quadratus lumborum, erector spinae.
- **Lumbodorsal (thoracolumbar) fascia:** three layers that fuse into one at the **tip of the 12th rib**, the standard entry point for retroperitoneoscopy.
- **Ribs:** access above the 12th rib risks the pleura; the intercostal bundle runs under the rib above.

### Fasciae and spaces (three embryological strata)

| Space | Boundaries | Contents |
|---|---|---|
| Anterior pararenal | Posterior peritoneum and anterior renal fascia | Colon, duodenum, pancreas |
| Perirenal | Anterior (Toldt) and posterior (Zuckerkandl) laminae of Gerota's fascia | Kidney, adrenal, perirenal fat, hilum, upper ureter, gonadal vessels |
| Posterior pararenal | Posterior lamina and transversalis fascia | Fat only |

- The laminae fuse laterally as the **lateroconal fascia**. The perirenal cone is **open inferiorly**, so urine, blood or pus track into the pelvis.
- Perirenal fat is fine and pale; pararenal fat is coarse and yellow-orange.

### Vessels, lymphatics and nerves

- **Aorta** enters at T12. Unpaired anterior branches supply the gut (coeliac, SMA (superior mesenteric artery), IMA (inferior mesenteric artery)); paired lateral branches supply adrenals, kidneys (L1–L2) and gonads; lumbar arteries go to the wall.

- At the hilum the **renal vein lies anterior to the artery**. The **left renal vein is long** (gonadal, adrenal and lumbar tributaries); the right is short.
- Lymph flows right to left; **cisterna chyli** at L1–L2.
- **Nerves at risk:** sympathetic trunk (medial to psoas), genitofemoral (on psoas), subcostal, iliohypogastric, ilioinguinal (on quadratus); hypogastric plexus (ejaculation).

### Principles of laparoscopic retroperitoneal surgery

1. **No natural cavity:** the working space must be created (finger and balloon dissection) before insufflation.
2. **Psoas is the key landmark:** it forms the floor. Keep it horizontal on screen; the kidney in Gerota's fascia lies anterior, and the peritoneum is pushed medially.
3. **Direct posterior route to the hilum:** the renal artery, posterior to the vein, is met first, allowing early arterial control.
4. **Peritoneum stays intact:** no bowel mobilisation or adhesiolysis; no spill of tumour, urine or pus into the peritoneum.
5. **Small space:** closely spaced ports, limited triangulation, lens smudging.
6. **Gas behaviour:** $CO_2$ (carbon dioxide) absorbs through fat, so hypercarbia and subcutaneous emphysema are commoner; haemodynamic swings are smaller than transperitoneal.

| Preferred for | Limitations |
|---|---|
| Adhesions, peritoneal dialysis, morbid obesity, pregnancy; posterior tumours, pyeloplasty, cyst surgery, renal biopsy | Small space, unfamiliar landmarks, harder suturing, learning curve |

- **Evidence:** randomised trials and meta-analysis: blood loss, complications and oncology similar to transperitoneal; the consistent advantage is **shorter operative time**.

### Salient steps: retroperitoneal laparoscopic nephrectomy

1. **Position:** full flank (90°), table flexed, axillary roll; surgeon behind the patient.
2. **Access (open Hasson):** 1.5–2.5-cm incision just below the 12th rib tip; split muscle, open the lumbodorsal fascia; finger confirms psoas behind and Gerota-covered lower pole in front.
3. **Space creation:** balloon dilator (commercial, or Gaur glove balloon with 600–800 mL saline) anterior to psoas, posterior to Gerota's fascia; blunt-tip balloon-cuff trocar seals the port.
4. **Ports:** two more trocars under vision, one off the 12th rib tip, one posterior and superior to the camera.
5. **Dissection:** along psoas, find the ureter and gonadal vein medially; lift ureter and lower pole to put the hilum on stretch; see arterial pulsation.
6. **Hilum:** clip or staple the **artery first**, then the vein; on the left a **lumbar vein** usually needs division.
7. **Mobilise:** upper pole (adrenal preserved or taken), then lateral and anterior attachments; radical = outside Gerota's fascia.
8. **Extract:** clip ureter, bag the specimen, remove via an extended port; close. - **Problems:** peritoneal tear (space collapses), emphysema, pneumothorax, vascular injury, flank hernia (intramuscular balloon), rhabdomyolysis (about 1%). **Bail-out:** open the peritoneum and convert.

**Diagram to draw:** transverse section showing the three spaces, Gerota's laminae and the balloon anterior to psoas.

> **Key points to remember:**
>
> - Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
> - Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
> - Psoas is the landmark; artery met first.
> - Main advantage: shorter operative time, peritoneum untouched.

## LAQ 7 Classify testicular tumors and describe management of seminoma

P3

**Type:** LAQ (25 marks)
**Source:** Practice Set 1 Paper 3, Q1 (stage I seminoma also answered in 2018-Summer Paper 2, Q2)

### Introduction

- Over 95% of testicular tumours are GCTs (germ cell tumours). Seminoma is 52–56% of GCNIS (germ cell neoplasia in situ)-derived GCTs, peaks in the fourth–fifth decade.
- **Risk factors:** cryptorchidism, previous contralateral GCT (twelvefold), family history, GCNIS, White race, infertility.
- All postpubertal GCTs except spermatocytic tumour carry **isochromosome 12p**.

### WHO (World Health Organization) classification (2016, retained 2022)

| Group | Tumours |
|---|---|
| **1. GCNIS-derived GCTs** | GCNIS; **seminoma**; **NSGCT (non-seminomatous GCT)**: embryonal carcinoma, postpubertal yolk sac tumour, choriocarcinoma and other trophoblastic tumours, postpubertal teratoma, teratoma with somatic-type malignancy; mixed GCT |
| **2. Non-GCNIS-derived GCTs** | **Spermatocytic tumour** (formerly spermatocytic seminoma); prepubertal teratoma, dermoid and epidermoid cyst; prepubertal yolk sac tumour |
| **3. Sex cord–stromal tumours** (<5%) | Leydig cell, Sertoli cell (including large cell calcifying), granulosa cell (adult, juvenile), fibroma–thecoma, mixed |
| **4. Mixed germ cell–sex cord stromal** | Gonadoblastoma (DSD [difference of sex development]) |
| **5. Others** | Lymphoma (commonest testicular tumour in older men), rete testis and collecting duct tumours, adnexal and paratesticular tumours, metastases |

- Any NSGCT element, however small, makes the tumour an NSGCT and is treated as such.
- **Seminoma pathology:** sheets of clear cells with lymphocyte-rich fibrovascular septa. Immunohistochemistry: CD117 and PLAP (placental alkaline phosphatase) positive, CD30 negative. Syncytiotrophoblasts (hCG [human chorionic gonadotropin]) in about 15%. "Anaplastic seminoma" is no longer recognised.
- **Spermatocytic tumour** (<1%): older men, no GCNIS, no i(12p), benign unless sarcomatous.

### Evaluation and staging

- Scrotal ultrasound; AFP (alpha-fetoprotein), β-hCG and LDH (lactate dehydrogenase) before and after orchidectomy. **A raised AFP means NSGCT**, whatever the histology.
- **Radical inguinal orchidectomy** with early cord control at the internal ring; never trans-scrotal. Offer sperm banking and prosthesis.
- CT (computed tomography) of chest, abdomen and pelvis; TNMS (tumour, node, metastasis, serum markers) staging.

- About 80% of seminomas are clinical stage (CS) I; 15–20% CS II (nodes IIA ≤2 cm, IIB 2–5 cm, IIC >5 cm).

## Management of seminoma by stage

**CS I** (cure by orchidectomy alone 80–85%; cancer-specific survival about 100% with any option)

| | **Surveillance (preferred)** | **Carboplatin** | **Radiotherapy** |
|---|---|---|---|
| Method | Examination, markers, CT for at least 5 years | 1 cycle **AUC (area under the curve) 7**, dose = 7 × (GFR [glomerular filtration rate] + 25) mg | **20 Gy** para-aortic strip |
| Relapse | 14–20% | About 5% (1 cycle) | 3–5% |
| Drawback | Late relapses (10–20% after 4 years) | Underdosing; late effects unknown | Second cancers, GI (gastrointestinal) toxicity |

- **Relapse risk factors:** tumour **>4 cm** and **rete testis invasion** (relapse 12%, 16%, 32% with 0, 1, 2 factors). LVI (lymphovascular invasion) is not a factor in seminoma.
- **Trials:** MRC TE19/EORTC (Medical Research Council/European Organisation for Research and Treatment of Cancer) 30982: one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE18: 20 Gy equals 30 Gy.
- Surveillance relapses are 84–100% retroperitoneal and almost all salvaged.

### CS IIA–B

- **Non-bulky (≤3 cm):** dog-leg radiotherapy with nodal boost (25–30 Gy in the chapter; EAU [European Association of Urology] uses 30 Gy for IIA and 36 Gy for IIB), or chemotherapy, or **primary RPLND (retroperitoneal lymph node dissection)** (SEMS trial).
- **Bulky (>3 cm) or multiple nodes:** BEP (bleomycin, etoposide, cisplatin) ×3 or EP (etoposide, cisplatin) ×4.

### CS IIC–III: chemotherapy by IGCCCG (International Germ Cell Cancer Collaborative Group)

| **Risk** | **Seminoma definition** | **5-year survival** | **Regimen** |
|---|---|---|---|
| Good (90%) | Any primary, **no NPVM (non-pulmonary visceral metastases)**, normal AFP, any hCG/LDH | 86% | BEP ×3 or EP ×4 |
| Intermediate (10%) | NPVM present | 72% | BEP ×4 |
| Poor | **None for seminoma** | – | – |
| - Carboplatin alone is inferior in advanced disease. | | | |

## Post-chemotherapy residual mass

```
Residual mass after chemotherapy ├─ <3 cm ──────────► observe (necrosis about 90%)
                                 └─ >3 cm ─► FDG-PET (fluorodeoxyglucose positron emission tomography) at ≥6 weeks
                                            ├─ negative ► observe
                                            └─ positive ► resection (full template PC-RPLND) or salvage
```

- Surgery is difficult because of desmoplasia; complete resection only 58–74%.

### Relapse and follow-up

- After surveillance or radiotherapy: radiotherapy for small retroperitoneal relapse, otherwise first-line BEP. After chemotherapy: VeIP (vinblastine, ifosfamide, cisplatin) or high-dose chemotherapy.
- Surveillance schedule: review every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly. CT detects >85% of relapses.

### Recent advances

- Serum **miR-371a-3p** microRNA: sensitive marker of residual disease and relapse.
- MRI (magnetic resonance imaging) for surveillance to cut radiation; primary RPLND for low-volume stage II.

**Diagram to draw:** WHO classification tree (GCNIS-derived vs non-GCNIS-derived vs sex cord–stromal) and the residual-mass flowchart.

> **Key points to remember:**
> - WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
> - CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
> - Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
> - No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

## Short answer questions (SAQ)

## SAQ 1 Surgical management of primary scrotal cancer

P1 · P4 **merged from 2**

*Asked as:* P1 SAQ "Surgical management of primary scrotal cancer" · P4 SAQ "Surgical management of primary scrotal cancer."

### Introduction and aetiology

- Rare. **SCC (squamous cell carcinoma)** is commonest. **Percivall Pott (1775)** linked it to chimney soot in chimney sweeps: the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus), chronic inflammation.
- **Other types:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Lymph drains to **superficial inguinal nodes, crossing the midline**: both groins at risk.

### Evaluation

- Painless nodule or ulcer on the anterolateral scrotum, often presenting late; **biopsy** confirms.
- Examine both groins (lymphatics cross the midline); FNAC (fine-needle aspiration cytology) of palpable nodes; CT (computed tomography) of abdomen and pelvis for staging.

### Staging (Lowe modification of Ray and Whitmore)

| Stage | Extent |
|---|---|
| A1 | Confined to scrotum |
| A2 | Invades testis, cord, penis, pubis or perineum |
| B | Inguinal nodes only |
| C | Pelvic nodes, no distant spread |
| D | Distant metastases |

### Management of the primary

1. **Wide local excision with a 2 cm margin**, full thickness including dartos; frozen sections for margins.
2. Testes and cord preserved unless invaded; hemi- or total scrotectomy for large tumours.
3. Stage A2: en bloc removal of involved testis, penis, perineum or pubis.
4. Positive margin: re-excise.

### Reconstruction (Chapter 87)

- Scrotal laxity usually allows primary closure.

| Defect | Management |
|---|---|
| <50%, no tension | Primary closure |
| <50%, tension | Scrotal advancement flap or secondary intention |
| >50% | Testes in thigh pockets (pouches); local or thigh flaps; later STSG (split-thickness skin graft) or flap |

### Management of nodes (extrapolated from penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; dynamic sentinel node biopsy or modified ILND for high-grade/invasive tumours.
- **Bulky or fixed:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.
- Routine prophylactic dissection is not advised.

### Other histologies

- **EMPD:** wide excision (up to 3 cm) guided by **mapping biopsies**; screen for associated GU (genitourinary) and other cancers.
- **Also know:** BCC, excision or Mohs surgery; melanoma, excision by Breslow depth ± sentinel node; sarcoma, wide excision.

**Diagram to draw:** elliptical wide excision with 2 cm margin and bilateral inguinal node drainage.

**Key points to remember:**

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

## SAQ 2 Varicocele: pathophysiology, controversies and surgical management

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 SAQ "Varicocele: controversies and present status of treatment" (asked as "Controversies in management of varicocele" and "Present status of varicocele treatment") · P1 SAQ "Varicocele – Pathophysiology" · P2 SAQ "Surgical management of bilateral varicocele"

### Definition and epidemiology

- Abnormal dilatation and tortuosity of the **pampiniform plexus** of the internal spermatic veins.
- About **15% of men, 35% with primary and 75–81% with secondary infertility** (P3: up to 80%). Appears after age 10, peaks at Tanner 3. About 90% left-sided.
- With sensitive tests most are bilateral; the clinical significance of subclinical ones is uncertain.
- **Venous anatomy:** anterior (internal spermatic), middle (deferential) and posterior (cremasteric) groups; the last two give return after repair and cause recurrence.

### Aetiology: why it develops

1. **Left-sided anatomy:** the left vein enters the left renal vein at a right angle, 8–10 cm higher than the right enters the IVC (inferior vena cava). The taller column overloads its valves. The right vein enters obliquely, which acts as a valve.
2. **Valve incompetence** or absence; continuous reflux predicts progression.
3. **Nutcracker:** left renal vein compressed between aorta and SMA (superior mesenteric artery).
4. **Collateral pathways**; **genetic** (4–8 times commoner in first-degree relatives); tall, thin habitus. - **Secondary:** an isolated **right-sided or non-decompressing** varicocele (one that does not decompress lying down) suggests a retroperitoneal or renal mass (tumour), so image the abdomen.

### Pathophysiology: how it damages the testis

| Mechanism | Explanation |
|---|---|
| **Heat (primary)** | Lost countercurrent heat exchange; the testis normally runs 2–4 °C below core |
| Oxidative stress | Raised reactive oxygen species cause sperm DNA fragmentation |
| Hypoperfusion and stasis | Hypoxia, toxin accumulation |
| Apoptosis | Germ cell loss |
| Leydig dysfunction | Lower testosterone than controls |

- The injury is **progressive and duration-dependent** (atrophy), and may affect **both** testes.

## Consequences

- **Testicular hypotrophy** (asymmetry above about 20% guides adolescent repair); poor semen; hypogonadism (lower testosterone); ache on standing (2–11%).
- Repair improves semen, DNA fragmentation, testosterone and pregnancy.

## Grading

| Grade | Finding |
|---|---|
| Subclinical | Doppler only (not treated) |
| I | Palpable with Valsalva only |
| II | Palpable, not visible |
| III | Visible |

## Controversies and present status of treatment (indications)

| Question / situation | Evidence | Current position |
|---|---|---|
| **Does repair improve fertility?** Infertile couple, **palpable** varicocele, abnormal semen | Madgar RCT (randomised controlled trial, 1995): pregnancy 44% versus 10% at 1 year; meta-analyses mixed | **Repair clinical** varicocele with **abnormal semen** in an infertile couple with normal female partner (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine]) |
| **Subclinical varicocele?** | No benefit for semen or pregnancy | **Do not treat** |
| **Normal semen?** | No proven benefit | Not for fertility |
| **Adolescent varicocele** | Testis volume is a surrogate; link to adult fertility unproven | Observe; repair for **testis asymmetry (hypotrophy) above 20%**, pain, or abnormal semen (Tanner 5 or 18 years) |
| **NOA (non-obstructive azoospermia)** | Small series: sperm returns to ejaculate in some | No definitive evidence; may be considered before micro-TESE (microsurgical testicular sperm extraction) |
| **Before IVF/ICSI** (in vitro fertilisation/intracytoplasmic sperm injection) | Repair may improve live-birth rate or allow less invasive ART (assisted reproductive technology) | Discuss, especially with oligozoospermia |
| **High DNA fragmentation** | Repair lowers fragmentation | Reasonable indication |
| **Hypogonadism** (low testosterone) | Repair raises testosterone in hypogonadal men | Emerging indication |
| **Pain** | Dull ache on standing; relief in most after microsurgery | Repair after other causes excluded |

- **Bilateral disease: do not repair subclinical** sides; repair each palpable side.

## Choice of technique

| Technique | Recurrence | Hydrocele | Comment / bilateral use |
|---|---|---|---|
| **Microsurgical subinguinal/inguinal** | Lowest, **0.5–1%** | **About 0** | **Best, preferred**: artery and lymphatics preserved; two small incisions, local or regional anaesthesia |
| Retroperitoneal (Palomo) | High with artery spared (15–25%) | High with mass ligation (about 7%) | Children (artery ligation acceptable) |
| Laparoscopic | 3–15% | Up to 12% | Both sides through same ports; general anaesthesia; serious morbidity possible |
| Embolisation / sclerotherapy | 4–25% (P2: 15–25%) | Nil | Outpatient; femoral (left) or jugular (right) access; technical failure |

## Microsurgical steps (each side)

1. Subinguinal incision over the external ring; deliver the cord.
2. Microscope and **micro-Doppler** to map the testicular artery; papaverine for spasm.
3. Ligate all internal and external spermatic veins (all veins except vasal); preserve **artery, lymphatics, vas and one set of vasal veins**.
4. Deliver the testis to ligate gubernacular and external spermatic veins (lower recurrence).
5. **Impulse test** confirms no missed vein.

## Bilateral varicocele: specific points

- **Never ligate both testicular arteries**: in adults this has caused azoospermia and atrophy. So artery-sparing microsurgery is safest bilaterally.
- Image for a renal or retroperitoneal mass if a varicocele is isolated right-sided or non-decompressing.
- Repair any coexisting hydrocele.

## Complications and outcome

- **Hydrocele** (lymphatic ligation; commonest after non-microsurgical repair); testicular artery injury (atrophy); recurrence (diagnose by upright colour Doppler showing reflux).
- **Predictors of poor result:** recurrence, artery ligation, hydrocele. Earlier repair recovers more.

## Diagrams to draw

- Left and right testicular veins draining into renal vein and IVC.
- Spermatic cord cross-section at microsurgery: artery, lymphatics and vas preserved, veins ligated.

**Key points to remember:**

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage; injury is progressive.
- Right or non-decompressing varicocele: image for a renal mass.
- Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
- Never ligate both testicular arteries.

## SAQ 3 Meatal stenosis and submeatal stricture in males

P1 · P2 **merged from 2**

*Asked as:* P2 SAQ "Management of Urethral meatal stenosis." · P1 SAQ "Treatment of submeatal stricture urethra in males"

### Definition

- **Meatal stenosis:** narrowing of the external urethral meatus below the normal calibre for age.
- **"Submeatal stricture"** is read as a stricture of the meatus and fossa navicularis (glanular urethra). In adults, meatal stenosis is usually the tip of a **fossa navicularis stricture**.

| Age | Normal calibre |
|---|---|
| Under 4 months | 6–10 Fr |
| 1–12 months | 10–12 Fr |
| 1–10 years | 12–14 Fr |
| Over 10 years | 14–18 Fr |

### Causes

- **Children:** almost always after **neonatal circumcision**: wet nappies irritate the exposed meatus (ammoniacal meatitis) and the ventral lips fuse into a thin membrane. Rarely congenital with hypospadias.
- **LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]):** main inflammatory cause in adults today; begins at the glans and meatus and progresses proximally.
- **Iatrogenic:** catheter or instrument trauma, TURP (transurethral resection of prostate).
- **Failed hypospadias repair.**

### Clinical features

- Thin, forceful stream deflected **upwards**; spraying; straining, frequency, incomplete emptying; pain at the start of voiding.

## Evaluation

- Child: watch the child void; calibrate with a bougie à boule. Uroflowmetry. Urethrography only for UTI (urinary tract infection) or suspected proximal lesion.
- Adult: uroflowmetry; **RUG (retrograde urethrogram) before any meatal surgery**, as a proximal stricture often coexists, especially in LS.
- Biopsy suspicious or atypical lesions: LS turns malignant (squamous cell carcinoma) in up to 8%.

## Treatment

| Option | Method | Indication / comment |
|---|---|---|
| **Topical steroid** | Clobetasol 0.05% once or twice daily for 1–2 months, then alternate days; gentle dilation and nappy hygiene in children | Meatitis and early narrowing in a child; early LS |
| **Dilation or self-calibration** | Meatal dilator lubricated with clobetasol | Elderly, unfit; seldom durable |
| **Ventral meatotomy (child)** | EMLA (eutectic mixture of local anaesthetics) cream in clinic, or general anaesthesia; parents spread the edges with a dilator tip 3 times daily for 7–10 days | Symptomatic stenosis in a child |
| **Ventral meatotomy (adult)** | Crush and incise ventrally; **three sutures** (apex and each side), mucosa to glans skin; dilate afterwards | Simple stenosis; in LS extend it with an inverted-V relaxing incision (extended meatotomy) |
| **Dorsal meatotomy** (Y-V plasty) | Excise the scar ridge, advance a flap | Scarred or recurrent; effective but cosmesis poorer |
| **Fossa navicularis reconstruction (flaps)** | **Jordan flap:** ventral urethrotomy through the stenosis; **transverse ventral penile skin island on dartos** transposed and inverted into the defect. **De Sy:** ventral longitudinal island advanced | Fossa navicularis involved, non-LS; excellent function and cosmesis; not in LS |
| **Graft** | **BMG (buccal mucosal graft)** inlay or onlay, one or two stages; glans wings developed | **LS: oral mucosa, never genital skin** |
| **Perineal urethrostomy** | Bypass | Pan-anterior LS, failed repairs, elderly |

- LS affects genital skin, so skin flaps recur; it spares the proximal urethra, so perineal urethrostomy works.

## Complications

- Bleeding, restenosis (commoner in LS), spraying, poor cosmesis.

## Diagrams to draw

- Ventral meatotomy: crush, incise ventrally, suture mucosa to glans at apex and both sides.
- Jordan flap: transverse ventral skin island on dartos inverted into the ventral meatotomy.

> **Key points to remember:**
> - Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
> - Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
> - Non-LS: meatotomy or Jordan/De Sy skin island flap.
> - LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

## SAQ 4 Perineal urethrostomy and eponymous urethroplasty techniques

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 SAQ "Eponymous urethroplasty techniques and perineal urethrostomy" · P1 SAQ "Perineal urethrostomy." · P2 SAQ "Perineal urethrostomy."

### Introduction: urethroplasty options

- Options: **EPA (excision and primary anastomosis)**, the gold standard for short bulbar strictures; **graft or flap onlay**; **staged repair**.
- **Principles:** grafts need a vascular bed; onlay beats tube; graft equals flap.

### Eponymous urethroplasty techniques

| Eponym | Technique | Use |
|---|---|---|
| **Johanson** | Two-stage: stricture laid open, edges sutured to skin (urethrostomy); tubularised later | LS (lichen sclerosus), failed hypospadias, infection |
| **Blandy** | Inverted-U scrotal or perineal skin flap into a ventral urethrotomy | Bulbar strictures; classic perineal urethrostomy (PU) |
| **Turner-Warwick** | Two-stage scrotal inlay; transpubic approach, omental wrap | Complex, posterior |
| **Webster** | Progressive perineal approach: mobilisation, crural separation, inferior pubectomy, rerouting | PFUI (pelvic fracture urethral injury) |
| **Barbagli** | **Dorsal onlay** BMG (buccal mucosal graft) spread-fixed to corpora | Bulbar |
| **Asopa** | **Dorsal inlay** graft through a ventral urethrotomy, no mobilisation | Penile and bulbar |
| **Kulkarni** | One-sided dorsolateral onlay; panurethral repair via perineum with penile invagination | Panurethral (LS) |
| **Orandi** | Ventral longitudinal penile skin island flap on dartos | Pendulous |
| **Quartey** | Showed skin island is a passenger on a fascial (dartos) flap | Basis of all genital skin flaps |
| **McAninch** | Circular fasciocutaneous penile skin flap | Long anterior strictures (up to 15 cm) |
| **Jordan** | Vessel-sparing EPA; ventral transverse island for fossa | Proximal bulbar; fossa navicularis |

- **Also know:** Monseur dorsal urethrotomy; Andrich–Mundy non-transecting anastomosis; Bracka two-stage BMG.

- **Diagram to draw:** bulbar cross-section showing dorsal onlay (Barbagli), dorsal inlay (Asopa) and ventral onlay.

### Perineal urethrostomy: definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses**, rather than reconstructs, a diseased anterior urethra.

### Indications

- **Complex anterior strictures:** LS panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk men** unwilling or unfit for staged reconstruction.
- **First stage** of a staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for penile or urethral cancer.
- Temporary: stricture after phalloplasty; Fournier gangrene; resectoscope access in morbid obesity.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU suits LS and is easy in LS.

### Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS raise restricture.

### Techniques

**1. Classical (Blandy) PU:** inverted-U perineal skin flap (a random-pattern advancement flap) advanced into a ventral bulbar urethrotomy (the opened bulbar urethra).

**2. BMG PU, steps:**

1. High lithotomy; inverted-U incision; split (divide) bulbospongiosus in the midline.
2. Free the spongiosum and **transect** it; **dorsal urethrotomy** into healthy mucosa.
3. Bring the healthy ventral urethra to perineal skin; calibrate proximally to **30 Fr**; close the distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin. - The graft prevents the circumferential scar that narrows a deep PU.

**Diagram to draw:** sagittal perineum with the inverted-U flap sutured into the opened bulbar urethra.

### Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or periodic dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Compared with staged reconstruction it is a shorter, single operation with less morbidity.
- High mid-term success and satisfaction in selected patients.

**Key points to remember:**

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU.
- PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly patients.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

## SAQ 5 Minimally invasive and robotic RPLND

P3 · P4 **merged from 2**

*Asked as:* P3 SAQ "Minimal invasive retroperitoneal lymph node dissection" · P4 SAQ "Robotic RPLND"

### Introduction

- **MI-RPLND (minimally invasive retroperitoneal lymph node dissection)** for testicular GCT (germ cell tumour): laparoscopic (1992) or robotic (2006). **It must duplicate open RPLND exactly.**

### Indications

- **Primary:** marker-negative stage I NSGCT (non-seminomatous GCT), especially high-risk IB; small-volume stage II NSGCT or seminoma (1–2 nodes ≤3 cm in the landing zone).
- **Postchemotherapy:** only a small solitary residual mass with little desmoplasia; **open** for large masses or vessel involvement.
- **Expanding indication:** primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy (SEMS, PRIMETEST).
- **Patient selection:** serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND.

### Templates and nerve sparing

- Boundaries: renal hila, ureters, ureter crossing the common iliacs; plus spermatic cord.
- **Primary:** extended **unilateral modified template** with nerve sparing (right: paracaval, precaval, interaortocaval; left: para-aortic, preaortic; both take nodes above the IMA (inferior mesenteric artery)). If nodes are enlarged, the dissection becomes bilateral.
- **Postchemotherapy:** bilateral full template.
- **Nerve sparing:** identify the sympathetic chains and L1–L4 postganglionic fibres to the hypogastric plexus below the IMA.
- **Diagram to draw:** right and left modified templates.

### Steps (robotic)

1. Positioning: **modified lateral decubitus** (unilateral) or **supine steep Trendelenburg with peritoneal hammock** (any template).
2. Mobilise colon; Kocherise duodenum (right); divide gonadal vein.
3. **Split-and-roll:** split over the great vessel from common iliac to renal vein; roll packets off.
4. **Divide lumbar vessels** to clear retrocaval and retroaortic tissue.

5. Clip lymphatics (left renal vein); bag each packet.

| Feature | Laparoscopic | Robotic |
|---|---|---|
| Retraction | Assistant | Fourth arm |
| Instruments | Straight | Wristed |
| Suturing, vessel control | Difficult | Easier |

### Outcomes

- Stay 1–3 days, blood loss mostly ≤200 mL; conversion about 5%; **antegrade ejaculation ≥90%** (robotic nerve sparing after primary surgery); chylous ascites as open. Node yields approach open (about 30).
- **Major bleeding:** pressure, raise pneumoperitoneum, convert if uncontrolled (open set ready).

### Concerns and guidelines

- Concerns: staging-only dissections, frequent adjuvant chemotherapy, unindicated surgery, catastrophic bleeding, follow-up <5 years; no randomised trials.
- **Guidelines:** AUA (American Urological Association): experienced surgeons may offer it; EAU (European Association of Urology): specialist centres only.

### Recent advances

- Serum miR-371a-3p for selection of stage I patients with occult nodes.
- RPLND for small-volume stage II seminoma (SEMS, PRIMETEST) to avoid chemotherapy and radiotherapy.

> **Key points to remember:**
> - Same template, split-and-roll, lumbar division and nerve sparing as open.
> - Best for chemotherapy-naive low-volume disease.
> - Conversion about 5%; ejaculation preserved ≥90%.
> - Expanding indication: primary RPLND for small-volume stage II seminoma.
> - Serum miR-371a-3p may select stage I patients with occult nodes.

## SAQ 6 Testicular microlithiasis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3a

### Definition

- **TM (testicular microlithiasis):** multiple small (<3 mm) bright echogenic foci **without acoustic shadowing** scattered in the testicular parenchyma on ultrasound.
- **Classic TM:** ≥5 microliths per testis (or per image); **limited TM:** <5, of no clinical significance.

### Pathology

- Laminated calcified bodies **inside seminiferous tubules**, formed on degenerated tubular cells that Sertoli cells fail to clear. Affected tubules are often obliterated.

### Prevalence

| Population | Prevalence |
|---|---|
| General population | 5–10% (population-based 4.2%, rising with age) |
| Multicentre review, >37,000 males | 2.9% |
| Boys with cryptorchidism | 3.5% |
| Previously cryptorchid testis vs contralateral | 14% vs 19% (no difference) |

### Associations

- Cryptorchidism, subfertility, **GCNIS (germ cell neoplasia in situ)** and TGCT (testicular germ cell tumour), Klinefelter syndrome, Down syndrome.

### Cancer significance

- Tumours were found in 4.64% with TM vs 0.33% without, but cryptorchidism was not accounted for.
- **TM is not clearly an independent risk factor** for TGCT. An EAU (European Association of Urology) paediatric panel review of several hundred children found only one cancer through puberty.
- **Exception:** contralateral TM in a man with previous GCT (germ cell tumour) signals a higher GCNIS risk.

### Management

- **Isolated TM, no risk factors:** reassure; **no further evaluation, no routine ultrasound follow-up**.
- **TM with risk factors** (previous GCT, cryptorchidism, infertility, atrophic testis): teach **monthly self-examination from puberty**; ultrasound only for a palpable change.
- **Biopsy:** not for TM alone. Contralateral biopsy at orchidectomy is considered when GCNIS risk is high: atrophic testis (<12 mL), cryptorchidism, age <40.
- Counsel carefully: incidental TM causes anxiety and unnecessary tests.

### Differential diagnosis on ultrasound

- Coarse calcification, "burned-out" tumour scar, stellate scar (these need orchidectomy if intratesticular), scrotal pearls, rete testis ectasia.

**Also know:** routine screening ultrasound for infertility is discouraged because of incidental TM; follow-up schedules remain undefined.

**Diagram to draw:** ultrasound sketch of a testis with scattered non-shadowing echogenic dots.

> **Key points to remember:**
> - Classic TM ≥5 microliths; limited <5.
> - Not an independent cancer risk factor in isolation.
> - Self-examination only with risk factors; no routine imaging.

## SAQ 7 Growing teratoma syndrome

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3a

### Definition

- GTS (growing teratoma syndrome; Logothetis, 1982): metastatic masses of an NSGCT (non-seminomatous germ cell tumour) **enlarge during or after chemotherapy while AFP (alpha-fetoprotein) and hCG (human chorionic gonadotrophin) are normal or falling**. Resection shows **mature teratoma only**.
- Uncommon: about 2–8% of metastatic NSGCT.

### Pathogenesis

- Teratoma is **histologically benign but genetically malignant** (i(12p), aneuploidy) and **chemoresistant**.
- Chemotherapy kills the sensitive elements (embryonal, yolk sac), so markers fall; the teratoma keeps growing, mostly by enlarging cysts. Chemotherapy-induced differentiation is a second theory.
- Main risk factor: teratoma in the orchidectomy specimen.

### Clinical features

- Retroperitoneum commonest; also lung, mediastinum, supraclavicular nodes.
- Often silent; or compression: ureteric obstruction, IVC (inferior vena cava) compression or thrombosis, bowel obstruction, back pain.

### Evaluation

- Markers before every cycle; **rising markers mean viable cancer, not GTS**.
- CT (computed tomography): enlarging, well-defined, **cystic** mass, sometimes with fat or calcification.
- **FDG-PET (fluorodeoxyglucose positron emission tomography) has no role**: teratoma is not FDG-avid.
- **Differential:** progressing viable GCT (germ cell tumour; markers rise); malignant transformation (somatic sarcoma or carcinoma); necrosis or fibrosis (mass shrinks).

### Management

- **Stop chemotherapy**; do not escalate to salvage chemotherapy (ineffective).
- **Prompt complete surgical resection**: PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection), with resection of every other site; adjacent vessel, kidney or bowel resection may be needed. Delay lets the mass become unresectable.
- Radiotherapy is ineffective.
- **Outcome:** favourable after complete resection (long-term survival 75–90% after resection of residual teratoma). Incomplete resection leads to regrowth, **malignant transformation** (rhabdomyosarcoma commonest) or **late relapse**.
- **Also know:** CDK4/6 (cyclin-dependent kinase 4/6) inhibitors studied for unresectable teratoma; lifelong CT and marker follow-up.

**Diagram to draw:** graph of mass size rising while marker curve falls during chemotherapy cycles.

> **Key points to remember:**
> - Enlarging mass plus normal or falling markers equals GTS.
> - Teratoma is chemoresistant and not FDG-avid.
> - Treatment is complete surgical resection, not more chemotherapy.

## SAQ 8 Prognostic index score in carcinoma penis

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3e

*Assumed meaning: prognostic factors, risk-group systems and nomograms in penile SCC (squamous cell carcinoma). No single universally used "prognostic index score" exists.*

### Introduction

- Prognosis rests on primary histology and stage and, above all, **inguinal node status**.

### 1. Nodal status (most important factor)

| Nodal finding | 5-year survival |
|---|---|
| Node negative | About 73% |
| ≤2 nodes, unilateral, no ENE (extranodal extension) | About 80% |
| >2 nodes | 0–50% |
| **ENE** or **pelvic nodes** | Poor (pelvic about 14%) |

- TNM (tumour-node-metastasis) 8th edition: pN1 (1–2 unilateral nodes) about 90% 3-year DSS (disease-specific survival); pN3 (ENE or pelvic) about one-third.

### 2. Primary tumour factors (predict occult nodes in cN0)

- **Depth:** corporal invasion → about 60% node positive.
- **Grade:** any grade III component = grade III.
- **LVI (lymphovascular invasion)** and **PNI (perineural invasion)**: strongest predictors; define pT1b.
- **Subtype:** verrucous and PeIN (penile intraepithelial neoplasia) do not metastasise; basaloid and sarcomatoid aggressive.

### 3. Risk-group systems

| System | Low | Intermediate | High |
|---|---|---|---|
| **Earlier EAU (European Association of Urology), after Solsona** | Tis, Ta, T1 G1 | T1 G2 | T2 or more, or G3 |
| **Current EAU–ASCO (American Society of Clinical Oncology) / TNM** | PeIN, Ta, **T1a** (nodes 0–18%) | — | **T1b or more** (one-third to over half) |

- Low risk → groin surveillance; high risk → **DSNB (dynamic sentinel node biopsy)** or ILND (inguinal lymph node dissection).

### 4. Nomograms and indices

- **Ficarra nomogram:** predicts nodal involvement (thickness, growth pattern, grade, LVI, corporal invasion, clinical node status).

- **Kattan nomogram:** predicts cancer-specific survival.
- A pathological **prognostic index** (grade + anatomical level of invasion + PNI; Cubilla group) has been proposed.
- **Status:** none externally validated; not used routinely.

### 5. Molecular markers

- **HPV (human papillomavirus)/p16 positive:** better DSS (not node status).
- Adverse: p53, PD-L1 (programmed death-ligand 1), Ki-67, high **NLR** (neutrophil-to-lymphocyte ratio), raised CRP (C-reactive protein).

### 6. Other

- Inguinal recurrence after observation: only about one-third survive 5 years. Size >5 cm worse.

**Diagram to draw:** flowchart: primary pathology → low or high risk → surveillance or DSNB/ILND.

> **Key points to remember:**
> - Node status is the strongest prognostic factor.
> - LVI, PNI, grade III define high-risk T1b.
> - Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

## SAQ 9 Testis sparing surgeries for CA testis

P4

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 4, Q3b

### Definition and rationale

- **Partial orchidectomy**: inguinal excision of the tumour, preserving the rest of the testis. The standard remains **radical inguinal orchidectomy**; sparing aims to keep androgen production and avoid lifelong testosterone.

### Indications

**Suspected germ cell tumour (GCT), all of:**

- Synchronous bilateral tumours or tumour in a **solitary testis**.
- Tumour **≤2 cm** (up to 2–3 cm, under 30% of testicular volume), preferably polar.
- Adequate preoperative testosterone.
- Not with a normal contralateral testis.

**Suspected benign or indeterminate lesions:**

- Small impalpable lesion (<10 mm) with normal markers: most benign (cyst, infarct, Leydig nodule), but 20–50% are small seminomas. Lesions <5 mm are rarely malignant.
- **Prepubertal**: teratoma and epidermoid cyst with normal AFP (alpha-fetoprotein): partial orchidectomy is the standard.

### Preoperative work-up

- Ultrasound of both testes; AFP, beta-hCG (human chorionic gonadotropin), LDH (lactate dehydrogenase); testosterone, LH (luteinising hormone); **sperm banking**; consent for

radical orchidectomy.

### Technique

1. Inguinal incision; early cord control with a soft clamp.
2. Deliver the testis; isolate with towels.
3. **Intraoperative ultrasound** localises the lesion.
4. Cold ischaemia optional if resection <30 minutes.
5. Incise the tunica parallel to the segmental vessels; excise the mass with a rim of normal tubules.
6. **Frozen section** (benign versus malignant) and biopsies of adjacent parenchyma for GCNIS (germ cell neoplasia in situ).
7. Close the tunica; fix the testis in the scrotum.

### Management of GCNIS

- **Radiotherapy about 18–20 Gy** to the remnant: prevents invasive GCT, keeps Leydig function, but causes sterility; up to 40% later need testosterone.
- German Testicular Cancer Study Group: no local recurrences with radiotherapy; most men who omitted it recurred.
- Radiotherapy may be deferred for paternity under close follow-up. Local recurrence → completion orchidectomy.

### Complications and follow-up

- Haematoma, atrophy, hypogonadism, local recurrence. Follow with self-examination, ultrasound, markers and testosterone.

**Diagram to draw:** testis with polar tumour excised with a rim of normal tissue.

> **Key points to remember:**
> - Solitary or bilateral tumour, ≤2 cm, normal testosterone.
> - Frozen section plus adjacent biopsies for GCNIS.
> - GCNIS: 18–20 Gy keeps Leydig function but sterilises.

## SAQ 10 Muscle assisted urethroplasty

P4

**Asked as:**

- Muscle assisted urethroplasty — *Winter 2023 Paper IV Q3b (SAQ)*
- Muscle assisted urethroplasty — *Practice Set 5 Paper IV Q3a (earlier version) (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3b)

*Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.*

### Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.

- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

### Muscle flaps used

| Flap | Supply | Use |
|---|---|---|
| **Gracilis** (muscle-only, axial, multiple pedicles) | Medial circumflex femoral branch, proximal thigh | Irradiated posterior urethra, rectourethral fistula, complex perineal repair |
| **Bulbospongiosus** | Perineal branches | Covers a ventral graft; support in bulbar repair |
| Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous) | Inferior epigastric | Large pelvic or perineal defects |

### Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

### Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

### Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

### "Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

**Diagram to draw:** gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

> **Key points to remember:**
> - Graft survival: imbibition then inosculation.
> - Muscle flap rescues a poor (irradiated) bed.
> - Gracilis is axial, robust, easy to tunnel.
> - Muscle-sparing: preserves bulbospongiosus and nerves.

## SAQ 11 Gracilis myocutaneous flap in Uro.

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 4, Q3d (gracilis-assisted urethroplasty also in 2023-Winter Paper 4, Q3b)

## Definition

- A pedicled **axial** flap of gracilis muscle, with or without skin, rotated from the medial thigh to bring **vascularised bulk** to the perineum, groin or vagina.

## Anatomy

| Feature | Detail |
|---|---|
| Origin–insertion | Inferior pubic ramus and body of pubis → medial proximal tibia (pes anserinus) |
| Blood supply | **Mathes–Nahai type II**: dominant pedicle from the **medial circumflex femoral artery** (profunda femoris), entering about 8–10 cm below the pubic tubercle; minor distal pedicles from the superficial femoral artery |
| Nerve | Anterior branch of the **obturator nerve** |
| Skin paddle | Reliable over the proximal two-thirds only |
| Function | Expendable (other adductors compensate) |

## Why it is used

- Brings blood supply into **irradiated or scarred** fields, fills dead space and **separates suture lines**. Near the perineum, easy to tunnel.

## Uses in urology

1. **Rectourethral fistula** (perineal): non-irradiated all closed; irradiated mostly closed, but more strictures and permanent colostomy.
2. **Complex or irradiated VVF (vesicovaginal fistula)** with tissue loss.
3. **Urethroplasty** in a poor bed: muscle as bed for BMG (buccal mucosal graft).
4. Extensive perineal or scrotal loss after **Fournier gangrene**.
5. **Groin cover** of femoral vessels after inguinal node dissection with skin excision.
6. Vaginal reconstruction after exenteration.

## Technique

1. Lithotomy; line from pubic tubercle to medial femoral condyle; gracilis lies just behind it.
2. Medial thigh incision; skin paddle over the proximal muscle if needed.
3. Divide the distal tendon; ligate minor pedicles; **preserve the dominant pedicle**.
4. Tunnel subcutaneously to the perineum; rotate without twisting.
5. Interpose and fix; drain the donor site.

## Complications

- Donor seroma, dehiscence, **distal skin-paddle necrosis**, thigh numbness; flap loss rare.
- **Also know:** alternatives: Martius, omentum, VRAM (vertical rectus abdominis myocutaneous), Singapore flap.

**Diagram to draw:** gracilis tunnelled on its proximal pedicle between urethra and rectum.

**Key points to remember:**

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
- Main uses: rectourethral fistula, complex VVF, Fournier, groin cover.

# Rapid Revision Sheet · Section 10

### Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma (LAQ)

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila (renal vessels), ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND (post-chemotherapy): full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum to ipsilateral superficial inguinal nodes; scrotum and penis drain to inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.
- Seminoma risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

### Grafts in urology and buccal mucosa (LAQ)

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, wet-adapted; thin vascular lamina propria gives excellent take.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.
- Bulbar success about 85–90%.

### Urethral stricture: etiology, classification, DVIU, management and failed VIU (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU (VIU) only once, for short bulbar strictures (<1.5–2 cm) at first attempt; do not repeat it.
- Santucci: about 8% long-term success; repeat DVIU futile.
- AUA: urethroplasty after failure; laser = cold knife; early urethroplasty, ladder abandoned.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- Optilume (ROBUST III): 74.6% vs 26.8% stricture-free at 1 year.

### Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy (LAQ)

- Nodal status is the strongest prognostic factor; ILND can cure.
- pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
- Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
- PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT [International Penile Advanced Cancer Trial] testing the sequence).
- Femoral triangle: ligament, sartorius, adductor longus.

- Flaps deep to Camper's; stay medial to femoral artery.
- Within 3 months; at least 8 nodes per side.
- VEIL: same template as open; fewer wound complications; equal nodal yield.
- VEIL not for bulky, fixed or skin-involving nodes.

**Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias (LAQ)**

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

**Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. (LAQ)**

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

**Classify testicular tumors and describe management of seminoma (LAQ)**

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

**Surgical management of primary scrotal cancer (SAQ)**

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

**Varicocele: pathophysiology, controversies and surgical management (SAQ)**

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.
- Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
- Never ligate both testicular arteries.

**Meatal stenosis and submeatal stricture in males (SAQ)**

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: steroid (clobetasol), buccal mucosa (never genital skin), perineal urethrostomy if extensive.

**Perineal urethrostomy and eponymous urethroplasty techniques (SAQ)**

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU.
- PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly patients.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

**Minimally invasive and robotic RPLND (SAQ)**

- Same template, split-and-roll, lumbar division and nerve sparing as open.
- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.
- Expanding indication: primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy.
- Robotic nerve sparing preserves antegrade ejaculation in ≥90% after primary surgery.
- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

**Testicular microlithiasis (SAQ)**

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

**Growing teratoma syndrome (SAQ)**

- Enlarging mass plus normal or falling markers equals GTS.
- Teratoma is chemoresistant and not FDG-avid.
- Treatment is complete surgical resection, not more chemotherapy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Prognostic index score in carcinoma penis (SAQ)**

- Node status is the strongest prognostic factor.
- LVI, PNI, grade III define high-risk T1b.
- Nomograms (Ficarra, Kattan) unvalidated; p16 favourable. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Testis sparing surgeries for CA testis (SAQ)**

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

**Muscle assisted urethroplasty (SAQ)**

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

**Gracilis myocutaneous flap in Uro. (SAQ)**

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
- Main uses: rectourethral fistula, complex VVF, Fournier, groin cover. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 11 · Stones, Obstruction & Trauma

## Long answer questions (LAQ)

### LAQ 1 Congenital PUJ obstruction, failed pyeloplasty and salvage procedures

P1 · P2 **merged from 3**

*Asked as:* P1 LAQ "Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management." · P1 SAQ "Management of failed pyeloplasty" · P2 SAQ "Salvage procedures for Complicated PUJO."

#### Definition

- PUJO (pelviureteric junction obstruction) is impaired flow of urine from the pelvis into the ureter. **Koff:** obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage". **Dilatation is not obstruction.**
- About 1 in 1500 live births; commonest cause of antenatal hydronephrosis. Male and left predominance; **bilateral in 10–40%**.

#### Pathogenesis

| Type | Lesion | Typical patient |
|---|---|---|
| **Intrinsic** (commonest) | Short adynamic segment: interrupted circular muscle, excess collagen, fewer pacemaker (interstitial Cajal-like) cells. The peristaltic wave cannot cross, so the pelvis empties poorly | Neonate or infant, antenatal finding |
| **Extrinsic** | **Lower-pole crossing vessel** anterior to the PUJ; the ureter kinks over it | Older child or adult with intermittent pain |
| Secondary | Kinking by a tortuous ureter in high-grade VUR (vesicoureteral reflux); fibroepithelial polyp; high insertion | Any age |

- **Crossing vessels** are persistent embryonic renal arteries. They occur at a fifth to a quarter of **normal** PUJs, so a vessel alone does not prove cause.
- **Also know:** proposed incomplete ureteric recanalisation; horseshoe kidney and duplex lower moiety associations.

#### Effects of obstruction (pathophysiology)

- Pelvic pressure rises; filtration and perfusion fall.
- **Immature kidney:** apoptosis and impaired nephrogenesis. **Mature kidney:** hypertrophy, then fibrosis (TGF-β [transforming growth factor-beta] signalling). Fibrosis, not dilatation, causes loss of function.
- **Also know:** concentrating defect, salt wasting; rarely renin-mediated hypertension.

#### Clinical features

- **Antenatal ultrasound** (commonest); infants: UTI (urinary tract infection), flank mass, failure to thrive.
- Older children and adults: **intermittent loin pain with vomiting (Dietl crisis)**, haematuria after minor trauma, stones, infection.

- **Associated anomalies (up to 50%):** contralateral PUJO, dysplasia or MCDK (multicystic dysplastic kidney), VUR (up to 40%).

## Diagnosis

### 1. Antenatal grading (APD, anteroposterior renal pelvic diameter)

| | Second trimester | Third trimester |
|---|---|---|
| Mild | 4 to <7 mm | 4 to <9 mm |
| Moderate | 7–10 mm | 9–15 mm |
| Severe | >10 mm | >15 mm |

- **UTD (urinary tract dilation) 2014 consensus:** six parameters (APD, central/peripheral calyces, parenchymal thickness and appearance, ureter, bladder). Prenatal A1 (low) or A2–3; postnatal **P1** (APD 10–<15 mm, central calyces), **P2** (≥15 mm or peripheral calyces or ureter), **P3** (abnormal parenchyma).
- **SFU (Society for Fetal Urology) grades 0–4:** 1 pelvis just seen; 2 pelvis plus major calyces; 3 all calyces, normal parenchyma; **4 parenchymal thinning**.

**2. Postnatal ultrasound** after **48 hours** (neonatal dehydration under-reads dilatation): large pelvis, rounded calyces, ureter not seen.

**3. MAG3 (technetium-99m mercaptoacetyltriglycine) diuretic renogram** ("well-tempered": hydrated, catheterised, furosemide)

- **DRF (differential renal function)** (normal 45–55%); drainage half-time <10 min normal, 10–20 equivocal, **>20 min obstructed** (adult chapter: >40 min).

**4. Others:** VCUG (voiding cystourethrogram) by UTI risk; MRU (magnetic resonance urography) or CT (computed tomography) angiography for vessels; **Whitaker test** (10 mL/min; >15–20 cm $H_2O$ obstructed) if equivocal.

## Primary management

**Conservative:** most resolve, about **one-third need surgery**; serial ultrasound and renogram.

### Indications for pyeloplasty

- **DRF <40%** or a **fall of >10%** on serial scans.
- **Increasing APD** or SFU 4 / P3 with poor drainage.
- **Breakthrough UTI**, symptoms, stones, hypertension.
- Bilateral or solitary kidney.

### Anderson–Hynes dismembered pyeloplasty: gold standard (~95% success)

1. Expose the PUJ; free the lower pole and look for a crossing vessel.
2. Stay sutures; transect the pelvis; reduce a redundant pelvis.
3. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a "no-touch" handle.
4. **Transpose** the PUJ anterior to any vessel (never divide it).
5. Watertight, tension-free, dependent anastomosis (5-0/6-0 absorbable); excise the segment.
6. Double-J or externalised stent (1–4 weeks equivalent); Foley 24–48 h with a double-J.

- **Route:** open preferred in infants; laparoscopic equals open with less pain; **robotic** (easier suturing, shorter learning curve) is now primary at most major centres.

- **Alternatives:** Foley Y-V plasty (high insertion); Culp spiral or vertical flap (long segment); **vascular hitch** (crossing vessel with normal PUJ); **endopyelotomy** (73% overall; **42% with vs 86% without a crossing vessel**), mainly for secondary PUJO.
- **Nephrectomy:** DRF **<15% in adults** (children: no agreed cut-off).

**Diagram to draw:** Anderson–Hynes steps: transection, lateral spatulation, transposition anterior to the crossing vessel, dependent anastomosis.

### Complications after pyeloplasty

- Urine leak, stent problems, UTI.

### Failed pyeloplasty: definition

- Persistent pain, UTI or stones; **obstructive diuretic renogram** (T½ >20 minutes); unchanged or increasing hydronephrosis; falling split function.
- Dilatation settles slowly; judge by renography.
- About 1 in 9 children need a secondary procedure. Most failures occur within 1–2 years (P2: within 2 years), but up to 30% occur later.

### Causes of failure

- **Ischaemia** (over-stripping) and **tension** at the anastomosis.
- **Missed crossing vessel**; inadequate excision of the narrow segment; non-dependent anastomosis; kinking.
- **Urine leak** or infection causing periureteric fibrosis.
- Missed distal stricture or high-grade VUR.

### Early failure (first days or weeks)

- Pain, fever, increasing dilatation: usually **oedema or clot**; temporary nephrostomy or stent, and most settle.

### Evaluation of late failure

- **MAG3 diuretic renogram:** obstruction (drainage) and split function.
- **CT angiography:** crossing vessel.
- Antegrade or retrograde pyelography (stricture length); **Whitaker test** if equivocal.
- Drain first (stent or nephrostomy) if infected or function is uncertain, then reassess.

### Salvage options

| Option | When | Results/notes |
|---|---|---|
| **Endopyelotomy** (antegrade or retrograde ureteroscopic, laser) | Short stricture, no significant crossing vessel, moderate hydronephrosis; first choice after failed open repair | About 70% after failed pyeloplasty; lateral incision; stent 4–6 weeks |
| **Redo pyeloplasty** (open, laparoscopic, robotic) | Most failures; long or dense stricture, crossing vessel; failed endopyelotomy | About 86–89% (P2: about 83–89%); best choice in most children; wide mobilisation of kidney and ureter; transperitoneal (scarred retroperitoneum) |
| **Ureterocalicostomy** | Scarred, small intrarenal pelvis; dilated lower-pole calyx | About 70% in redo series; **resect** lower-pole parenchyma (not nephrotomy) |

| Option | When | Results/notes |
|---|---|---|
| **Buccal mucosa onlay graft** | Long scarred PUJ/UPJ (ureteropelvic junction) or proximal ureter, dense scar | Robotic with omental wrap; failure about 15–25% (CORRUS) |
| **Appendiceal onlay flap** | Right-sided proximal stricture | Pedicled on appendiceal artery |
| Flaps (spiral, vertical), downward nephropexy | Gain length | Combine with redo repair |
| **Ileal ureter** | Long defect / long ureteric loss | Contraindicated if creatinine >2 mg/dL, bladder dysfunction, IBD (inflammatory bowel disease), radiation |
| **Autotransplantation** | Solitary or poor contralateral kidney; other repairs impossible; long ureteric loss | About 90% long term |
| **Nephrectomy** | Split function <15% with normal contralateral kidney; multiple failures | Definitive; laparoscopic |

- Rule of thumb: failed pyeloplasty: try endourology first if anatomy suits; failed endopyelotomy: redo pyeloplasty. Redo pyeloplasty is the most reliable.
- **Also know:** robotic near-infrared fluorescence (indocyanine green, ICG) to identify the ureter and perfusion; Davis intubated ureterotomy (historical).

### Principles of redo surgery

- Transperitoneal route in children (retroperitoneum is scarred).
- Ureteric catheter; wide mobilisation of kidney and ureter; **excise all scar**; spatulate; **tension-free, watertight** anastomosis over a stent; wrap with omentum or perirenal fat.
- **Also know:** near-infrared fluorescence shows perfusion; downward nephropexy.

**Diagram to draw:** ureterocalicostomy: lower-pole parenchyma excised, spatulated ureter sewn to lower calyx.

### Follow-up

- After primary pyeloplasty: renogram and ultrasound after stent removal.
- After redo: stent out at 4–6 weeks; renograms for 2–3 years (late failures).

### Recent advances

- Robotic redo with buccal graft and ICG fluorescence; pyramidal thickness ≤3 mm predicts surgery; urinary biomarkers.

**Key points to remember:**

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

## LAQ 2 Urolithiasis: etiopathogenesis, medical management and prevention of recurrence

P1 · P2 · P4 **merged from 3**

*Asked as:* P4 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" / "Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS" · P2 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" (also P2 SAQ "Describe medical management of renal calculi") · P1 LAQ "Discuss pathophysiology of renal stones and also describe how to prevent its recurrence"

### Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to **50% within 5 years** (another source: about half recur within 5–10 years). Removal is surgical; recurrence is metabolic. Recurrent formers need a **metabolic evaluation** and targeted therapy.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20% (calcium oxalate with or without apatite about 80% in all), uric acid and struvite about 7% each, cystine 1–3%.
- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy inside the kidney.

### Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Anatomical (stasis)**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

### Physicochemistry: state of saturation

| State | Concentration product (CP) | What happens |
|---|---|---|
| Undersaturated | Below Ksp (solubility product) | No crystals; stones may dissolve |
| **Metastable** | Between Ksp and Kf (formation product) | No spontaneous nucleation, but growth on existing crystals; inhibitors act here. **Zone of therapy** |
| Unstable | Above Kf | Spontaneous nucleation, growth, aggregation |

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

### Steps of stone formation

1. **Supersaturation:** low volume; high calcium, oxalate or uric acid; low citrate.

2. **Nucleation:** homogeneous, or **heterogeneous** on debris or other crystals (needs less energy).
3. **Growth and aggregation:** aggregates, not single crystals, can occlude a tubule within the 5–7 minute nephron transit (**free-particle theory**).
4. **Retention (fixed-particle theory): Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) point to a secondary cause: enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria.

### Inhibitors, promoters and matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm–Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

### Pathophysiology by urinary abnormality

| Abnormality (definition) | Mechanism |
|---|---|
| **Hypercalciuria** (commonest; >200 mg/day, or >4 mg/kg/day) | Absorptive (gut), renal leak (secondary high PTH [parathyroid hormone]), resorptive (primary hyperparathyroidism): Pak classification |
| Hyperoxaluria (>40 mg/day) | **Primary:** hepatic enzyme defect (PH1, AGXT gene); **enteric:** malabsorbed fat binds calcium, freeing oxalate; dietary |
| Hyperuricosuria (>600 mg/day) | Urate promotes calcium oxalate nucleation |
| **Hypocitraturia** (<320 mg/day) | Distal RTA, diarrhoea, hypokalaemia, animal protein (acid load, intracellular acidosis) |
| **Uric acid stones** | **Low urine pH (<5.5)** more than excess uric acid; pKa 5.5 |
| **Cystine** | Autosomal recessive defect of tubular cystine reabsorption |
| **Struvite** | **Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate and carbonate apatite; biofilm |

### Medical management: aims

- Relieve pain, dissolve suitable stones and **prevent recurrence** (up to 50% at 5 years).

### Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory. Urgent drainage if infection with obstruction.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm** after migration from the kidney: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

### Dissolution (chemolysis)

- **Uric acid** stones (radiolucent) dissolve with oral potassium citrate to alkalinise (EAU [European Association of Urology]: pH 7.0–7.2 for dissolution); allopurinol if hyperuricosuric.
- **Cystine:** alkalinise to pH 7.0–7.5 with thiol drugs. **Struvite:** acetohydroxamic acid occasionally dissolves.

## Prevention of recurrence

### 1. Screen every first-time former (AUA [American Urological Association], EAU)

- History, serum electrolytes, creatinine, calcium, uric acid; urine pH and culture; **stone analysis**; imaging. This sorts patients into high and low risk.

### 2. General measures (every former)

- **Fluid for urine ≥2.5 L/day** (drink about 3 L). **Borghi 1996**: urine >2 L cut 5-year recurrence (12.1% vs 27%; recurrence halved).
- **Normal calcium**, low sodium (≤2.3 g/day), less animal protein, more fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension]-type diet). **Borghi 2002**: this diet halved recurrence compared with a low-calcium diet. Restricting calcium raises oxalate absorption.
- Limit high-oxalate foods (spinach, nuts, chocolate); weight loss.
- Citrus juices (lemonade, orange) are adjuncts only.

### 3. Metabolic evaluation that guides therapy (high-risk or recurrent formers, children) (see 2015-Summer Paper 2, Q2)

- **Stone analysis** (infrared spectroscopy or X-ray diffraction); serum chemistry: serum calcium, uric acid, creatinine, electrolytes, PTH if calcium is high-normal.
- One or **two 24-hour urines** on usual diet, at least 4 weeks after an episode: volume, pH, calcium, oxalate, citrate, uric acid, sodium, potassium, creatinine, cystine. Therapy targeted to these results reduces stone events.
- Targets: volume >2.5 L, calcium <250 mg (women) or <300 mg (men), oxalate <40 mg, citrate >320 mg, uric acid <600 mg.

### 4. Selective drug therapy

| Abnormality | Drug and dose | Rationale and notes |
|---|---|---|
| Hypercalciuria | **Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg | Distal calcium reabsorption plus volume contraction. Add potassium citrate: thiazide lowers potassium, and hypokalaemia causes hypocitraturia |
| Primary hyperparathyroidism | Parathyroidectomy | Cures resorptive hypercalciuria |
| Hypocitraturia, distal RTA, chronic diarrhoea | **Potassium citrate 30–60 mEq/day** | Alkali load reduces tubular (proximal) citrate reabsorption; avoid urine pH ≥7 |
| Hyperuricosuria / hyperuricosuric calcium oxalate stones | Allopurinol 300 mg/day | Watch hypersensitivity; reduce dose in renal impairment |
| Uric acid stones | Potassium citrate to **pH 6.0–6.5** (avoid >7.0: calcium phosphate); allopurinol only for hyperuricosuria or failed alkalinisation | A pH disease: pKa 5.5; dissolution possible |
| Enteric hyperoxaluria | Low fat and oxalate; **calcium with meals**; liquid potassium citrate; magnesium; cholestyramine | Calcium binds oxalate in the gut |

| Abnormality | Drug and dose | Rationale and notes |
|---|---|---|
| Primary hyperoxaluria type 1 | Pyridoxine; **lumasiran** (RNA [ribonucleic acid] interference against glycolate oxidase) | ILLUMINATE-A: 84% vs 0% reached near-normal urine oxalate |
| Cystinuria | Urine ≥3 L/day (another source: >3 L); low sodium and protein; potassium citrate to pH **7.0–7.5**; then **tiopronin** (or D-penicillamine) | Aim cystine <250 mg/L; thiols form soluble mixed disulfides |
| Struvite | **Complete clearance**; culture-specific antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily (another source: two to three times daily) if surgery impossible | Urease inhibitor; risk of thromboembolism |

- **Also know:** recurrent calcium stones with a normal work-up: thiazide or potassium citrate (AUA: even without abnormality); citrus juices are adjuncts.
- **Trials**: Ettinger 1988 (chlorthalidone) and Barcelo 1993 (potassium citrate) reduced new stones. **NOSTONE (2023)**: hydrochlorothiazide 12.5–50 mg did not reduce recurrence, which questions the older trials.

**5. Monitoring and follow-up**

- Repeat 24-hour urine within **6 months** of starting therapy, then yearly; reanalyse any new stone.
- Serum potassium 1–2 weeks after starting a thiazide or citrate; imaging for new stones.

**Diagram to draw:** saturation zones (Ksp, Kf) and Randall plaque on a renal papilla.

**Diagram to draw:** flowchart from stone analysis and 24-hour urine to abnormality to selective drug therapy.

## Recent advances in prevention

- **ROKS (Recurrence of Kidney Stone) nomogram** predicts individual risk.
- **Also know:** NOSTONE (2023) negative; citrus juices are adjuncts; *Oxalobacter formigenes* loss; lumasiran for PH1.

## Recent advances in RIRS instrumentation

| Advance | Principle | Benefit or limitation |
|---|---|---|
| **Digital flexible ureteroscopes** | CMOS (complementary metal-oxide semiconductor) chip at the tip; LED (light-emitting diode) light | Sharper image that does not degrade; tip larger (~8.5 Fr); no more durable |
| **Single-use scopes** (LithoVue, 2016) | 7.7 Fr tip, ≥270° deflection, new each case | No repair or reprocessing; cheaper below ~99 cases a year; for lower pole and difficult cases |
| **Thulium fibre laser** (about 1940 nm) | Long pulse, very high frequency | Fine dust, minimal retropulsion; faster than Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) at low settings |
| **Pulse-modulated (Moses) Ho:YAG** | Pulse shaped through a vapour bubble | Less retropulsion, faster dusting |

| Advance | Principle | Benefit or limitation |
|---|---|---|
| Access sheaths (12/14 Fr) | Outflow channel | Lower intrarenal pressure and temperature; severe ureteric injury in ~1 in 8 |
| **Suction sheaths and FANS** (flexible and navigable suction access sheath) | Vacuum at the sheath; flexible tip follows the scope into calyces | Aspirates dust; lower pressure; better stone-free rates in early studies |
| **Intrarenal pressure monitoring** | Sensor in the wire or scope (LithoVue Elite) | High pressure causes backflow, SIRS (systemic inflammatory response syndrome) and sepsis; safe limits not yet defined |
| **Robotic flexible ureteroscopy** (Avicenna Roboflex) | Seated console controls the scope | Ergonomics, less radiation to surgeon |

- **Laser safety**: tissue damage above 43 °C, so keep power ≤10 W in the ureter and ≤25 W in the kidney, with continuous cool irrigation.
- **Also know:** dusting versus fragment-and-basket; nitinol tipless baskets; anti-retropulsion devices; Monarch robotic platform for PCNL (percutaneous nephrolithotomy) access.

**Key points to remember:**

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

## LAQ 3 Metabolic evaluation in stone disease

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Discuss metabolic evaluation in urinary calculus disease." / "Describe metabolic evaluation in stone disease" / "Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi" · P1 SAQ "Metabolic Evaluation in Stone Disease"

### Introduction and rationale

- Stone disease is a chronic metabolic disorder: first-time formers have about a **50% risk of recurrence within 10 years** (another source: up to 50% recur within 5–10 years).
- Evaluation finds a treatable cause and directs targeted therapy; therapy matched to 24-hour urine results reduces stone events. The **ROKS (Recurrence of Kidney Stone) nomogram** estimates individual risk.

### Physicochemical basis (why we measure these)

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention, often on **Randall plaque**.
- In the **metastable zone** (between solubility and formation products) crystals grow on existing nuclei but do not form spontaneously; above it nucleation is spontaneous.
- **Promoters** (calcium, oxalate, uric acid, low volume) raise supersaturation; **inhibitors** (citrate, magnesium, Tamm–Horsfall protein) raise the supersaturation needed to crystallise. The 24-hour urine measures both.

### Who needs evaluation

- **Screening evaluation** for every first-time former (AUA [American Urological Association], EAU [European Association of Urology]).
- **24-hour urine for high-risk and recurrent formers:** recurrent formers, **children**, solitary kidney, bilateral or large burden, GI (gastrointestinal) disease, bariatric surgery or bypass, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, family history; uric acid, cystine or infection stones. Bilateral staghorns are high risk by definition.

### Step 1: Screening evaluation (every first-time former)

| Component | What is assessed |
|---|---|
| History | Colic, UTI (urinary tract infection) and recurrent UTI, previous stones, fluid and diet; GI disease or bariatric surgery, gout, sarcoidosis, family history; lithogenic drugs: calcium, vitamins C and D, acetazolamide, topiramate, steroids, triamterene, indinavir |
| Blood | Creatinine, sodium, potassium (electrolytes), calcium, uric acid, bicarbonate (if RTA suspected); **PTH (parathyroid hormone)** if calcium is high or high-normal |
| Urine | Urinalysis: pH, crystals (hexagonal = cystine), pyuria; culture; cyanide–nitroprusside cystine screen, confirmed by quantitative 24-hour cystine |
| **Stone analysis** | Infrared spectroscopy or X-ray diffraction: uric acid points to metabolic syndrome, cystine to a genetic cause, struvite to infection |
| Imaging | NCCT: burden predicts future events; uric acid usually <500 HU (Hounsfield units) |

- Screening is a **triage**: low risk gets general advice; high risk goes on to 24-hour urine.

### Imaging in evaluation for urolithiasis

| Imaging | Role |
|---|---|
| **NCCT** (non-contrast CT) | Gold standard: sensitivity 97%, specificity 95%. Size, site, burden, density (uric acid 200–400 HU; calcium oxalate 600–1200 HU), skin-to-stone distance |
| Low-dose NCCT (≤3 mSv) | Equal accuracy for stones ≥3 mm |
| Ultrasound | First line in pregnancy and children |
| KUB (plain X-ray) | Follow-up of radio-opaque stones only |
| CT urography | Anatomy before PCNL (percutaneous nephrolithotomy) |
| MAG3 (mercaptoacetyltriglycine) renogram or DMSA (dimercaptosuccinic acid) scan | Split function |

### Step 2: Extended metabolic evaluation (24-hour urine)

- **One or two 24-hour urines** on the usual diet, at least 4 weeks after the episode, stone-free and infection-free. Creatinine checks completeness (men about 20–25, women about 15–20 mg/kg/day).
- Measure volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine (cystine if suspected).

| Parameter | Desirable per day | Abnormality and meaning |
|---|---|---|
| Volume | **>2.5 L** | Low volume: commonest, easiest to fix |
| Calcium | <250 mg women, <300 mg men (or <4 mg/kg) | **Hypercalciuria**: absorptive (PTH suppressed), renal leak (PTH high, serum calcium normal), resorptive (primary hyperparathyroidism) |
| Oxalate | **<40 mg** | Dietary; **enteric** (fat malabsorption, gastric bypass: fat binds calcium, freeing oxalate; often 60–70 mg); **primary** (hepatic, often >100 mg, childhood; type 1 (AGXT gene) ~70%, type 2 (GRHPR), type 3 (HOGA1)) |
| Citrate | **>320 mg** | Hypocitraturia: distal RTA, diarrhoea, hypokalaemia, high animal protein |
| Uric acid | <600 mg (labs: <750 women, <800 men) | Hyperuricosuria promotes calcium oxalate stones |
| pH | 5.8–6.2 | <5.5 uric acid (metabolic syndrome); >6.5–7 calcium phosphate, RTA, infection |
| Sodium | <100 mmol | High sodium raises calcium excretion |
| Cystine | Negative screen | Cystinuria; keep <250 mg/L |

- **Uric acid stones** depend mainly on pH: below the pKa (5.5) uric acid is undissociated and insoluble.
- **Infection stones**: urease-splitting organisms, pH >7, struvite.
- **Also know:** magnesium, potassium, phosphate; hypomagnesiuria; RSS (relative supersaturation) is calculated from these and guides therapy; medullary sponge kidney and calyceal diverticulum (stasis plus metabolic defects); 2,8-dihydroxyadenine stones (APRT [adenine phosphoribosyltransferase] deficiency).

### Interpreting hypercalciuria (Pak classification)

| Type | Defect | Serum calcium | PTH | Fasting urine calcium |
|---|---|---|---|---|
| Absorptive (20–40%) | Increased gut absorption | Normal | Normal or suppressed | Normal |
| Renal leak (5–8%) | Tubular calcium loss | Normal | High (secondary) | High |
| Resorptive (3–5%) | Primary hyperparathyroidism | High | High | High |

- Fast and calcium-load subtyping is no longer advised, because it does not change treatment.

### Special tests

- **Distal RTA:** urine pH >6, low citrate, calcium phosphate stones. Incomplete forms: oral **ammonium chloride load** (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria:** autosomal recessive transport defect (cystine, ornithine, lysine, arginine); quantitative 24-hour cystine; solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
- **Primary hyperoxaluria:** genetic testing (type 1, AGXT gene, about 70%).

### Evaluation algorithm

`Stone episode → screening evaluation Low risk → general advice High risk → 24-h urine ×1–2 → targeted therapy → repeat within 6 months → yearly`

**Diagram to draw:** supersaturation zones (undersaturated, metastable, unstable) and the evaluation flowchart (first stone, screening, low risk with general advice or high risk with 24-hour urine and targeted therapy).

### Treatment linked to abnormality

- **General**: urine >2.5 L, sodium ≤2.3 g/day, normal calcium, less animal protein, more fruit and vegetables.

| Abnormality | Treatment |
|---|---|
| Low volume | Fluid for urine >2.5 L/day |
| Hypercalciuria | **Thiazide** (hydrochlorothiazide 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate; plus sodium restriction |
| Primary hyperparathyroidism | **Parathyroidectomy** |
| Hypocitraturia, distal RTA | **Potassium citrate** 30–60 mEq/day |
| Enteric hyperoxaluria | Low fat and oxalate; **calcium with meals**; magnesium |
| Primary hyperoxaluria type 1 | Pyridoxine, **lumasiran** (RNAi [ribonucleic acid interference]); liver ± kidney transplant |
| Hyperuricosuria | **Allopurinol** 300 mg/day |
| Uric acid stones | **Potassium citrate** to urine pH 6.0–6.5 |
| Cystinuria | Urine ≥3 L/day; pH **7.0–7.5**; **tiopronin** |
| Struvite | Complete clearance; AHA (acetohydroxamic acid) if surgery impossible |

- **Borghi 1996**: urine >2 L/day cut 5-year recurrence (12% vs 27%). **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low calcium. **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence.

### Special groups and follow-up

- **Children:** all evaluated (spot ratios if not toilet-trained; consider Dent disease, monogenic causes). **Pregnancy:** defer 24-hour urine (physiological hypercalciuria).
- **Follow-up (AUA):** repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug side effects (potassium on thiazide or citrate); re-analyse every new stone. Adherence is poor; citrus juices are only adjuncts.

### Recent advances

- **Lumasiran** (RNA interference) for primary hyperoxaluria type 1; **NOSTONE (2023):** hydrochlorothiazide did not reduce recurrence; ROKS nomogram.

### Staghorn calculus (bilateral staghorn question)

- **Definition:** a branched stone filling the renal pelvis and extending into the calyces. **Partial:** pelvis and some calyces; **complete:** pelvis and all major calyces.
- **Composition:** classically struvite (magnesium ammonium phosphate) with carbonate apatite. Bacterial **urease** (Proteus, Klebsiella, Ureaplasma) splits urea into ammonia, so urine pH rises above 7.2 and struvite crystallises inside a bacterial biofilm. Fewer than half of modern staghorns are infection stones; calcium phosphate, cystine and uric acid also occur.
- **Why treat:** untreated staghorns destroy the kidney and cause sepsis, XGP (xanthogranulomatous pyelonephritis) and death.

### Staghorn: work-up before surgery

- NCCT with coronal and 3D views: burden, calyceal anatomy, colon and spleen.
- **Renal function:** eGFR (estimated glomerular filtration rate) and split function, because both kidneys are at risk.
- **Infection:** culture-specific antibiotics before surgery; drain an obstructed or infected kidney (nephrostomy through a calyx usable later for PCNL). Culture the stone.
- **Guy's stone score (2011)** predicts stone-free rate:

| Grade | Stone |
|---|---|
| I | Single stone in mid or lower pole or pelvis, simple anatomy |
| II | Single upper-pole stone; multiple stones in simple anatomy; or single stone in abnormal anatomy |
| III | Multiple stones in abnormal anatomy; calyceal diverticulum; partial staghorn |
| IV | Staghorn; or any stone in spina bifida or spinal injury |

- **Also know:** STONE score, CROES nomogram, S-ReSC score.

### Staghorn: treatment options (AUA, EAU)

- **Goal: complete clearance**, because each fragment is an infected nidus; also clear infection and preserve function.

| Option | Role |
|---|---|
| **PCNL** | **First line.** Multiple tracts if needed; upper-pole access reaches pelvis, lower pole and ureteropelvic junction. Flexible nephroscopy raises clearance; second-look for residue |
| **ECIRS** (endoscopic combined intrarenal surgery) | PCNL (usually supine) plus retrograde flexible ureteroscopy: fewer tracts, less bleeding, higher clearance |
| Staged RIRS (retrograde intrarenal surgery) | Selected partial staghorns; patients who cannot stop anticoagulants |
| SWL (shock wave lithotripsy) | Not as monotherapy; adjunct for residual fragments |
| **Anatrophic nephrolithotomy** (Boyce–Smith) | Incision along Brödel's avascular line, artery clamped, cold ischaemia, infundibula reconstructed. Now rare; when endoscopy cannot clear the stone in a reasonable number of sessions |
| **Nephrectomy** | Poorly functioning kidney (split below about 10–20%) that is infected or painful; XGP |

## Bilateral staghorns: sequence

``` Sepsis or obstruction on either side? Yes -> drain + antibiotics -> treat that side once sterile No -> treat the better-functioning (or symptomatic) kidney first

|

NCCT: stone-free? -> second look if residue

|

Other kidney after recovery (selected: same-session bilateral PCNL, or PCNL + contralateral RIRS) ```

- **Why the better kidney first:** it secures renal function early; some start with the simpler side. No trial defines the order.
- **Same-session bilateral surgery:** feasible at experienced centres if the first side was short and uncomplicated; saves an anaesthetic but risks bilateral bleeding or obstruction.
- Check creatinine after each side. Remove a non-functioning kidney only after the other side is cleared.

## Staghorn: complications and follow-up

- PCNL risks: bleeding (transfusion about 1%; angioembolisation), sepsis, pleural, colon, spleen or liver injury.
- NCCT for stone-free status; treat residual fragments >4 mm. Surveillance cultures.
- 24-hour urine even in struvite formers, then targeted therapy; urine ≥2.5 L/day.
- **AHA** 250 mg three times daily, a urease inhibitor, only when surgery is exhausted. Avoid if creatinine >2.5 mg/dL or in pregnancy; causes DVT (deep vein thrombosis), haemolytic anaemia.

**Diagram to draw:** staghorn with an upper-pole PCNL tract and a retrograde flexible scope (ECIRS).
```

**Key points to remember:**

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
- NCCT gold standard (97%/95%).
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

## LAQ 4 Retroperitoneal fibrosis: diagnosis and (medical) management

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis." · P1 SAQ "Retroperitoneal fibrosis" · P2 SAQ "Retro peritoneal fibrosis medical management"

### Introduction

- RPF (retroperitoneal fibrosis) is a rare inflammatory fibrosis (fibrous plaque) encasing the aorta, IVC (inferior vena cava) and ureters, centred on the **distal aorta at L4–L5**.
- Incidence about 1 per 200,000–500,000 a year; age 40–60; male:female 2–3:1.

### Etiology

| Idiopathic (about 70%) | Secondary (about 30%) |
|---|---|
| Part of **chronic periaortitis**, a large-vessel vasculitis | **Drugs** (commonest identifiable cause): methysergide and other ergots, β-blockers, phenacetin |
| Proposed antigen: **ceroid** (oxidised lipid in plaque) | **Malignancy:** lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma |
| Probably autoimmune: membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis; ANA (antinuclear antibody) positivity predicts relapse | Radiotherapy; asbestos |
| Part of multifocal fibrosclerosis: sclerosing mediastinitis and cholangitis, orbital pseudotumour, Riedel thyroiditis | Infection: tuberculosis, actinomyces, gonorrhoea, schistosomiasis |

- Many idiopathic cases are now classed as **IgG4 (immunoglobulin G4)-related disease** (raised serum IgG4, IgG4-positive plasma cells, storiform fibrosis).

### Pathology

- **Gross:** flat, tan, dense plaque centred on the **distal aorta at L4–L5**, wrapping the ureters; may invade ureter or psoas.
- **Histology changes with stage:**
- Early (active): collagen with capillary proliferation and lymphocytes, plasma cells and fibroblasts (steroid-responsive).
- Late (burnt-out): acellular, avascular sheets of collagen.
- **Trap:** malignant RPF looks identical except for small islands of tumour cells, so take **deep, multiple biopsies**.
- **Mechanism of obstruction:** extrinsic compression and loss of ureteric peristalsis; the lumen stays patent.

### Clinical features

- **Dull lower back or flank pain** (commonest); weight loss, anorexia, nausea, malaise, fever, hypertension.
- About half present with **uraemia** or oliguria/anuria; symptoms last 4–6 months before diagnosis.

- **Extension:** IVC (leg oedema, DVT [deep venous thrombosis]); renal vein (gross haematuria); spermatic cord and scrotum. Examination is usually unrevealing.

## Diagnostic evaluation

### Laboratory

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein) in half to two-thirds; leucocytosis; anaemia; raised creatinine (renal failure); ANA; serum IgG4.

### Imaging

| Test | Findings and role |
|---|---|
| CT (computed tomography) / CT urography | Homogeneous soft-tissue mass enveloping aorta, IVC and ureters; **medial deviation** of proximal and mid-ureters with smooth tapering |
| Retrograde pyelography | When renal function is poor |
| MRI (magnetic resonance imaging) | Low T1; **high T2 signal means active disease**; enhancement falls with treatment |
| **PET-CT (positron emission tomography)** | **Most sensitive test of activity**; guides biopsy and follow-up |
| Renography | Split function (repair or nephrectomy) |

- **Biopsy** (CT-guided or at ureterolysis) if atypical or malignancy is suspected; may be omitted with no malignancy history, no lymphadenopathy and typical imaging.

## Treatment algorithm

```
RPF with hydronephrosis -> Relieve obstruction (stents; nephrostomy if anuric or septic) -> Treat post-obstructive diuresis -> Stop culprit drug; exclude malignancy -> Active disease: steroids (+/- second-line drugs) -> Failure or burnt-out fibrosis: ureterolysis
```

## Initial management

- Emergent **stents or PCN (percutaneous nephrostomy)**; nephrostomy if critically ill, uraemic and anuric with little urine. Replace fluid during **post-obstructive diuresis**.
- **Stop the culprit drug;** treat any secondary cause.

## Medical management

| Agent | Role and regimen |
|---|---|
| **Prednisone** (first line) | About 60 mg daily (about 1 mg/kg), tapered to 5 mg, total course up to 2 years. About 80% respond; relapse on tapering 25–50% |
| Mycophenolate mofetil (with steroids), azathioprine, cyclophosphamide, ciclosporin, colchicine | Second line or steroid-sparing |
| Tamoxifen; medroxyprogesterone | Thought to inhibit fibroblasts |
| **Rituximab** | Refractory or IgG4-related disease |

- Steroids work on the cellular, active phase: pain and symptoms settle in days, ESR falls and the mass shrinks. Active inflammation predicts response; burnt-out fibrosis responds poorly.

- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen, so steroids remain first line.
- Drainage plus medical therapy clears obstruction in about two-thirds; only about **15% need ureterolysis**.

### Surgical ureterolysis

- For failed medical therapy, burnt-out (fibrotic) disease or tissue diagnosis. **Bilateral**, even if hydronephrosis is unilateral; open, laparoscopic or robotic.
- Steps: 1. Pre-place stents to identify the ureters. 2. Mobilise the colon; take deep biopsies for frozen section. 3. Start at the distal, non-dilated ureter. 4. "Split and roll": right-angle clamp between ureter and plaque; incise the fibrosis over it. 5. Close any ureterotomy with absorbable suture. 6. **Separate the ureter from the fibrotic bed so it cannot be re-encased.**

| Protection method | How |
|---|---|
| Intraperitonealisation | Ureter lateralised; peritoneum closed behind it |
| Lateral retroperitoneal placement (lateralisation) | Ureter displaced laterally (equivalent results) |
| **Omental wrap** | Omentum split on the gastroepiploic arteries, wrapped round each ureter; adds blood supply; best for extensive disease |

- Postoperative steroids may reduce recurrence.
- **Outcomes:** open about 94% stent-free at 1 year; laparoscopic about 90%; robotic with omental wrap about 86%.

### Other options

- Long-term metallic stents (unfit); autotransplantation (rarely needed) if ureterolysis is impossible; nephrectomy for an unrecoverable (dead) kidney.

### Follow-up and monitoring

- Symptoms, ESR/CRP, creatinine, ultrasound; periodic CT/MRI or PET-CT for activity; remove stents once the ureters drain.
- Watch for relapse and steroid side effects (diabetes, osteoporosis).

**Diagram to draw:** periaortic plaque at L4–L5 drawing both ureters medially, with an omental wrap.

> **Key points to remember:**
> - Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
> - Medially deviated ureters; PET-CT best shows activity; biopsy deep.
> - Drain first, then prednisone (about 80% respond).
> - Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
> - Rituximab for IgG4-related or refractory disease.
> - About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

## LAQ 5 Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Classification and treatment of lower track trauma in # pelvis" (also P2 SAQ "Pelvic # assessment and management" and P2 SAQ "Early surgical intervention in urethral injuries") · P3 LAQ "Describe early and late management of PFUD injuries"

*PFUD is taken as pelvic fracture urethral distraction defect, i.e. PFUI (pelvic fracture urethral injury). Lower urinary tract trauma means bladder and urethral injury in pelvic fracture.*

### Introduction

- Pelvic fracture accompanies **83–95% of bladder ruptures**, yet only **5–10% of pelvic fractures** injure the bladder. Urethral injury (PFUI) occurs in about **10% of male** and up to 6% of female pelvic fractures.
- 80–94% have other major (associated) injuries, which cause the deaths; long-term disability comes from wrong management.

### Mechanism

- **EPR (extraperitoneal rupture):** pelvic disruption (shearing) tears the bladder at its fascial attachments, or bone spicules lacerate it.
- **IPR (intraperitoneal rupture):** a blow on a **full bladder** bursts the dome, the weakest part, covered by peritoneum.
- **PFUI:** the posterior urethra is fixed to the pubis (puboprostatic ligaments, perineal membrane), so displacement shears it, usually at the **bulbomembranous junction**. In prepubertal boys the tear may extend into the prostate and bladder neck.
- Highest risk: straddle fracture of all four pubic rami, vertically unstable fractures, symphyseal diastasis.
- The sphincter is usually avulsed but stays functional, so most men stay continent after repair.
- The gap fills with scar: an obliterative **distraction defect**, not a true stricture.

### Classification

#### Bladder: AAST (American Association for the Surgery of Trauma) grades I–V

| Grade | Injury |
|---|---|
| I | Contusion, intramural haematoma, partial-thickness laceration |
| II | Extraperitoneal wall laceration <2 cm |
| III | Extraperitoneal ≥2 cm, or intraperitoneal <2 cm |
| IV | Intraperitoneal ≥2 cm |
| V | Laceration extending into bladder neck or ureteric orifice (trigone) |

- Practical: contusion, EPR (simple or complex), IPR, combined.

#### Urethra: Goldman I–V

| Type | Injury |
|---|---|
| I | Posterior urethra stretched, intact |

| Type | Injury |
|---|---|
| II | Disruption above the urogenital diaphragm (perineal membrane) |
| III | Disruption of membranous urethra through the urogenital diaphragm, extending below the membrane into the bulb (commonest) |
| IV | Bladder neck injury extending into urethra (IVa: bladder-base injury mimicking IV) |
| V | Pure anterior urethral injury |

- **Campbell-Walsh practical grading: partial injury; minor distraction** (a cystoscope passes retrogradely into the bladder); **major distraction** (a cystoscope cannot pass into the bladder). Practically partial vs complete disruption. Stricture risk and need for urethroplasty rise from partial to major.
- Anterior urethra: contusion, incomplete or complete.

### Assessment and clinical features

- **ATLS (Advanced Trauma Life Support)** resuscitation first; associated injuries take priority; pelvic binder and stabilisation.
- Bladder: **gross haematuria (nearly all)**, suprapubic pain, inability to void, free fluid.
- Urethra: **triad: blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate and butterfly perineal haematoma are unreliable.
- **Vaginal and rectal examination**; vaginal examination in every female.

### Investigations

- **Blood at the meatus mandates RUG (retrograde urethrogram) before any catheter** (AUA [American Urological Association]). Method: 16 Fr catheter tip in fossa navicularis, 3 mL balloon, penile traction, oblique position, 20 mL contrast, film during injection. Extravasation with bladder filling means partial injury; without filling, complete. Females: urethroscopy.
- **Cystography:** absolute indication: **gross haematuria + pelvic fracture** (rupture in ~29%). Relative: gross haematuria alone; microscopic haematuria with fracture.
- **CT cystography:** retrograde fill with **350 mL** of 2–4% contrast; no drainage film needed.
- Plain cystogram: scout, full-bladder and **drainage films** (posterior leaks otherwise missed).
- Clamping the catheter after IV contrast is inadequate.
- **EPR:** flame-shaped pelvic contrast. **IPR:** contrast outlines bowel loops.

### Bladder injury: management

- **Uncomplicated EPR:** 22 Fr urethral catheter 2–3 weeks; cystogram before removal; repair if leak persists beyond 4 weeks.
- **Immediate repair:** IPR; penetrating injury; bladder neck injury; bone fragments in the bladder; rectal or vaginal injury; poor drainage or clots; open pelvic fixation or laparotomy for other reasons.
- IPR: open or laparoscopic two-layer closure; inspect ureteric orifices; cystogram at 7–10 days. At laparotomy, repair EPR from inside the bladder, leaving the pelvic haematoma undisturbed.
- Suprapubic tube adds nothing over urethral catheter (AUA). Antibiotics only around IPR repair.
- Rectal or vaginal injury: separate the walls, no overlapping suture lines, interpose tissue.

## Posterior urethra (PFUI): early management

- Key decision: **immediate repair versus deliberate delay**. Clean, localised injuries are repaired at once; PFUI is drained and reconstructed later.

```
Blood at meatus → RUG Partial → one gentle catheter attempt (confirm position)
Complete → SPC (suprapubic catheter): percutaneous 16 Fr or open (if bladder
repair needed) ± brief primary realignment if stable (flexible cystoscopy,
antegrade + retrograde) Bladder neck/rectal injury → immediate repair Delayed
(2–3 months) → up-and-down-o-gram → perineal excision and primary anastomosis
(corporal split → pubectomy → rerouting)
```

1. ATLS resuscitation; associated injuries take priority.
2. **SPC (suprapubic cystostomy): the standard of care.** Percutaneous (peel-away sheath, 16 Fr Foley; avoid small "punch" tubes) or open through a lower midline incision, which allows bladder inspection (needed if bladder repair is required). - SPC is safe with pelvic plating (AUA Urotrauma guideline); place and tunnel it high, away from hardware.
3. **Primary realignment (PR)** (AUA Urotrauma option): a brief attempt in a stable patient, acutely or within days, by combined antegrade and retrograde flexible cystoscopy over a guidewire (avoid prolonged attempts). Catheter out at 4–6 weeks; keep the SPC and remove it 7–14 days later if voiding well.
4. Partial tear: one gentle catheter attempt allowed.
5. **Immediate open exploration/repair** only for **bladder neck or rectal injury**, or when laparotomy is needed anyway. - **Also know:** female urethral injuries are usually repaired early; in boys tears may extend into the bladder neck.

## SPC versus primary realignment

| | **SPC + delayed urethroplasty** | **Primary realignment** |
|---|---|---|
| Advantage | Simple; haematoma untouched | Partial tears may heal; aligned ends ease repair |
| Drawback | Complete injuries need urethroplasty | Most still form a 1–2 cm stenosis |
| Evidence | Prospective data: **no significant difference** in need for urethroplasty | Modern PR does not add ED (erectile dysfunction) or incontinence |

## Anterior urethra: early management

| **Injury** | **Early management** |
|---|---|
| Contusion, incomplete straddle tear | Urethral catheter alone |
| Major straddle injury | SPC (standard); primary realignment for lesser injuries; delayed anastomotic urethroplasty |
| **Low-velocity gunshot or stab** | **Immediate primary repair:** debride minimally, spatulate, tension-free anastomosis over a catheter |
| High-velocity or close-range shotgun | SPC, staged repair |
| **Penile fracture with urethral tear** | **Immediate repair:** partial tear oversewn; complete tear debrided and anastomosed; catheter 2–3 weeks |

- Penile fracture: urethral injury in about 20% of Western and 3% of Asian series; **intraoperative flexible cystoscopy** before catheterisation. Limit spongiosal debridement.

## Late management of PFUI

**Timing:** delayed repair at 2–3 months (P3: after **3 months**), once the scar is stable and the patient is ambulant.

### Evaluation

- **"Up-and-down-o-gram":** cystogram plus RUG while the patient tries to void, to measure the gap length; antegrade flexible cystoscopy via the SPC tract if the bladder neck stays closed / does not open. Bladder neck appearance does **not** predict its later function.
- **MRI (magnetic resonance imaging)** for complex cases: gap length, prostatic displacement (cephalad displacement predicts pubectomy).

**Endoscopic:** DVIU (direct vision internal urethrotomy) only for short, incomplete stenosis. "Cut-to-the-light" or repeated DVIU for gaps ≥1 cm fails (false passages).

### Perineal excision and primary anastomosis (EPA; delayed anastomotic urethroplasty): treatment of choice

1. Exaggerated lithotomy (under 5 hours); sounds via meatus and SPC tract.
2. Midline or lambda incision; mobilise the bulbar urethra up to the penoscrotal junction (not beyond: chordee).
3. Divide the urethra at the scar; excise **all** fibrosis until healthy prostatic mucosa appears over the sound (identify the prostatic apex).
4. Spatulate both ends to 30 Fr; 8–10 interrupted 4-0 absorbable sutures; tension-free mucosa-to-mucosa anastomosis over a silicone catheter.

**Progressive perineal approach** (each step shortens the path to bridge a longer gap):

| Step | Manoeuvre |
|---|---|
| 1 | Bulbar urethral mobilisation (enough for most) |
| 2 | Separation of the corporal bodies (corporal splitting, intracrural space) |
| 3 | Inferior wedge pubectomy (Kerrison rongeur) |
| 4 | Supracrural rerouting of the urethra around one corpus |

- **Abdominoperineal or transpubic approach:** urethrorectal fistula, failed repair, bladder neck injury, children.
- **Postoperative:** voiding study at 21–28 days; cystoscopy at about 6 months.

## Outcomes and complications

- Urethroplasty success above 95%. Restenosis 5–15%; often DVIU-treatable because the scar is already excised.
- **ED** up to 82%, mostly from the injury (cavernous nerves, pudendal arteries; nerve, arterial, venous). **Incontinence <4%** (uncommon; bladder neck).
- Missed IPR: ascites, acidosis, azotaemia (peritoneal reabsorption).
- Fistula, abscess, neurogenic voiding dysfunction; follow up ≥1 year.
- **Also know:** chordee, penile shortening, ejaculatory dysfunction, lithotomy neuropathy, penile Doppler before repair in impotent men.

**Diagram to draw:** pelvis with EPR (flame-shaped) vs IPR on cystogram; RUG showing partial vs complete posterior urethral disruption; sagittal pelvis showing the bulbomembranous gap and the four progressive perineal steps.

> **Key points to remember:**
>
> - Blood at meatus: RUG before catheter.
> - Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
> - EPR: catheter 2–3 weeks; IPR: immediate repair.
> - PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
> - Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
> - Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
> - Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
> - ED mostly from the injury; incontinence <4%.

## LAQ 6 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management

P2

**Asked as:**

- Traumatic renal vascular parenchymal injuries : diagnosis and management. — *Summer 2017 Paper II Q2 (LAQ)*
- Diagnosis and Treatment of renal trauma — *Practice Set 9 Paper II Q1 (LAQ)*
- Grade 4 Renal trauma. — *Practice Set 3 Paper II Q3c (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 9 Paper II Q1: also answered in 2017-Summer Paper 2, Q2; grade IV in Practice Set 3 Paper 2, Q3c) (Practice Set 3 Paper II Q3c: renal trauma also answered in 2017-Summer Paper 2, Q2)

*Assumed meaning: diagnosis and management of traumatic renal parenchymal and renal vascular injuries.*

### Introduction

- Kidney is the **commonest urological organ injured** (up to 5% of all trauma); **blunt injury causes 80–90%**; firearms cause most penetrating injuries.
- **Rapid deceleration** stretches the kidney at its fixed points (hilum and UPJ, ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle or UPJ avulsion.
- Main renal artery injury is rare (about 0.05% of blunt trauma).

### Classification: AAST (American Association for the Surgery of Trauma) 2018

| Grade | Injury |
|---|---|
| I | Subcapsular haematoma and/or contusion, no laceration |
| II | Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation |
| III | Laceration >1 cm without collecting-system injury; **any vascular injury or active bleeding contained within Gerota's fascia** |
| IV | Laceration into collecting system with extravasation; renal pelvis laceration and/or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction (thrombosis) |
| V | **Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney |
| - Advance one grade for multiple injuries, up to grade III. Change from 1989: main vessel injury moved from IV to V; pseudoaneurysm and AVF (arteriovenous fistula) now grade IV. | |

### Definition: AAST 2018, grade IV in detail

- Laceration into the **collecting system with urinary extravasation**.
- **Renal pelvis laceration and/or complete UPJ (ureteropelvic junction) disruption.**
- **Segmental artery or vein injury** (pseudoaneurysm, AVF [arteriovenous fistula]).
- **Active bleeding beyond Gerota's fascia.**
- **Segmental or complete infarction** from thrombosis without active bleeding.
- Change from 1989: main-vessel injury moved to grade V; segmental vascular and UPJ injuries added to grade IV.

## Clinical diagnosis

### History and examination

- Mechanism is the most important (key) history; mechanism matters. Signs: flank bruise or tenderness, **fractured ribs 9–12**, flank or low chest wound.

### Haematuria

- Test the **first urine sample**. Significant: gross haematuria, or microscopic (microhaematuria) (>5 RBC [red blood cells] per high-power field, hpf) with **shock (SBP [systolic blood pressure] <90 mmHg)**.
- Degree does **not** match severity; it may be **absent in pedicle injury** and UPJ avulsion (in about half of UPJ injuries), and in grade IV.

## Indications for imaging (AUA (American Urological Association) and EAU (European Association of Urology))

1. Blunt trauma with gross haematuria.
2. Blunt trauma with microhaematuria and **SBP (systolic blood pressure) <90 mmHg** at any time.
3. Rapid deceleration (high-speed collision, fall), even without haematuria.
4. Penetrating trauma with likely renal injury, any haematuria.
5. Children: image liberally (hypotension appears late). - Microhaematuria without shock or deceleration (microhaematuria alone without shock): observe without imaging.

## Imaging

- **Contrast-enhanced CT (computed tomography)** is the gold standard. Arterial/nephrographic phase (~80 s) shows bleeding; **delayed excretory phase** (5–10 min / minutes) shows urinary extravasation (urine leak).
- **CT red flags:**

| Finding | Suggests |
|---|---|
| Medial haematoma | Pedicle (vascular) injury |
| Medial extravasation, no contrast in distal ureter (without distal ureteric contrast) | Renal pelvis or UPJ avulsion |
| Absent enhancement, cortical rim sign | Main renal artery occlusion |
| Active contrast blush (vascular contrast blush) | Arterial bleeding needing AE (angioembolisation) |

| Finding | Suggests |
|---|---|
| Haematoma rim >4 cm (another source: >3.5–4 cm), medial laceration | High risk of intervention; bleeding likely to need intervention (≥2 of rim, blush, medial laceration predict intervention) |
| - **One-shot IVP (intravenous pyelography)** on table (2 mL/kg, one film at 10 min) in the unstable patient: confirms a **functioning contralateral kidney**. FAST (focused assessment with sonography for trauma) is non-specific. Renal vein injury is poorly seen on CT. | |

## Management

`Unstable despite resuscitation → laparotomy (± damage control) (or angioembolisation / AE if team ready, else laparotomy) Stable → stage with CT Grade I–III → non-operative Grade IV–V → non-operative in selected; AE for active bleeding Active blush / pseudoaneurysm / AVF → superselective AE Extravasation → observe (90% heal); stent ± nephrostomy if urinoma enlarges, fever, obstruction, or UPJ/pelvis injury`

- **Non-operative**: bed rest until gross haematuria clears, serial haemoglobin (**every 4–6 h**); success >90% in blunt injury and selected penetrating ones. **Non-operative success** about 85% for grade IV–V even in penetrating series; most failures need only AE, stent or nephrostomy.
- Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- Urinary extravasation heals spontaneously in **76–90%** (another source: up to 90%).
- **Angioembolisation**: superselective coils for active bleeding, pseudoaneurysm, AVF or delayed bleeding; repeat AE often succeeds.
- **Reimaging** (reimage / repeat CT at ~48 hours) for grade IV–V with deep laceration (deep lacerations), or new fever, pain or falling haematocrit (routine reimaging debated).

## Indications for exploration

| Absolute | Relative |
|---|---|
| Haemodynamic instability, no or transient response | Extravasation with devitalised parenchyma (>25%; lowers the threshold) |
| Expanding or pulsatile retroperitoneal haematoma | Arterial thrombosis |
| Suspected pedicle avulsion | Associated colon or pancreatic injury |
| UPJ avulsion | Incomplete staging |

## Operative technique (operative steps)

- Midline transabdominal (laparotomy); life-threatening injuries first.
- **Vascular control before opening Gerota's fascia**: lift small bowel; incise retroperitoneum over the aorta above the inferior mesenteric artery → find the left renal vein crossing the aorta → loop vein and both arteries; then reflect colon, open Gerota's fascia.
- **Renorrhaphy**: debride → suture-ligate vessels → watertight collecting-system closure → close defect over bolster → omental cover → drain. Polar injury: partial nephrectomy.

- **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, resuscitate in ICU (intensive care unit), relook (return) at 24–48 h.

### Renovascular injury

- **Main artery thrombosis** (intimal tear): unilateral with normal other kidney → observe (revascularisation rarely succeeds). **Solitary or bilateral → revascularise** (warm ischaemia ideally <120 min); endovascular stent for (or stent an) intimal flap.
- **Vein**: left renal vein can be ligated near the IVC (inferior vena cava) (gonadal, adrenal collaterals); right cannot. Repair with 5-0 vascular suture; segmental veins may be ligated.

### Complications

- **Delayed bleeding** (within 3 weeks; usually within 21 days): AE (embolise).
- Urinoma or abscess: stent, nephrostomy, drain.
- **Hypertension**: Goldblatt (arterial stenosis), **Page kidney** (haematoma compression), segmental ischaemia, AVF.
- Hydronephrosis, loss of function; **also know:** AKI (acute kidney injury).

### Follow-up

- Blood pressure, urinalysis, creatinine; EAU: DMSA (dimercaptosuccinic acid) scan for salvaged grade III–V kidneys; imaging if new symptoms.

**Diagram to draw:** left renal vein crossing the aorta, with vessel loops on the vein and artery. For grade IV: kidney with laceration into a calyx, urinoma and segmental pseudoaneurysm.

> **Key points to remember:**
> - AAST 2018: segmental vessel injury grade IV; main vessel grade V.
> - Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
> - Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
> - Delayed CT phase is essential; blush means AE.
> - Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
> - Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
> - Revascularise only a solitary kidney or bilateral injuries.

## LAQ 7 ESWL: mechanism, indications, factors affecting outcome, complications and current status

P2

**Asked as:**

- Describe indication, mechanism of stone fragmentation and current status of ESWL? — *Summer 2019 Paper II Q2 (LAQ)*
- Describe factors affecting outcome of ESWL. Discuss complications of ESWL and strategies to avoid them. — *Summer 2021 Paper II Q1 (LAQ)* (Mechanism, indications and current status are answered in 2019-Summer Paper 2, Q2.)
- Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. — *Practice Set 13 Paper II Q2 (LAQ)* (also answered in 2021-Summer Paper 2, Q1; mechanism and indications in 2019-Summer Paper 2, Q2)

**Type:** LAQ (25 marks)

### Introduction

- ESWL (extracorporeal shock wave lithotripsy), now SWL, breaks stones with focused acoustic shock waves generated outside the body (generated at F1 and focused on the stone at F2). First human use 1980 (Dornier HM1).
- A shock wave is a short pulse: peak positive pressure 30–100 MPa, then a negative tail of about −10 MPa.
- Stones break by spallation, shear, squeezing and **cavitation**. Cavitation also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

### Components of a lithotripter

- Energy source at F1 (first focus); focusing system; coupling (water or gel); fluoroscopy and/or ultrasound to place the stone at F2. A focal zone wider than the stone improves breakage.

| Energy source | Principle | Advantages | Disadvantages |
|---|---|---|---|
| Electrohydraulic (HM3) | Underwater spark; ellipsoid reflector | Most effective | Electrode wear; variable shocks |
| Piezoelectric | Ceramic elements on a spherical dish | Long life; little pain | Low energy; tiny focal zone |
| Electromagnetic (modern machines) | Coil moves a metal membrane; lens focuses | Consistent; high energy at F2 | — |

### Mechanisms of stone fragmentation

- Each shock creates stress inside the stone. Above a critical value, cracks form; repeated shocks make them grow and join (**fatigue**).

| Mechanism | How it works |
|---|---|
| **Spallation** | Wave reflects and inverts at the back surface of the stone; tension breaks off the distal face |
| **Shear and tear (superfocusing)** | Waves inside the stone reflect and focus; cracks form where shear peaks |

| Mechanism | How it works |
|---|---|
| **Quasistatic squeezing** | Wave travels faster in stone than fluid; a ring of pressure compresses the stone |
| **Dynamic squeezing** | Shear waves at the stone edges combine; strongest force in initial breakage |
| **Cavitation** | Negative phase forms bubbles; collapse erodes the surface. Excess bubbles shield the stone and injure tissue |

## Indications

- **Renal (AUA/EAU):** <2 cm outside the lower pole and lower pole <1 cm (SWL or URS [ureteroscopy]); EAU: 10–20 mm equal to endourology; >20 mm only after PCNL (percutaneous nephrolithotomy).
- **Ureteric:** any level, best <10 mm; emergency SWL within 48 hours works well. Patients avoiding anaesthesia and stents; children.
- **Poor choices:** >2 cm or staghorn; lower pole >1 cm (**Lower Pole I**: stone-free 21% SWL vs 91% PCNL); cystine, brushite, calcium oxalate monohydrate; infection stones.

## Contraindications

- **Absolute:** pregnancy; uncorrected coagulopathy or (ongoing) antithrombotics; untreated UTI (urinary tract infection); obstruction distal to the stone; nearby arterial aneurysm (aneurysm near the stone; aortic >4 cm); severe uncontrolled (untreated) hypertension.
- **Relative:** obesity or skeletal deformity preventing targeting; pacemaker.

## Factors affecting (influencing) outcome / predictors of success

### 1. Stone factors

| Factor | Favourable | Unfavourable |
|---|---|---|
| Size (most important) | <10 mm | Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy) |
| Density on NCCT (non-contrast computed tomography / non-contrast CT [computed tomography]) | <900 HU (Hounsfield units) clears best | >1000 HU predicts failure |
| Composition | Calcium oxalate dihydrate, uric acid | Cystine, brushite, calcium oxalate monohydrate (dense, homogeneous) |
| Impaction (ureter) | Recent, mobile | Chronic impaction (soft-tissue ring of thick ureteral wall around the stone on CT, unchanged position): may be the strongest predictor |
| Burden and site | Single, renal pelvis or upper calyx | Multiple sites; lower pole >1 cm |

- Infection stones and stones in several sites: prefer a more efficient modality.

### 2. Renal anatomical factors

- **Lower pole (lower pole anatomy):** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR **21% SWL vs 91% PCNL**. A steep infundibulopelvic

angle and a long (>10 mm), narrow (<5 mm) infundibulum reduce clearance.
- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

**3. Patient factors**

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.
- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

**4. Machine and technique factors (technique to optimise results)**

1. **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
2. Treat infection; stop antithrombotics; control blood pressure. No routine stent: it does not improve clearance.
3. **Analgesia:** good analgesia, GA (general anaesthesia) or sedation keeps the stone in focus, because stone movement reduces breakage.
4. **Coupling:** gel as a mound, free of air; air pockets in the gel cut fragmentation by up to two-fifths.
5. **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks (bubbles clear between shocks), so less shielding and renal injury.
6. **Ramping:** start at low energy and step up (ramping energy up also helps); this causes protective vasoconstriction.
7. Positioning (prone for mid-ureteric stones over the sacrum), continuous imaging (monitor with imaging), experienced team. Little gain beyond two sessions (more than two sessions add little).
8. **Adjuncts:** α-blockers (faster fragment clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

## Complications

| Group | Complication | Comment |
|---|---|---|
| Fragment-related | Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments | Steinstrasse rises several-fold with burden >2 $cm^2$ |
| Infective | UTI (urinary tract infection), pyelonephritis, urosepsis (sepsis) | Infection stones, stents |
| Renal tissue injury | Haematuria (transient, common); **subcapsular or perirenal haematoma** (minority; rarely embolisation) | Risk: age, hypertension, obesity, diabetes, coronary disease, thrombocytopenia, anticoagulation, high energy, many shocks, narrow-focus machines |
| Other organs | Liver, spleen, pancreas, colon / bowel, lung (children), skeletal muscle | Very rare |
| Cardiac | Arrhythmia during treatment | Pause or ECG (electrocardiogram) gating |

| Group | Complication | Comment |
|---|---|---|
| Skin | Petechiae, bruising | Minor |

- **Refuted long-term effects** (**not** proven): new hypertension, diabetes, decline in (loss of) renal function and infertility.

### Strategies to avoid complications

1. **Select.** Absolute contraindications as above. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI, bowel reconstruction). Stop antithrombotics before and for the day and (about) 48 hours after. Control blood pressure; reprogramme / check pacemakers.
3. **Protect the kidney.** Slow rate; **slow ramping** (ramp over at least 4 minutes) (induces protective renal vasoconstriction and lowers haematoma risk); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

### Current status

- **Declining use worldwide.** URS has grown with small digital scopes and lasers, giving higher single-session stone-free rates. Newer lithotripters are also less effective than the HM3.
- **SWL vs URS:** SWL has fewer complications, no stent or general anaesthesia, lower cost, but more retreatment. Guidelines still list it first-line for small renal and ureteric stones.

### Recent advances

- **Burst wave lithotripsy:** bursts of 5–100 ultrasound cycles at 4–8 MPa; less cavitation; done awake (no anaesthesia); uniform fragments.
- **Ultrasonic propulsion** moves fragments to help them pass.

**Diagram to draw:** ellipsoid reflector with F1 (spark) and F2 (stone), stone and gel coupling, with spallation and squeezing arrows on the stone; box of failure predictors.

> **Key points to remember:**
>
> - Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
> - Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
> - Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
> - Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
> - Steinstrasse risk rises with burden >2 $cm^2$; no routine stent.
> - Use is declining as URS grows; burst wave lithotripsy is coming.

## LAQ 8 Management of upper ureteric stones

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 2 Paper 2, Q2 (SWL also answered in 2019-Summer Paper 2, Q2 and 2021-Summer Paper 1, Q1; flexible URS in 2019-Summer Paper 2, Q3b)

### Introduction

- **Upper (proximal) ureter:** UPJ (ureteropelvic junction) to the upper border of the sacroiliac joint.
- Proximal stones pass least often, so many need intervention. Dangers: renal damage and sepsis.

### Evaluation

- **NCCT (non-contrast CT)**: gold standard; gives size, density (HU, Hounsfield units), SSD (skin-to-stone distance) and impaction. KUB (kidney–ureter–bladder radiograph) for SWL follow-up; creatinine, urine culture, coagulation.

### 1. Emergency: drain first

- **AUA (American Urological Association) indications for urgent intervention:** infection with obstruction, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Obstructed infected kidney: drain only** (double-J stent or PCN, percutaneous nephrostomy; equal for sepsis); definitive treatment later. PCN suits large stones (stent fails if transverse diameter ≥9.5 mm).

### 2. Observation and MET (medical expulsive therapy): limits

| Spontaneous passage | Rate |
|---|---|
| By size | 1–4 mm 78%; 5–7 mm 60%; 8–10 mm 39% |
| By site | UVJ (ureterovesical junction) 92%; distal 75%; mid 60%; **proximal 48%** |

- **Observation** suits small (<5–6 mm) uncomplicated stones: NSAIDs (non-steroidal anti-inflammatory drugs) first.
- **MET (alpha-blockers, tamsulosin or silodosin):** reduce spasm so a larger urine bolus pushes the stone. Benefit is clear only for **distal stones 5–10 mm**; meta-analysis shows **no benefit for proximal stones**. SUSPEND (2015) showed no overall benefit.
- Trial of passage is limited to **4–6 weeks**, with imaging.

### 3. Choice of active treatment

- Factors: stone (size, density, impaction), patient (habitus, anticoagulation, preference), surgeon and equipment.

**EAU (European Association of Urology) recommendation for proximal ureteric stones**

| Size | First choice | Alternative |
|---|---|---|
| <10 mm | SWL or URS (ureteroscopy) | – |
| >10 mm | URS (retrograde or antegrade) | SWL |
| Large (>15–20 mm), impacted | PCNL with antegrade URS | Laparoscopic ureterolithotomy |

**SWL versus URS**

| | SWL (shock wave lithotripsy) | URS |
|---|---|---|
| Stone-free rate (AUA/EAU meta-analysis, approximate) | <10 mm about 90%; >10 mm about 70% | <10 mm about 80%; >10 mm about 80% |
| Sessions | More retreatment | Single session; higher SFR at 2 weeks, equal at 3 months |
| Anaesthesia | Sedation or none | Usually general |
| Complications | Fewer | Perforation, stent pain |
| Anticoagulation | Must stop | Laser URS can continue |

- **SWL fails** with size >10–15 mm, impaction (CT "ring" sign), HU >1000, SSD >10 cm, cystine or COM (calcium oxalate monohydrate) stones.
- **SWL technique:** supine; 60–90 shocks/min; ramp energy; no routine pre-stent. Contraindicated in pregnancy, bleeding disorder, untreated UTI (urinary tract infection), distal obstruction, uncontrolled hypertension.

### 4. Ureteroscopy for proximal stones

- **Flexible URS** preferred: it follows a stone that migrates into the kidney.
- Safety wire; access sheath (12/14 Fr); tight ureter: pre-stent and return after 2–4 weeks.
- **Laser:** holmium:YAG (2120 nm) or thulium fibre laser; dusting or fragmenting; keep **≤10 W in the ureter** to avoid thermal injury.
- **Retropulsion** into the kidney: reduced by low-energy settings and antiretropulsion devices (Stone Cone, NTrap); pneumatic lithotripsy works poorly above the brim.
- **Push-back:** stone deliberately pushed into the renal pelvis with a catheter, then treated by SWL, RIRS (retrograde intrarenal surgery) or PCNL; now largely replaced by in-situ flexible URS.
- Stent only for perforation, impaction, infection or a planned second stage.

### 5. Antegrade approach (PCNL with antegrade URS)

- Indications: large (>15–20 mm), hard or **impacted proximal stones**, failed retrograde access, urinary diversion, transplant, concomitant renal stones.
- **Upper or mid-pole posterior calyx** gives a straight line to the UPJ; flexible nephroscope or antegrade URS down the dilated ureter; high SFR.

### 6. Laparoscopic (or robotic) ureterolithotomy

- Indications: large impacted stones, failed SWL/URS, where endourology is unavailable, or with concomitant UPJ obstruction (pyeloplasty).
- Steps: retroperitoneal or transperitoneal access; identify the bulge; **clamp the ureter above the stone** to prevent migration; longitudinal ureterotomy; remove; double-J stent; close with 4-0 absorbable sutures; drain.
- High single-stage SFR; risk of urine leak. **Open ureterolithotomy** is now rare.

### Complications of treatment

| Complication | Points |
|---|---|
| Perforation | 0–4%; stop, stent about 4 weeks |

| Complication | Points |
|---|---|
| **Avulsion** | 0.06–0.5%; **proximal third most at risk**; nephrostomy and delayed reconstruction |
| Stricture | 3–6%; image all after URS |
| Sepsis | About 5% after URS |
| SWL | Steinstrasse, haematoma, colic |

**Also know:** pregnancy: conservative, then stent or URS (no SWL).

### Recent advances

- Single-use digital scopes; pressure sensors; thulium fibre laser; robotic URS; **BWL (burst wave lithotripsy)**.

**Diagram to draw:** algorithm for proximal stone: drain if infected; <10 mm SWL or URS; >10 mm URS; large impacted antegrade or laparoscopic.

> **Key points to remember:**
>
> - Proximal stones pass in only 48%; MET does not help proximal stones.
> - <10 mm: SWL equals URS; >10 mm: URS first (EAU).
> - Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
> - Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

## LAQ 9 Uric acid calculi: etiopathogenesis, medical management and allopurinol

P2

**Asked as:**

- Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, — *Practice Set 9 Paper II Q2 (LAQ)*
- Allopurinol — *Practice Set 10 Paper II Q3a (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

### Introduction

- Uric acid stones form **8–10%** of stones in the USA and are increasing with obesity and diabetes. They are **radiolucent** and can be **dissolved medically**.

### Etiopathogenesis

#### 1. Why humans form uric acid stones

- Humans and apes lack hepatic **uricase**, which converts uric acid to allantoin (10–100 times more soluble). Blood and urine uric acid are therefore high.

#### 2. Physical chemistry: pH is the key

- Uric acid is a weak acid, **pKa about 5.35–5.5**. At pH 5.5 half is free uric acid and half is urate.
- Free uric acid is poorly soluble (about 96–100 mg/L); sodium urate is about 20 times more soluble. At **pH 6.5, over 90% is ionised**.
- **No natural inhibitor** of uric acid crystallisation is known, so stone formation depends almost entirely on supersaturation.

#### 3. The three determinants

| Determinant | Causes | Importance |
|---|---|---|
| **Low urine pH (<5.5)** | Insulin resistance, obesity, type 2 diabetes, gout, high animal-protein diet, chronic diarrhoea, ileostomy | **Most important**; most formers have normal uric acid excretion |
| **Low urine volume** | Hot climate or occupation, diarrhoea, dehydration | Concentrates uric acid |
| **Hyperuricosuria (>600 mg/day)** | Purine-rich diet, gout, myeloproliferative disorders, tumour lysis, uricosuric drugs, inherited defects | Least often the driver in idiopathic cases |

#### 4. Why urine is too acid: insulin resistance

- Insulin normally stimulates **ammonia production** from glutamine in the proximal tubule and the $Na^+/H^+$ exchanger NHE3.
- Insulin resistance reduces ammonium excretion. Acid is then buffered as titratable acid, so urine pH falls.
- **Renal lipotoxicity:** fat accumulates in the proximal tubule, which then burns fatty acids instead of glutamine, so ammoniagenesis falls further.
- Endogenous acid production is also higher. Uric acid stones are thus a **renal manifestation of the metabolic syndrome**; about a third of stones in diabetics are uric acid.

**5. Aetiological categories**

- **Congenital:** HGPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency (Lesch–Nyhan), PRPP (phosphoribosylpyrophosphate) synthetase overactivity, hereditary renal hypouricaemia (URAT1, the urate transporter).
- **Idiopathic:** low urine pH without gout ("gouty diathesis").

**6. Uric acid and calcium stones**

- In **hyperuricosuric calcium oxalate stone disease**, urate promotes calcium oxalate by removing inhibitors, by "salting out" and by **epitaxy** (urate crystals act as a nidus).
- **Also know:** ammonium acid urate stones (laxative abuse, diarrhoea, ileostomy); theobromine (cocoa) inhibits uric acid crystallisation in vitro.

## Diagnosis in brief

- Radiolucent on X-ray; CT **200–400 HU (Hounsfield units)**; **dual-energy CT** identifies uric acid.
- Urine pH persistently <5.5; serum uric acid; 24-hour urine; screen for diabetes.

## Rationale of medical management

**Principle:** uric acid stones are a **"pH disease" more than a uric acid disease**. Raising pH turns insoluble uric acid into soluble urate. Diluting urine lowers supersaturation. Lowering uric acid production helps only when excretion is high.

**1. Fluids**

- Urine output **at least 2.5 L/day** lowers uric acid concentration.

**2. Diet and lifestyle**

- Limit non-dairy **animal protein** (it gives purines and an acid load; dietary purines provide about 30% of urinary uric acid).
- More fruit and vegetables (alkali load). Weight loss, exercise and glucose control correct the underlying insulin resistance.

**3. Urinary alkalinisation (mainstay)**

- **Potassium citrate 30–60 mEq/day** in two or three doses. Citrate is metabolised to bicarbonate, raising urine pH.
- **Target pH 6.0–6.5** for prevention (EAU [European Association of Urology]: 6.2–6.8); **avoid pH >7.0**, which risks calcium phosphate stones.
- **Why potassium, not sodium:** sodium alkali raises urinary calcium and can cause calcium stones. Sodium bicarbonate or sodium citrate is used if potassium is unsafe (renal impairment, hyperkalaemia).
- Patients **monitor urine pH with dipsticks**.

**4. Dissolution (oral chemolysis)**

- Pure uric acid stones dissolve over weeks to months. **EAU: pH 7.0–7.2**; stent or nephrostomy if obstructed or infected; monitor by imaging.

**5. Xanthine oxidase inhibitors**

- **Allopurinol 300 mg/day** lowers uric acid production. It is **not first-line**, because low pH, not hyperuricosuria, is the usual cause.
- Indications: hyperuricosuria with hyperuricaemia (e.g. gout), recurrence despite alkalinisation (AUA, American Urological Association), hyperuricosuric calcium oxalate

stones.

- Reduce the dose in renal impairment. Risks: rash; **allopurinol hypersensitivity syndrome** (2–8% have reactions; may be fatal); Stevens–Johnson syndrome.
- **Febuxostat** is an alternative. **Rasburicase** (recombinant uricase) prevents uric acid nephropathy in tumour lysis.

**Landmark trial: Ettinger (1986)**, randomised placebo-controlled trial: allopurinol more than halved recurrence in hyperuricosuric, normocalciuric calcium oxalate stone formers.

## Allopurinol (in detail)

### Introduction and mechanism

- A **purine (hypoxanthine) analogue** that inhibits **xanthine oxidase (xanthine dehydrogenase)**. This blocks hypoxanthine → xanthine → uric acid, lowering serum and urinary uric acid.
- Metabolised in the liver to **oxypurinol**, the active metabolite. It has a long half-life and is cleared by the kidney, so the dose is cut in renal impairment.

### Uses in urology

| Indication | Rationale |
|---|---|
| **Recurrent calcium oxalate stones with hyperuricosuria** (>600 mg/day) and normal urinary calcium | Urate "seeds" calcium oxalate (heterogeneous nucleation, salting out, inhibitor adsorption). **Ettinger (1986) RCT (randomised controlled trial): recurrence more than halved**; AUA (American Urological Association) recommends it |
| Uric acid stones | **Not first line:** low urine pH is the main cause, so alkalinise with potassium citrate to pH 6.0–6.5. Allopurinol is added for hyperuricosuria, gout or failure of alkalinisation |
| 2,8-DHA (dihydroxyadenine) stones (APRT (adenine phosphoribosyltransferase) deficiency) | Blocks adenine → DHA; febuxostat more effective |
| Tumour lysis prophylaxis; gout; HPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency, PRPS (phosphoribosylpyrophosphate synthetase) overactivity | Prevents urate nephropathy |

### Dose

- Adult **300 mg/day** (start 100 mg, titrate); child 4–10 mg/kg/day (maximum 300 mg). Reduce for low creatinine clearance.

### Adverse effects

- Rash (commonest), myalgia, gastrointestinal upset; gout flare on starting; interstitial nephritis.
- **Hypersensitivity in 2–8%**, from skin reactions to **allopurinol hypersensitivity syndrome** (fatal in up to 27%) and Stevens-Johnson syndrome. Risk is higher with high dose, renal failure, diuretics and statins. **HLA-B*58:01** carriers (some Asian populations) have higher risk.
- **Xanthine stones** (radiolucent): rare, mainly with high doses in Lesch-Nyhan syndrome.

**Drug interactions**

- **Azathioprine and 6-mercaptopurine:** xanthine oxidase metabolises them, so toxicity (marrow suppression) rises. Reduce their dose sharply (important after transplant).

**Alternatives**

- **Febuxostat:** non-purine xanthine oxidase inhibitor, no dose change in mild to moderate renal impairment. Carries a cardiovascular-death warning.
- **Also know:** rasburicase (urate oxidase) for tumour lysis.

**Diagram to draw:**

- Sigmoid curve of uric acid solubility against urine pH, marking pKa 5.5 and target 6.0–6.5.
- Purine pathway (hypoxanthine → xanthine → uric acid) showing the xanthine oxidase block.

> **Key points to remember:**
>
> - Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
> - Insulin resistance reduces ammoniagenesis, so urine is acid.
> - Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
> - Allopurinol only for hyperuricosuria or failed alkalinisation.
> - Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
> - Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
> - Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

## LAQ 10 Urinary crystals

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 11 Paper 2, Q2

### Introduction

- Crystals are precipitated urinary solutes seen on microscopy of the sediment. They occur in both normal and diseased states.
- Their value lies in suggesting stone type, metabolic disease, infection or drug toxicity. **Cystine is the one crystal that is never normal.**

### How crystals form (physicochemistry)

- Urine is normally **supersaturated** with calcium oxalate and phosphate. Crystals form only when the concentration product passes a critical level.

| Saturation zone | What happens |
|---|---|
| Undersaturated (below Ksp, solubility product) | No crystals; stones may dissolve |
| Metastable (Ksp to Kf, formation product) | No spontaneous nucleation; growth on existing crystals; inhibitors effective |
| Unstable (above Kf) | Spontaneous nucleation, growth and aggregation |

- **Nucleation** in vivo is mostly heterogeneous (on cell debris or other crystals). Crystals then grow, aggregate and are retained, for example on **Randall plaque**.
- **Inhibitors:** citrate, magnesium, pyrophosphate, Tamm–Horsfall protein, nephrocalcin, osteopontin. No inhibitor exists for uric acid or cystine, so these depend on pH and concentration.
- **Determinants of crystalluria:** urine volume, solute excretion, pH, temperature and inhibitors.

### Examination of urine for crystals

- Fresh midstream specimen, examined within 2 hours at body or room temperature. **Refrigeration precipitates crystals** that may not redissolve.
- Dipstick pH (physiological range 4.5–8.0); centrifuged sediment; polarised light helps identify crystals.
- Phosphaturia (cloudy alkaline urine) **clears with acetic acid**; pyuria does not.
- Candida mimics calcium oxalate and red cells; budding and hyphae separate it.

### Classification by urine pH and shape

| Urine pH | Crystal | Shape | Clinical significance |
|---|---|---|---|
| Acidic | **Uric acid** | Rhomboid or lemon-shaped plates, rosettes; yellow-brown | Uric acid stones (pH <5.5; pKa 5.35), gout, tumour lysis |
| Acidic | Amorphous urates | Pink "brick-dust" granules | Usually normal (concentrated, cooled urine) |

| Urine pH | Crystal | Shape | Clinical significance |
|---|---|---|---|
| Acidic | **Cystine** | **Hexagonal** colourless plates | Always abnormal: cystinuria |
| Acidic or alkaline | **Calcium oxalate dihydrate** | **Envelope** (octahedral) | Common, often normal; hypercalciuria |
| Acidic or alkaline | Calcium oxalate monohydrate | Dumbbell or oval | Hyperoxaluria; ethylene glycol poisoning |
| Alkaline | **Triple phosphate (struvite)** | **Coffin-lid** prisms | Urease-producing infection (Proteus); staghorn stones |
| Alkaline | Amorphous phosphate | White granules | Benign phosphaturia |
| Alkaline | Calcium phosphate (brushite, apatite) | Needles or rosettes; amorphous | RTA (renal tubular acidosis), hyperparathyroidism |
| Alkaline | Ammonium biurate | "Thorny apple" spheres | Old specimens; ammonium urate stones (diarrhoea, laxative abuse, ileostomy) |
| Any | Cholesterol | Notched flat plates | Nephrotic syndrome, lipiduria |

- **Rare metabolic crystals:** 2,8-dihydroxyadenine, brown spiculated spheres (APRT [adenine phosphoribosyltransferase] deficiency, radiolucent stones); xanthine (hereditary xanthinuria, high-dose allopurinol); tyrosine needles and leucine spheres (severe liver disease).

### Drug crystals

- **Two routes:** the drug itself precipitates (direct), or it changes urine chemistry (indirect).

| Drug | Crystal and features | Prevention |
|---|---|---|
| **Indinavir** (protease inhibitor) | Needles or fan-shaped plates; crystalluria in about two-thirds of new users; stones 4–13%; radiolucent, often invisible on CT | Hydration; more soluble below pH 5.5 |
| Atazanavir | Highest stone risk of newer antiretrovirals | Hydration |
| Sulfonamides (sulfadiazine) | "Sheaves of wheat" | Hydration, alkalinise |
| Acyclovir (high-dose intravenous) | Birefringent needles; acute kidney injury | Slow infusion, hydration |
| Amoxicillin, ciprofloxacin (high dose) | Needles; worse in alkaline urine | Hydration |
| Methotrexate (high dose) | Crystal nephropathy | Alkalinise, hydration |
| Triamterene, ephedrine, guaifenesin, silicate antacids | Stones of drug or metabolite | Stop the drug |

- **Indirect:** loop diuretics and steroids (hypercalciuria, calcium stones); acetazolamide and topiramate (alkaline urine, hypocitraturia, calcium phosphate); cytotoxics (uric acid); laxative abuse (ammonium urate).

### Clinical significance

1. **Diagnosis of stone type** before stone analysis is available; crystals in a stone former help direct metabolic work-up.

2. **Recurrence risk:** persistent crystalluria in repeated early-morning samples, and large or aggregated crystals, suggest active stone disease.
3. **Infection:** struvite crystals with urine pH >7.2 signal a urease-producing organism.
4. **Acute kidney injury:** uric acid (tumour lysis), calcium oxalate monohydrate (ethylene glycol) and drug crystals cause intratubular obstruction.
5. **Monitoring therapy:** absence of cystine or uric acid crystals confirms adequate alkalinisation. - Most crystalluria in healthy people is transient and harmless.

### Management principles

- Repeat on a fresh, warm specimen; confirm pH; 24-hour urine metabolic evaluation and stone analysis.
- **Fluids** for urine output above 2–2.5 L/day.
- **pH manipulation:** alkalinise (potassium citrate) for uric acid, cystine, methotrexate and sulfonamides; avoid alkali in calcium phosphate and struvite.
- Treat the cause: antibiotics and stone clearance for struvite; stop or change the drug; specific therapy (thiols for cystine, allopurinol for uric acid).

**Diagram to draw:** sketches of hexagon (cystine), envelope (calcium oxalate), coffin lid (struvite), rhomboid (uric acid), sheaf of wheat (sulfonamide).

> **Key points to remember:**
> - Cystine hexagons are always pathological.
> - Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
> - Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
> - Refrigerated urine precipitates crystals: examine fresh within 2 hours.

## LAQ 11 Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

P3

**Type:** LAQ (25 marks)
**Source:** 2015-Summer Paper 3, Q1

### Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.

### Types and materials

| Feature | Details |
|---|---|
| Size | 3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm |
| Polymers | Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain |
| **Metallic (Resonance)** | Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis |
| Others | Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental) |
| Designs | Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit) |

**Also know:** anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

**Diagram to draw:** double-J stent in situ with both coils and side holes.

### Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

### Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register.

- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear encrustation; II circular encrustation; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

### Ureteroscopy

| | Semirigid | Flexible |
|---|---|---|
| Tip | 7 Fr or less | 4.5–7.5 Fr; 3.6 Fr channel |
| Use | Distal and mid ureter | Upper ureter, kidney |
| Deflection | None | Over 270° |

- Segura's dictum: always a safety wire. Tight ureter: stent and return after **2–4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **TFL (thulium fibre laser, about 1940 nm)**. **Dusting** (low energy, high frequency) is faster but leaves fines; **fragment and basket** confirms stone clearance and gives stone for analysis. Keep pressure and temperature low with outflow.

**Steps of URS**: (1) sterile urine, antibiotic dose; (2) cystoscopy and **safety guidewire**; (3) semirigid inspection, dilate only if needed; (4) access sheath over a working wire under fluoroscopy (for flexible URS); (5) low-pressure irrigation, laser, basket; (6) inspect the ureter on withdrawal; (7) decide on stent.

- **Also know:** anti-retropulsion devices; digital and single-use scopes; URS suits patients who cannot stop anticoagulants.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

### Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU (European Association of Urology)** agrees.
- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

```
Uncomplicated URS -> no stent (or short string stent) Complicated -> double-J: injury 2-6 wk; second stage until re-look
```

### Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

**Key points to remember:**

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

## LAQ 12 Laparoscopic pyeloplasty.

P3

**Type:** LAQ (25 marks)
**Source:** 2016-Summer Paper 3, Q2

### Introduction

- **PUJO (pelviureteric junction obstruction)** is impaired flow from pelvis to ureter that, if untreated, causes progressive renal damage (Koff).
- Causes: **intrinsic** adynamic segment (abnormal circular muscle), **extrinsic** crossing lower-pole vessel, high insertion; secondary (failed repair, stone).
- **Anderson–Hynes (1949) dismembered pyeloplasty** is the gold standard by any route. First laparoscopic pyeloplasty **1993 (Schuessler)**; robotic **2002**. Success **over 90–95%**, equal to open, with less morbidity.

### Indications

- Symptoms (pain, Dietl's crisis), infection, stones, hypertension (rare).
- Impaired split function or **fall of 10% or more** on serial renography; rising APD (anteroposterior diameter); obstructed solitary kidney or bilateral disease.
- Children: most antenatal hydronephrosis resolves; about one-third need surgery.
- **Contraindications**: uncorrected coagulopathy, untreated UTI (urinary tract infection), cardiopulmonary disease barring pneumoperitoneum.
- **Nephrectomy instead** if split function under 15% in adults with a normal other kidney.

### Preoperative evaluation

- **USG (ultrasonography)**: large pelvis, small calyces, ureter not seen.
- **MAG3 (mercaptoacetyltriglycine) diuretic renogram**: split function and an obstructive curve that does not wash out after furosemide. Usual teaching: washout half-time under 10 minutes normal, 10–20 minutes equivocal, over 20 minutes obstructed (Ch 93 quotes over 40 minutes). Split function under 40% supports repair.
- **CT (computed tomography) angiography**: anatomy, **crossing vessels**, stones.
- **Retrograde pyelogram** at surgery: confirms the level, excludes ureteric stricture.
- **Whitaker test** if equivocal: perfusion 10 mL/min; pelvic pressure over 15–20 cm H2O means obstruction.
- Sterile urine. Drain beforehand (stent or nephrostomy) only for infection, azotaemia or severe pain.

### Approaches

| | **Transperitoneal** | **Retroperitoneal** |
|---|---|---|
| Space | Large, familiar landmarks | Small, created by balloon |
| Bowel | Colon reflected (left: **transmesocolic** window) | Not entered; leak stays extraperitoneal |
| Use | Commonest; easier suturing | Avoids bowel; harder orientation |

| | Transperitoneal | Retroperitoneal |
|---|---|---|
| - One paediatric randomised trial favoured retroperitoneal (shorter stay); overall outcomes similar. | | |

### Steps (transperitoneal Anderson–Hynes)

1. Cystoscopy, retrograde pyelogram, catheter.
2. 45–60° flank; pneumoperitoneum 12–15 mmHg in adults (under 12 mmHg in children); 3–4 ports, camera at umbilicus.
3. Reflect colon medially, or open the mesocolon over the pelvis.
4. Find the ureter; trace it to the PUJ with its periureteric tissue; minimal diathermy. Look for a **crossing vessel**.
5. Hitch stitch on the pelvis; transect the pelvis above the PUJ; **reduce** a redundant pelvis if needed.
6. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a handle (no-touch).
7. **Transpose** the PUJ in front of any crossing vessel.
8. Anastomose the spatulation apex to the most dependent pelvis with 4-0 absorbable (5-0/6-0 in infants): back wall, stent, front wall, then close the pelvis. Watertight, tension-free, funnel-shaped.
9. **Antegrade double-J** over a wire; clamp the catheter: urine from the stent side holes confirms the bladder coil.
10. Excise the narrow segment; drain near (not on) the suture line.

### Crossing vessel

- It lowers endopyelotomy success (42% vs 86%), so pyeloplasty is preferred. **Never divide** a lower-pole vessel (segmental infarction).
- **Vascular hitch (Hellström)**: vessel moved up onto the pelvis, only if the PUJ is intrinsically normal and peristalsing.

### Stenting and postoperative care

- Double-J for **4–6 weeks** (children 1–4 weeks: no difference). Externalised stent in infants avoids a second anaesthetic.
- Foley for 24–48 hours (bladder pressure refluxes up the stent); drain out when dry. If drain output rises after Foley removal, replace the Foley.
- **MAG3 renogram** after stent removal, then ultrasound; follow for years.

### Robotic pyeloplasty

- 3D vision, wristed suturing, tremor filtration, **shorter learning curve**. Now first-line at major centres.
- Success 94.7–100%, the same as laparoscopy; advantage is ergonomic, at higher cost.

### Outcomes

- Laparoscopic: operating time 2–4 hours; complications 2–16%; conversion 0–5.5%; stay about 2.7 vs 4.2 days (open); success 87–99%.
- **Up to 30% of failures appear after 2 years**, so follow-up must be long.

### Complications

- Urine leak and urinoma, stent symptoms or migration, UTI, haematuria, ileus, bowel or vessel injury, port-site hernia, recurrent obstruction.

### Management of failure

`Early (clot, oedema) -> nephrostomy, wait Persistent obstruction -> CT angiography (missed vessel?) short, no vessel -> endopyelotomy (about 70%) otherwise -> redo laparoscopic/robotic pyeloplasty (about 89% after failed open; buccal graft onlay) small intrarenal pelvis -> ureterocalicostomy poor function -> nephrectomy`

**Also know:** Y-V plasty (high insertion); Culp spiral flap (long narrowing); stones removed through the pyelotomy with a flexible scope; safe in horseshoe and pelvic kidneys.

### Recent advances

- Single-port robot (2018); indocyanine green fluorescence for redo cases; buccal mucosa ureteroplasty.

**Diagram to draw:** Anderson–Hynes steps: transection, lateral spatulation, transposition over the crossing vessel, dependent anastomosis.

> **Key points to remember:**
>
> - Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
> - Transpose over a crossing vessel; never divide it.
> - Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
> - Robotic equals laparoscopic in outcome, with easier suturing.

## LAQ 13 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury

P3

**Asked as:**

- Iatrogenic ureteric injury — *Practice Set 2 Paper III Q1 (LAQ)* (upper ureteric injury also answered in 2023-Winter Paper 3, Q2)
- Discuss management of ureteric injuries occurring during gynec surgeries — *Practice Set 4 Paper III Q1 (LAQ)* (also answered in Practice Set 2 Paper 3, Q1)
- Discuss Etiology, Diagnosis and Management of Upper ureteric injury. — *Winter 2023 Paper III Q2 (LAQ)*

**Type:** LAQ (25 marks)

*Assumed scope: surgical (iatrogenic) injury at any level of the ureter, in gynaecological, colorectal, vascular and urological surgery. Assumed meaning (upper ureteric injury): injury to the upper (proximal) ureter, from the PUJ (pelviureteric junction) to the pelvic brim, from external trauma or surgery.*

### Introduction

- Ureteric trauma is uncommon: 1–2.5% of genitourinary trauma. **Iatrogenic injury is commoner overall** than external trauma.
- **The proximal ureter is the commonest site of external injury. The distal ureter is the commonest (usual) iatrogenic site** in non-urological surgery.
- Gynaecological surgery has historically caused about half of all iatrogenic ureteric injuries; hysterectomy is the main culprit.
- It is mainly a problem of **recognition** (the core problem): thermal, ligation and devascularisation injuries are invisible at operation and declare themselves days to weeks later. Injury is often **missed** and declares itself later as urinoma, sepsis, stricture or fistula.
- Late diagnosis brings urinoma, sepsis, AKI (acute kidney injury), fistula and renal loss, and needs nephrostomy about ten times more often.

### Aetiology and incidence (EAU (European Association of Urology) trauma guideline data)

#### 1. External trauma

- **Penetrating** (most): gunshot wounds up to about 90%. Blast damages the intramural blood supply up to 2 cm from the wound.
- **Blunt** (up to about 40%): rapid deceleration avulses the ureter at the **PUJ**; linked to lumbar transverse process fractures.
- **Over 90% have associated injuries** (small bowel commonest, colon, vessels, kidney 10–28%). Mortality 6–9% reflects these.

#### 2. Iatrogenic

| Source | Key facts (upper-ureter relevance) |
|---|---|
| **Gynaecological** | Historically about half. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery highest (1.7–4.3%). Endometriosis and placenta accreta raise risk |

| Source | Key facts (upper-ureter relevance) |
|---|---|
| Colorectal | 0.15–10%; laparoscopic equals open |
| Colorectal, spinal, retroperitoneal surgery | Ligation, transection, cautery |
| **Urological / Ureteroscopy** | Ureteroscopy is the commonest (iatrogenic) cause in some series: abrasion, perforation 0.2–2%, avulsion under 0.5% (another source: <0.3%) (proximal third most at risk); stricture about 3% (another source: 0.3–3%). Risk: proximal stone, impacted stone, long operation, large sheath, inexperience. Also radical prostatectomy, partial nephrectomy or ablation (late proximal stricture) |
| Partial nephrectomy, thermal ablation | Proximal stricture months later (lower-pole and hilar tumours) |
| Vascular (aortic) surgery | Aorto-iliac bypass: transient hydronephrosis 12–20%; up to 85% unrecognised (missed initially). Devascularisation. Rare ureteroarterial fistula (chronic stent plus pelvic surgery or radiation) |

### Gynaecological incidence (EAU trauma guideline data)

| Procedure | Ureteric injury (%) |
|---|---|
| Caesarean section (emergency) | 0.01–0.06 |
| Vaginal hysterectomy | 0.02–0.5 |
| Abdominal hysterectomy | 0.03–2.0 |
| Minimally invasive hysterectomy | 0.13–6.0 (laparoscopic = robotic, about 0.9) |
| Prolapse surgery | **1.7–4.3 (highest)** |

- **Danger sites in the pelvis (applied anatomy):** pelvic brim near (at) the infundibulopelvic ligament; **under the uterine artery** ("water under the bridge"); beside the uterosacral ligament; vaginal fornix / vaginal cuff near the UVJ (ureterovesical junction).
- Up to a third are recognised during open surgery, fewer during laparoscopy; the uterosacral region is hardest to see.
- **Mechanisms:** ligation, crush, laceration, transection, avulsion, thermal injury, **devascularisation**. About 10% of women get a large share of distal ureteric blood from the uterine artery (the uterine artery supplies much of the distal ureter), so some (ischaemic) injuries are unavoidable.
- **Risk factors:** previous surgery, adhesions, malignancy, radiotherapy, heavy bleeding, long operation, learning curve; endometriosis (encasement, adhesions, medial deviation), placenta accreta.
- **URS (ureteroscopy) risks:** long operation, 12/14-Fr sheath, impacted or proximal stone, inexperience.

## Classification: AAST (American Association for the Surgery of Trauma) ureter scale

| Grade | Injury (description) |
|---|---|
| I | Haematoma or contusion, no devascularisation (without devascularisation) |
| II | Laceration <50% of circumference |
| III | Laceration ≥50% |

| Grade | Injury (description) |
|---|---|
| IV | Complete transection, <2 cm devascularised (with <2 cm devascularisation) |
| V | Avulsion, >2 cm devascularised (with >2 cm devascularisation) |

- Advance one grade for bilateral injury, up to grade III.

## Prevention

- See (identify) the ureter at the **pelvic brim** (crossing the common iliac artery) and trace it before ligating pedicles. Good haemostasis and exposure.
- **Prophylactic stents** (or lighted stents) help recognition in high-risk (complex) cases, but do not clearly reduce (prevent) injury. **ICG (indocyanine green)** instilled into the ureter (intraureteric) makes it fluoresce (fluoresces) for hours.
- **Cystoscopy** after minimally invasive hysterectomy and prolapse surgery, with oral phenazopyridine or IV (intravenous) dye to show efflux.
- URS: safety wire; ≤14-Fr sheath in an unstented ureter; fragment before basketing; "stop, stent, stage".

## Diagnosis and recognition

### Suspicion (trauma)

- Penetrating flank or abdominal wound; high-energy deceleration.
- **Haematuria is absent in 25–45%**, so its absence does not exclude injury.

### Intraoperative (best outcome)

- Inspect the ureter (at laparotomy); IV (intravenous) indigo carmine or methylene blue; cystoscopy for efflux (jets), with retrograde (on-table) pyelogram or stent. Check the other ureter and bladder. About 11% are missed at trauma laparotomy.
- Methylene blue is an MAO (monoamine oxidase) inhibitor: avoid with SSRIs/SNRIs (selective serotonin / serotonin–noradrenaline reuptake inhibitors; serotonin toxicity), G6PD (glucose-6-phosphate dehydrogenase) deficiency and pregnancy.

### Delayed presentation (most cases) / postoperative

- Mean diagnosis about day 10: fever, ileus, flank pain, leucocytosis, rising creatinine, rising drain output, drain or wound urine leak, urinoma, anuria (bilateral), or **ureterovaginal fistula** (UVF): normal voiding plus constant leak, because the other kidney fills the bladder, usually within 4 weeks.

### Imaging and tests

| Test | Role |
|---|---|
| **CT (computed tomography) urography with 10-minute delayed images** (both ureters) | Best test in stable patients. Signs: extravasation, non-opacified distal ureter, delayed nephrogram, urinoma |
| **Retrograde pyelography** | Most sensitive; shows site; allows stenting at once |
| Antegrade nephrostogram | When retrograde fails or gap is wide |
| One-shot IVU (intravenous urography) | Non-diagnostic in 33–100%; never rely on it |

- **Drain-fluid (or vaginal fluid) creatinine** above (far above) serum creatinine confirms urine.
- **Double-dye test** separates VVF (vesicovaginal fistula) from ureterovaginal fistula: oral phenazopyridine plus intravesical methylene blue. Orange pad = UVF; blue = VVF; both =

both.

## Management

### Principles

- **Stability decides timing; level and length decide the operation.**
- Repair: mobilise sparing (preserve) adventitia; debride to bleeding tissue (edges); **spatulated, tension-free, watertight, stented** repair (anastomosis) with fine absorbable suture; drain; retroperitonealise; omentum (omental wrap) for thermal, radiation, blast or vascular injury; non-tunnelled reimplant.
- **No bowel interposition or autotransplantation acutely.**

**By timing** ``` Recognised at surgery -> repair at once (stable patient) Unstable -> ligate + later nephrostomy, or exteriorised stent Recognised late -> try retrograde stent (or nephrostomy + antegrade stent)

|-- succeeds -> stent 4-8 weeks, then image |-- fails -> nephrostomy; early repair if <=7 days, otherwise delayed repair after about 6 weeks Uncontrolled sepsis / collections -> drain; repair regardless of timing Ureteric injury suspected

|-- Unstable -> damage control: ligate + later PCN, | or exteriorised stent / cutaneous ureterostomy |-- Stable, found at surgery -> immediate repair over stent |-- Found late / postoperatively -> Attempt retrograde stent (+/- PCN, drain) -> Fails -> PCN; delayed definitive repair Suspected late injury -> retrograde stent (or nephrostomy + antegrade stent)

```
|-- succeeds -> stent 4-8 weeks, then image
|-- fails -> nephrostomy
     |-- found <=7 days -> early repair
     |-- later -> repair after about 6 weeks
```

Sepsis / uncontrolled collections -> drain; repair regardless of timing ```

### At surgery, by injury

| Injury | Action |
|---|---|
| Suture through wall, minor thermal | Remove suture; stent |
| Partial tear | Close transversely (transverse closure, 4-0/5-0) over stent, 4–8 weeks |
| Ligation | Remove ligature (deligate); stent if viable, excise and repair if not |
| Transection, lower ureter (usual) | **Ureteroneocystostomy ± psoas hitch** (95–100% success) |
| Long distal gap | **Boari flap** + hitch (gaps up to 12 cm) |
| Transection above pelvic brim | Ureteroureterostomy |
| Unstable patient | Ligate + later nephrostomy, or exteriorised stent |

### Repair by grade

- **Grade I–II (contusion, partial):** double-J stent for about 6 weeks; or close a laceration transversely over a stent. Excise non-viable contused ureter.
- **Grade III–V:** formal reconstruction.

### By level (complete injury)

| Level | Repair |
|---|---|
| Upper | Ureteroureterostomy; ureteropyelostomy (PUJ (pelviureteric junction) avulsion); ureterocalicostomy |
| Mid | Ureteroureterostomy ± downward nephropexy; TUU (transureteroureterostomy) when delayed |
| Lower (commonest iatrogenic) | **Ureteroneocystostomy, non-tunnelled ± psoas hitch** (95–100% success); **Boari flap** for gaps up to 12 cm |
| Long loss (delayed only) | Ileal ureter or Yang–Monti; buccal graft ureteroplasty; autotransplantation (last resort) |

### Options for the upper ureter

| Procedure | Key points |
|---|---|
| **Ureteroureterostomy** | Commonest acute repair; short defects (2–3 cm); spatulate 5–6 mm, 180° apart; 5-0 absorbable over double-J; up to 90% success; leak 10–24%, stricture up to 12% |
| **Ureteropyelostomy** (reimplant into pelvis) | PUJ avulsion; spatulated |
| **Ureterocalicostomy** | Pelvis or PUJ destroyed; lower-pole parenchyma resected |
| **Renal descensus** | Mobilise kidney downwards; gains up to 8 cm |
| **TUU (transureteroureterostomy)** | Delayed; long loss when the bladder cannot reach; 90–97% success. Avoid with stones, RPF (retroperitoneal fibrosis), TB (tuberculosis), urothelial cancer or radiation |
| **Ileal ureter / Yang–Monti** | Long or panureteric loss; 81–100% success. Avoid if creatinine >2 mg/dL, bladder dysfunction, inflammatory bowel disease, radiation enteritis. Acidosis about 20% |
| Appendiceal interposition | Right side; about 92% short-term success |
| **Buccal mucosa graft ureteroplasty** | Delayed long strictures; onlay with omental wrap, often robotic; about 87% success |
| **Autotransplantation** | Last resort before nephrectomy; 5–10% renal units lost |
| Nephrectomy | Rarely; unstable with unreconstructable kidney, or poor function |

### Iatrogenic endoscopic injury

- **URS perforation:** stop the procedure and place a **stent** about 4 weeks (± nephrostomy), image later; later reconstruction if stricture develops. **Avulsion:** nephrostomy, delayed reconstruction; avoid nephrectomy.

### Fistulae

- **Ureterovaginal fistula (UVF):** stent first (best / ideally within 2 weeks; most heal); if stenting fails, reimplant with psoas hitch or Boari flap (success ≥90%). Early and delayed repair do equally well.
- **Ureteroarterial fistula:** endovascular covered stent.
- **Also know:** TUU (transureteroureterostomy), ileal ureter, buccal graft ureteroplasty and autotransplantation for long delayed defects; robotic reconstruction (repair) for delayed repair matches open; avoid nephrectomy.

### Complications and follow-up

- Urinoma, abscess, leak, fistula, stricture, loss of renal function.
- Persistent leak: check stent position, take drain off suction, catheterise the bladder, then nephrostomy.
- **Follow-up:** ultrasound or diuretic renogram at 4–12 weeks and (at) 1 year; strictures (late stenosis) can be silent and may appear up to 30 months.

### Medicolegal

- Disclose and document; malpractice claims most often follow gynaecological surgery.

**Diagram to draw:** psoas hitch with non-tunnelled reimplant; ureter's relation to the uterine artery (ureter passing under the uterine artery); spatulated ureteroureterostomy over a double-J stent.

> **Key points to remember:**
>
> - Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
> - Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
> - AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
> - CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
> - Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
> - Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
> - Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

## LAQ 14 Energy sources for breaking stones in urology

P4

**Asked as:**

- Describe energy sources for breaking stones in Urology — *Practice Set 1 Paper IV Q2 (LAQ)*
- Describe energy sources for braking stones in uro — *Practice Set 5 Paper IV Q1 (LAQ)*

**Type:** LAQ (25 marks)
(also covered in 2025-Winter Paper 4, Q2 and Q3b; also answered in Practice Set 1 Paper 4, Q2)

*Read as "energy sources for breaking stones in urology".*

### Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

### Classification

| Group | Sources |
|---|---|
| Extracorporeal | SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational) |
| Intracorporeal mechanical | Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic |
| Intracorporeal electrical | EHL (electrohydraulic lithotripsy) |
| Intracorporeal light | Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY |

### 1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

| Generator | Mechanism | Strength | Weakness |
|---|---|---|---|
| Electrohydraulic (original HM3, 1980) | Underwater spark at F1 of an ellipsoid reflector | Very effective | Electrode life <40,000 shocks; focus drifts |
| Electromagnetic (modern standard) | Coil pushes a metal membrane; lens or reflector focuses | Consistent; high energy at F2; long life | — |
| Piezoelectric | Many ceramic elements on a spherical dish | Long life; little pain | Small focal zone, low energy |

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.
- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.

- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

### 2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

### 3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

### 4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

### 5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

### 6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

| | **Ho:YAG** | **TFL** |
|---|---|---|
| Wavelength | **2120 nm** (often quoted 2100) | 1940 nm (water absorption peak) |
| Frequency | Up to about 120 Hz | Up to about 2000 Hz |
| Peak power | High | Low → less retropulsion |
| Fibre | ≥200 µm | 50–150 µm (more flow and deflection) |
| Product | Fragments and dust | Finer dust |

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

### Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.

- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

### Choice by site

| Site | Usual energy |
|---|---|
| Ureter | Laser (semirigid or flexible); pneumatic distally; SWL |
| Kidney | SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser |
| Bladder | Laser, pneumatic, EHL, mechanical |

### Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

**Diagram to draw:** SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

> **Key points to remember:**
> - SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
> - Holmium 2120 nm, photothermal, breaks every composition.
> - TFL 1940 nm: low peak power, finer dust, less retropulsion.
> - EHL is poorly focused, so it is not first-line in the ureter.

## Short answer questions (SAQ)

## SAQ 1 Flexible ureteroscopy / RIRS and accessories

P1 · P2 · P3 **merged from 3**

*Asked as:* P2 SAQ "RIRS (Retrograde intrarenal surgery)" / "RIRS" / "Flexible Ureteroscopy." / "Flexible URS" · P1 SAQ "Flexible Ureterorenoscopy" · P3 SAQ "Accessories for RIRS"

### Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy (fURS; flexible ureteroscopy) inside the kidney; it inspects and treats the whole upper tract retrogradely by the natural route.
- Marshall first reported it in 1964; a working channel and active deflection made it therapeutic. Active deflection, miniaturised digital scopes and the holmium laser made it practical.

### Instrument

| Feature | Details |
|---|---|
| Tip | 4.5–7.5 Fr, tapered (shaft larger) (digital tips up to about 8.5 Fr) |
| Working channel | **3.6 Fr** (laser fibre, basket), shared by accessories and irrigation |

| Feature | Details |
|---|---|
| Deflection | **≥270° both ways** (reaches lower pole); lower pole angle averages about 140° (up to 175°) |
| **Fibreoptic** | Smallest tip; image degrades as fibres break (black spots); repair about every 15 cases |
| **Digital** ("chip-on-tip", CCD [charge-coupled device] or CMOS [complementary metal-oxide semiconductor] chip) | Better image; larger tip (about 8.5 Fr); not more durable |
| **Single-use** (LithoVue, 2016) | 7.7 Fr; no repair, reprocessing or sterilisation cost; cheaper below about 99 cases a year |

## Indications

- **Stones:** renal stones **<2 cm**; lower pole <1 cm (or 1–2 cm / larger as an alternative to PCNL [percutaneous nephrolithotomy]); patients who must stay on anticoagulants (first choice when anticoagulants cannot be stopped); failed SWL (shock wave lithotripsy); obesity; bleeding disorders; staged treatment of larger stones.
- **Others:** diagnosis and laser ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; short strictures and endopyelotomy; calyceal diverticulum; infundibular stenosis; essential haematuria; foreign bodies.

## Technique

1. Sterile urine; single-dose antibiotic; α-blocker or **pre-stenting** if the ureter is tight.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Ureteral access sheath (UAS)** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure and temperature and allows repeated passes, but causes ureteric injury in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) or thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket. Keep power ≤10 W in the ureter and ≤25 W in the kidney (heat above 43 °C damages tissue).
5. Lower pole stones: relocate the stone to an upper calyx with a basket.
6. Inspect the ureter on withdrawal; stent unless uncomplicated. **Omit stent** if the case was uncomplicated with normal anatomy, normal renal function, no ureteric injury and no second stage (AUA [American Urological Association]). - **Difficult (unfavourable) lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU [European Association of Urology] factors against SWL; they also hinder fURS).

## Accessories

### Guidewires (0.018–0.038 inch, 145–260 cm)

| Type | Construction | Use | Weakness |
|---|---|---|---|
| Hydrophilic | Nitinol core, slippery when wet | Bypassing impacted stones, tortuous ureter | Displaces easily |
| PTFE (polytetrafluoroethylene)-coated | Steel or nitinol | Safety and working wire | Damages scope channel on backloading |
| **Hybrid** | Floppy hydrophilic tip, stiff PTFE shaft | Both roles | — |

- **Safety wire** keeps access and allows stenting after injury (AUA, EAU).
- **Also know:** open-ended or dual-lumen catheter (pyelogram, second wire); stiff wire for the sheath; balloon dilators.

**UAS (ureteral access sheath)**

- Hydrophilic sheath with tapered obturator; usually **12/14 Fr** (inner/outer); tip below the PUJ (pelviureteric junction).
- **Pros:** lower intrarenal pressure and temperature; repeated passes; less scope wear.
- **Cons:** ureteric injury in nearly half, severe in about **1 in 8** (less if pre-stented); no proven fall in infection.
- **Suction sheaths and FANS (flexible and navigable suction access sheath):** aspirate dust and keep pressure low.

**Retrieval devices**

- **Nitinol tipless baskets** (about 1.5–2.4 Fr): kink-resistant, open flush in calyces; relocate lower-pole stones. Small baskets allow more flow but are fragile. Graspers release easily.

**Laser and fibres**

- Ho:YAG or thulium fibre laser; small-core fibres (about 200 µm) keep deflection and flow; **≤10 W in the ureter**.

**Irrigation and antiretropulsion**

- Gravity, pressure bag, hand or foot pumps (hand pumps: more fever); pressure sensors emerging.
- Stone Cone, NTrap, BackStop gel: mainly ureteric stones.

**Stents**

- **Double-J**; AUA: may omit after uncomplicated URS (ureteroscopy).

## Outcomes

- Stone-free rate is higher than SWL for renal stones <2 cm but falls with size and lower pole location. Larger stones may need staged procedures.
- Reported rates depend on the imaging used (CT strictest).

## Complications

- Stent symptoms (commonest); **sepsis about 5%**; stricture about 3%; perforation (a few per cent); bleeding or subcapsular haematoma <1%; **avulsion <1%** (nephrostomy, delayed reconstruction); scope damage.

## Recent advances

- Single-use digital scopes; **thulium fibre laser** and pulse-modulated (Moses) holmium (less retropulsion, finer dust); **suction access sheaths and FANS**; intrarenal pressure sensors (pressure-sensing scopes); robotic flexible ureteroscopy; AI (artificial intelligence) navigation.

**Diagram to draw:** flexible scope through an access sheath over a wire, deflected into the lower pole calyx, with a tipless basket in the lower calyx.

**Key points to remember:**

- 3.6 Fr channel; ≥270° deflection.
- 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
- Sepsis about 5%, stricture about 3%.
- Hydrophilic wire passes; PTFE wire stays.
- Nitinol tipless basket for calyces.

## SAQ 2 Renal tubular acidosis (distal and type 2)

P1 · P2 · P3 **merged from 3**

*Asked as:* P2 SAQ "Renal tubular acidosis" · P1 SAQ "Distal Renal tubular acidosis." · P3 SAQ "Type 2 Renal tubular acidosis?"

### Definition

- **RTA (renal tubular acidosis)**: hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular $H^+$ secretion or $HCO_3^-$ reabsorption.
- **Distal (type 1) RTA:** the collecting duct cannot secrete $H^+$ despite systemic acidosis. Result: **hypokalaemic, hyperchloraemic, normal anion gap metabolic acidosis** with **urine pH >6**: the only acidosis with alkaline urine.
- **Proximal (type 2) RTA:** failure of the proximal tubule to reclaim bicarbonate, causing hyperchloraemic, normal anion gap metabolic acidosis.

### Comparison of types

| Feature | Type 1 (distal) | Type 2 (proximal) | Type 4 (hyperkalaemic) |
|---|---|---|---|
| Defect | $H^+$ secretion (α-intercalated cell) | Proximal $HCO_3^-$ reabsorption | Aldosterone deficiency or resistance |
| Urine pH in acidosis | **Always >6** | <5.5 at steady state | Low |
| Potassium | Low | Low | **High** |
| Serum $HCO_3^-$ | Low, may be very low | 15–18 mEq/L (P3: 14–18 mEq/L) | Mildly low |
| Urine calcium | High | Normal or mildly high | Low |
| Urine citrate | **Profoundly low** | Near normal | – |
| Stones | **Common: up to 70%; calcium phosphate, nephrocalcinosis** | Uncommon | Uncommon |
| Causes | Sjögren, lupus, inherited | Fanconi | Diabetes, obstruction |
| Treatment | **Potassium citrate** 1–2 mEq/kg/day | Alkali 10–15 mEq/kg/day plus potassium | Fludrocortisone, low-potassium diet, stop offending drugs |

### Distal (type 1) RTA

#### Normal acidification and the defect

- **α-intercalated cells** secrete $H^+$ by apical $H^+$-ATPase; bicarbonate exits by the basolateral $Cl^-/HCO_3^-$ exchanger (AE1). Normally urine pH can fall to 4.5.
- In distal RTA, potassium is secreted in place of $H^+$, causing hypokalaemia.

**Causes**

| Inherited | Acquired (secondary) |
|---|---|
| **SLC4A1** (AE1): mostly dominant | Sjögren syndrome, SLE (systemic lupus erythematosus) |
| **ATP6V1B1, ATP6V0A4** ($H^+$-ATPase): recessive, **sensorineural deafness** | Obstructive uropathy, pyelonephritis, medullary sponge kidney |
| **CA2**: mixed proximal–distal (carbonic anhydrase II deficiency) | Hypercalciuria, hyperparathyroidism, amphotericin B |

**Why stones form (urological importance)**

- **Stones in up to 70%**: **calcium phosphate** stones and **nephrocalcinosis**. Stones lead to diagnosis in up to half.
- **Profound hypocitraturia** (most important): acidosis increases proximal citrate reabsorption.
- **Hypercalciuria:** bone buffering releases calcium. **Fixed alkaline (high) urine pH** precipitates calcium phosphate.

**Clinical features**

- Stones, nephrocalcinosis, weakness (hypokalaemia), osteomalacia or rickets; children fail to thrive (growth failure).
- **Incomplete distal RTA:** normal bicarbonate, but urine pH cannot fall below 5.5 after an acid load.

## Proximal (type 2) RTA

**Pathophysiology**

- The proximal tubule reclaims **85–90% of filtered bicarbonate** (luminal carbonic anhydrase; exit by the basolateral Na–$HCO_3$ cotransporter).
- **Self-limiting:** bicarbonate wasting lowers plasma $HCO_3^-$ until the filtered load drops below the lowered threshold; the intact distal tubule then copes, giving a **steady state at about 14–18 mEq/L (P2: 15–18 mEq/L) with urine pH <5.5**.
- **Hypokalaemia** from distal sodium bicarbonate delivery; worsens on alkali.

**Causes**

| Isolated | Generalised (Fanconi syndrome) |
|---|---|
| Carbonic anhydrase inhibitors (acetazolamide, **topiramate**) | **Cystinosis** (commonest in children), Lowe, Wilson, galactosaemia |
| Carbonic anhydrase II deficiency (mixed) | **Ifosfamide, cisplatin**, aminoglycosides, valproate |
| | Myeloma, amyloidosis, lead |

- **Fanconi:** glycosuria, aminoaciduria, phosphaturia, uricosuria.

**Clinical features**

- Children: growth failure, **rickets**; adults: osteomalacia, weakness.

- **Stones uncommon**: urinary citrate stays near normal and urine is acid at steady state. Topiramate and acetazolamide are exceptions (alkaline urine, calcium phosphate stones).

### Type 4 (hyperkalaemic) RTA

- Hyperkalaemia, low urine pH; diabetes, obstructive uropathy. Amiloride and trimethoprim cause hyperkalaemic distal RTA.

### Diagnosis

- Bloods: low bicarbonate, high chloride, low potassium (types 1 and 2); urine pH >6 on a fasting morning sample in distal RTA; urine can acidify below 5.5 in proximal RTA.
- **UAG (urine anion gap) = ($Na^+$ + $K^+$) − $Cl^-$** reflects urinary ammonium: **positive** (low ammonium excretion) in renal causes; negative in GI (gastrointestinal) loss such as diarrhoea.
- **Ammonium chloride load** (0.1 g/kg orally, hourly urine pH for about 6 h): **failure to fall below 5.5** confirms distal RTA. The **furosemide–fludrocortisone test** is a better-tolerated alternative.
- **Bicarbonate loading:** urine pH rises; **fractional bicarbonate excretion >15%** indicates proximal RTA. Look for Fanconi features; drug history.
- 24-hour urine: low citrate, high calcium (distal); ultrasound for nephrocalcinosis.

### Treatment

- **Distal RTA: potassium citrate** (about 1–2 mEq/kg/day in divided doses): corrects acidosis, hypokalaemia and hypocitraturia, reduces calcium excretion and stone formation. Sodium alkali is less suitable because sodium raises urinary calcium. Treat the cause; manage stones endourologically.
- **Proximal RTA:** treat the cause; **large alkali doses (about 10–15 mEq/kg/day)** as potassium citrate or mixed citrate, replacing potassium; thiazide; phosphate and vitamin D for rickets.
- **Type 4:** fludrocortisone, low-potassium diet, stop offending drugs.

### Urological relevance

- Obstruction causes hyperkalaemic distal RTA, reversible by relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).
- Potassium citrate beats sodium alkali, which raises calcium excretion.
- **Also know:** CKD (chronic kidney disease) acidosis; amiloride and trimethoprim (hyperkalaemic distal RTA); mixed type (carbonic anhydrase II deficiency); medullary sponge kidney; laxative abuse (ammonium urate stones); inherited forms (recessive with deafness); growth failure in children.

**Diagram to draw:** nephron with defect sites; α-intercalated cell with $H^+$-ATPase, AE1 and the stone mechanism; proximal tubular cell with carbonic anhydrase.

**Key points to remember:**

- Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
- $NH_4Cl$ load: pH fails to fall below 5.5 in distal RTA.
- Potassium citrate is the treatment for distal RTA.
- Type 2 (proximal): bicarbonate wasting; $HCO_3^-$ 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
- Fractional $HCO_3^-$ excretion >15%; stones uncommon in type 2.
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause.

## SAQ 3 PCNL patient positioning: prone vs supine

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each." · P2 SAQ "Describe various pt positions in PCNL"

### Introduction

- PCNL (percutaneous nephrolithotomy) position decides calyceal access, working space, airway safety and simultaneous retrograde access.

### Indications for PCNL

- Renal stones **>2 cm**, staghorn, lower pole >1 cm, calyceal diverticulum, failed retrograde route or urinary diversion, UPJ (ureteropelvic junction) obstruction.

### Positions

| Position | Set-up | Advantages | Disadvantages |
|---|---|---|---|
| **Prone** (Goodwin, 1955) | Lithotomy for ureteric catheter, then turned prone; chest and pelvic bolsters; horizontal bolsters at nipple line and ASIS (anterior superior iliac spine) bring the kidney down more than vertical ones; arms ≤90°, face and knees padded | **Posterior calyx and upper pole** access; large working area; multiple tracts; familiar | Turning (repositioning); airway hard to reach; pressure and positioning injuries; retrorenal colon commoner prone; needs split-leg table for ECIRS |
| **Prone split-leg / prone-flexed** | Legs apart for flexible cystoscopy; hip flexion opens the costovertebral space | ECIRS (endoscopic combined intrarenal surgery) prone | Needs split-leg table; prone cystoscopy learning curve |
| **Supine** (Valdivia-Uría, 1987) | Ipsilateral flank at table edge on a lumbar bolster; arm across chest | **Airway retained**; no repositioning; single position, shorter time; **gravity drains fragments**, low intrarenal pressure; safer anaesthesia | Smaller working area; collapsed system; anterior calyces; longer tract; mobile kidney; harder upper pole |

| Position | Set-up | Advantages | Disadvantages |
|---|---|---|---|
| **Galdakao-modified supine Valdivia** (Scoffone) | Supine flank tilt with ipsilateral leg extended and other leg in lithotomy | **Simultaneous retrograde access (ECIRS)** in one position | Longer, more horizontal tract |
| **Bart's flank-free modified supine** (Bach) | Bolster under the thorax, flank free | More flank space | Fewer data |
| **Lateral decubitus (flank)** | Kidney-up flank position | Obese, kyphoscoliosis, poor respiratory reserve | Unfamiliar fluoroscopic orientation |

## Technique: prone vs supine

| Feature | Prone | Supine |
|---|---|---|
| Puncture and tract | Posterior calyx puncture under fluoroscopy or ultrasound; tract more vertical; large working area | Puncture in posterior axillary line; tract more horizontal and longer; kidney more mobile; air outlines anterior calyces |
| Drainage | Against gravity; higher pelvic pressure | Gravity drains fragments, low intrarenal pressure |
| Anaesthesia | Airway access difficult; turning needed | Airway retained; no repositioning |

## Positioning safety (prone)

- Bolster shins so toes float; pad knees; arms ≤90° to the torso (brachial plexus); axillary and pectoral rolls; padded face and eyes.

## Specific complications

- **Prone:** ocular injury, neuropraxia (brachial plexus), pressure injury, endotracheal tube dislodgement while turning.
- **Supine: colon injury** (if puncture too anterior), access failure to upper pole, kidney displacement during dilatation.
- **Both:** bleeding (transfusion now about 1%), pleural injury above the 12th rib, sepsis, bowel, spleen or liver injury.

## Choosing a position

- **Prone:** standard anatomy, upper pole or multiple posterior tracts.
- **Supine variants:** obesity, cardiopulmonary disease, kyphosis, ECIRS, ileal conduit or continent channel, combined ureteric stone.
- **Spinal deformity:** individualise; pad every pressure point.
- **Evidence:** stone-free rates and complications similar; supine often shortens operating time; choice is mainly surgeon preference.

**Diagram to draw:** axial CT cross-section showing prone (vertical) and supine (horizontal) tracts.

**Key points to remember:**

- Prone: posterior and upper pole access, large working area.
- Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
- Horizontal bolsters drop the kidney more.
- Similar outcomes; choose by anatomy and experience.

## SAQ 4 Renal autotransplantation

P1 · P3 **merged from 2**

*Asked as:* P3 SAQ "Auto-transplantation of Kidney?" / "Role of auto transplant of kidney in Urology." / "Renal Autotransplantation" · P1 SAQ "Autotransplantation"

### Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in **1963** for proximal ureteric injury.
- **Principle:** bringing the kidney to the bladder replaces the ureter without bowel; the bench gives a cold, bloodless field.

### Role (indications)

| Group | Examples |
|---|---|
| **Ureteric loss (commonest)** | Long-segment loss or stricture; long iatrogenic or traumatic loss; repeated failed repairs (**delayed setting only, never acute**) |
| **RPF (retroperitoneal fibrosis)** | When ureterolysis fails or is impossible and function is satisfactory |
| **Tumour** | Total ureterectomy for UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy in a solitary kidney; left RCC (renal cell carcinoma) needing suprarenal caval resection (new drainage for the right kidney) |
| **Vascular** | Complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome |
| **Pain** | Selected loin pain–haematuria syndrome |

- **Preferred over ileal ureter** when the contralateral kidney is absent or poor, or bowel is unsuitable (creatinine >2 mg/dL, inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

### Procedure

CT (computed tomography) angiography, renogram, ureteric study → harvest as for live donor nephrectomy (open/laparoscopic/robotic): maximal vessel length, preserve ureteric blood supply → cold perfusion on bench ± ex vivo repair → Gibson incision, extraperitoneal iliac fossa → renal vein to external iliac vein (end-to-side); renal artery to external iliac (end-to-side) or internal iliac (end-to-end) → drainage: pyelovesicostomy (± Boari flap),

ureteroneocystostomy, or native ureter left in continuity → stent, catheter, drain

- Laparoscopic harvest plus Gibson implantation: less analgesia, faster recovery; **robotic autotransplantation** emerging.

### Complications and outcome

- Arterial or venous **thrombosis**, bleeding, urine leak, anastomotic stricture, lymphocele, infection.
- **Renal unit loss 5–10%** (lower in transplant centres); major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.**

### Follow-up

- Doppler ultrasound early; renogram and blood pressure later.

### Alternatives

- Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti, buccal ureteroplasty, renal descensus.

**Diagram to draw:** kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

> **Key points to remember:**
> - For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
> - Delayed setting only; bench surgery for complex vascular or tumour cases.
> - Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
> - Unit loss 5–10%; success about 90%.

## SAQ 5 Stones in pregnancy

P2 · P3 **merged from 2**

*Asked as:* P2 SAQ "Management of Stones in pregnancy." / "Urolithiasis in pregnancy" · P3 SAQ "Management of lower ureteric stone of size less than 1 cm in pregnancy?"

### Background

- 1 in 200 to 1 in 1500 pregnancies; renal colic is the **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Why calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Raised citrate, magnesium and urine flow protect, so stone rates barely change.
- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

### Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; **transvaginal US** for the distal ureter; a rise in Doppler **RI (resistive index)** suggests acute obstruction.

- Second line: **MRI (magnetic resonance imaging)** (stone as a filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy** (under 50 mGy).

### Management algorithm

```
Symptomatic stone in pregnancy (including lower ureteric stone <1 cm) ├─ sepsis, solitary kidney, renal failure, uncontrolled pain │ → urgent stent or nephrostomy (+ antibiotics) ├─ pain controlled, no infection → observe (50–80% pass) └─ failed observation → ureteroscopy + holmium laser or temporary stent/nephrostomy until delivery
```

### Conservative management

- Hydration, paracetamol, small doses of morphine (opioids).
- **NSAIDs (non-steroidal anti-inflammatory drugs) are contraindicated** (ductus arteriosus closure in the third trimester).
- **MET (medical expulsive therapy)** with tamsulosin is off-label.

### Ureteroscopy

- **URS (ureteroscopy) with holmium laser:** first-line intervention and definitive treatment for a non-complex stone in any trimester; ideal for a distal (lower ureteric) stone; second trimester is the safest window.
- Neuraxial anaesthesia, lateral tilt, minimal fluoroscopy (lead shield beneath the patient, US guidance), fetal monitoring.
- Third-trimester URS may raise preterm labour risk.

### Temporising drainage

- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents about **every 4 weeks** (P3: every 4–6 weeks; 4–6 weeks in series), nephrostomy about every 3 weeks; definitive treatment after delivery.

### Other modalities

- **Asymptomatic renal stones:** observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** until after delivery except in expert centres.

### After delivery

- Clear residual stones; **metabolic evaluation** postpartum.

**Diagram to draw:** the management flowchart above.

> **Key points to remember:**
> - US first; MRI or low-dose CT second (≤50 mGy).
> - Observation first; no NSAIDs.
> - URS with laser is definitive; SWL contraindicated.
> - Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

## SAQ 6 Ureterocalycostomy

P2 · P3 **merged from 2**

*Asked as:* P3 SAQ "Indication, procedure and complications of Uretero-Calycostomy?" · P2 SAQ "Uretero-calycostomy."

### Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

### Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

### Procedure (open, laparoscopic or robotic)

`Flank position (45–60°); pre-placed double-J stent → mobilise ureter with generous periureteral tissue → mobilise kidney; partial lower-pole nephrectomy to expose the calyx widely (not simple nephrotomy) → close (ligate) the pelvic outlet; spatulate ureter laterally → apex stitch to lateral calyx wall; second at 180° → interrupted mucosa-to-mucosa sutures, all placed then tied → capsule closed over cut surface, away from anastomosis → cover with perinephric fat, peritoneum or omentum; drain`

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.

### Postoperative care and follow-up

- Stent out at 4–6 weeks; urogram about 1 month later, then renogram.
- Later endoscopy needs **antegrade upper-pole access**.

### Complications and outcome

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, loss of the renal unit.
- Success about **70%** in the largest (mostly redo) series; better in small paediatric (robotic) series. The access route does not affect outcome.

### Alternatives

- Endopyelotomy, redo pyeloplasty, flap repair, Davis intubated ureterotomy, ileal ureter, buccal graft, autotransplantation, nephrectomy (poor kidney); calicovesicostomy for a low-lying kidney.

**Diagram to draw:** kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
> - Stricture is the main failure; success about 70% in redo series.

## SAQ 7 Tubeless PCNL

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Tubeless PCNL" · P4 SAQ "Tubeless PCNL"

### Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

### Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

### Selection

| Suitable | Keep a nephrostomy |
|---|---|
| Complete clearance, single tract | Pyonephrosis or infected stone |
| Minimal bleeding | Significant bleeding |
| Intact collecting system | Significant perforation of (injury to) the collecting system |
| Supracostal access, bilateral cases (shown safe) | Residual stone needing a second look |

### Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent through the urethra, or a string.
3. Withdraw the sheath under vision; the **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue in the tract did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

### Evidence

- Meta-analyses of randomised trials: shorter stay and operative time, and less pain.
- The pain benefit is over large tubes (**>16 Fr**); there is **no benefit over small 8–9 Fr tubes**; small tubes (≤10 Fr) give similar comfort without more bleeding.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

### Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

### Current position

- Choose tubeless, small tube (≤10 Fr) or stent case by case; both stents and small tubes are acceptable (AUA, American Urological Association). Pairs well with mini-PCNL.

**Diagram to draw:** kidney with sealed tract and double-J stent.

> **Key points to remember:**
> - Tubeless = stent only; totally tubeless = neither.
> - Benefit is mainly over large (>16 Fr) tubes.
> - Keep a tube for bleeding, perforation, infection or second look.

## SAQ 8 Mini-PCNL and newer PCNL techniques

P3 · P4 **merged from 2**

*Asked as:* P4 SAQ "Miniperc" / "Newer PCNL techniques and nomenclature of miniaturised PCNL" · P3 SAQ "Mini Perc."

### Definition and why miniaturise

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.
- In PCNL, bleeding rises with tract size. Smaller tracts mean less bleeding and pain, at the cost of longer operating time and poorer vision.

### Nomenclature (by tract or outer sheath size)

| Name | Tract size | Features |
|---|---|---|
| Standard | 24–34 Fr | Ultrasonic or pneumatic lithotripsy; graspers |
| **Mini** | 15–22 Fr | First "mini-perc": Jackman 1998, 13 Fr; irrigation washes fragments out ("vacuum-cleaner" effect) |
| **Super-mini** | About 10–14 Fr | Sheath with **active suction** of fragments |
| **Ultra-mini** | 11–14 Fr | 3 Fr inline telescope in 11–13 Fr sheath |
| **Micro-perc** | <10 Fr | Single-step "all-seeing needle" with optics; laser only, fragments pass spontaneously |

- **Problem:** names overlap and cut-offs are contested (paediatric chapter: mini <20–24 Fr), so many authors report the **actual sheath size in Fr**.
- The **working channel**, not sheath size, decides which instruments pass: the smallest systems take only a laser fibre.

### Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Exit: often **tubeless** or small nephrostomy (≤10 Fr: less pain).

### Newer PCNL techniques

1. **Supine PCNL and ECIRS:** Galdakao-modified supine; simultaneous retrograde flexible ureteroscopy watches puncture and dilatation; fewer tracts.
2. **Ultrasound-guided access:** less radiation; shows pleura and viscera.
3. **Single-step or balloon dilatation.**
4. **Suction sheaths:** lower pressure, faster clearance.
5. **Lasers:** high-power and pulse-modulated holmium, TFL (thulium fibre laser) for dusting through small tracts.
6. **Flexible nephroscopy** after bulk removal: higher stone-free rate.
7. **Tubeless or small (≤10 Fr) nephrostomy:** less pain.

- **Also know:** robotic or artificial intelligence-assisted puncture under study.

### Indications and contraindications

- Renal stones **1–2 cm**, **lower pole stones** >1 cm (alternative to RIRS [retrograde intrarenal surgery]), failed SWL (shock wave lithotripsy) or RIRS, **paediatric stones**, calyceal diverticular stones; selected larger stones or extra staghorn tracts.
- **Contraindications**: as for PCNL (untreated infection, uncorrected coagulopathy, pregnancy).

### Advantages over standard PCNL

- Less bleeding and haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; stone-free rates non-inferior to standard PCNL.
- Children: a 24–30 Fr tract in a child equals about a 72 Fr tract in an adult, so miniaturisation matters most here.

### Disadvantages

- **Longer operative time**, poorer vision, slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a tight sheath, especially when the scope nearly fills it, raises pelvic pressure and causes pyelovenous backflow, **fever and sepsis**. Keep outflow free, use low-pressure irrigation, or suction sheaths.

### Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding (fewer transfusions), shorter stay, longer operation; same complications.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears against a 24 Fr tract**.
- Mini-PCNL versus RIRS, large matched study: similar stone-free and retreatment rates; RIRS shorter stay.

**Diagram to draw:** scale bar of tract sizes compared (micro, ultra-mini, mini, standard).

**Key points to remember:**

- Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
- Working channel decides instruments; suction lowers pressure.

## SAQ 9 Newer methods of stone analysis

P3 · P4 **merged from 2**

*Asked as:* P3 SAQ "Newer methods of stone analysis" / "Newer techniques of stone analysis" · P4 SAQ "Newer methods of stone analysis and their significance in management of calculus disease."

### Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former (every first stone); **repeat if stones recur on treatment** (evolving metabolic condition), or on early recurrence after clearance.
- Composition points to the cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.
- Composition can change over time (e.g. calcium oxalate converting to brushite), so re-analyse recurrent stones.

### Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**, so it cannot separate COM (calcium oxalate monohydrate, whewellite) from COD (dihydrate, weddellite), or apatite from brushite.
- High error rates; misses rare stones (2,8-DHA, drugs); needs a large sample.

### Newer methods

| Method | Principle | Comment |
|---|---|---|
| **FTIR (Fourier-transform infrared) spectroscopy** | Each compound absorbs a characteristic infrared spectrum | Commonest reference method; tiny sample; detects proteins and drugs |
| **XRD (X-ray diffraction)** | Crystal lattice diffracts X-rays in a unique pattern | Gold standard for crystals; misses amorphous material |
| **Polarising microscopy** | Crystal optics (birefringence) | Cheap; needs an experienced observer (observer-dependent) |
| **Morpho-constitutional analysis (Daudon)** | Stereomicroscopy of surface and section, plus FTIR | Subtypes point to cause (e.g. type Ia COM: low urine volume; Ic: primary hyperoxaluria) |
| **SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)** | Surface structure and elements | Research |

| Method | Principle | Comment |
|---|---|---|
| **Micro-CT (micro-computed tomography)** | 3-D non-destructive structure | Research |
| **Raman spectroscopy** | Scattered light spectrum | Research |

### In vivo prediction

- **DECT (dual-energy computed tomography):** two tube energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid stones** with near-perfect accuracy; it estimates composition in vivo, mainly uric acid versus calcium stones.
- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; >1000 HU predicts SWL failure; cystine and brushite are dense.
- **Endoscopic recognition** of morphology (lost when stones are dusted). Dusting loses the specimen: collect fragments.

### Significance in management

- **Uric acid:** oral dissolution by urinary alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy.
- **Brushite:** can convert from calcium oxalate after SWL or infection; avoid over-alkalinisation.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency; treat with allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

**Diagram to draw:** FTIR spectrum outline and COM (dumbbell) versus COD (envelope) crystal shapes.

> **Key points to remember:**
> - FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
> - DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
> - Composition guides dissolution, SWL choice and prevention.
> - Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

## SAQ 10 Radionuclide studies and diuretic renography in PUJ obstruction

P1

**Asked as:**

- Radionuclide studies for PUJ obstruction. — *Summer 2021 Paper I Q3e (SAQ)*
- Radionuclear studies in PUJ obstruction — *Practice Set 4 Paper I Q3e (SAQ)*
- Diureteric renography — *Practice Set 2 Paper I Q3f (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 4: also answered in 2021-Summer Paper 1, Q3e. Practice Set 2: PUJ obstruction renography also in 2021-Summer Paper 1, Q3e; tracers: see Practice Set 2 Paper 1, Q3e)

*"Diureteric" is read as diuretic renography.*

## Introduction / principle

- **Dilatation is not obstruction.** PUJ (pelviureteric junction) obstruction is impaired drainage that damages the kidney. **Diuretic renography** is the functional standard (functional gold standard for PUJ obstruction) and best follow-up test (best follow-up study).
- A capacious pelvis retains tracer passively and furosemide washes it out; an obstructed system cannot empty even under diuresis.

## Tracers

| Tracer | Handling | Use |
|---|---|---|
| $^{99m}$Tc-**MAG3** (technetium-99m mercaptoacetyltriglycine) | Tubular secretion | **Agent of choice** (tracer of choice): split function and drainage; good images (good curves) even with poor function |
| $^{99m}$Tc-**DTPA** (diethylenetriaminepentaacetic acid) | Glomerular filtration | GFR (glomerular filtration rate); poor in renal impairment; an alternative to MAG3 |
| $^{99m}$Tc-**DMSA** (dimercaptosuccinic acid) | Cortical retention | Static split function and scars; no drainage data |

## "Well-tempered" diuretic renogram (standard protocol)

| Element | How | Why |
|---|---|---|
| **Hydration** | Oral, or IV (intravenous) fluid in children (about 15 mL/kg) | Dehydration mimics obstruction |
| **Bladder catheter** | Urethral catheter | Removes full-bladder (back-pressure) and reflux (VUR, vesicoureteral reflux) effects |
| **Furosemide** | 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children | Maximal diuresis |
| **Timing** | **F+20** (classic; 20 min after tracer, when the pelvis is full; Campbell authors use F+15), **F0** (with tracer) or **F−15** (15 min before, so the system is already in diuresis) | Diuretic before filling (early diuretic into an unfilled system) misleads; always state timing |
| **Regions of interest** | Regions of interest over cortex (for split) and pelvis (dilated pelvis for drainage); then upright images | Wrong placement gives spurious values |

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- First study after about 4–6 weeks of age (immature function).

## Outputs

- **DRF (differential renal function)** from counts at 1–3 min (1–3 minute counts); normal 45–55%.
- **Drainage half-time (T½, half-clearance time)** after furosemide, from maximal filling, not from injection:

| T½ after furosemide | Interpretation |
|---|---|
| <10 min (minutes) | Not obstructed |
| 10–20 min (minutes) | Equivocal (indeterminate) |

| T½ after furosemide | Interpretation |
|---|---|
| >20 min (minutes) | Obstructed |

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity remaining / left at 40 min) raises sensitivity; output efficiency; NORA (normalised residual activity); a **biphasic curve** and poor upright (post-void) drainage predict progression.

### Indications

- PUJ (pelviureteric junction) obstruction, antenatal hydronephrosis, megaureter, after pyeloplasty, transplant ureter.

### Indications for pyeloplasty from renography

- Consider surgery for:
- **DRF <40%**, **fall in DRF** on serial comparable scans (**>10%** on serial scans; ≥5% in children; 10% in older series), **T½ >20 min** (minutes) or upright retention; also (poor drainage with) symptoms, infection, increasing dilatation.
- Adult DRF <15%: likely non-salvageable; nephrostomy and reassess, or nephrectomy.

### Pitfalls

- Dehydration, early diuretic (mistimed diuretic), full bladder, wrong ROI (region of interest), very poor function, very capacious pelvis.
- One Campbell chapter calls T½ >40 min obstructed; children have no agreed (paediatric T½) cut-off (false positives up to 44%; neonatal false-positive rates reach 44%); one Campbell chapter uses **T½ >40 min** in adults, so **serial trends and curve shape** matter / decide.

### Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal; spontaneous drainage means success.
- Equivocal (cases) → **Whitaker test** (>15–20 cm $H_2O$ obstructed) or MRU (magnetic resonance urography).

**Diagram to draw:** normal, obstructed and dilated non-obstructed renogram curves (O'Reilly curves I, II, IIIa and IIIb) with furosemide arrow.

> **Key points to remember:**
> - MAG3 is the tracer of choice; hydrate and catheterise.
> - Well-tempered: hydrate, catheter, standard diuretic dose and timing.
> - Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
> - DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
> - Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

## SAQ 11 What are nanobacteria? Describe their role in urolithiasis.

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 1, Q3e

### Definition

- **Nanobacteria**, now called **CNPs (calcifying nanoparticles)**, are spherical particles far smaller than bacteria with a **calcium apatite shell**, described in bovine and human serum by **Kajander and Çiftçioglu (1998)** as *Nanobacterium sanguineum*.
- **Their living nature is disputed.** Proposed as the smallest self-replicating bacteria, but later work showed "replication" is mineral nucleation, and that they are **mineralo-protein complexes** (apatite with serum proteins such as fetuin-A). Their bacterial DNA proved to be contamination. Most now regard them as **non-living**.

### Claimed properties

| Property | Claim |
|---|---|
| Size | Pass through bacteria-retaining filters |
| "Growth" | Very slow, in serum-containing medium |
| Shell | Apatite; resistant to heat and gamma irradiation |
| Detection | Electron microscopy, von Kossa and Hoechst staining, specific antibodies, culture |
| Susceptibility | Tetracycline and EDTA (ethylenediaminetetraacetic acid) inhibit them |

### Proposed role in urolithiasis

- **Nidus theory:** CNPs in blood or urine are filtered or excreted, deposit apatite in renal tubular cells and interstitium, and act as a **nucleating nidus** for crystal growth.
- **Link to Randall plaque:** idiopathic calcium oxalate stones grow on **Randall plaque**, an interstitial **apatite** deposit that starts in the basement membrane of the thin loop of Henle and erodes through the papilla. CNPs are **associated with**, but **not proven to cause**, Randall plaque (Campbell, Chapter 95).
- Early studies found apatite nanoparticles in a large majority of stones; rat studies reported renal calcification after injection.
- Also implicated in atherosclerotic calcification, polycystic kidney cysts and chronic prostatitis.

### Therapeutic implications (unproven)

- Tetracycline-based regimens with EDTA were tried in stones and chronic prostatitis/chronic pelvic pain syndrome; no RCT (randomised controlled trial) evidence supports routine use.

### Current position

- Not in stone guidelines; pathogenesis is explained by supersaturation, inhibitors, Randall plaque and ductal plugs.

**Diagram to draw:** renal papilla with interstitial apatite (Randall plaque) eroding through urothelium, CNPs as nidus, and calcium oxalate stone attached.

**Key points to remember:**

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

## SAQ 12 Tear drop bladder

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 1, Q3a

### Definition

- A **tear-drop (pear-shaped or inverted-pear) bladder** is a bladder that is symmetrically **compressed from both sides** and elongated upwards on cystogram, IVU (intravenous urogram) or CT (computed tomography).
- It is a radiological sign of a **bilateral perivesical mass or collection**, not a disease itself.

### Causes

| Group | Causes |
|---|---|
| **Collections** | **Pelvic haematoma** (pelvic fracture: commonest in trauma); urinoma (extraperitoneal leak); bilateral lymphocele (after lymphadenectomy or transplant); pelvic abscess |
| **Fat** | **Pelvic lipomatosis** |
| **Muscle** | **Iliopsoas hypertrophy** (muscular young men) |
| **Vascular** | Dilated pelvic venous collaterals (IVC [inferior vena cava] occlusion); bilateral iliac artery aneurysms |
| **Nodes and tumour** | Pelvic lymphadenopathy (lymphoma, metastases); pelvic retroperitoneal fibrosis; pelvic tumours |

### Pelvic lipomatosis (the classic cause)

- Overgrowth of benign, non-encapsulated fat in the perivesical and perirectal spaces; mainly in men, more in Black men.
- **Presentation:** LUTS (lower urinary tract symptoms), suprapubic pain, constipation; often incidental.
- **Imaging:**
- Plain film: pelvic lucency. IVU: elevated, elongated pear-shaped bladder; **medial deviation of both ureters**; hydronephrosis.
- Barium enema: straightened, elongated rectosigmoid.
- **CT or MRI (magnetic resonance imaging):** fat density is diagnostic.
- **Cystoscopy:** high bladder neck, difficult passage; often **cystitis glandularis**, which can progress to adenocarcinoma. Biopsy and surveillance are needed.
- **Management:** observe if no obstruction. Upper-tract obstruction → stent or nephrostomy, ureteric reimplantation or urinary diversion. Weight loss rarely helps; surgical excision of fat is difficult.

### Evaluation of a tear-drop bladder

- History: trauma, pelvic surgery or transplant, malignancy.
- **CT** shows the cause (blood, fat, fluid, muscle, nodes, vessels). Ultrasound for collections; aspiration (creatinine for urinoma, lymph for lymphocele).

### Management

- Treat the cause: pelvic haematoma usually observed; lymphocele drained or marsupialised (laparoscopic fenestration); urinoma drained and leak repaired.

**Diagram to draw:** cystogram outline of a bladder squeezed into a tear drop by bilateral pelvic haematoma.

> **Key points to remember:**
> - Tear-drop bladder = bilateral extrinsic compression.
> - Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
> - Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

## SAQ 13 Retrograde endopyelotomy

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 1, Q3f

### Definition and principle

- Endoscopic **full-thickness incision of the UPJ (ureteropelvic junction)** from inside the ureter, through to peripelvic fat, followed by stenting so the ureter heals with a wider lumen (Davis "intubated ureterotomy" concept, 1943).
- **Retrograde ureteroscopic holmium laser endopyelotomy** is now the standard endourological route: direct vision without a percutaneous tract.

### Indications and contraindications

| Favourable | Unfavourable / contraindicated |
|---|---|
| Functionally significant UPJO (UPJ obstruction); short segment | **Long stenotic segment** |
| **Secondary UPJO** (failed pyeloplasty) | **Upper-tract stones** (treat percutaneously instead) |
| Moderate hydronephrosis, fair function | **Crossing vessel**, **gross hydronephrosis**, poor function |
| - Pre-operative CT (computed tomography) angiography or Doppler to look for crossing vessels. | |

### Technique

1. General anaesthesia; retrograde pyelogram; hydrophilic safety wire into the pelvis.
2. Access sheath; flexible ureteroscope to the UPJ (semirigid in some women). Dilate a narrow intramural ureter or pre-stent.
3. Holmium laser, 200-µm fibre, 0.8–1.2 J, 10–15 Hz.
4. **Lateral incision** (no crossing vessels laterally), deepened into fat and extended into normal ureter and pelvis.

5. Optional balloon dilation to 24 Fr; DJ (double-J) stent and Foley catheter.
6. Stent out after some weeks; **diuretic renogram 4 weeks later**; follow up for 24–36 months.

### Results

- Holmium series 70–80% at 5 years (range 44–89%). Excluding crossing vessels >4 mm raises success to about 90%.
- Crossing vessel present versus absent: 42% versus 86% (antegrade series).
- Still below pyeloplasty (about 95%), but most cost-effective endourological option.

### Complications

- Urine leak, stent migration, infection, bleeding (lateral cut spares vessels), stricture (rare with small scopes); ureteroarterial fistula reported.
- **Also know:** antegrade (percutaneous) endopyelotomy for UPJO with stones; cautery wire balloon (more haemorrhage); balloon dilation alone (about 42% success); failure then needs pyeloplasty.

**Diagram to draw:** UPJ with lateral full-thickness incision and crossing vessel lying anterior/posterior.

> **Key points to remember:**
> - Lateral full-thickness incision over a stent.
> - Crossing vessel and gross hydronephrosis predict failure.
> - Success 70–80%, below pyeloplasty (about 95%).

## SAQ 14 Post-obstructive diuresis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3b

### Definition

- POD (post-obstructive diuresis): marked polyuria after relief of obstruction, as the kidney sheds retained salt, water and urea.
- No agreed rate; a common working definition is **>200 mL/h for 2 consecutive hours or >3 L in 24 hours**. Up to 300 mL/h is common.
- **Physiological** (appropriate, self-limited) versus **pathological** (continues after euvolaemia, causing hypovolaemia and electrolyte loss).

### Incidence and setting

- Unilateral obstruction: **about 2%** (the other kidney compensates). Bilateral or solitary kidney: **about 50%**.
- Commonest after catheterising **chronic retention with renal impairment**; also bilateral ureteric obstruction and PUV (posterior urethral valves).

### Mechanisms

1. **Transporter dysregulation:** down-regulation of $Na^+/K^+$-ATPase and aquaporin-2 → concentrating defect (nephrogenic DI (diabetes insipidus)); loss of medullary tonicity.

2. **Volume expansion:** on relief, raised ANP (atrial natriuretic peptide), low aldosterone and sympathetic tone → salt and water loss.
3. **Osmotic diuresis** from retained urea.

### Risk factors

- Bilateral obstruction, raised creatinine, oedema, heart failure, hypertension, confusion.
- After valve ablation: oligohydramnios, preterm birth, raised creatinine.

### Clinical features

- Starts within hours of relief; most settle within 24–48 hours.
- Danger signs: postural hypotension, tachycardia, weight loss, hypernatraemia (free-water loss), low potassium, magnesium, phosphate.

### Management

``` Relieve obstruction (catheter / nephrostomy)

|

Monitor hourly urine output, vitals, weight

|

Alert, can drink -> free oral fluids (most resolve)

|

Output persists / cannot drink / unstable -> admit; electrolytes every 4-6 h -> intravenous 0.45% saline: about 75% of urine output -> replace K, Mg, phosphate; adjust to urine sodium/osmolality ```

- **Do not overcorrect:** replacing every millilitre sustains the diuresis.
- **Gradual bladder decompression has no role**: it does not prevent haematuria or diuresis.
- PCN (percutaneous nephrostomy) for upper-tract obstruction; hydration alone for physiological POD.
- **Also know:** decompression haematuria; neonatal PUV concentrating defect; no approved drug therapy.

**Diagram to draw:** graph of urine output against time after catheterisation.

> **Key points to remember:**
> - Unilateral about 2%, bilateral about 50%.
> - Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
> - Replace about 75% of output with 0.45% saline; do not overcorrect.
> - Gradual decompression has no role.

## SAQ 15 Stone complexity scores

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3b
```

### Purpose

- Grade stone and kidney complexity, chiefly to **predict stone-free rate (SFR) after PCNL (percutaneous nephrolithotomy)**. Used for counselling, choosing the surgeon, auditing and comparing series.

### Main scores

| Score | Variables and scoring | Predicts |
|---|---|---|
| **Guy's Stone Score** (Thomas, 2011) | **Grade I:** single stone in mid or lower pole, or pelvis, simple anatomy. **II:** single upper-pole stone, or multiple stones with simple anatomy, or single stone with abnormal anatomy. **III:** multiple stones with abnormal anatomy, calyceal diverticulum stone, or partial staghorn. **IV:** complete staghorn, or any stone in spina bifida or spinal injury | SFR on KUB (kidney, ureter, bladder radiograph); not complications |
| **S.T.O.N.E. nephrolithometry** | **S**ize (area), **T**ract length (≤100 or >100 mm), **O**bstruction (hydronephrosis), **N**umber of involved calyces, **E**ssence (density, ≤950 or >950 HU [Hounsfield units]); total 5–13 | SFR, blood loss, operative time, stay; needs CT (computed tomography) |
| **CROES nomogram** (Clinical Research Office of the Endourological Society) | Stone burden, location, staghorn, number, prior treatment, **annual centre volume** | SFR; only multicentre derivation, but SFR on KUB |
| **S-ReSC** (Seoul National University Renal Stone Complexity) | Number of **9 sites** involved on CT: pelvis, upper and lower major calyces, anterior and posterior minor calyces of the three groups; score 1–9 | SFR; not complications |
| **RUSS** (Resorlu–Unsal Stone Score) | 1 point each: stone >20 mm; lower-pole stone with infundibulopelvic angle <45°; stones in more than one calyx; abnormal anatomy; score 0–4 | SFR after **RIRS** (retrograde intrarenal surgery) |

### Limitations

- Each score added what the previous lacked: anatomy, then CT depth and density, then centre factors, then calyceal distribution.
- Mostly predict clearance, **not complications**. Meta-analysis: heterogeneous; no single generalisable score.
- **Also know:** Triple D score predicts SWL (shock wave lithotripsy) success, not PCNL.

**Diagram to draw:** kidney outline showing the 9 S-ReSC sites.

> **Key points to remember:**
> - Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
> - S.T.O.N.E.: five variables, score 5–13.
> - CROES includes centre volume; RUSS is for RIRS.

## SAQ 16 Calyceal diverticulum – classification and management

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3c

## Definition

- Cystic cavity in the renal cortex, **lined by transitional epithelium**, connected to a calyx (or pelvis) by a **narrow neck**. Fills passively with urine.
- About 4.5 per 1000; equal in children and adults; no side or sex preference.

## Classification

| | Type I | Type II |
|---|---|---|
| Frequency | Commoner | Less common |
| Connects to | Minor calyx (usually upper pole) | Renal pelvis or major calyx |
| Size, symptoms | Small, usually silent | Larger, often symptomatic |

## Aetiology

- **Congenital** (favoured): persistent third- and fourth-generation ureteric bud branches.
- **Acquired:** cortical abscess draining into a calyx, infundibular stenosis, trauma, reflux.

## Clinical features

- Stasis: **stones in up to 40%**, milk of calcium, infection, pain, haematuria.

## Evaluation

- Ultrasound mimics a simple cyst. **CT (computed tomography) urography with delayed images shows contrast pooling**; milk of calcium layers and shifts with position.
- Retrograde pyelography maps the neck before intervention.
- **Differential:** hydrocalycosis (dilated major calyx from infundibular obstruction), megacalycosis, simple cyst.

## Management

- **Asymptomatic:** observe with ultrasound.
- **Treat** pain, recurrent infection, haematuria, abscess, symptomatic stones.

| Approach | Technique | Best for |
|---|---|---|
| **Ureteroscopy** | Find the ostium, laser-incise or balloon-dilate the neck, remove stones | Upper or mid-pole, small stones |
| **PCNL** (percutaneous nephrolithotomy) | Direct puncture onto stone, clear stones, fulgurate lining, dilate or create neck | Posterior, lower-pole, larger stones; long-term results good |
| **Laparoscopic or robotic** | Unroof, remove stones, close neck, obliterate cavity, pack with fat | Anterior, thin-parenchyma, failed endoscopy |
| Partial nephrectomy | Excision | Rare; destroyed segment |

- Endoscopy keeps the cavity draining by widening the neck; laparoscopy abolishes it by closing the neck.
- **Also know:** SWL (shock wave lithotripsy) gives symptom relief but low stone clearance; fulguration of lining is debated.

**Diagram to draw:** kidney with upper-pole diverticulum, narrow neck to calyx, stone inside.

**Key points to remember:**

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

## SAQ 17 Pelvic lipomatosis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3e

### Definition

- Rare benign **overgrowth of non-encapsulated mature fat** in the perivesical and perirectal spaces, compressing the bladder, ureters, rectum and pelvic veins.
- Mostly men aged 20–60; associated with obesity and hypertension. Cause unknown; obesity alone does not explain it (weight loss rarely shrinks the fat). Chronic inflammation and venous obstruction are proposed.

### Clinical features

- LUTS (lower urinary tract symptoms), haematuria, recurrent UTI (urinary tract infection), suprapubic or perineal pain, constipation.
- Leg oedema or venous thrombosis from venous compression.
- **Hydronephrosis and renal failure** from ureteric compression in a substantial minority.
- **Proliferative cystitis (cystitis glandularis)** is common; it can transform to **adenocarcinoma**.

### Evaluation

- **IVU (intravenous urography) or cystogram:** elevated, elongated **"pear-" or "teardrop-shaped" bladder**; medially displaced pelvic ureters; high prostate.
- **Barium enema:** elongated, narrowed rectosigmoid.
- **Ultrasound:** echogenic pelvic fat; hydronephrosis.
- Medial deviation of **both** pelvic ureters also occurs in retroperitoneal fibrosis and after abdominoperineal surgery.
- **CT (computed tomography) or MRI (magnetic resonance imaging): diagnostic**; homogeneous fat density surrounding pelvic organs, separates it from tumour.
- **Cystoscopy:** elevated trigone, long prostatic urethra (rigid scope may be difficult); biopsy for cystitis glandularis.
- **Differential of pear-shaped bladder:** pelvic haematoma, lymphadenopathy, liposarcoma, iliopsoas hypertrophy, venous collaterals, pelvic fibrosis.

### Management

- **Surveillance** (often progressive): ultrasound for hydronephrosis, renal function, periodic cystoscopy for cystitis glandularis.
- Weight loss, antibiotics and steroids give limited or no benefit; COX-2 (cyclo-oxygenase-2) inhibitors for cystitis glandularis are case reports only.
- **Ureteric obstruction:** stents or nephrostomy; ureteric reimplantation (difficult, dense fat).

- **Advanced or failed cases:** urinary diversion; cystectomy if adenocarcinoma or unmanageable bladder.
- Surgical excision of fat is difficult: no tissue planes, heavy bleeding.
- Resect cystitis glandularis transurethrally and treat obstruction or inflammation.

**Diagram to draw:** cystogram outline of pear-shaped bladder with medially deviated ureters.

> **Key points to remember:**
> - Pear-shaped bladder with medial ureters; CT shows fat.
> - Cystitis glandularis common; adenocarcinoma risk.
> - Watch for hydronephrosis; diversion if obstruction progresses.

## SAQ 18 Cystinuria

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 21 SAQ Sheet, Q3a

### Definition and genetics

- **Autosomal recessive** defect of proximal tubular (and jejunal) reabsorption of the dibasic amino acids **COLA: cystine, ornithine, lysine, arginine**. Only cystine is insoluble, so only cystine forms stones.
- Transporter subunits and ICC (International Cystinuria Consortium) types:

| Gene | Subunit | Chromosome | Type |
|---|---|---|---|
| SLC3A1 | rBAT (heavy) | 2p | A |
| SLC7A9 | $b^{0,+}$AT (light) | 19q | B |
| Both | — | 2 and 19 | AB |

- Fractional excretion of cystine rises from about 0.4% to up to 100%. Incidence 1 in 1000 to 1 in 17,000; **1–3% of adult stones, up to 10% of paediatric stones**; males more severe.

### Pathophysiology

- **Pure supersaturation disease**: no specific inhibitor. Solubility is pH dependent: about 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9.
- Coexisting hypocitraturia (commonest), hyperuricosuria or hypercalciuria cause mixed calcium–cystine stones.

### Clinical features

- First stone at mean about 12 years; recurrent, bilateral, staghorn stones; CKD (chronic kidney disease) from repeated surgery (about a quarter of adults have eGFR [estimated glomerular filtration rate] <60).

### Diagnosis

- **Hexagonal crystals** in urine (always abnormal); cyanide–nitroprusside screen; 24-hour urinary cystine (normal <60 mg/day/1.73 $m^2$; homozygotes often >400).
- Stone: yellow, waxy, faintly radiopaque (sulphur); **resists SWL (shock wave lithotripsy)**.

### Management

- **Fluids:** urine output **≥3 L/day**, including at night.
- **Diet:** sodium ≤2300 mg/day; limit animal protein (not in children).
- **Alkalinise to pH 7.0–7.5** with potassium citrate 30–60 mEq/day; avoid >7.5 (calcium phosphate stones) and sodium alkali (raises cystine excretion).
- **Thiols** if cystine stays >250 mg/L: thiol–cystine mixed disulfide is soluble.
- **Tiopronin** (choice): 300 mg twice daily, up to 1200 mg/day.
- **D-penicillamine:** 250 mg/day, titrated; rash, proteinuria, nephrotic syndrome, pancytopenia.
- Captopril: weak evidence.
- Monitor blood count, liver tests, urine protein.
- **Surgery:** ureteroscopy or PCNL (percutaneous nephrolithotomy) with laser; nephron-sparing approach.

**Diagram to draw:** proximal tubule cell with rBAT/$b^{0,+}$AT transporter and hexagonal crystal.

> **Key points to remember:**
> - Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
> - Hexagonal crystals; target cystine <250 mg/L.
> - Urine ≥3 L, pH 7.0–7.5, then tiopronin.

## SAQ 19 Trans Uretero Ureteric anastomoses

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3a

### Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

### Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

### Contraindications

| Absolute | Relative (disease that may involve both ureters) |
|---|---|
| Donor ureter too short to reach without tension | Stone disease, **urothelial carcinoma** |
| Diseased recipient ureter | Retroperitoneal fibrosis, pelvic radiation |

| Absolute | Relative (disease that may involve both ureters) |
|---|---|
| | Chronic pyelonephritis, genitourinary tuberculosis |
| - Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk). | |

### Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

### Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

### Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

**Diagram to draw:** donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

> **Key points to remember:**
> - End-to-side, spatulated, tension-free; cross above the IMA.
> - Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
> - Risk: unilateral injury becomes bilateral.

## SAQ 20 Medical expulsive therapy (MET) in urinary calculi

P2

**Asked as:**

- Medical expulsive therapy in urinary calculi. — *Summer 2021 Paper II Q3d (SAQ)*
- Medical expulsive Rx for urinary calculi. — *Practice Set 11 Paper II Q3e (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2021-Summer Paper 2, Q3d)

## Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

## Rationale: natural history

- Spontaneous passage depends on **width**, size and site:

| Width | Spontaneous passage | Location | Spontaneous passage |
|---|---|---|---|
| 0–3 mm | 98% | Vesicoureteric junction (small stones) | 92% |
| 5 mm | 65% | Distal ureter | 75% |
| >6.5 mm | 9% | Proximal ureter | 48% |

## Mechanism

- **α1 receptors** (mainly **α1A and α1D**) are abundant in (dominate in) the distal ureter and VUJ (vesicoureteric junction). Blockade reduces spasm and the force and frequency of peristalsis (peristaltic force and frequency), so a larger urine bolus pushes the stone / builds higher pressure above the stone.
- Others: calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) relieve pain but do not aid passage.**

## Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses but twice the retrograde ejaculation); alfuzosin, doxazosin.

## Evidence

| Study | Result |
|---|---|
| **SUSPEND (Pickard, 2015):** 1136 patients; tamsulosin, nifedipine or placebo | No difference in need for intervention at 4 weeks (19% vs 20%) |
| **STONE (Meltzer, 2018):** stones <9 mm (up to 9 mm) | Tamsulosin no better than placebo |
| Meta-analyses and a large distal-stone RCT (randomised controlled trial) | Benefit for **distal stones >5 mm** |

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm** (smaller ones pass anyway; larger rarely pass).

## Indications and guideline position (AUA [American Urological Association] and EAU [European Association of Urology])

- AUA and EAU: offer MET for **ureteric stones <10 mm**, especially distal, when immediate intervention is not needed (EAU: distal >5 mm).
- **After SWL (shock wave lithotripsy)** for faster fragment clearance (to clear fragments). Children: off-label but effective.

## Contraindications (need urgent drainage instead; drain urgently)

- Infection with obstruction or sepsis, solitary or bilateral obstruction, renal deterioration (impairment), refractory pain or vomiting.

### Protocol

- NSAID analgesia; counsel on off-label use, dizziness, hypotension, retrograde ejaculation.
- Periodic KUB (kidney–ureter–bladder radiograph) or ultrasound follow-up; **intervene after 4–6 weeks** if not passed.

**Diagram to draw:** ureter with α1 receptors concentrated distally (at the distal ureter and VUJ [vesicoureteric junction]) and a stone above it (above the vesicoureteric junction).

> **Key points to remember:**
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

## SAQ 21 Physiology of UPJ in relation to urine transport

P2

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 2, Q3f

*UPJ = ureteropelvic (pelviureteric) junction.*

### Anatomy relevant to function

- The UPJ is the apex of a funnel-shaped pelvis; its spiral smooth muscle is continuous with calyces and ureter.

### Pacemaker and electrical activity

- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain and only modulate.
- **Primary pacemaker: atypical smooth muscle cells** at the pelvicalyceal border; they fire spontaneously (HCN, hyperpolarisation-activated cyclic nucleotide-gated, channels).
- **ICC-like (interstitial cells of Cajal-like) cells** (c-Kit positive) in the pelvis and UPJ relay the signal to typical smooth muscle and act as back-up pacemakers.
- Signal spreads through **gap junctions**. Calcium entry gives a plateau action potential; calcium–calmodulin activates myosin light-chain kinase, causing contraction.

### Bolus formation and transport

1. Urine fills the pelvis at low pressure; the UPJ is closed.
2. Pelvic contraction pushes urine through the opening UPJ into the upper ureter.
3. The UPJ closes behind it, forming a **bolus**.
4. Ureteric walls **coapt** behind the bolus; the wave (2–6 per minute, 20–80 cm $H_2O$) carries it to the bladder. - At **low flow**, the UPJ acts as a filter: several pelvic contractions occur for each ureteric contraction. - As **flow rises**, ureteric frequency reaches 1:1, then bolus volume rises. - At **very high flow**, boluses merge into a continuous column and the ureter dilates **without obstruction**.

### Effect of obstruction

- **Acute:** stronger, more frequent contractions.
- **Chronic:** dilatation, walls fail to coapt, peristalsis becomes ineffective; raised pelvic pressure lowers GFR (glomerular filtration rate).

- **Intrinsic UPJO (UPJ obstruction):** abnormal muscle, excess collagen, fewer ICC-like cells, so an **aperistaltic segment** cannot transmit the wave.
- **Also know:** endotoxin and progesterone reduce peristalsis. Diuretic renogram and Whitaker test (chapter: pressure difference >12 cm $H_2O$ suggests obstruction; classic teaching: <15 unobstructed, >22 obstructed) assess transport.

**Diagram to draw:** pelvis, UPJ and ureter showing bolus formation and coaptation.

> **Key points to remember:**
> - Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
> - Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
> - Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

## SAQ 22 Complications of PCNL, including bleeding and persistent haematuria after PCNL

P2

**Asked as:**

- Complications of PCNL — *Practice Set 2 Paper II Q3e (SAQ)*
- What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!? — *Practice Set 6 Paper II Q3b (SAQ)*

**Type:** SAQ (10 marks)
(mini-PCNL complications in 2022-Summer Paper 1, Q1) (PCNL complications also in Practice Set 2 Paper 2, Q3e)

### Introduction

- PCNL (percutaneous nephrolithotomy) is the standard for renal stones >2 cm. Most complications arise from **access and tract dilatation**.
- Graded by the **Clavien–Dindo** system (I: no intervention; II: drugs or transfusion; III: intervention; IV: ICU [intensive care unit]; V: death).
- Bleeding is the commonest serious complication of PCNL. Transfusion was historically ~7%, now nearer 1% with smaller tracts. Angioembolisation is needed in <1%.

### Incidence (systematic review, Seitz 2012, quoted by EAU [European Association of Urology])

| Complication | Rate |
|---|---|
| Fever | 10.8% |
| Transfusion | 7% (nearer 1% with modern and mini tracts) |
| Thoracic | 1.5% |
| Sepsis | 0.5% |
| Organ injury | 0.4% |
| Angioembolisation | 0.4% |
| Death | 0.05% |

## Main complications

| Complication | Cause | Prevention and management |
|---|---|---|
| **Acute bleeding** | Infundibular or segmental vessel injury | Fornix puncture; place sheath, tamponade with balloon or **Kaye catheter**, clamp nephrostomy; transfuse |
| **Delayed bleeding** | Pseudoaneurysm, AV (arteriovenous) fistula | **Selective angioembolisation** |
| **Infection, SIRS (systemic inflammatory response syndrome), sepsis** | Infected stone, high intrarenal pressure | Sterile urine, culture-specific antibiotics, low-pressure irrigation; stop and drain if pus |
| **Pleural** (pneumothorax, hydrothorax) | Supracostal access, highest above 11th rib | Subcostal where possible, puncture in expiration; chest drain |
| **Colon** | Retrorenal colon, left lower pole | CT review; extraperitoneal: double-J stent and withdraw nephrostomy into colon (colostomy tube) |
| **Spleen, liver** | Upper-pole access, organomegaly | Ultrasound guidance; conservative or surgery |
| **Collecting system** | Perforation, extravasation | Stop, drain; later infundibular stenosis |
| **Fluid absorption** | Long operation, perforation | **Normal saline**; hyponatraemia, overload |

**Also know:** venous air embolism, DVT (deep vein thrombosis) and pulmonary embolism, neuropraxia and eye injury (prone), residual fragments, loss of renal function (multiple tracts), urinoma, nephrocutaneous fistula, retained hardware.

## Risk factors

- Staghorn, multiple tracts, supracostal access, large tract, long operation, CKD (chronic kidney disease), diabetes, positive urine culture, inexperience.
- **Bleeding risk factors:** infundibular or upper-pole puncture, multiple or large tracts, staghorn stone, long operation, antithrombotic drugs, coagulopathy, diabetes.

## Reducing risk

- **Tranexamic acid** reduces transfusion (Cochrane).
- Ultrasound-guided access, mini-PCNL, ECIRS (endoscopic combined intrarenal surgery).
- **Bleeding prevention:** puncture through the **calyceal fornix** along a posterior calyx (Sampaio: no arterial injury); correct clotting; mini-tracts.

## Bleeding in PCNL

### Causes

| Timing | Cause |
|---|---|
| **Intraoperative** | Parenchymal or venous tract bleeding; infundibular artery injury; through-and-through puncture; posterior segmental artery injury (upper-pole access); torn infundibulum or pelvis |
| **Early postoperative** | Tract bleeding after sheath removal; clot retention |

| Timing | Cause |
|---|---|
| **Delayed (about 1 week)** | **Pseudoaneurysm**, **AV (arteriovenous) fistula** |

**Mechanism of delayed bleeding**

- **Pseudoaneurysm:** clot seals an arterial tear, then ruptures intermittently.
- **AV fistula:** artery and vein heal together; the thin vein cannot withstand arterial pressure.

**Management at day 6 (persistent haematuria)**

- At day 6, assume an **arterial lesion** until proved otherwise. `Resuscitate: IV access, crossmatch, transfuse; Hb, clotting → Stop anticoagulants; catheter, irrigate clots → Stable, mild: bed rest, clamp nephrostomy, serial Hb → Persistent or recurrent bleeding or Hb fall → CT angiography (arterial blush) or direct angiography → SUPERSELECTIVE ANGIOEMBOLISATION (coils) → Fails or unstable after embolisation → open exploration, partial or total nephrectomy`
- **Angiography** diagnoses >90% and treats in the same sitting. It may miss intermittent bleeds; non-selective embolisation loses renal tissue.
- Clot colic or obstruction: double-J stent or nephrostomy.
- Follow-up: Hb and creatinine; check for infection and abscess.

**Intraoperative ladder (for completeness)**

- Pressure on tract → nephrostomy clamped (tamponade) → balloon reinflation or **Kaye tamponade catheter** → angioembolisation.

**Diagram to draw:** kidney cross-section with posterior calyx, Brödel line, pleura and retrorenal colon; angiogram sketch: pseudoaneurysm before and after coiling.

> **Key points to remember:**
> - Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
> - Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
> - Early bleeding venous: tamponade.
> - Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
> - Treatment: superselective angioembolisation.
> - Supracostal access increases pleural injury; colon injury at left lower pole.

## SAQ 23 Steinstresse.

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 7 Paper 2, Q3e

*Assumed meaning: steinstrasse.*

### Definition

- **Steinstrasse** (German, "stone street"): a column of stone fragments in the ureter that fails to pass and may obstruct. Typically after **SWL (shock wave lithotripsy)**; occasionally after PCNL (percutaneous nephrolithotomy) or URS (ureteroscopy).
- Incidence is low overall (about 4–7% after SWL, EAU [European Association of Urology]) but rises several-fold with stone burden **>2 cm²**.

### Risk factors

- Large burden (>2 cm), renal pelvic stones, hard stones that break into large fragments, high energy, impaired ureteric drainage.

### Classification (Coptcoat)

| Type | Composition |
|---|---|
| I | Small fragments only |
| II | Larger lead fragment (4–5 mm) followed by small fragments |
| III | Large fragments throughout |

### Clinical features

- Asymptomatic (silent obstruction), colic, nausea; fever and sepsis if infected; anuria and AKI (acute kidney injury) in a solitary kidney. Most lie in the distal ureter.

### Evaluation

- KUB (kidney, ureter, bladder) X-ray, ultrasound, or non-contrast CT (computed tomography); creatinine, urine culture, blood counts.

### Management

```
Steinstrasse ├─ infection / sepsis → urgent PCN (percutaneous nephrostomy) + antibiotics ├─ asymptomatic, no obstruction → MET + analgesia, imaging every 1–2 weeks └─ failure, pain, obstruction: ├─ lead fragment (type II/III) → SWL to lead fragment └─ URS with laser lithotripsy (most definitive)
```

- **MET (medical expulsive therapy):** α-blockers (tamsulosin) improve fragment clearance and reduce pain.
- **PCN** is preferred in infected obstruction: drains without manipulating the ureter, and fragments often pass once the kidney is decompressed. A stent is an alternative.
- Retained fragments with poor progress: URS clears the column.

### Prevention

- Choose PCNL or RIRS (retrograde intrarenal surgery) for large burdens.
- Energy ramping, slower shock rate; MET after SWL.
- **Stent before SWL:** may reduce steinstrasse with large stones but does not improve stone-free rate and adds stent symptoms. The chapters disagree: Chapter 97 says it may reduce steinstrasse; Chapter 98 advises against stenting at SWL.

**Diagram to draw:** ureter with lead fragment and trailing column (Coptcoat types I–III).

> **Key points to remember:**
> - Risk rises several-fold with burden >2 cm².
> - Coptcoat I–III by fragment size.
> - Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

## SAQ 24 Treatment of inferior calyceal calculus

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 2, Q3b

## Introduction

- Lower pole is the **commonest site** of renal stones. Gravity-dependent position makes fragments hard to clear, so **SWL (shock wave lithotripsy) works poorly** here.

## Factors in decision

- **Stone:** size, density (<900 HU [Hounsfield units] favours SWL), composition.
- **Anatomy unfavourable for SWL clearance:** acute **infundibulopelvic angle**, long and narrow infundibulum, skin-to-stone distance >10 cm.
- **Patient:** obesity, anticoagulants (favour URS [ureteroscopy]), pregnancy, preference.
- Evaluation: NCCT (non-contrast CT), urine culture, renal function.

## Treatment by size

| Size | Options |
|---|---|
| **Asymptomatic, small** | Observation acceptable; about half have an event in 5 years. Treat if growing, symptomatic, infected or occupational |
| **<10 mm** | SWL or **RIRS** (retrograde intrarenal surgery). AUA (American Urological Association): either |
| **10–20 mm** | EAU (European Association of Urology): SWL only if favourable anatomy; otherwise **RIRS or PCNL** (percutaneous nephrolithotomy) |
| **>10 mm (AUA)** | **PCNL** (highest SFR [stone-free rate]) or URS; SWL not first-line |
| **>20 mm** | PCNL first; staged RIRS an alternative |

## Evidence

- **Lower Pole I (Albala 2001):** stones >1 cm: SFR 91% PCNL vs 21% SWL.
- Lower Pole II: stones ≤1 cm, URS vs SWL: no significant SFR difference.

## Technique points

- **RIRS:** flexible ureteroscope (deflection up to 270°); relocate the stone to an upper calyx with a basket before laser (holmium or thulium fibre) lithotripsy; access sheath.
- **PCNL:** direct lower pole posterior calyx puncture; mini-PCNL reduces bleeding (transfusion near 1%).
- **SWL adjuncts:** percussion, diuresis, inversion therapy and alpha-blockers improve fragment clearance.
- **Also know:** lower pole calyceal diverticulum or infundibular stenosis favours PCNL; residual fragments >4 mm after URS need treatment.

## Follow-up

- Imaging for residual fragments; metabolic evaluation if recurrent.

**Diagram to draw:** infundibulopelvic angle and infundibular length/width on an IVU (intravenous urogram) tracing.

> **Key points to remember:**
> - Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
> - <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
> - Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
> - Reposition stone during RIRS.

## SAQ 25 Primary Hyperoxaluria

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 10 Paper 2, Q3f

### Definition

- Rare **autosomal recessive** errors of **hepatic glyoxylate metabolism**: glyoxylate is not converted to glycine and is oxidised to oxalate, an end product cleared only by the kidney.
- Hyperoxaluria = urinary oxalate >40 mg/day; in PH (primary hyperoxaluria) often **>100 mg/day**.

### Types

| | PH1 (~70%) | PH2 (~10%) | PH3 (~10%) |
|---|---|---|---|
| Enzyme (gene) | **AGT** (alanine-glyoxylate aminotransferase; **AGXT**), liver peroxisome, **pyridoxine (B6) cofactor** | GRHPR (glyoxylate reductase/hydroxypyruvate reductase) | HOGA1 (4-hydroxy-2-oxoglutarate aldolase), mitochondria |
| Urine marker | **Glycolate** | **L-glycerate** | 4-hydroxy-2-oxoglutarate |
| Course | **Most severe**; ESKD (end-stage kidney disease) in 2nd–3rd decade | Milder, can reach ESKD | Childhood stones; rarely ESKD |

### Pathophysiology

- High urinary oxalate → calcium oxalate crystals in tubules → **nephrocalcinosis** and recurrent stones.
- **Vicious circle:** falling GFR (glomerular filtration rate) reduces oxalate clearance → plasma oxalate rises → **systemic oxalosis** (bone, retina, heart, vessels, skin).

### Clinical features

- Spectrum from **infantile oxalosis with renal failure** to occasional adult stones; some diagnosed only after a transplant fails.
- Clues: childhood stones, nephrocalcinosis, family history, **calcium oxalate monohydrate** stones.

### Diagnosis

`24-h urinary oxalate ↑ (child: spot oxalate:creatinine ratio) → urinary glycolate / L-glycerate / 4-hydroxy-2-oxoglutarate → genetic testing (AGXT, GRHPR, HOGA1) all normal → secondary (enteric, dietary) hyperoxaluria`

- Plasma oxalate when GFR is low; screen siblings and related living donors.

### Management

- **Hyperhydration** (about 3 L/m²/day), **potassium citrate**, normal calcium intake.
- **Pyridoxine** trial in all PH1: responsive mainly with the Gly170Arg mutation; it restores AGT activity and peroxisomal targeting.
- **Lumasiran** (RNA interference against glycolate oxidase, which cuts glyoxylate supply): **ILLUMINATE-A**: 84% vs 0% reached urinary oxalate ≤1.5 × upper limit of normal. **Also know:** nedosiran.
- Stones: ureteroscopy or PCNL (percutaneous nephrolithotomy); limit SWL (shock wave lithotripsy).
- **ESKD:** intensive dialysis, then **combined liver–kidney transplant for PH1** (only the liver replaces the enzyme; a lone kidney would be re-damaged). Kidney alone usually suffices for PH2.

**Diagram to draw:** hepatic glyoxylate pathway showing the AGT block (PH1) and the lumasiran target (glycolate oxidase).

> **Key points to remember:**
> - PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
> - Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
> - Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

## SAQ 26 Fracture penis

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3c

### Definition and mechanism

- **Penile fracture** is a tear of the **tunica albuginea** with rupture of the corpus cavernosum, in the **erect** penis.
- Abnormal bending of the erect penis causes an abrupt rise in intracavernosal pressure beyond tunical strength → **transverse tear**.
- The tunica (inner circular, outer longitudinal layers) is **thinnest ventrolaterally**, so most tears are ventral or lateral, proximal to mid shaft.
- **Causes:** intercourse (commonest; rear-entry), masturbation, rolling over; self-inflicted "taqaandan" (Middle East); collagenase injection for Peyronie disease.
- **Bilateral tears in about 10%**; urethral injury about 20% in the West vs about 3% in Asia and the Middle East.

### Clinical features

- **Cracking or popping sound → pain → rapid detumescence → swelling and discoloration.**
- **"Eggplant deformity"** when Buck's fascia is intact; if torn, haematoma spreads to scrotum, perineum and suprapubic area.
- Penis deviates **away from the tear**; tunical defect may be palpable.
- **Urethral injury:** blood at meatus, haematuria, retention.

### Investigations

- **Diagnosis is clinical** (AUA [American Urological Association] Urotrauma Guideline).
- **Ultrasound** for equivocal cases (localises the tear); **MRI** (magnetic resonance imaging) if still equivocal (100% negative predictive value). Cavernosography is discouraged.
- Urethra: **intraoperative flexible cystoscopy** (AUA and EAU [European Association of Urology] preference) or RUG (retrograde urethrogram).
- **"False fracture":** dorsal vein or artery rupture.

### Management

- **Immediate surgical repair** (less curvature, ED [erectile dysfunction] and nodules than conservative care). 1. Incision: ventral penoscrotal or lateral over the defect; subcoronal degloving if site unclear (preputial ischaemia risk). 2. Evacuate haematoma; find the tear (artificial erection helps). 3. Close the tunica with **interrupted 2-0/3-0 absorbable sutures**; avoid debriding erectile tissue. 4. Urethra: partial → close over catheter; complete → tension-free repair; catheter 2–3 weeks.
- Abstinence about 1 month. Delay up to 7 days does not worsen outcome. After collagenase injection without urethral injury, conservative care is acceptable.

### Complications

- About 20%: nodule, curvature, ED, painful erection, infection, stricture or fistula.

**Diagram to draw:** cross-section of penis showing tunical tear, Buck's fascia and haematoma spread.

> **Key points to remember:**
> - Snap, detumescence, eggplant deformity.
> - Urethral injury about 20% (West); suspect if bilateral.
> - Immediate exploration and interrupted absorbable tunical closure.

## SAQ 27 Ovarian vein syndromes

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3e

### Definition

- **Ovarian vein syndrome** (Clark, 1964) is extrinsic compression of the **right ureter** by a dilated or thrombosed **right ovarian vein**, causing intermittent hydroureteronephrosis. It is a **controversial** entity; many cases are now attributed to other causes.
- The plural also covers ovarian vein thrombosis and pelvic congestion.

### Anatomy (basis)

- Ovarian veins run in the infundibulopelvic ligaments. The **right** enters the IVC (inferior vena cava) obliquely; the **left** enters the left renal vein at a right angle.
- Gonadal vessels **cross anterior to the ureter**; ovarian veins dilate greatly in pregnancy.

### Types

| Syndrome | Mechanism | Features |
|---|---|---|
| **Ovarian vein (ureteric) syndrome** | Dilated right ovarian vein compresses the ureter at the crossing | Multiparous women; right flank pain (premenstrual or in pregnancy); recurrent UTI (urinary tract infection) |
| **Ovarian vein thrombosis** | Postpartum hypercoagulability and venous stasis; mostly right-sided | Fever and right lower-quadrant or flank pain after delivery; may obstruct ureter; risk of pulmonary embolism |
| **Pelvic congestion syndrome** | Incompetent, refluxing ovarian veins with pelvic varices; sometimes secondary to nutcracker (left renal vein compression) | Chronic pelvic pain worse on standing, dyspareunia, post-coital ache |

### Controversy

- Right-sided dilatation in pregnancy is mostly **physiological hydronephrosis** (progesterone plus uterine compression; stops at the pelvic brim). Venous compression is rarely proven, so diagnosis is by **exclusion**.

### Investigations

- Ultrasound; **CT urography** or IVU (intravenous urography): right hydroureter to a smooth oblique extrinsic indentation, normal ureter below.
- **Diuretic renogram** to prove obstruction; retrograde pyelography.
- CT or MR (magnetic resonance) venography, Doppler; **gonadal venography** (gold standard for reflux).
- **Exclude:** stone, retrocaval ureter, iliac vessel crossing, retroperitoneal fibrosis, endometriosis.

### Management

- **Asymptomatic:** observe. **Obstruction:** stent or nephrostomy if infected or pregnant; definitive **ureterolysis with ligation and division of the ovarian vein** (open or laparoscopic).
- **Thrombosis:** anticoagulation and antibiotics.
- **Pelvic congestion:** progestogens or GnRH (gonadotropin-releasing hormone) analogues; **coil or sclerotherapy embolisation**; ligation.

**Diagram to draw:** right ovarian vein crossing anterior to the right ureter, with proximal hydroureteronephrosis.

> **Key points to remember:**
> - Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
> - Postpartum ovarian vein thrombosis: right side, anticoagulate.
> - Treatment: ureterolysis with vein division, or embolisation.

## SAQ 28 Post traumatic bladder neck injury: acute and delayed management?

P3

**Type:** SAQ (10 marks)
**Source:** 2021-Summer Paper 3, Q3c

### Introduction

- The bladder neck is the **internal sphincter**. It is torn mainly in **pelvic fracture**, often with prostatic urethral, rectal or vaginal injury; **commoner in children**.
- Missed injury causes **incontinence, fistula and stricture**.

### Diagnosis

- Pelvic fracture with gross haematuria, retention or blood at the meatus; **vaginal examination** in females.
- **RUG (retrograde urethrogram)**, then **retrograde CT or plain cystography** (350 mL).
- Grading: AAST (American Association for the Surgery of Trauma) bladder **grade V** = laceration into the bladder neck or ureteric orifice; Goldman type IV = bladder rupture extending into the proximal urethra.

### Acute management

- Resuscitate; if unstable, damage control (drainage), repair later.
- **Bladder neck injury needs immediate repair** (AUA (American Urological Association) and EAU (European Association of Urology)), unlike isolated posterior urethral distraction (suprapubic tube, delayed repair). `Lower midline incision → anterior cystotomy → identify ureteric orifices (stent if near) → minimal debridement; re-approximate neck to prostatic urethra, layered absorbable sutures over a 22 Fr catheter → rectal or vaginal tear: separate layers, no overlapping suture lines, interpose omentum (colostomy for rectum) → remove bone spicules; drain ± suprapubic tube`
- Catheter 2–3 weeks; cystogram before removal.

### Delayed management (3–6 months)

| Problem | Management |
|---|---|
| Associated urethral distraction | Cystogram + RUG + endoscopy; **anastomotic urethroplasty first**, not combined with bladder neck surgery |
| Bladder neck contracture | Endoscopic incision or resection; open Y-V plasty if refractory |
| **Incontinence** (both sphincters damaged) | Urodynamics; **AUS (artificial urinary sphincter)**; bladder neck reconstruction (Young–Dees–Leadbetter) or fascial sling; last resort **bladder neck closure + Mitrofanoff** ± augmentation |
| Fistula | Layered repair with tissue interposition |
| - Bladder neck appearance on imaging **does not predict** continence; restore continuity first, then treat incontinence. | |
| - Continence is poor whenever the bladder neck is involved (paediatric series: incontinence up to 20% overall). | |

**Diagram to draw:** pelvic fracture with bladder neck tear and repair over a catheter.

**Key points to remember:**

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

## SAQ 29 Boari flap

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3d

### Definition

- A **tubularised flap of anterior bladder wall**, based posterolaterally on the side of the lesion, swung up to bridge a distal or mid-ureteric gap. Described in dogs 1894, humans 1947; laparoscopic 2001, robotic 2008.

### Reconstructive ladder (distal ureter)

| Option | Gap bridged |
|---|---|
| Ureteroneocystostomy | Distal 4–5 cm |
| + Psoas hitch | Adds up to 5 cm (total 6–10 cm) |
| **Boari flap (+ hitch)** | **10–15 cm**; spiral flap may reach pelvis (right) |
| Downward nephropexy | Adds up to 5–8 cm |
| - Beyond: TUU (transureteroureterostomy), ileal ureter, autotransplantation. | |

### Indications

- Long distal or mid-ureteric loss: gynaecological injury (commonest), stricture, failed reimplantation, distal TCC (transitional cell carcinoma) ureter, tuberculosis.

### Contraindications

- **Small, contracted or poorly compliant bladder** (radiation, tuberculosis); uncorrected outlet obstruction or neurogenic bladder. Not usually for acute unstable trauma.
- Preoperative: urodynamics or cystogram for capacity; antegrade and retrograde studies for length.

### Technique

1. Mobilise bladder; divide contralateral superior vesical pedicle if needed.
2. Mark flap on the **non-distended** bladder: **base ≥4 cm, apex ≥3 cm, length:base ≤3:1**; length = gap + 3 cm if tunnelled.
3. Raise flap; **psoas hitch** its base (sutures superficial, parallel to fibres: spare genitofemoral and femoral nerves).
4. Anastomose spatulated ureter: **non-refluxing** 3 cm submucosal tunnel, or **refluxing** direct (apex about 3 times ureteric diameter).
5. Stent (double-J); tubularise in two layers 2-0 absorbable; drain, catheter 7–14 days. - Laparoscopic and robotic: transperitoneal, refluxing anastomosis, results excellent.

### Complications

- **Recurrent stricture** (commonest; ischaemia or tension), reduced capacity with frequency and urgency, leak, nerve palsy, iliac vessel injury, pseudodiverticulum.
- eGFR (estimated glomerular filtration rate) fell in 44% within 2 years after reimplantation: counsel.
- **Alternative:** bladder advancement flap, similar reach and simpler.

**Diagram to draw:** bladder outline with oblique flap (4 cm base, 3 cm apex), hitched and tubularised to ureter.

> **Key points to remember:**
> - Bridges 10–15 cm; combine with psoas hitch.
> - Base ≥4 cm, length:base ≤3:1, measure bladder empty.
> - Contraindicated in small contracted bladder.

## SAQ 30 Struvite stones.

P4

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 4, Q3a

### Definition and epidemiology

- **Infection stones:** magnesium ammonium phosphate hexahydrate ($MgNH_4PO_4 \cdot 6H_2O$), usually mixed with **carbonate apatite**.
- 5–15% of stones (falling); female:male ~2:1; highest in India and Pakistan. Risk: neurogenic bladder, spinal injury, diversion, catheters, obstruction, diabetes, elderly.

### Pathogenesis

```
Urease-producing bacteria Urea + H2O -> 2NH3 + CO2 NH3 -> NH4+ (urine pH > 7.2) CO2 -> carbonate; phosphate -> PO4(3-) + Mg -> struvite; + Ca -> carbonate apatite
```

- Sterile urine has no urease, so infection is essential. Ammonia damages the GAG (glycosaminoglycan) layer; bacteria live inside the stone in a **biofilm**, so antibiotics cannot sterilise a retained stone.
- **Organisms: Proteus mirabilis** (commonest), Klebsiella, Pseudomonas, Providencia, Staphylococcus, **Ureaplasma** (easily missed). E. coli is rarely urease-positive.

### Clinical features

- Recurrent UTI (urinary tract infection), fever, flank pain, haematuria; often few symptoms. Grow fast into **staghorn** calculi. Untreated: pyonephrosis, XGP (xanthogranulomatous pyelonephritis), sepsis, renal loss.

### Evaluation

- Urine pH >7; culture (urine and stone cultures often differ). NCCT (non-contrast computed tomography): burden and anatomy; DMSA (dimercaptosuccinic acid) or renogram for split function. Stone analysis; 24-h urine (many have hypercalciuria or hypocitraturia).

### Management

- **Complete stone clearance** is the goal: each fragment is an infected nidus.
- Culture-specific antibiotics before surgery; drain an obstructed infected kidney first.
- **PCNL (percutaneous nephrolithotomy): first line** for staghorns (AUA, American Urological Association). Add flexible nephroscopy, ECIRS (endoscopic combined intrarenal surgery) or second look. SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery) for small stones only.
- Anatrophic nephrolithotomy rarely; nephrectomy for a poorly functioning kidney.
- **Prevention:** surveillance cultures, suppressive antibiotics, fluids, correct stasis or metabolic defects.
- **AHA (acetohydroxamic acid)** 250 mg three times daily (urease inhibitor), only when surgery is exhausted. Side effects in up to 30% (haemolytic anaemia, DVT (deep vein thrombosis)); avoid if creatinine >2.5 mg/dL or pregnancy.
- **Also know:** hemiacidrin irrigation (historical); matrix stones (radiolucent, Proteus).

**Diagram to draw:** urease cascade; staghorn with PCNL tract.

> **Key points to remember:**
> - Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
> - Proteus mirabilis commonest; Ureaplasma easily missed.
> - Complete clearance with PCNL; AHA only when surgery is not possible.

## SAQ 31 Role of open and laparoscopic stone surgeries in modern era of endourology.

P4

**Type:** SAQ (10 marks)
**Source:** 2021-Summer Paper 4, Q3f

### Introduction

- SWL (shock wave lithotripsy), URS (ureteroscopy) and PCNL (percutaneous nephrolithotomy) treat almost all stones. Open surgery is now a small minority in centres with full endourology, but remains in resource-limited settings.
- **AUA (American Urological Association)/Endourological Society 2016:** consider laparoscopic, robotic or open surgery **only when SWL, URS and PCNL fail or are expected to fail**; not first line.

### Indications

| Category | Examples |
|---|---|
| Endourology failed or unlikely | Very complex staghorn needing many PCNL sessions; failed multiple procedures |
| Anatomical problem treated together | **PUJ (pelviureteric junction) obstruction + stone:** pyeloplasty with pyelolithotomy; anterior calyceal diverticulum; infundibular stenosis |
| Difficult access | Pelvic or ectopic kidney (laparoscopy-assisted PCNL); morbid obesity; skeletal deformity |

| Category | Examples |
|---|---|
| Poor renal function | **Nephrectomy** for a non-functioning kidney; partial nephrectomy for a non-functioning pole |
| Large impacted ureteric stone | Laparoscopic ureterolithotomy after failed URS/SWL |
| Concomitant surgery | Other open or laparoscopic procedure |
| Bladder | Very large stone; with open prostatectomy, diverticulectomy or augmentation; small bladder; urethral stricture |

### Procedures

- **Pyelolithotomy:** simple, extended (Gil-Vernet, sinus dissection) or coagulum.
- **Anatrophic nephrolithotomy (Smith and Boyce):** for complex staghorn. Clamp renal artery; cool with ice slush; give mannitol; incise along **Brödel's avascular line**; remove stone; repair infundibula; close.
- Radial nephrotomy, ureterolithotomy, cystolithotomy.

### Laparoscopic and robotic surgery

- Transperitoneal or retroperitoneal pyelolithotomy and ureterolithotomy; robotic anatrophic nephrolithotomy reported.
- Best for large pelvic stone in an extrarenal pelvis: high single-stage clearance, less bleeding than PCNL. Drawbacks: longer operation, calyceal stones need a flexible scope.

| Feature | Open | Laparoscopic/robotic |
|---|---|---|
| Morbidity | Large incision, pain, long stay | Less pain, shorter stay |
| Kidney | Ischaemic loss of function (anatrophic) | Similar repair, less wound morbidity |
| Later surgery | Adhesions make redo difficult | Fewer |

### Complications

- Bleeding, urinary leak, infection, loss of renal function, residual fragments.

**Diagram to draw:** Brödel's line incision for anatrophic nephrolithotomy.

> **Key points to remember:**
> - Open/laparoscopic only when endourology fails or is unsuitable.
> - Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
> - Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

## SAQ 32 Diet in stone disease

P4

**Asked as:**

- Diet in stone disease. — *Winter 2023 Paper IV Q3f (SAQ)*
- Diet in stone disease. — *Practice Set 5 Paper IV Q3f (earlier version) (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3f)

## Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

## General measures

| Measure | Target | Mechanism and evidence |
|---|---|---|
| **Fluid** | Urine **≥2.5 L/day** (about 3 L intake), spread through the day | Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years |
| **Normal calcium** | 1000–1200 mg/day, with meals | Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet |
| **Low sodium** | Salt about 4–5 g/day | Sodium increases urinary calcium, lowers citrate |
| **Limit animal protein** | 0.8–1.0 g/kg/day | Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria |
| **Fruit and vegetables** (DASH [Dietary Approaches to Stop Hypertension] diet) | Daily | Alkali and citrate; DASH linked to fewer stones |
| **Weight** | BMI (body mass index) 18–25 | Obesity: low urine pH, more oxalate and uric acid |

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**
- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

## Stone-specific diet

| Stone or abnormality | Diet |
|---|---|
| **Hyperoxaluria** | Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals |
| **Enteric hyperoxaluria** | Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium |
| **Uric acid** | Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss |
| **Cystine** | Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children) |
| **Struvite** | Infection control; diet has little role |

## Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

**Diagram to draw:** "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

**Key points to remember:**

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

## SAQ 33 ECIRS (endoscopic combined intrarenal surgery)

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3b

### Definition

- **ECIRS:** PCNL (percutaneous nephrolithotomy) and retrograde flexible ureteroscopy performed **at the same time, in one position**, by two surgeons.
- Term and technique popularised by **Scoffone (2008)** in the **GMSV (Galdakao-modified supine Valdivia) position**. Combined access dates back to the 1980s (Lawson's retrograde needle placement, 1983).

### Rationale

- Each route covers the other's blind spots: the retrograde scope guides and watches the puncture, and reaches calyces the rigid nephroscope cannot.
- Result: one tract, better clearance.

### Positioning

| Position | Features |
|---|---|
| **GMSV** (preferred) | Supine; flank on a bolster at the table edge; ipsilateral leg extended, contralateral leg in lithotomy |
| Bart's modified supine | Variant of supine |
| Prone split-leg | Easier posterior calyces; needs a split-leg table |

- **Supine advantages:** airway access, gravity drainage of fragments, no turning, simultaneous retrograde access.

### Technique

1. Cystoscopy; UAS (ureteral access sheath) placed; flexible ureteroscope maps the stones and chooses the target calyx.
2. Ultrasound or fluoroscopic puncture **watched endoscopically** (double bull's-eye technique).
3. Wire grasped and brought out: through-and-through access.
4. Tract dilated under vision (balloon) and sheath placed.
5. Antegrade lithotripsy and suction; the retrograde scope **relocates stones** from inaccessible calyces into the nephroscope's view, or lasers them.
6. Final check with both scopes; stent or tubeless exit. - Mini-ECIRS uses a mini-PCNL tract.

### Indications

- Staghorn and complex multi-calyceal stones, combined renal and ureteric stones, anomalous kidneys, failed previous treatment, urinary diversion, obese or cardiopulmonary patients (supine).

### Advantages (meta-analysis)

- Single position, **fewer tracts, less bleeding**, fewer ancillary procedures, **higher stone-free rate**; less fluoroscopy; the UAS keeps intrarenal pressure low.

### Disadvantages

- Two surgeons and two sets of equipment; cost; longer set-up; ureteric injury from the sheath; less working space in supine.

**Diagram to draw:** GMSV position with nephroscope from the flank and flexible ureteroscope from below in the same calyx.

> **Key points to remember:**
> - Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
> - Retrograde scope guides the puncture and relocates stones.
> - Fewer tracts, less bleeding, higher stone-free rate.

# Rapid Revision Sheet · Section 11

### Congenital PUJ obstruction, failed pyeloplasty and salvage procedures (LAQ)

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

### Urolithiasis: etiopathogenesis, medical management and prevention of recurrence (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

### Metabolic evaluation in stone disease (LAQ)

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride ($NH_4Cl$) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
- NCCT gold standard (97%/95%).
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

### Retroperitoneal fibrosis: diagnosis and (medical) management (LAQ)

- Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
- Medially deviated ureters; PET-CT best shows activity; biopsy deep.
- Drain first, then prednisone (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease.
- About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

**Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management (LAQ)**

- Blood at meatus: RUG before catheter.
- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
- Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
- ED mostly from the injury; incontinence <4%.

**Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management (LAQ)**

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Delayed CT phase is essential; blush means AE.
- Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
- Revascularise only a solitary kidney or bilateral injuries.

**ESWL: mechanism, indications, factors affecting outcome, complications and current status (LAQ)**

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 $cm^2$; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

**Management of upper ureteric stones (LAQ)**

- Proximal stones pass in only 48%; MET does not help proximal stones.
- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

**Uric acid calculi: etiopathogenesis, medical management and allopurinol (LAQ)**

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.

- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

**Urinary crystals (LAQ)**

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

**Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)**

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

**Laparoscopic pyeloplasty. (LAQ)**

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

**Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury (LAQ)**

- Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
- Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

**Energy sources for breaking stones in urology (LAQ)**

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

**Flexible ureteroscopy / RIRS and accessories (SAQ)**

- 3.6 Fr channel; ≥270° deflection.
- 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
- Sepsis about 5%, stricture about 3%.

- Hydrophilic wire passes; PTFE wire stays.
- Nitinol tipless basket for calyces.

**Renal tubular acidosis (distal and type 2) (SAQ)**

- Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
- $NH_4Cl$ load: pH fails to fall below 5.5 in distal RTA.
- Potassium citrate is the treatment for distal RTA.
- Type 2 (proximal): bicarbonate wasting; $HCO_3^-$ 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
- Fractional $HCO_3^-$ excretion >15%; stones uncommon in type 2.
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause.

**PCNL patient positioning: prone vs supine (SAQ)**

- Prone: posterior and upper pole access, large working area.
- Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
- Horizontal bolsters drop the kidney more.
- Similar outcomes; choose by anatomy and experience.

**Renal autotransplantation (SAQ)**

- For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

**Stones in pregnancy (SAQ)**

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

**Ureterocalycostomy (SAQ)**

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
- Stricture is the main failure; success about 70% in redo series.

**Tubeless PCNL (SAQ)**

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

**Mini-PCNL and newer PCNL techniques (SAQ)**

- Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).

- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
- Working channel decides instruments; suction lowers pressure.

**Newer methods of stone analysis (SAQ)**

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
- DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.
- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

**Radionuclide studies and diuretic renography in PUJ obstruction (SAQ)**

- MAG3 is the tracer of choice; hydrate and catheterise.
- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
- Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

**What are nanobacteria? Describe their role in urolithiasis. (SAQ)**

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

**Tear drop bladder (SAQ)**

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

**Retrograde endopyelotomy (SAQ)**

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

**Post-obstructive diuresis (SAQ)**

- Unilateral about 2%, bilateral about 50%.
- Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
- Replace about 75% of output with 0.45% saline; do not overcorrect.
- Gradual decompression has no role.

**Stone complexity scores (SAQ)**

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

**Calyceal diverticulum – classification and management (SAQ)**

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

**Pelvic lipomatosis (SAQ)**

- Pear-shaped bladder with medial ureters; CT shows fat.
- Cystitis glandularis common; adenocarcinoma risk.
- Watch for hydronephrosis; diversion if obstruction progresses.

**Cystinuria (SAQ)**

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Trans Uretero Ureteric anastomoses (SAQ)**

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
- Risk: unilateral injury becomes bilateral.

**Medical expulsive therapy (MET) in urinary calculi (SAQ)**

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

**Physiology of UPJ in relation to urine transport (SAQ)**

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

**Complications of PCNL, including bleeding and persistent haematuria after PCNL (SAQ)**

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.
- Supracostal access increases pleural injury; colon injury at left lower pole.

**Steinstresse. (SAQ)**

- Risk rises several-fold with burden >2 $cm^2$.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

**Treatment of inferior calyceal calculus (SAQ)**

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.

- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

**Primary Hyperoxaluria (SAQ)**

- PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
- Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
- Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

**Fracture penis (SAQ)**

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

**Ovarian vein syndromes (SAQ)**

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

**Post traumatic bladder neck injury: acute and delayed management? (SAQ)**

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

**Boari flap (SAQ)**

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

**Struvite stones. (SAQ)**

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

**Role of open and laparoscopic stone surgeries in modern era of endourology. (SAQ)**

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

**Diet in stone disease (SAQ)**

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

**ECIRS (endoscopic combined intrarenal surgery) (SAQ)**

- Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
- Retrograde scope guides the puncture and relocates stones.
- Fewer tracts, less bleeding, higher stone-free rate. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 12 · Bladder Cancer & Diversion

## Long answer questions (LAQ)

### LAQ 1 Intestinal segments in urology and orthotopic neobladder

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Use of intestinal segments in urology" / "Various uses of intestines in Uro" / "Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder." (also P2 SAQ "Neobladder") · P1 SAQ "Instestinal segments in urology"

#### Introduction

- Bowel is used to divert urine (conduit, continent pouch, neobladder), store urine, carry it to the skin, enlarge the bladder (augmentation), replace the ureter, make catheterisable channels and line a neovagina.
- Choice depends on renal function, previous surgery or irradiation, bowel disease, mesenteric length and the reconstruction needed.

#### Uses at a glance

| Use | Segment and operation |
|---|---|
| **Incontinent diversion** | **Ileal conduit** (10–15 cm ileum, 10–15 cm from the ileocaecal valve); colon conduit (transverse) after pelvic radiation; jejunal conduit only if nothing else (avoid) |
| **Continent cutaneous pouch** | Indiana (right colon, plicated ileum as channel), Mainz I, Kock (ileal nipple valves); with appendix (Mitrofanoff) or tapered ileum channel |
| **Orthotopic neobladder** | Studer (40–44 cm ileum plus afferent limb), Hautmann (~70 cm ileum, W shape); about 50–70 cm detubularised ileum joined to the urethra |
| **Rectal reservoir** | Ureterosigmoidostomy, Mainz II: now rare (acidosis, cancer) |
| **Augmentation cystoplasty** | "Clam" ileocystoplasty (20–40 cm ileum; ~25 cm ileum), sigmoid, caecum; gastrocystoplasty in renal failure. For refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation) |
| **Ureteric replacement** | **Ileal ureter**, isoperistaltic ileum, for long defects; Yang–Monti tubes |
| **Catheterisable channels** | **Mitrofanoff** appendicovesicostomy; **Yang–Monti** ileal tube (1–2 cm ileum retubularised transversely) when the appendix is absent; MACE (Malone antegrade continence enema) for bowel |
| **Neovagina** | **Sigmoid** preferred (wide lumen, mobile mesentery, less stenosis); ileum if pelvis irradiated; MRKH (Mayer–Rokitansky–Küster–Hauser), cloaca, gender-affirming vaginoplasty |

- Ileal ureter contraindications: creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.

## Choice of segment

| Segment | Electrolyte disturbance | Advantages / use | Drawbacks |
|---|---|---|---|
| **Stomach** | Hypochloraemic, hypokalaemic metabolic alkalosis | Secretes $H^+$ and $Cl^-$ (acid); less mucus; usable for renal failure, short gut, dense adhesions | **Haematuria–dysuria syndrome**, ulcers |
| **Jejunum** | Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis** | Outside radiation field | Severe salt loss; avoid |
| **Ileum** | **Hyperchloraemic metabolic acidosis** | Default: mobile, compliant, constant (reliable) blood supply; default conduit, ureteric replacement | B12 and bile-salt loss; obstruction ~10% |
| **Colon** | Hyperchloraemic metabolic acidosis, hypokalaemia | Easy antireflux tunnel; transverse colon outside (spared by) the pelvic radiation field | Diarrhoea if ileocaecal valve used; obstruction ~5% |
| Appendix | — | Mitrofanoff catheterisable channel | — |

## Technical principles of using bowel

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Assess renal function. Mark the stoma site before surgery when a stoma is possible, through the rectus.
- **Mesentery:** transilluminate, preserve a major mesenteric arcade, wide base; clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths point at splenic flexure, Sudeck point at rectosigmoid).
- **Bowel anastomosis:** well-vascularised, no tension, no spillage, close the mesenteric window.
- **Detubularisation** (Kock): opening the bowel on its antimesenteric border and refolding it **abolishes coordinated (peristaltic) contractions** and gives more volume for the same surface area; a **sphere** gives the largest volume for its surface area; by Laplace's law (pressure ∝ wall tension/radius) a larger radius means lower pressure, so pressure stays low.

## Ureterointestinal anastomoses

| Technique | Type | Stricture |
|---|---|---|
| **Bricker** | Refluxing, end to side | 4–22% (mean 6%) |
| **Wallace** | Refluxing, ureters joined end to end | ~3% (lowest) |
| Leadbetter–Clarke, Goodwin | Nonrefluxing (antireflux) tunnel in colon | 8–14% |

- Principles: spatulate, preserve periureteric tissue, tension-free, mucosa to mucosa, soft stents (leak 3–5% without). Bring the bowel to the ureter, not the ureter to the bowel.
- Refluxing (Bricker, Wallace; stricture about 3–6%) versus antireflux tunnels (more strictures).
- Fix and retroperitonealise the anastomosis. The left ureter passes behind the sigmoid mesentery; avoid angulation under the IMA (inferior mesenteric artery), a common stricture site.
- Reflux with normal unobstructed ureters has not been shown to harm the adult kidney; antireflux tunnels stricture more. Obstruction, not low-pressure reflux, damages kidneys.

### Stomas

- **Protruding (nipple, "rosebud") stoma** for appliance wearers: less stenosis, better appliance fit. **Flush stoma** for continent pouches that are catheterised.
- **Loop-end (Turnbull) stoma** for obese patients with a short mesentery. Site is marked before surgery, through the rectus muscle.

### Selecting conduit versus continent diversion

- **Renal function needed for continent diversion:** creatinine <2 mg/dL or GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria; normal liver function; ability to self-catheterise.
- **Contraindications to continent diversion:** poor renal or hepatic function, inability or unwillingness to self-catheterise, inflammatory bowel disease or short gut. A neobladder also needs a tumour-free urethra and a competent sphincter.
- A conduit is simplest and fastest; continent options improve body image but carry more reoperations and metabolic problems.

### Complications of bowel in the urinary tract

**Metabolic** (worse with longer segment, longer contact time, poorer renal function)

- **Hyperchloraemic acidosis (ileum, colon):** $NH_4^+$ (ammonium) replaces $Na^+$ on the $Na^+/H^+$ exchanger, coupled with $Cl^-/HCO_3^-$ exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.
- **Hypokalaemia:** worse with colon; replace potassium before correcting acidosis (paralysis risk).
- **Jejunal syndrome:** treat with sodium chloride and thiazide. **Gastric alkalosis:** H2 blockers or PPI (proton-pump inhibitor).
- **Altered sensorium:** hyperammonaemia (drain the pouch/diversion, lactulose), hypomagnesaemia.
- **Drug reabsorption:** phenytoin, methotrexate; drain pouches during chemotherapy.
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Nutrition:** B12 deficiency (give parenteral B12 after 5–7 years of ileal diversion), bile-salt diarrhoea, short-gut effects.

**Other**

- Infection (bacteriuria in about three-quarters of conduits; treat only symptoms or urea-splitting organisms, or pure Proteus or Pseudomonas).
- Mucus.
- **Stones:** colon conduit 3–4%, ileal conduit 10–12%, continent (caecal) reservoirs ~20% (mucus, staples).
- **Augment perforation (pouch rupture):** surgical emergency; regular CIC prevents it.
- Ureterointestinal stricture (lifelong risk), renal deterioration, bowel obstruction, stomal stenosis and parastomal hernia.
- **Secondary malignancy:** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years). Neovagina: stenosis, mucus discharge, adenocarcinoma reported.

## Orthotopic neobladder: definition and principles

- An **orthotopic** detubularised bowel (usually ileal) reservoir joined to the native urethra after radical cystectomy, when the urethra is tumour-free, to allow **volitional voiding** per urethra without a stoma. 80–90% of men and about 75% of women are medically eligible.
- **Principles:** 1. **Reliable outlet:** intact rhabdosphincter (pudendal nerve), healthy, tumour-free urethra. 2. **Low pressure:** complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area. 3. **Adequate capacity:** ≥500 mL when mature (44 cm double-folded ileum: <200 mL early, **500–600 mL** by 1 year). 4. Unobstructed upper tracts.

## Neobladder: selection and contraindications

- **Absolute contraindications:** tumour at the **urethral margin** (positive urethral margin on frozen section), urethral stricture, poor renal function (eGFR [estimated glomerular filtration rate] <35–40 mL/min), severe liver disease, inability to self-catheterise.
- **Relative:** prostatic stromal invasion (men); bladder-neck or anterior vaginal tumour (women); poor sphincter; prior radiation (irradiated pelvis); frailty. Age, obesity and CIS (carcinoma in situ) alone are not bars.

## Neobladder techniques compared

| Design | Segment and shape | Ureters | Advantage | Drawback |
|---|---|---|---|---|
| **Studer** | 40–44 cm ileum, double-folded U + afferent limb | Refluxing, isoperistaltic limb | Simple, no staples, short ureters; stricture <3% | Reflux on cystogram (benign) |
| **Hautmann W** | About 70 cm ileum, W | Refluxing (since 1997) | Largest early capacity, early continence | More late retention, more absorption |
| T pouch | 44 cm + 15 cm T limb | Antireflux flap valve | Accepts dilated or short ureters | More reoperations than Studer |
| Abol-Enein | 40 cm ileum, W | Serous-lined tunnel | Durable antireflux | Needs long, non-dilated ureters |
| Camey II | 65 cm ileum, U | Le Duc antireflux tunnel | Early detubularised design | Le Duc strictures 7–29% |
| Hemi-Kock | 61 cm ileum | Intussuscepted nipple | Good early results | Nipple stenosis, staple stones |
| Padua (VIP) | About 42 cm ileum, funnel | Le Duc | Anatomical funnel; robotic | — |
| Mainz III | 10–15 cm caecum + 20–30 cm ileum (ileocaecal) | Submucosal tunnel | Good compliance | Loses ileocaecal valve |
| Sigmoid (Reddy) | 35 cm sigmoid colon, U | Tunnelled | Spares ileum (short bowel) | Poor compliance, urgency |

- **Hautmann vs Studer:** larger initial capacity gives earlier continence but more late retention and absorption.
- **Studer steps (brief):** isolate ileum 15–20 cm proximal to the ileocaecal valve; open distal 40–44 cm antimesenterically; sew into a U and fold into a sphere; stented ureters end-to-side to the afferent limb; dependent point to urethra.

- **USC (University of Southern California) RCT (randomised controlled trial):** Studer vs T pouch, no difference in renal function at 3 years; T pouch needed more secondary procedures.

### Neobladder outcomes

| Outcome | Result |
|---|---|
| Day continence | 80–90% (3–6 months) |
| Night incontinence | 20–50% (no sphincter reflex, mucosal water secretion) |
| Retention needing CIC | Men 4–10%; women 20–30%: men leak, women fail to empty |
| Urethral recurrence (men) | About 7% at 5 years |

### Neobladder complications

- **Early:** urine leak, ileus, infection, mucus plugging.
- **Late: hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, incisional hernia, urethral recurrence.

### Care and follow-up

- Neobladder: mucus irrigation, bicarbonate; voiding training (sit, relax, strain), timed voiding, night alarm. Annual metabolic panel, B12, upper tracts; urethral cytology if CIS.
- Lifelong for all bowel reconstructions: electrolytes, bicarbonate, renal function, B12, upper-tract imaging (ultrasound or CT) for stricture and stones; endoscopy of colonic or augmented segments as above.

### Recent advances

- Robotic intracorporeal diversion (complications similar to open in experienced hands); robotic intracorporeal neobladder comparable to open; tissue-engineered bladder (experimental) and alternatives to bowel (ureterocystoplasty, autoaugmentation) remain limited.
- **Also know:** neobladder uptake is low despite 80–90% of men being eligible.

**Diagram to draw:** ileal conduit (loop) with mesenteric arcade and Wallace anastomosis.

**Diagram to draw:** Studer U-pouch with afferent limb, ureters and urethral anastomosis.

> **Key points to remember:**
> - Ileum and colon: hyperchloraemic acidosis ($NH_4Cl$ absorption).
> - Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
> - Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
> - Wallace has the lowest stricture rate; detubularisation gives low pressure.
> - Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
> - Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
> - Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
> - Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
> - Day continence 80–90%, night 20–50%; women retain (20–30%).

# LAQ 2 Bladder-preserving treatment of TCC

P2 · P4 **merged from 2**

*Asked as:* P2 LAQ "Classification of bladder cancer and bladder-preserving treatment of TCC" · P4 LAQ "Discuss bladder preservation surgery for TCC of urinary bladder."

*Read as: classification of bladder cancer, and bladder-preserving treatment of TCC (transitional cell carcinoma, urothelial carcinoma), including TURBT (transurethral resection of bladder tumour), radical TURBT, partial cystectomy and trimodality therapy.*

## Classification of bladder cancer

### 1. Histological

| Type | Share | Note |
|---|---|---|
| **Urothelial carcinoma** | ~90% | Variants: micropapillary, plasmacytoid, sarcomatoid, nested; squamous or glandular differentiation |
| Squamous cell carcinoma | ~5% | Schistosomiasis, catheter, stones |
| Adenocarcinoma | ~1–2% | Urachal (dome), exstrophy |
| Small cell / others | <1% | Neuroendocrine; sarcoma |

### 2. Grade (WHO 2004/2016, World Health Organization)

- PUNLMP (papillary urothelial neoplasm of low malignant potential), low-grade and high-grade carcinoma. WHO 1973 (G1–G3) is still reported by some.
- Low-grade (FGFR3 pathway) recurs but rarely progresses. High-grade and CIS (carcinoma in situ; TP53/RB pathway) progress.

### 3. Stage: TNM 2017

| T | Definition |
|---|---|
| Ta / Tis | Non-invasive papillary / flat CIS |
| T1 | Lamina propria |
| T2a / T2b | Inner / outer half of muscle |
| T3a / T3b | Perivesical fat, microscopic / macroscopic |
| T4a / T4b | Prostatic stroma, uterus, vagina / pelvic or abdominal wall |

- N1 one true-pelvic node; N2 ≥2 pelvic nodes; N3 common iliac; M1a distant nodes; M1b organs.

### 4. Clinical groups

- **NMIBC (non-muscle-invasive, ~70%)**: Ta, Tis, T1. Risk groups: low (solitary primary low-grade Ta <3 cm), intermediate, high (high-grade, T1 or CIS).
- **MIBC (muscle-invasive)**: T2–T4a. Metastatic.
- **Molecular:** luminal papillary versus basal/squamous subtypes.

## Why bladder preservation (introduction)

- Standard for MIBC (muscle-invasive bladder cancer) is cisplatin NAC (neoadjuvant chemotherapy) plus RC (radical cystectomy) with PLND (pelvic lymph node dissection).

- RC is morbid and patients are often elderly. Bladder preservation suits **(1) unfit patients** and **(2) fit, motivated patients with limited disease**.
- **Why single modalities fail:** they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%; occult nodes about 25%).

### Options

| Option | Use / status |
|---|---|
| Radical TURBT alone | Not a primary treatment |
| **Partial cystectomy + PLND** | Highly selected solitary tumour **<3 cm** in mobile bladder (dome), **no CIS**, 1–2 cm margin, frozen sections; **urachal adenocarcinoma** |
| RT (radiotherapy) alone | Inferior; quasi-palliative |
| Chemotherapy alone | Inferior; not recommended |
| **TMT (trimodality therapy)** | **Best-studied, guideline-accepted** |

### TURBT for NMIBC: the standard

- Complete resection with **detrusor muscle in the specimen**; separate base biopsies; cold-cup biopsy of suspicious areas.
- **En bloc resection** (loop or laser) keeps architecture, gives more detrusor and fewer perforations.
- **Repeat TURBT within 6 weeks** for all T1, incomplete resection or no muscle. Residual tumour in ≥40%.
- Then intravesical chemotherapy or BCG (bacillus Calmette–Guérin) by risk group.

### Radical TURBT for MIBC

- Rationale: about 10% are pT0 at cystectomy. Only for tumour limited to superficial muscle, with a negative re-resection.
- About a third later progress to MIBC on long follow-up; selected series report 5-year CSS (cancer-specific survival) about 82%. Not a primary treatment.

### Partial cystectomy

| Ideal candidate | Contraindication |
|---|---|
| Small (**<3 cm**), **solitary** tumour in a mobile part of bladder (dome) | **CIS**, multifocal tumour |
| No CIS, no hydronephrosis | Trigone or bladder neck tumour |
| 1–2 cm margin achievable away from the ureteric orifice and trigone; adequate residual capacity | Small bladder; prostatic urethral involvement |
| **Urachal adenocarcinoma** (treatment of choice: en bloc dome, urachus and umbilicus) | |

#### Technique (steps):

1. NAC (neoadjuvant cisplatin chemotherapy) if cisplatin-eligible.
2. PLND (always).
3. Cystotomy away from tumour.
4. Full-thickness excision with perivesical fat, **frozen-section margins**.

5. Ureteric reimplantation if needed.
6. Watertight closure; irrigation (prevents seeding); drain.
7. Cystogram before catheter removal.

**Outcomes:** 5-year OS (overall survival) 67–70%; intact bladder 65–74%; salvage RC up to a quarter. Late recurrence, so lifelong follow-up.

### Trimodality therapy (TMT)

**Components:** (1) **maximal, visibly complete TURBT**; (2) **RT** to bladder ± pelvic nodes; (3) concurrent **radiosensitising chemotherapy**.

**Selection criteria:**

- cT2 (selected cT3–4a), small solitary tumour.
- **No hydronephrosis, no extensive CIS (no CIS)**, visibly complete TURBT.
- Good bladder function; motivated, compliant patient.
- Only 6–19% of MIBC meet strict criteria.

**Radiosensitising regimens:**

| Regimen | Evidence |
|---|---|
| **Mitomycin C + 5-FU (fluorouracil)** | **BC2001**: 2-year locoregional DFS (disease-free survival) **67% vs 48%** vs RT alone, no excess late toxicity |
| Cisplatin (± 5-FU or paclitaxel) | MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols |
| Gemcitabine | Cisplatin-ineligible |
| Carbogen + nicotinamide (BCON) | Hypoxia modification |

**Radiation dose and schedules:** 64–66 Gy in 32 fractions or **55 Gy in 20** (non-inferior).
`Split course (MGH/RTOG): TURBT → chemoRT ~40 Gy → cystoscopy + biopsy CR → consolidation to 60–66 Gy Residual tumour → immediate cystectomy Continuous (current): TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr → cystoscopy + biopsy and imaging at 3–4 months ├─ Complete response → surveillance (NMIBC recurrence: TURBT + BCG) └─ Persistent / recurrent MIBC (invasive) → salvage cystectomy (10–30%)`

**Response assessment:** cystoscopy with biopsy of tumour site, cytology, CT (computed tomography)/MRI (magnetic resonance imaging). **CR (complete response)** leads to surveillance. Non-response leads to salvage RC.

**Salvage cystectomy:**

- Needed in **10–30%** for persistent or recurrent muscle-invasive disease; it is the designed safety net.
- NMIBC (non-muscle-invasive) recurrence: TURBT and intravesical BCG.

**Outcomes:**

- 5-year CSS 50–84%, OS 36–74%.
- MGH/RTOG: CR 69%; 5-year DSS (disease-specific survival) 71%; 80% of 5-year survivors keep the bladder.
- Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and similar CSS (81% vs 84%).

- Low late toxicity; good quality of life.

### Comparison with radical cystectomy

| | Radical cystectomy | Bladder preservation (TMT) |
|---|---|---|
| Staging | Pathological | Clinical only |
| Nodes | Removed | Irradiated or not treated |
| Function | Diversion; sexual dysfunction | Native bladder; RT (bowel/bladder) toxicity |
| Follow-up | Upper tract, urethra | **Intensive cystoscopy: 3-monthly for 2 years, to 10 years**, plus CT |
| Evidence | Standard | No completed RCT (SPARE closed for poor accrual) |

### Follow-up after preservation

- Cystoscopy every 3–4 months for 2 years, 6-monthly in years 3–4, then yearly to 10 years.
- CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5 years; cytology.

### Recent advances

- **Immunotherapy + chemoradiation:** PD-1/PD-L1 (programmed death-1 / ligand-1) inhibitors with TMT under trial (e.g. SWOG/NRG S1806 with atezolizumab; KEYNOTE-992 with pembrolizumab); results awaited.
- Hypofractionation 55 Gy/20 non-inferior (BC2001–BCON meta-analysis: same toxicity).
- Bladder sparing after complete response to NAC (± checkpoint inhibitor) with biomarker selection (DNA repair gene mutations, circulating tumour DNA); RETAIN: about half recurred.
- Robotic partial cystectomy; molecular subtypes.
- **Also know:** RT alone is quasi-palliative; chemotherapy alone not recommended.

**Diagram to draw:** TNM T-stage diagram of the bladder wall, with the TMT flowchart from TURBT to chemoradiation, response check, surveillance or salvage cystectomy.

> **Key points to remember:**
> - WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
> - Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
> - Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
> - TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
> - BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
> - Salvage cystectomy 10–30%; follow up the bladder for 10 years.

## LAQ 3 Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic

P2 · P4 **merged from 2**

*Asked as:* P4 LAQ "Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion" · P2 LAQ "Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction."

*Read as: principles of continence in continent diversion, orthotopic neobladder versus non-orthotopic (heterotopic) urinary diversion after radical cystectomy, and the Mitrofanoff principle in urinary and GI (gastrointestinal, bowel) reconstruction, i.e. antegrade continence enema channels.*

### Introduction, definition and types

- A continent diversion stores urine in a bowel reservoir and empties it **at will**: by voiding through the native urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous pouch**).
- **Neobladder definition**: an **orthotopic** reservoir of detubularised bowel (usually ileum) joined to the native urethra after radical cystectomy, allowing **volitional voiding** without a stoma.
- Evolution: ileal conduit (Bricker, 1950), then continent cutaneous (Kock, 1982), then orthotopic (Camey, 1979; Hautmann, 1988; Studer, 1989).
- **Ileal conduit** (Bricker, 1950) remains the commonest diversion worldwide; **ureterosigmoidostomy** is the oldest continent diversion.
- AUA (American Urological Association)/SUO (Society of Urologic Oncology) and EAU (European Association of Urology): counsel every patient on all options. **80–90% of men and ~75% of women** are medically eligible for a neobladder, but far fewer receive one.

| Group | Examples | How urine is held and emptied |
|---|---|---|
| Orthotopic | Studer, Hautmann W, T pouch, Padua, Mainz III | Native rhabdosphincter; voids by relaxing and straining |
| Incontinent cutaneous | Ileal conduit; transverse colon, sigmoid or jejunal conduit | Stoma and appliance |
| Continent cutaneous | Indiana pouch; right colon pouch with appendix (Mitrofanoff) or Yang–Monti channel; Kock (historical) | Self-catheterisation through a valved channel |
| Rectal | Ureterosigmoidostomy, Mainz II (sigma-rectum pouch) | Anal sphincter; urine mixed with stool |

### Principles of continence

- **Continence is a pressure balance:** low reservoir pressure must stay below adequate outlet resistance. Failure of either side causes leakage.

#### A. Low-pressure, adequate-capacity reservoir

- **Detubularisation** (opening bowel on its antimesenteric border) and cross-folding abolish coordinated peristaltic contractions.
- **Spherical shape:** greatest volume for a given surface area. By **Laplace's law** ($P = 2T/R$; wall tension = pressure × radius), a larger radius gives lower pressure for the same wall

tension. Result: a low-pressure reservoir.

- **Capacity ≥500 mL** when mature. Bowel stretches with time (ileum more than colon); 44 cm of double-folded ileum holds <200 mL at first and 500–600 mL within a year.
- Ileum is most compliant; colon has pressure waves.

**B. Reliable outlet**

- **Orthotopic:** intact **rhabdosphincter** (pudendal nerve), healthy urethra, careful apical dissection, nerve sparing. Voiding is by Valsalva and pelvic floor relaxation. Night leakage (20–50%) reflects loss of the guarding reflex.
- **Continent cutaneous:** a stable efferent channel that is easy to catheterise and has a durable, dependable continence mechanism (see continent cutaneous mechanisms below).

**C. Protect the upper tracts and metabolism**

- **Unobstructed ureterointestinal anastomosis** to protect the kidneys.
- **Enough renal function** (eGFR, estimated glomerular filtration rate, about ≥35–40 mL/min) to clear the acid load reabsorbed by bowel mucosa.
- Ability to self-catheterise.

**Studer as an example of the principles: orthotopic low-pressure ileal reservoir** (**Studer, 1989**): detubularised ileal reservoir with a long **isoperistaltic afferent limb**, joined to the urethra. **Detubularisation** stops coordinated contractions, so pressure stays low; a **sphere** (Laplace) gives most volume for surface area. The **isoperistaltic afferent limb** limits reflux (damps reflux during Valsalva voiding) and accepts **short ureters**. No staples in the reservoir. Continence depends on the **rhabdosphincter**.

## Choice of bowel segment

| Segment | Points |
|---|---|
| **Ileum** (default) | Most compliant, lowest pressure; mesentery reaches the urethra; mucosa atrophies, so absorption falls |
| Ileum + caecum (Mainz) | Near-ileal compliance; loses ileocaecal valve |
| Colon (right, sigmoid) | Less distensible, pressure waves, so more night leakage; used when ileum is short or diseased |
| Stomach | Secretes acid: helps in renal impairment; haematuria–dysuria; rarely used |

## Reflux versus obstruction

- Voiding by Valsalva raises pressure equally throughout, so reflux into a low-pressure pouch is benign, while **every antireflux valve can obstruct** (Le Duc tunnel stricture 7–29%; late Kock nipple stenosis).
- Direct refluxing anastomosis (Wallace, Leadbetter): stricture about 3–6%.
- **USC (University of Southern California) RCT (randomised controlled trial)**, 484 patients: Studer vs T pouch, no difference in renal function or infection at 3 years; the antireflux T pouch gave no renal benefit and needed more secondary procedures (reoperations).
- Studer: reflux on cystogram, yet kidneys preserved in about 95%.

## Continent cutaneous mechanisms

| Mechanism | Example | How it works | Weakness |
|---|---|---|---|
| **Nipple valve** (intussuscepted) | Kock (historical), Mainz I | Reservoir pressure compresses the intussuscepted segment | Same pressure effaces it; failure ≥15%, reoperation about one-third; late nipple stenosis |
| **Flap valve**: narrow tube (appendix) tunnelled submucosally or in taenia | **Mitrofanoff** (Penn, right colon pouch), Yang–Monti ileal tube if no appendix, Kropp | Narrow tube in a submucosal tunnel is coapted as pressure rises | No pop-off; stomal stenosis |
| **Imbricated ileocaecal valve + tapered ileum** | **Indiana** pouch | Continence zone lies at the plicated valve | Bowel frequency, B12 loss |
| - Used when continence is wanted but the urethra is unusable; umbilical stoma preferred. | | | |
| - **Also know** (native outlet): Kropp, Pippi Salle, slings, AUS (artificial urinary sphincter), bulking, bladder neck closure. | | | |

## The Mitrofanoff principle

- **Mitrofanoff (1980):** any supple, narrow tube implanted in a **submucosal tunnel** of a low-pressure reservoir, with its other end at skin, gives a **continent catheterisable channel**.
- As the reservoir fills, pressure transmitted through the mucosa flattens the tunnelled segment against the muscle backing: a **flap valve**. Continence lies in the tunnel, not in the conduit.
- Appendix: only a **2 cm tunnel** suffices because of its small calibre.

### Conduits

| Conduit | Points |
|---|---|
| **Appendix** (first choice) | Natural tube, minimal harvest morbidity; needs length ≥5–6 cm and ≥12 Fr; caecal cuff or **Mitchell caecal extension** adds length |
| **Yang–Monti** | 2–3 cm ileum opened on the antimesenteric border and retubularised transversely over 12 Fr; double or spiral Monti for obese; twice the revisions |
| Tapered ileum | Long enough; kinks if redundant |
| Ureter, fallopian tube | More stenosis |
| **Casale** bladder flap | No bowel; consumes bladder capacity |

### Technique (appendicovesicostomy)

1. Mobilise caecum; detach appendix with a caecal cuff, preserving mesoappendix.
2. Spatulate tip; tunnel submucosally (posterolateral or anterior bladder wall), intra- or extravesically.

3. Bring the base to the umbilicus or right iliac fossa with skin flaps, without tension or twist; keep it short and straight; fix the bladder to the abdominal wall.
4. Catheterise after each step. Channel catheter 2–4 weeks; then CIC every 4 hours by day.

### Mitrofanoff applications in urinary reconstruction

- **Neurogenic bladder** (spina bifida, spinal injury), often with augmentation, when urethral CIC is difficult (wheelchair-bound, sensate urethra).
- After bladder neck reconstruction or closure; exstrophy; valve bladder.
- **Continent cutaneous diversion** after cystectomy: Penn and right colon pouch (appendix in taenia; up to 100% continence); Yang–Monti if no appendix.
- Unusable urethra: complex stricture, urethrectomy.

### Mitrofanoff applications in GI reconstruction: MACE

- **MACE (Malone antegrade continence enema; Malone, 1990):** appendix brought to skin as an appendicocaecostomy with an antireflux tunnel (flap valve) so stool does not leak.
- Antegrade washouts empty the colon, giving **faecal continence** in neurogenic bowel (spina bifida), anorectal malformations, cloaca and refractory constipation.
- Options: Monti-MACE; **split appendix** for both MACE and urinary channel; caecostomy button. Yang–Monti for the urinary channel when the appendix is absent or used for MACE.

### Results and complications of continent catheterisable channels

- **Continence >95%** for flap valves; nipple valves about 85%.
- **Stomal stenosis is the commonest problem**; also false passage, channel polyps or granulation, pouch stones, revision (suprafascial or subfascial).
- Missed CIC: no pop-off, so **reservoir rupture** or upper tract damage.

### Patient selection

| | Orthotopic only | All continent diversions |
|---|---|---|
| Absolute contraindications | Tumour at the urethral margin (frozen section); significant urethral stricture | Severe renal (eGFR <40 mL/min) or hepatic dysfunction (liver failure); physical or mental inability to self-catheterise |
| Relative | Poor external sphincter; bladder-neck or anterior vaginal wall tumour (women); prostatic stromal invasion (men); irradiated sphincter (irradiated pelvis) | Frailty, poor cognition, social problems, prior radiation or bowel resection |

- **Renal function**: creatinine <1.7–2.2 mg/dL or eGFR **>35–40 mL/min** (below about 35–40 mL/min = severe renal impairment; another source: <40 mL/min), because the kidney must excrete acid reabsorbed from stored urine. Formal tests: urine pH ≤5.8 after ammonium chloride, osmolality ≥600 mOsm/kg after water deprivation.
- **Urethral recurrence**: in men the strongest predictor is prostatic stromal invasion (prostatic urethral tumour raises urethral recurrence risk); in women, bladder-neck or anterior vaginal wall tumour. **Intraoperative frozen section** of the urethral margin decides. CIS (carcinoma in situ) or multifocality alone do not exclude a neobladder.
- **Age and obesity alone are not contraindications**; comorbidity, cognition and support decide. A conduit suits the frail or unsupported patient.

- Continent cutaneous pouch: continence wanted but urethra unusable.
- Mark a conduit stoma site in everyone: up to ~10% change plan intraoperatively.

### Orthotopic versus non-orthotopic: comparison

| Feature | Orthotopic neobladder | Ileal conduit | Continent cutaneous | Ureterosigmoidostomy |
|---|---|---|---|---|
| Appliance | None | Bag | None; small stoma | None |
| Body image | Best | Worst | Good | Good |
| Complexity | High | Lowest | Highest | Moderate |
| Continence | Day 80–90%; night incontinence 20–50% | Not applicable | High (Indiana, Mitrofanoff) | Depends on anal sphincter |
| Catheterisation | 10–20% of men, ~30% of women | None | Always | None |
| Bowel used | Long ileal segment (Studer 54–56 cm) | 10–15 cm ileum | Right colon + terminal ileum | None isolated |
| Typical late problems | Night leakage, retention, acidosis | Stomal stenosis (20–24%), parastomal hernia, renal deterioration | Stomal stenosis, stones, difficult catheterisation | Severe acidosis, pyelonephritis, cancer at the anastomosis |

### Neobladder types

| Design | Segment | Ureters |
|---|---|---|
| **Studer** | 40–44 cm ileum + afferent limb | Refluxing, isoperistaltic limb |
| **Hautmann W** | About 70 cm ileum | Refluxing; largest capacity |
| T pouch | 44 cm + T limb | Antireflux flap valve |
| Camey II; Padua | Ileum | Antireflux (Le Duc) |
| Mainz III; sigmoid | Ileocaecal; colon | Tunnelled |

| | Studer | Hautmann |
|---|---|---|
| Ileum | 54–56 cm, U + afferent limb | ~70 cm folded in a W |
| Initial capacity | Smaller early capacity | Larger: earlier continence |
| Late | Less late retention | More retention, reabsorption |

### Studer technique (operative)

- Brief: isolate ileum 15–25 cm from the ileocaecal valve; open the distal part on the antimesenteric border, sew into a U, fold into a sphere; end-to-side ureteric anastomoses to the afferent limb; anastomose the dependent point to the urethra. 1. **Isolate 54–56 cm of terminal ileum**, ending **15–25 cm** above the ICV (ileocaecal valve) (another source: **15–20 cm proximal to the ICV**); restore bowel continuity. 2. **Reservoir: distal 40–44 cm** (two 20–22-cm limbs) opened on the antimesenteric border; sew the two limbs (medial edges)

into a **U** (back wall). 3. **Afferent limb:** leave the proximal **10–15 cm** tubular as the afferent limb (another source: **about 15 cm**; original 20 cm; 10–12 cm in later figures); close its end. 4. **Fold the U in the opposite direction** (across) and close the anterior wall into a **sphere**. 5. **Ureters:** spatulated **end-to-side, refluxing** ureteroileal anastomoses to the afferent limb, interrupted 4-0, **stented**. 6. **Urethral anastomosis** at the most dependent point with pre-placed sutures; catheter and suprapubic tube.

- Capacity rises from <200 to **500–600 mL** in a year.
- **Intracorporeal robotic Studer** follows the same steps.

### Postoperative care, voiding training and follow-up

- Mucus irrigation; oral bicarbonate; cystogram before catheter removal.
- **Voiding training**: void **sitting**, relax the pelvic floor, strain gently; **timed voiding** every 2 hours, extended to 3–4 hours as capacity grows (under 200 mL to 500–600 mL in a year). **Night alarm.**
- Lifelong follow-up: annual metabolic panel, B12, upper-tract imaging; urethral wash cytology if CIS (carcinoma in situ).

### Current status of orthotopic neobladder

**1. Preferred where suitable:** closest to a native bladder. AUA/SUO and EAU: counsel every cystectomy patient on all options and refer to an experienced surgeon.

**2. Under-used.** 80–90% of men and about 75% of women are eligible, yet only about 9% of US patients receive any continent diversion, and this is falling; continent diversion in RAZOR 21%, iROC 11%. Causes: comorbid patients, surgeon training, intracorporeal robotics.

**3. Designs in use. Studer** (40–44 cm U plus isoperistaltic afferent limb) and **Hautmann W** (about 70 cm; refluxing since 1997) dominate; T pouch or serous-lined tunnel when antireflux is wanted.

**4. Oncological safety.**

- Men: urethral recurrence about 7% at 5 years; main risk is prostatic stromal involvement. The neobladder seems protective (meta-analysis HR (hazard ratio) 0.44).
- Women: risk lies with bladder-neck tumour; no skip lesions, so preserve the urethra. **Frozen section** of the urethral margin decides in both sexes.
- Local recurrence <10%, rarely disabling the pouch; nodal disease is not a bar. Selected **post-radiation** patients do well.

**5. Function (outcomes)**

| Outcome | Result |
|---|---|
| Day continence | 80–90% (most by 3–6 months); another source (Studer): 90–98%; about 90% |
| Night incontinence | 20–50% (up to half leak): no sphincter (guarding) reflex, mucosal water secretion, nocturnal diuresis |
| Retention needing CIC | Men 4–10%; **women 20–30%** (pouch angulation): men leak, women retain |
| Self-catheterisation | **10–20% of men, ~30% of women** |

**6. Complications.** Early morbidity equals conduit. Late: hyperchloraemic metabolic acidosis, vitamin B12 deficiency, mucus, ureteroileal stricture, stones, pouch rupture, pouch–vaginal fistula (5–10%), hernia.

7. **Robotic intracorporeal neobladder:** comparable to open; long-term data lacking.
8. **Quality of life:** no diversion proven superior; body image favours neobladder.

- **Also know:** nerve sparing and uterine preservation aid continence.

### Metabolic complications (bowel in the urinary tract)

| Segment | Disturbance |
|---|---|
| Ileum, colon | **Hyperchloraemic metabolic acidosis** (ammonium chloride absorbed); hypokalaemia, worse with colon |
| Jejunum | Hyponatraemic, hypochloraemic, hyperkalaemic acidosis |
| Stomach | Hypochloraemic, hypokalaemic alkalosis |

- Acidosis is severe in most ureterosigmoidostomies; most colonic (Mainz) pouches need alkali. Treat with sodium bicarbonate or citrate, and replace potassium as well.
- **Also know:** osteomalacia; vitamin B12 deficiency and bile-salt diarrhoea after ileal loss (liver B12 stores last 3–5 years); stones (ileal conduit 10–12%, continent caecal reservoirs ~20%); bacteriuria; drug reabsorption (methotrexate, phenytoin); ammoniagenic encephalopathy; growth delay in children.
- **Secondary cancer**: ureterosigmoidostomy 6–29% after 10–20 years (yearly endoscopy from 5 years); ileal conduit and ileal neobladder lowest.

### Surgical complications

- **Early (≤90 days)**: ileus, urine leak (long pouch suture lines; drain percutaneously), infection, mucus retention; not more than with a conduit.
- **Late**: ureteroileal stricture ~3–6% (Wallace or direct end-to-side) (Studer/neobladder: **<3%**); neobladder retention, urethral stricture, rare pouch rupture; ventral and parastomal hernia; **hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, stones, pyelonephritis (commonest late, Studer), urethral recurrence.
- Continent cutaneous channels: stomal stenosis commonest (see results above).

### Quality of life

- No randomised trial. Body image favours the neobladder, urinary function the conduit; no diversion is superior on HRQoL (health-related quality of life).

### Recent advances

- RARC (robot-assisted radical cystectomy): **RAZOR (2018)** non-inferior 2-year progression-free survival (open diversion in both arms); **iROC**: intracorporeal diversion gave more days alive and out of hospital at 90 days.
- Intracorporeal neobladders and nerve- or organ-sparing cystectomy are increasing.

**Diagrams to draw:**

- Studer neobladder and ileal conduit side by side; Studer U-pouch with afferent limb, ureters and urethral anastomosis; ileal segment marked (afferent limb plus 40–44-cm U), then the folded spherical pouch with afferent limb.
- Flap valve (tunnelled appendix collapsing as pressure rises) beside a nipple valve.

**Key points to remember:**

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL; 500–600 mL by 1 year) against a reliable outlet.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel; Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

## LAQ 4 Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy

P2 · P4 **merged from 2**

*Asked as:* P2 LAQ "Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types" · P4 LAQ "Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?"

### Introduction

- MIBC (muscle-invasive bladder cancer; TCC, transitional cell carcinoma, now called urothelial carcinoma) is cT2–T4a disease.
- Nearly half of patients treated by RC (radical cystectomy) alone develop metastases from occult micrometastases, so cure needs local plus systemic therapy, planned by an MDT (multidisciplinary team).
- Cure improves with **systemic therapy before surgery** and a **meticulous lymph node dissection** (accurate staging, possible therapeutic value).
- UTUC (upper tract urothelial carcinoma) behaves alike, but RNU (radical nephroureterectomy) removes a kidney, changing chemotherapy timing.

### Staging

- Complete TURBT / Re-TURBT (transurethral resection of bladder tumour) with detrusor muscle; **MRI (magnetic resonance imaging) with VI-RADS** (Vesical Imaging-Reporting and Data System; ≥3 suggests invasion); CT (computed tomography) urogram and chest CT; bone scan if symptomatic.
- Assess fitness and cisplatin eligibility.

### Management algorithm

```
cT2–T4a N0 M0 ├─ Cisplatin-eligible → cisplatin NAC (neoadjuvant chemotherapy) │ → RC + bilateral PLND (pelvic lymph node dissection) ├─ Cisplatin-ineligible → RC + PLND directly │ (pT3–4 or N+ after either → adjuvant nivolumab or chemotherapy) ├─ Selected, motivated → trimodality therapy (TMT) └─ Unfit → chemoradiation / palliative radiotherapy cN+ → induction chemotherapy → consolidation RC in responders M1 → systemic therapy
```

**1. Cystectomy pathway: radical cystectomy (standard of care for cT2–T4a N0 M0)**

- **Cisplatin-based NAC:** ddMVAC (dose-dense methotrexate, vinblastine, doxorubicin, cisplatin) or GC (gemcitabine–cisplatin) ×4 (details below).
- **Cisplatin-ineligible:** direct cystectomy.
- **RC + standard bilateral PLND:** obturator, external and internal iliac nodes. **SWOG S1011:** extended PLND gave no survival benefit (details below).
- Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases.
- Do not delay beyond **12 weeks** from diagnosis (worse survival). Diversion: ileal conduit, orthotopic neobladder or continent pouch.
- **Adjuvant:** cisplatin chemotherapy if no NAC; **nivolumab** for pT3–4 or N+ (details below).

**2. Bladder preservation: TMT (trimodality therapy)**

- Maximal (visibly complete) TURBT + RT (radiotherapy) (64 Gy, or 55 Gy in 20 fractions) + radiosensitiser. **BC2001:** mitomycin C + 5-FU (fluorouracil) improved 2-year locoregional disease-free survival (67% vs 48%).
- Best candidates: cT2, solitary tumour, **no CIS (carcinoma in situ), no hydronephrosis**, good bladder function.
- 5-year cancer-specific survival 50–84%; salvage cystectomy 10–30%.
- **Partial cystectomy:** solitary dome or urachal tumour, no CIS.
- **Also know:** chemotherapy alone or radical TURBT alone is not recommended.

**3. Metastatic disease**

- First line: **EV (enfortumab vedotin) + pembrolizumab** (EV-302: about doubled overall survival (OS) vs platinum). Alternatives: platinum chemotherapy then **avelumab maintenance** (JAVELIN Bladder 100); erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

## Neoadjuvant chemotherapy (NAC) in MIBC

### Rationale

- Treats micrometastases early, while tumour burden is low; better tolerated before surgery, so more patients complete it.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

**Eligibility:** cT2–T4a N0 M0 and cisplatin-fit. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, grade ≥2 hearing loss, grade ≥2 neuropathy, NYHA (New York Heart Association) class III–IV failure.

| Trial | Regimen | Result |
|---|---|---|
| **SWOG 8710** (Southwest Oncology Group) | MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC vs RC | pT0 38% vs 15%; median survival 77 vs 46 months (P4: 5-year OS 57% vs 43%); **P = 0.06** |
| **BA06 30894** (EORTC/MRC, European Organisation for Research and Treatment of Cancer/Medical Research Council) | CMV (cisplatin, methotrexate, vinblastine) ×3 | 16% relative reduction in mortality; 10-year OS 36% vs 30% |
| **ABC meta-analysis** (Advanced Bladder Cancer) | Cisplatin combinations | OS HR (hazard ratio) 0.86: **5% absolute 5-year survival gain** |
| **VESPER** | ddMVAC ×6 vs GC ×4 | ddMVAC better than GC: organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity |
| **NIAGARA** | Perioperative durvalumab + GC, then adjuvant durvalumab, vs GC | Better event-free survival (HR 0.68) and OS (HR 0.75) |

- **Guidelines (EAU, European Association of Urology; AUA, American Urological Association; NCCN, National Comprehensive Cancer Network):** cisplatin-based NAC for all eligible patients; ddMVAC 3–6 cycles or GC 4 cycles.

- **No survival benefit:** single-agent cisplatin, carboplatin regimens, neoadjuvant radiotherapy.
- **Cisplatin-ineligible:** proceed straight to RC; neoadjuvant pembrolizumab (PURE-01) and atezolizumab (ABACUS) are investigational.
- The chapter calls neoadjuvant immunotherapy unproven; NIAGARA has since made perioperative durvalumab + GC an option.

## Adjuvant therapy in MIBC

| | Neoadjuvant | Adjuvant |
|---|---|---|
| Selection | Clinical stage (understaging risk) | True pathological stage, but still overtreats |
| Delivery | Better tolerated | Post-RC morbidity; a quarter never receive it |
| Evidence | Level 1 | Weaker: small, early-closed trials |

### Adjuvant chemotherapy

- Trials were small or closed early. **ABC 2022 update:** cisplatin-based adjuvant chemotherapy improves 5-year OS by about 6%. Benefit greatest in node-positive and extravesical disease.
- Use: cisplatin-naive patients (no NAC) with **pT3–4 or N+**: ddMVAC or GC.

### Adjuvant immunotherapy

- **CheckMate 274:** nivolumab for 1 year after RC (pT3–4a or N+, with or without prior NAC): median disease-free survival **20.8 vs 10.8 months**. It also included UTUC.
- **IMvigor010** (atezolizumab): negative overall (included pT2); benefit in ctDNA (circulating tumour DNA)-positive patients.
- Use nivolumab if cisplatin-ineligible or declining chemotherapy, or with ypT3 or worse after NAC.
- **Also know:** adjuvant pelvic radiotherapy (Cairo trials) improves pelvic control in pT3–4; ctDNA-guided therapy is emerging.

## Lymphadenectomy in MIBC

- **Primary drainage:** obturator, internal and external iliac, presacral. **Secondary:** common iliac, para-aortic, paracaval.
- Node positivity: pT2 8–26%, pT3 33–50%, pT4 36–80%.

| Template | Extent |
|---|---|
| **Standard** | Bilateral external and internal iliac and obturator nodes; genitofemoral nerve laterally, node of Cloquet caudally, up to the common iliac bifurcation |
| **Extended** | Adds common iliac and presacral nodes, up to the aortic bifurcation |
| Super-extended | Up to the inferior mesenteric artery |

- **Evidence:** German phase III (limited vs extended): no survival benefit. **SWOG S1011** (most had NAC): median nodes 24 vs 39, **no difference** in disease-free survival (HR 1.10) or OS; more grade ≥3 events. So **standard bilateral PLND** is the standard of care.
- **Node count:** fewer than 10 nodes doubled the risk of death (SWOG 8710 reanalysis). **Also know:** lymph node density, extranodal extension (prognostic).

- Unexpected nodes at RC: frozen section, proceed if resectable within the pelvis. Clinically node-positive: chemotherapy first, then RC in responders.

### Neoadjuvant and adjuvant therapy in UTUC

- **Why neoadjuvant:** CKD (chronic kidney disease) is common and GFR falls after RNU, losing cisplatin eligibility. Give platinum **while both kidneys work**.
- Biopsy under-stages, so select by high-risk features (high grade, invasion on imaging, hydronephrosis).
- **EA8141** (phase II): accelerated MVAC, or gemcitabine–carboplatin if CrCl (creatinine clearance) 30–49; complete response 14%, below ypT2N0 60%.
- **AUA/SUO (Society of Urologic Oncology):** neoadjuvant platinum is a standard option for high-risk UTUC, especially if postoperative CrCl will fall below 50.
- **Adjuvant: POUT** (phase III, pT2–T4 or N+ after RNU): gemcitabine–platinum vs surveillance; disease-free survival HR 0.45, 3-year 71% vs 46%. Weaker with carboplatin, favouring neoadjuvant cisplatin. Adjuvant nivolumab (CheckMate 274) also included UTUC.

### Lymphadenectomy in UTUC

- Node positivity 12–25%. Lymph drains **right to left and top to bottom**, so the template depends on side and level.

| Primary site | Right | Left |
|---|---|---|
| Renal pelvis, upper two-thirds ureter | Hilar, paracaval, retrocaval, **interaortocaval** | Hilar, para-aortic |
| Lower third ureter | Ipsilateral common, external, internal iliac, obturator, presacral | Same |

- Value: pN0 better than pNx; ≥8 nodes improved disease-specific survival above pT1. Indicated for high-grade or invasive disease; robotic yield equals open. Risk: chylous leak.

### Urinary diversion options

- Ileal conduit (simplest, commonest), continent cutaneous pouch, or **orthotopic neobladder**. No diversion is proven superior for QoL (quality of life).

### Neobladder: principles

1. **Reliable outlet:** intact rhabdosphincter, tumour-free urethra (frozen section).
2. **Low pressure:** complete **detubularisation** and cross-folding abolish peristalsis; a **sphere** gives most volume (Laplace).
3. **Capacity:** 44 cm of folded ileum holds <200 mL at first, **500–600 mL by 1 year**. - **Contraindications:** positive urethral margin, stricture, eGFR (estimated GFR) <35–40 mL/min, liver failure, inability to self-catheterise. Age, obesity and nodes are not bars.

### Critical evaluation of neobladder types

| Type | Bowel; ureteric technique | Strengths | Weaknesses |
|---|---|---|---|
| **Studer** | 40–44 cm ileum + isoperistaltic afferent limb; refluxing | Simple, no staples, takes short ureters; stricture <3%; good long-term function | Night incontinence; reflux on cystogram (harmless) |

| Type | Bowel; ureteric technique | Strengths | Weaknesses |
|---|---|---|---|
| **Hautmann W** | About 70 cm ileum; refluxing since 1997 | Largest capacity, early continence; stricture fell 9.5% → 1% | More retention, acidosis, B12 loss |
| **T pouch** (Stein–Skinner) | 44 cm + tunnelled T-limb flap valve | Antireflux; accepts dilated ureters | Complex; more secondary procedures (USC [University of Southern California] RCT) |
| **Abol-Enein serous tunnel** | 40 cm W, serosal troughs | Antireflux; stricture about 3.8% | Long ureters needed; no dilated ureters |
| **Hemi-Kock** | Intussuscepted nipple | Good early results | Late nipple stenosis, stones; abandoned |
| **Camey II; Padua (VIP)** | 65 cm or 42 cm ileum; Le Duc tunnel | Early designs | Le Duc stricture 7–29% |
| **Mainz III; right colon; sigmoid (Reddy)** | Colon ± ileum | When ileum short or diseased | High-pressure contractions, poor compliance, more leakage |

### Critical points

- **Reflux vs obstruction:** Valsalva voiding pressurises the whole system equally, so reflux is benign; every antireflux valve can obstruct. **USC RCT** (484 patients): Studer = T pouch for renal function and infection; T pouch needed more secondary procedures. Hence **Studer and Hautmann** are the most popular.
- **Segment:** ileum is most compliant; colon and stomach give high pressures.
- **Capacity trade-off:** a larger pouch gives earlier continence but more retention and metabolic load.
- **Robotic intracorporeal neobladder:** comparable outcomes; long-term data awaited.

## Neobladder outcomes, complications and follow-up

- **Day continence 80–90%; night 20–50%.** Retention: men 4–10%, **women 20–30%**.
- **Late:** hyperchloraemic metabolic acidosis, B12 deficiency, mucus, stones, ureteroileal stricture, pouch rupture, urethral recurrence (men about 7% at 5 years).
- **Follow-up:** blood gas, B12, upper-tract imaging, urethral cytology if CIS.

**Diagrams to draw:** bladder wall T-staging (T2a, T2b, T3, T4) beside the standard PLND template; standard vs extended pelvic templates, and right vs left UTUC templates; Studer pouch with afferent limb and urethral anastomosis.

**Key points to remember:**

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

# LAQ 5 Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation

P2 · P3 · P4 **merged from 3**

*Asked as:* P4 LAQ "Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy" · P2 SAQ "Intravesical therapy for TCC of bladder" · P3 SAQ "Immediate post TURBT intravesical Therapy for TCC"

## Introduction

- NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) is 70–80% of bladder cancers. "Urothelial carcinoma" now replaces "transitional cell carcinoma" (TCC).
- Two problems: **recurrence** (40–60%) and **progression** (about 6% in low-grade Ta to 17% in high-grade T1). Intravesical therapy follows complete TURBT (transurethral resection of bladder tumour).
- **Rule:** chemotherapy reduces recurrence only. BCG (bacille Calmette-Guérin) is the only agent that also reduces progression.

## Risk-adapted use (AUA [American Urological Association]/SUO)

| Risk | Criteria | Therapy |
|---|---|---|
| Low | LG (low-grade) solitary (primary) Ta ≤3 cm; PUNLMP | Single post-op instillation only |
| Intermediate | LG Ta recurrent, >3 cm or multifocal; LG T1; HG (high-grade) Ta ≤3 cm | 6-week chemotherapy or BCG induction; maintenance 1 year |
| High | HG T1; HG Ta large, multifocal or recurrent; any CIS; LVI (lymphovascular invasion); variant histology | BCG induction + maintenance 3 years |

- EAU (European Association of Urology) adds a "very high" group; EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables.

## 1. Single immediate instillation after TURBT

### Rationale

- After TURBT, NMIBC recurs in 40–60%. Many early recurrences come from **floating cells implanting** on the raw resection bed. A single chemotherapy instillation (SI) kills them before implantation.

### Timing

- Implantation occurs within hours, so instil **as early as possible: within 6 hours** (P3: ideally within 2–6 h, in theatre or recovery). Guidelines allow up to 24 h, but MMC given at 24 h lost its benefit (no benefit at 24 hours).

### Agents

| Agent | Dose | Note |
|---|---|---|
| **MMC (mitomycin C)** | 40 mg in 20–40 mL | Alkylating; skin rash |
| **Gemcitabine** | 2 g in 100 mL, 1 h | Standard in USA; mild toxicity |
| Epirubicin | 50 mg | Chemical cystitis |
| - **Also know:** pirarubicin works; thiotepa does not. | | |

**Technique:** exclude perforation; instil via catheter; retain 1–2 h; drain.

**Evidence**

- **Sylvester 2016 individual-patient meta-analysis:** about 35% relative reduction in recurrence; **no benefit** with >1 recurrence/year or EORTC recurrence score ≥5.
- **SWOG (Southwest Oncology Group) S0337:** gemcitabine 2 g versus saline in suspected low-grade NMIBC: 4-year recurrence 35% vs 47%; no excess toxicity.
- No effect on progression.

**Who benefits**

- **Low risk** (primary solitary low-grade Ta ≤3 cm): SI alone; greatest benefit.
- Intermediate risk: SI then adjuvant course. High risk: BCG follows.

**Contraindications**

- **Suspected perforation**, extensive resection, bleeding needing irrigation: withhold.
- **Never BCG immediately**: intravasation of live bacteria causes sepsis and death.

**Complications**

- Irritative symptoms (commonest), chemical cystitis, skin desquamation (MMC), contracted bladder; serious harm mainly after perforation.
- **Also know:** continuous saline irrigation may also reduce recurrence.

## 2. Adjuvant chemotherapy

- **MMC** (alkylating), 40 mg in 20 mL weekly ×6. Optimise it (Au, 2001): empty bladder, fluid restriction, sodium bicarbonate (alkalinised urine), concentrated dose. Skin rash.
- **Gemcitabine** (nucleoside analogue): minimal absorption, well tolerated. **Anthracyclines** (epirubicin, doxorubicin) cause chemical cystitis.
- Intermediate risk: 6-week induction, then monthly maintenance for 1 year.
- **Gem/Doce:** gemcitabine 1 g, drain, then docetaxel 37.5 mg; weekly ×6, then monthly for 2 years. Used in BCG-unresponsive disease and during BCG shortage (details under combination therapy).

## 3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin**. Cytokines follow (IFN-γ [interferon-gamma], IL [interleukin]-2, IL-12, TNF-α [tumour necrosis factor-alpha]). A **Th1 (T-helper 1)** response gives cell-mediated tumour kill; TRAIL release is important.
- **Technique:** 1 vial in 50 mL saline. Start **2–4 weeks** after TURBT (re-epithelialisation). Urinalysis first; retain 1–2 hours.
- **SWOG schedule (Lamm, 2000):** 6 weekly induction doses, then 3 weekly doses at 3 and 6 months, then every 6 months to 3 years. Median recurrence-free survival 76.8 vs 35.7 months.
- **EORTC 30962:** full dose; 1 year for intermediate, 3 years for high risk. **NIMBUS:** reduced frequency was inferior.
- CIS complete response up to 84%; BCG beats MMC only with maintenance. Only maintenance BCG reduces progression.
- **Contraindications:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk intravasation); previous BCG sepsis; total incontinence; immunosuppression (now relative). Relative: active UTI (urinary tract infection), liver disease, previous TB (tuberculosis).

## BCG toxicity

| Grade | Features | Management |
|---|---|---|
| 1 | Cystitis, fever <38.5 °C, <48 h | Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs) |
| 2 | Severe or >48 h; persistent fever | Culture; isoniazid + rifampicin; resume at ½–⅓ dose |
| 3 | Sepsis, organ involvement | Stop BCG; isoniazid + rifampicin + ethambutol (triple therapy); steroids only with cover |

- BCG is resistant to **pyrazinamide**. BCG sepsis 0.5%.

## BCG failure

- **Adequate BCG:** ≥5 of 6 induction + ≥2 of 3 maintenance doses.
- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite adequate BCG, or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Also know:** refractory, intolerant, exposed, delayed relapse.

## BCG-unresponsive options

- **Timely radical cystectomy** is the standard (EAU, AUA).
- If unfit or refusing:
- **Pembrolizumab** (KEYNOTE-057): IV every 3 weeks; CIS complete response 41%.
- **Nadofaragene firadenovec:** adenovirus with interferon-α2b gene, every 3 months; CIS complete response 53%.
- **N-803:** IL-15 superagonist with BCG; CIS complete response 71%.
- Gem/Doce; valrubicin; **TAR-200** (gemcitabine-releasing device); cretostimogene (trials).

## Device-assisted therapy

- **EMDA (electromotive drug administration):** current increases MMC tissue delivery 4–7-fold. Sequential BCG + EMDA-MMC improved outcomes in T1.
- **Chemohyperthermia:** RITE (radiofrequency, 42 °C); HIVEC (heated recirculating MMC).
- **Also know:** UGN-102 MMC gel; interferon with low-dose BCG.

## Combination intravesical therapy

### Rationale

- Single agents fail: chemotherapy reduces **recurrence** only; BCG reduces **progression** but fails in about a third.
- Combinations add **different mechanisms**, overcome resistance, and are useful during **BCG shortage** and in **BCG-unresponsive** NMIBC.

### 1. Sequential chemotherapy (doublets)

| Regimen | Method | Results |
|---|---|---|
| **Gem/Doce** (gemcitabine–docetaxel) | Gemcitabine 1 g in 50 mL (60–90 min), drain, then docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years | High-grade recurrence-free survival about 65% at 1 year, 52% at 2 years; CIS complete response 60% at 1 year; **84% avoid cystectomy** |
| Gem/MMC | Gemcitabine 1 g, then MMC 40 mg | Durable recurrence-free about 38% |

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral sodium bicarbonate and antiemetics aid tolerance. No level 1 evidence; phase III BRIDGE compares it with BCG in BCG-naive patients.

**2. Chemotherapy with BCG**

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA-MMC** in T1: lower recurrence and progression than BCG alone. A later trial was non-significant.
- Chemohyperthermia (RITE, radiofrequency-induced thermochemotherapeutic effect) potentiates BCG.

**3. BCG with immunomodulators**

- **BCG + interferon-α (α2b):** no gain over BCG; allows BCG dose reduction to one-third.
- **N-803 (nogapendekin alfa)** = an IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS: complete response 71%, 45% at 12 months. FDA (Food and Drug Administration)-approved in 2024 (chapter: "under review").

**4. BCG with checkpoint inhibitors (BCG-naive high risk)**

- **CREST** (subcutaneous sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free or disease-free survival over BCG alone.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

**5. Device-assisted and novel combinations**

- **Chemohyperthermia** (MMC at 42 °C: RITE; HIVEC, hyperthermic intravesical chemotherapy) increases uptake.
- **TAR-200:** gemcitabine-releasing intravesical device (± cetrelimab); very high early complete response in BCG-unresponsive CIS (SunRISe-1).
- **Cretostimogene** (oncolytic adenovirus) + pembrolizumab: about two-thirds complete response at 12 months.

**Toxicity of combinations**

- Additive cystitis; immune-related events with checkpoint inhibitors.

**Diagrams to draw:** BCG mechanism (fibronectin, cytokines, Th1, kill) with SWOG schedule timeline; Gem/Doce sequence timeline (instil, dwell, drain, instil); floating cells implanting on the resection bed and killed by instillation.

**Key points to remember:**

- Single instillation within 6 h (at most 24 h); never BCG immediately; withhold if perforation; best for low-risk solitary tumours.
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Only maintenance BCG reduces progression; SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

## LAQ 6 CA bladder, etiology, newer investigations and treatment.

P2

**Type:** LAQ (25 marks)
**Source:** 2016-Summer Paper 2, Q1

### Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

### Aetiology (risk factors)

| Factor | Key facts |
|---|---|
| **Smoking** | 30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting |
| **Occupation** (5–10%) | Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps) |
| Environment | **Arsenic** in water (West Bengal, Bangladesh, Taiwan) |
| Radiation | Pelvic radiotherapy 1.5–4 times; latency 15–30 years |
| Drugs | **Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting) |
| Chronic inflammation | *Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma |
| Genetic | First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion |
| - **Also know:** obesity raises risk; diet has no proven role. | |

### Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

| TNM (tumour, node, metastasis), 2017 | Extent |
|---|---|
| Ta / Tis | Papillary non-invasive / flat CIS |
| T1 | Lamina propria |
| T2a / T2b | Inner / outer half of muscle |
| T3a / T3b | Perivesical fat, microscopic / macroscopic |
| T4a / T4b | Prostatic stroma, uterus, vagina / pelvic or abdominal wall |
| N1 / N2 / N3 | One pelvic node / several / common iliac |

### Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.

- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris System** reporting).

### Newer investigations

| Test | Principle | Value |
|---|---|---|
| **Blue-light cystoscopy (PDD, photodynamic diagnosis)** with HAL (hexaminolevulinate) | Porphyrins in tumour fluoresce red under blue light | About one-third more CIS; recurrence benefit inconsistent |
| **NBI (narrow-band imaging)** | 415 and 540 nm light absorbed by haemoglobin highlights vessels | More tumours; many false positives |
| **Urine biomarkers** | NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder | More sensitive, less specific than cytology; **none replaces cystoscopy** |
| **mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)** | Muscle-line integrity scored 1–5 | **≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT |
| **En bloc TURBT (transurethral resection of bladder tumour)** | Tumour removed in one piece with detrusor | Better staging and muscle sampling, fewer perforations; recurrence similar |
| - **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases. | | |

### Treatment

**1. TURBT**: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

**2. NMIBC by risk (AUA (American Urological Association))**

| Risk | Definition | Treatment |
|---|---|---|
| Low | Solitary low-grade Ta ≤3 cm; PUNLMP | Single instillation; surveillance |
| Intermediate | Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm | Chemotherapy or BCG; maintenance 1 year |
| High | High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion | **BCG induction + 3-year maintenance**; early cystectomy if very high risk |

| Risk | Definition | Treatment |
|---|---|---|
| - **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression. | | |
| - **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200. | | |

**3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0**

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.
- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

**4. Metastatic**: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

## Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

**Diagram to draw:** bladder wall layers with Ta, Tis, T1–T4 marked.

> **Key points to remember:**
>
> - Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
> - Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
> - High-risk NMIBC: BCG with 3-year maintenance.
> - MIBC: cisplatin NAC + cystectomy; trimodality in selected.

## LAQ 7 Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman

P2

**Asked as:**

- Describe in detail the management of T1G3 TCC of urinary bladder? — *Summer 2019 Paper II Q1 (LAQ)*
- Discuss management of T1G3 bladder tumor — *Practice Set 7 Paper II Q3a (SAQ)*
- Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13 — *Practice Set 4 Paper II Q1 (LAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 7 Paper II Q3a: also answered in 2019-Summer Paper 2, Q1) (Practice Set 4 Paper II Q1: T1G3 management also answered in 2019-Summer Paper 2, Q1)

*Practice Set 4 Paper II Q1: "C13" is read as CIS (carcinoma in situ): T1 N0 M0 urothelial carcinoma with concurrent CIS in a 25-year-old woman.*

### Introduction and risk category

- T1G3 is high-grade (HG) urothelial carcinoma (the current name for TCC, transitional cell carcinoma) that invades (invading) the lamina propria but not the muscularis propria (not muscle). TCC is now urothelial carcinoma. CIS is flat and always high grade.
- T1 is 20–30% of NMIBC (non-muscle-invasive bladder cancer) and its most dangerous form: recurrence >80%, progression to muscle invasion 30–50%.
- Grade, not stage, drives progression. So every HG T1 is **high risk** (AUA [American Urological Association]/SUO [Society of Urologic Oncology] and EAU [European Association of Urology]).
- **T1 with CIS is high-risk NMIBC** (AUA and EAU). CIS is the second strongest progression factor after grade. Untreated CIS progresses to muscle invasion in 40–83%.
- **EAU 2021 very high risk:** T1 high grade (HG) with CIS plus at least one of age >70, multiple papillary tumours, or diameter ≥3 cm. In this woman (25-year-old), age does not count, so number and size decide. LVI (lymphovascular invasion) and variant histology also make it very high risk.
- **Young woman:** long life expectancy, fertility, pregnancy and sexual function shape every choice.

### Risk factors for progression

- **Adverse factors:** concurrent CIS (carcinoma in situ), LVI (lymphovascular invasion), variant histology, size >3 cm, multifocality, prostatic urethral involvement, residual T1 at re-resection.

| Factor | Significance |
|---|---|
| Concurrent CIS (carcinoma in situ) | Second most important factor after grade |
| Depth of invasion (substaging) | Invasion deep to the muscularis mucosae is worse; T1a/T1b proposed, no agreed method |
| LVI (lymphovascular invasion) | Suggests occult spread |

| Factor | Significance |
|---|---|
| Variant histology | Micropapillary (any amount), plasmacytoid, sarcomatoid, nested |
| Size >3 cm, multifocal, recurrent | More residual tumour and progression |
| Prostatic urethra, diverticulum | Poor BCG contact; diverticulum lacks muscle |
| Residual T1 at re-TURBT | Progression up to 80% |
| HG T1 at 3 months after BCG | Progression about 82% |
| - **Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; hydronephrosis suggests muscle invasion. | |

### Evaluation

- Cystoscopy, urine cytology (usually positive in CIS), CT (computed tomography) urography (upper tract tumour <5%, more with CIS), bimanual examination under anaesthesia (including vagina).
- Smoking and occupational history.
- **Pregnancy test** before CT and BCG (bacille Calmette-Guérin); expert pathology review.

### Step 1: Complete, good-quality TURBT (transurethral resection of bladder tumour)

- Resect all visible tumour **with detrusor muscle** in the specimen; send the base separately.
- Blue-light cystoscopy or NBI (narrow-band imaging) finds more CIS (**blue-light cystoscopy** finds about one-third more CIS). **En bloc resection** (where possible) keeps orientation for substaging.
- Biopsy the prostatic urethra if tumour is at the bladder neck or trigone, or with CIS.
- Biopsy red velvety areas and the **bladder neck** (relevant to a later neobladder).

### Step 2: Re-TURBT (restaging) in all T1

- Within **2–6 weeks** (within 6 weeks), before BCG (bacille Calmette-Guérin).
- **Why:** residual tumour in ≥40%; upstaging to muscle invasion in 25–30% (about half if no muscle in the first specimen). **Residual T1: progression up to 80%.** Re-TURBT improves recurrence- and progression-free survival: Divrik's RCT (randomised controlled trial) showed better recurrence-free and progression-free survival.

```
HG T1 on TURBT → re-TURBT at 2–6 weeks ├─ ≥T2 → neoadjuvant chemotherapy + cystectomy ├─ residual T1 / very high risk → offer early radical cystectomy └─ T0, Ta or CIS → BCG induction + 3-year maintenance
```

- Practice Set 7 version: HG T1 → re-TURBT; T0, Ta, CIS, T1 → BCG (bacille Calmette-Guérin); very high risk → offer early radical cystectomy.
- Practice Set 4 version: T1 + CIS → re-TURBT within 6 weeks; T0/Ta/CIS → BCG induction + 3-year maintenance.

### Step 3a: BCG – standard for most patients (bladder preservation)

- **Mechanism:** BCG attaches to (binds) **fibronectin**, triggers cytokines (IFN-γ [interferon-gamma], IL [interleukin]-2) and a **Th1 (T-helper 1)** immune (cytokine) response that kills tumour. It is the **only intravesical agent that reduces progression**.

- **Schedule:** start 2–4 weeks after TURBT, never immediately (intravasation and sepsis); induction weekly ×6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** 3 weekly doses at 3 and 6 months, then every 6 months (6-monthly) to **3 years**. Median recurrence-free survival 76.8 vs 35.7 months. **EORTC 30962:** full dose for 3 years in high risk.
- **CIS:** tumour-free up to 84%, about half durable; persistent CIS at 3 months: second course gives a further 30–50% complete response.
- BCG is live: avoid pregnancy and breastfeeding during treatment.
- **Contraindications (avoid with):** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis common; BCG sepsis needs isoniazid, rifampicin, ethambutol and steroids (antituberculous drugs and steroids).

### Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Patients who progress under surveillance do worse than those presenting with muscle invasion (progressors do worse than primary muscle-invasive disease). Immediate cystectomy gives about **92% 10-year disease-free survival**.
- **AUA:** offer to fit patients with persistent HG T1 on re-resection, or T1 with CIS, LVI or variant histology.
- **EAU very-high-risk group (2021 model; risk factors: age >70, multiple papillary tumours, diameter ≥3 cm):** T1 HG with CIS plus ≥1 risk factor; T1 HG without CIS but with all three; also LVI, variant histology, CIS of the prostatic urethra (prostatic urethral CIS), persistent T1 at re-TURBT. Offer immediate cystectomy; BCG if declined or unfit.
- **Offer for (Practice Set 4):** very high risk, residual HG T1, LVI, variants, diffuse CIS, or patient preference.
- Also: diverticular tumour, tumour not controllable endoscopically.

### Step 3c: Cystectomy tailored to a young woman

- Standard female operation is anterior exenteration (bladder, urethra, anterior vagina, uterus, cervix, ovaries).
- **Organ-sparing** (≤cT2, bladder neck uninvolved): keep the **uterus and ovaries** (fertility, hormones, avoids surgical menopause; ovaries also make testosterone). Keep the **vagina and neurovascular tissue** (vaginal dissection no more dorsal than 2 and 10 o'clock) for sexual function and continence. Oncological safety of sparing is not fully known.
- **Orthotopic neobladder** suits her: needs a negative bladder-neck biopsy and **urethral frozen section** (negative bladder neck means negative urethra; CIS does not predict urethral involvement). Day continence 80–90%; **retention 20–30% in women**, so she must accept CIC (clean intermittent self-catheterisation). See Practice Set 4 Paper 2, Q3e.
- **Fertility:** if uterus or ovaries must go, offer oocyte or embryo cryopreservation first. Pregnancy after a neobladder is possible; caesarean uses a high midline incision to avoid the pouch.

### Step 4: BCG failure (BCG-unresponsive disease)

- **Adequate BCG for T1:** ≥5 of 6 induction doses.
- **BCG-unresponsive:** HG T1 at the 3-month check after induction; persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within 6 months (Ta/T1) or 12 months

(CIS) of the last dose.

- **Standard:** timely radical cystectomy (EAU, AUA).
- **Unfit or refusing (declined):** pembrolizumab (KEYNOTE-057: CIS complete response 41%); nadofaragene firadenovec (53%); N-803 with BCG; gemcitabine–docetaxel; TAR-200; trials.
- **Also know:** BCG-intolerant; delayed relapse may respond to BCG rechallenge.

### Follow-up (AUA/SUO high risk)

- Bladder preserved: cystoscopy and cytology every 3–4 months (3-monthly) for 2 years, 6-monthly for 2 years, then yearly for life.
- Upper-tract imaging yearly for 2 years (regular upper-tract imaging).
- Positive cytology with normal cystoscopy: check prostatic urethra and upper tracts.
- After neobladder: B12, metabolic panel, upper tracts.

### Recent advances

- Checkpoint inhibitors with BCG in BCG-naive disease (trials); urine markers predicting BCG response; TAR-200.

**Diagram to draw:** bladder wall layers showing CIS, Ta, T1a/T1b (above and below the muscularis mucosae) and T2.

> **Key points to remember:**
> - Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
> - BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
> - T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
> - Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
> - BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
> - Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

## Short answer questions (SAQ)

### SAQ 1 Haemorrhagic cystitis: diagnosis and management

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Diagnosis and management of hemorrhagic cystitis" · P3 SAQ "Management of Haemorrhagic cystitis."

### Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after other sources are excluded (cystoscopy, cytology, upper tract imaging).

### Causes and mechanism

| Cause | Mechanism |
|---|---|
| **Cyclophosphamide, ifosfamide** | Liver forms **acrolein**, excreted in urine, which sloughs urothelium; onset within 48 h |

| Cause | Mechanism |
|---|---|
| **Pelvic radiation** (about 5%; 6 months to 10 years later) | **Obliterative endarteritis**, ischaemia, fibrosis, fragile neovessels |
| **Viral** | **BK polyomavirus**; adenovirus 11 and 35; after bone marrow or renal transplant |
| Others | Bacterial infection; busulfan, thiotepa; penicillins; amyloidosis |

## Diagnosis

- History: drugs, radiotherapy, transplant, anticoagulants. Severity: microscopic, gross, clots, clot retention.
- Blood count, coagulation, renal function, urine culture, viral PCR (polymerase chain reaction) if transplanted.
- **Exclude bladder cancer first:** cytology, cystoscopy with biopsy, upper tract imaging (CT urography). Cyclophosphamide and radiation are also risk factors for bladder cancer.

## Prevention

- **Mesna** binds and inactivates acrolein; hyperhydration with forced diuresis, CBI (continuous bladder irrigation).

## Management (escalate by severity; stability decides the branch)

`Resuscitate, transfuse, correct coagulopathy, treat infection → large-bore 3-way catheter + CBI → cystoscopy under anaesthesia: clot evacuation, fulguration, biopsy → intravesical alum 1% → bleeding after 48 h: hyperbaric oxygen (+ nephrostomies if bleeding continues) → formalin 1%, then 4% (cystogram first) → cystectomy + conduit diversion (Unstable at any stage: internal iliac embolisation)`

## Intravesical agents

| Agent | Mechanism and method | Caution / key point |
|---|---|---|
| **Alum 1%** (first line) | Protein precipitation, vasoconstriction; 200–300 mL/h; no anaesthesia | Avoid in renal failure (aluminium toxicity) |
| Aminocaproic acid | Antifibrinolytic CBI, 200 mg/L | Only in a clot-free bladder; thromboembolism |
| Prostaglandins (carboprost), silver nitrate, hyaluronate (second-line options) | Vasoconstriction, coagulation, GAG (glycosaminoglycan) repair | Silver nitrate: cystogram |
| **Formalin 1–4%** | Protein fixation; **80–90% success**; under anaesthesia, instilled by gravity for 10–15 min | **Cystogram to exclude reflux** (occlude ureters if present); fibrosis, contracted (small) bladder, ureteric stricture |

## Other measures

- **HBOT** (hyperbaric oxygen therapy): 100% oxygen at 2–3 atmospheres, 30–40 sessions; raises tissue oxygen and drives neovascularisation; 80–90% response, but many relapse.

**RICH-ART** trial: improved urinary symptom scores in radiation cystitis.

- **Nephrostomies:** divert urokinase-rich urine if bleeding continues.
- **Angioembolisation:** selective anterior division of internal iliac; spare the posterior division (gluteal pain); initial control about 90%. Used if unstable.
- **Cystectomy with conduit diversion:** last resort; remove the bladder (pyocystis risk).
- **Viral:** supportive (hydration, irrigation); cidofovir (intravenous or intravesical; evidence anecdotal).
- **Also know:** tranexamic acid and laser ablation are unproven; oral pentosan polysulfate in children.

**Diagram to draw:** stepwise escalation ladder.

> **Key points to remember:**
> - Acrolein (prevent with mesna); radiation endarteritis; BK virus.
> - Exclude bladder cancer.
> - Alum first; formalin only after cystogram.
> - Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

## SAQ 2 BCG therapy and BCG failure

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "BCG therapy in Bladder Cancer" · P2 SAQ "BCG failure"

### Introduction

- BCG (bacille Calmette-Guérin), a live attenuated mycobacterium (Morales, 1976), is standard for high-risk NMIBC (non-muscle-invasive bladder cancer) and the only intravesical agent that reduces both **recurrence and progression**.

### Mechanism

- BCG binds **fibronectin** on the urothelium; cytokines (IFN-γ [interferon-gamma], IL-2 [interleukin-2] and TNF-α [tumour necrosis factor-alpha]) and immune cells follow.
- A **Th1 (T-helper 1)** response gives cell-mediated tumour kill. It needs repeated stimulation, which is why **maintenance** is essential.

### Indications

- **High-risk NMIBC:** HG (high-grade) T1, large/multifocal/recurrent HG Ta, **CIS (carcinoma in situ)** (initial complete response up to 84%).
- **Intermediate risk:** option. Not for low risk.

### Technique and schedule

- Start **2–4 weeks after TURBT (transurethral resection of bladder tumour)**; urinalysis first; atraumatic catheter; retain **1–2 hours**.
- **Induction:** weekly × 6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** weekly × 3 at 3 and 6 months, then every 6 months to **3 years** (recurrence-free survival 76.8 vs 35.7 months).
- **EORTC (European Organisation for Research and Treatment of Cancer) 30962:** intermediate risk, full dose for 1 year; high risk, full dose for 3 years. **NIMBUS:** reduced

frequency was inferior.

### Contraindications

- **Absolute:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk **intravasation**); previous BCG sepsis; total incontinence; immunosuppression (now debated).
- **Relative:** UTI (urinary tract infection), liver disease, prior TB (tuberculosis).

### Toxicity

| Grade | Features | Management |
|---|---|---|
| 1 | Cystitis, fever <38.5 °C, <48 h | Symptomatic |
| 2 | Severe or >48 h | Isoniazid + rifampicin; resume at ½–⅓ dose |
| 3 | Sepsis, organ involvement | Stop BCG; add ethambutol; steroids only with cover |
| - BCG is resistant to **pyrazinamide**. | | |

### BCG failure: definitions

- "Failure" was poorly defined; consensus categories now guide treatment.

| Category | Definition | Implication |
|---|---|---|
| **Adequate BCG** | ≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction; for T1, ≥5 of 6 induction alone | Threshold for "unresponsive" |
| **BCG-unresponsive** | Persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within **6 months (Ta/T1) or 12 months (CIS)** of last dose; **HG T1 at first 3-month check** | More BCG futile; cystectomy |
| **BCG-resistant** | CIS or HG Ta at 3 months after induction | Reinduction: 30–50% respond |
| Delayed relapse | HG relapse beyond these windows | Rechallenge may work |
| **BCG-intolerant** | Toxicity prevented adequate BCG | Salvage chemotherapy |

- **Why BCG fails:** inadequate dosing; PD-L1 (programmed death-ligand 1) upregulation on tumour cells.

### Evaluation of BCG failure

- Cystoscopy, cytology, **re-TURBT** and biopsies to exclude invasion.
- **Extravesical disease:** prostatic urethral biopsy (24–30%) and CT urography (upper tract 0.8–25%).

### Management of BCG failure

1. **Timely radical cystectomy** is standard (EAU [European Association of Urology], AUA [American Urological Association]): HG T1 at 3 months progresses in 82%; CIS failure in about 50%.
2. **Bladder sparing** (unfit or unwilling):

| Option | Key result (CIS complete response, CR) |
|---|---|
| **Pembrolizumab** IV (KEYNOTE-057) | 41% at 3 months, 19% at 12 months; grade 3–4 toxicity 13% |
| **Nadofaragene firadenovec** (adenovirus carrying IFN-α2b [interferon alpha-2b] gene) every 3 months | 53% at 3 months, 24% at 12 months |
| **Gemcitabine–docetaxel** sequential | 60% at 1 year; 84% avoid cystectomy (retrospective) |
| **N-803 + BCG** (IL-15 [interleukin-15] superagonist) | 45% at 12 months |

- Salvage CR rates halve by 12 months.
- **Also know:** BCG-refractory (older term); gemcitabine, gemcitabine–mitomycin, valrubicin, chemohyperthermia, EMDA (electromotive drug administration), chemoradiation; TAR-200; cretostimogene.

### Recent advances

- PD-1/PD-L1 (programmed death-1/ligand-1) inhibitor + BCG in BCG-naive high risk improved event-free survival (CREST, POTOMAC).
- TAR-200 gemcitabine-releasing device.

**Diagrams to draw:** SWOG schedule timeline; flowchart: HG recurrence after BCG, was BCG adequate?, unresponsive, then cystectomy or bladder sparing.

> **Key points to remember:**
> - Only BCG reduces progression; maintenance is essential.
> - Never immediately after TURBT; start at 2–4 weeks.
> - Adequate BCG = 5/6 induction + 2/3 maintenance.
> - Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
> - Timely cystectomy is standard; salvage CR rates halve by 12 months.

## SAQ 3 Malignancy in augmented bladder and after urinary diversion

P1 · P4 **merged from 2**

*Asked as:* P4 SAQ "Secondary tumours after urinary diversion" · P1 SAQ "Cancer in augmented bladder"

### Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more; usually decades). Distinct from recurrent urothelial cancer.
- Bladder augmentation (enterocystoplasty) is mainly for neurogenic bladder. **Earliest reported tumour 4 years after surgery**; tumours after augmentation are often **aggressive and advanced at diagnosis**.

### Risk by diversion (German registry, Kälble 2011)

| Diversion | Risk | Surveillance |
|---|---|---|
| **Ureterosigmoidostomy** | 2.6% (6–29% in long-term series); mean latency about 26 years | Regular colonoscopy **from 5 years** |
| Augmentation cystoplasty | 1.6% | Endoscopy, often from 10 years (see surveillance below) |
| Colonic neobladder | 1.3% | — |
| Ileocaecal pouch | 0.14% | If symptomatic |
| Ileal neobladder | 0.05% | No routine |
| Ileal conduit | 0.02% | No routine |

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

### Risk factors

- **Gastrocystoplasty** (highest among augments), **exstrophy**, immunosuppression after transplant, smoking, chronic infection, long latency.
- The underlying disease may itself carry risk: bladder cancer in spina bifida is about 2–4%, mostly stage III or IV, and non-gastric augmentation did not appear to increase that risk.

### Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis. Ureterosigmoidostomy is the extreme model: urothelium beside colonic mucosa bathed in faeces.
- **N-nitrosamines (N-nitroso compounds)** from bacterial action on urinary nitrates in stagnant, infected urine (probably enhancers, not sole cause).
- Chronic inflammation at the anastomosis releases growth factors (EGF, epidermal growth factor) that drive proliferation; ornithine decarboxylase; abnormal mucin; perianastomotic urothelium is genetically unstable.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

### Pathology

- Mostly **adenocarcinoma**, usually at the anastomosis (ureterocolic or **enterovesical**); also adenomatous polyps, urothelial carcinoma (aggressive, often metastatic), signet-ring and other types, sarcoma.
- **Nephrogenic adenoma:** benign; probably the commonest tumour after cystoplasty.

### Clinical features

- **Gross haematuria**, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent UTI (urinary tract infection), pelvic pain, new incontinence, mass.
- Neurogenic patients may notice symptoms late.

### Surveillance and prevention

- **Ureterosigmoidostomy:** regular colonoscopy from year 5.

- **Augmentation:** chapters differ: Campbell bladder-surgery chapter advises **endoscopy yearly from 10 years**; paediatric chapter: no consensus, not shown cost-effective (Mayo group: unnecessary). Higher-risk groups benefit most.
- Investigate any haematuria promptly: cystoscopy with biopsy, cytology, CT (computed tomography) urography; biopsy and staging.
- **Prevention:** prefer ileal segments; counsel patients; regular clean intermittent catheterisation, daily bladder irrigation (less stasis, mucus and stones), treat urea-splitting infection, stop smoking, lifelong (transitional) follow-up.

### Management

- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total (radical) cystectomy with **excision of the augment**, lymphadenectomy and new urinary diversion (conduit or continent pouch).
- Chemotherapy as for histology; prognosis poor because of late stage.
- **Also know:** after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance. Stones, perforation and metabolic acidosis also occur after augmentation.

**Diagrams to draw:** ureterosigmoid anastomosis with tumour at the urothelial–colonic junction; clam ileocystoplasty with tumour at the enterovesical suture line.

> **Key points to remember:**
> - Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
> - Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
> - Excise defunctioned anastomoses.
> - Gastric segment, exstrophy, immunosuppression: highest risk in augments.
> - Haematuria needs cystoscopy; treatment is cystectomy.

## SAQ 4 Optical coherence tomography in carcinoma bladder

P2 · P4 **merged from 2**

*Asked as:* P4 SAQ "Optical coherence tomography in carcinoma bladder" · P2 SAQ "What is optical coherence tomography. What is its potential role in management of CA bladder?"

### Introduction

- **WLC (white-light cystoscopy)** is the gold standard but subjective: many biopsies are negative and small lesions are missed. WLC misses flat lesions and cannot show depth.
- **OCT (optical coherence tomography)** is an **"optical biopsy"** technique listed among enhanced cystoscopic methods for NMIBC (non-muscle-invasive bladder cancer): real-time cross-sectional imaging of the bladder wall. It works like ultrasound but uses light.

### What is OCT: principle

- OCT is the **optical analogue of B-mode ultrasound**. **Near-infrared light** (about 1300 nm) is shone on tissue, and the echo delay of back-scattered light is measured instead of sound, using **low-coherence interferometry** (back-scattered light measured against a reference beam).
- The delay of reflected light gives depth. Scanning across builds a cross-sectional image.

- **Resolution about 10–20 µm** (near histology). **Depth about 2–3 mm** (P2: only **1–2 mm**): urothelium, lamina propria and superficial muscle.
- First described in 1991 (ophthalmology).

### Technique

- A fibre-optic probe passes through the cystoscope's working channel.
- WLC or blue light finds the lesion; the probe is placed **in contact** with it and read at once (real time).

### Image interpretation

| Tissue | OCT appearance |
|---|---|
| Normal | Three layers: thin, dark, low-scatter urothelium; bright lamina propria with clear boundary; darker muscularis visible below |
| Inflammation | Layers preserved but thickened |
| Papillary tumour (Ta) | Thickened urothelium, papillary pattern; lamina propria boundary intact |
| Invasive (T1) | Layers lost; **loss (break) of the urothelium–lamina propria boundary** = invasion |

### Potential role in carcinoma bladder

1. **Diagnosis:** tell tumour from inflammation at suspicious areas; separate benign from malignant flat lesions in real time; fewer unnecessary biopsies.
2. **Staging aid: Ta vs T1** (lamina propria invasion) during TURBT (transurethral resection of bladder tumour); helps in diverticula and thin walls.
3. **Resection margins:** checks margins and completeness of TURBT.
4. **Surveillance** of NMIBC, including after BCG (bacille Calmette-Guérin).
5. **Upper tract:** through the ureteroscope, OCT correlates well with final grade and stage of UTUC (upper tract urothelial carcinoma).

### Evidence and limitations

- Small studies show **high sensitivity but lower (moderate) specificity**. Inflammation, scar, recent TURBT and post-BCG change mimic tumour (false positives). Adding OCT to fluorescence (blue-light) cystoscopy improves specificity.
- Tiny field of view: needs WLC or PDD (photodynamic diagnosis) to guide it.
- Too shallow to judge (deep) muscle invasion; cannot grade (no cellular detail); cost; learning curve.
- Still **investigational**; not recommended routinely by guidelines.

### Other enhanced cystoscopy techniques

| Technique | Principle |
|---|---|
| **PDD (blue light, HAL, hexaminolevulinate)** | Tumour porphyrins fluoresce red; detects about one-third more CIS (carcinoma in situ) |
| **NBI (narrow-band imaging)** | 415/540 nm light absorbed by haemoglobin; shows vessels; many false positives |
| **CLE (confocal laser endomicroscopy)** | Cell-level surface images after fluorescein |

| Technique | Principle |
|---|---|
| - **Also know:** Raman spectroscopy. | |

**Diagram to draw:** OCT image showing the three normal bladder layers beside a tumour with loss of layering.

> **Key points to remember:**
> - Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
> - Loss of layering suggests tumour; boundary loss suggests T1.
> - Role: Ta vs T1; cannot assess deep muscle.
> - High sensitivity, moderate specificity, false positives from inflammation; investigational.

## SAQ 5 Bladder diverticulum.

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3c

### Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

### Classification

| Feature | Congenital (primary) | Acquired (secondary) |
|---|---|---|
| Age | Children (<10 years) | Men >60 years |
| Cause | Weak detrusor at the ureteric hiatus | **BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder |
| Number, site | Solitary; **paraureteral (Hutch)** | Often multiple, anywhere |
| Bladder wall | Smooth | Trabeculated |
| Associations | VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes | Stones, cancer |
| - **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum. | | |

### Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.

- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

### Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

### Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.
- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.
- **Tumour inside**: partial or radical cystectomy.

### Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.
- Complications: ureteric injury, leak.

**Diagram to draw:** paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

## SAQ 6 Intravesical gene therapy.

P2

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 2, Q3a

### Concept

- A gene is instilled into the bladder so that urothelial and tumour cells make an anticancer product, or the vector itself kills them.
- **Why the bladder suits it:** direct contact, little systemic exposure, response checked by cystoscopy and cytology.

- **Barrier:** the glycosaminoglycan layer resists vectors → **transduction enhancers** (Syn3; DDM, n-dodecyl-β-D-maltoside).
- **Indication:** high-risk **BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer)** in patients unfit for or declining radical cystectomy. BCG-unresponsive: persistent or recurrent CIS (carcinoma in situ) or high-grade Ta/T1 after adequate BCG, or relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.

## Agents

| Agent | Vector, mechanism | Key data |
|---|---|---|
| **Nadofaragene firadenovec** | **Replication-incompetent adenovirus type 5** carrying the **interferon-α2b gene**, with **Syn3** surfactant. Urothelium keeps secreting interferon → apoptosis (TRAIL, TNF-related apoptosis-inducing ligand) and immunity, overcoming the short dwell of interferon protein | 75 mL every **3 months**, 1-hour dwell. CIS: **complete response 53% at 3 months, 24% at 12 months**; papillary high-grade RFS (recurrence-free survival) 44% at 12 months; serious events 2%. **FDA (Food and Drug Administration) approved December 2022**, first bladder gene therapy |
| **Cretostimogene grenadenorepvec (CG0070)** | **Replication-competent oncolytic adenovirus**: E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity | Two 6-week inductions + maintenance; single-arm phase III BOND-003 in BCG-unresponsive CIS: high CR (complete response) rate; combination with pembrolizumab under study |
| **Inodiftagene vixteplasmid (BC-819)** | Plasmid: **diphtheria toxin A** gene under tumour-active **H19** promoter | Investigational |

## Administration and toxicity

- Catheter instillation, patient turned during dwell; biosafety precautions.
- Mostly grade 1–2: **bladder spasm, urgency, discharge around catheter, fatigue**.

## Place in practice

- Alternatives: radical cystectomy (standard), **pembrolizumab** (KEYNOTE-057), gemcitabine–docetaxel, **N-803** (interleukin-15 superagonist) + BCG.
- Limits: responses wane (median CR duration about 10 months); single-arm trials; cost.

**Also know:** non-viral IL-12 (interleukin-12) plasmid (EG-70); early adenoviral attempts failed through poor transduction.

**Diagram to draw:** adenovirus transducing a urothelial cell → interferon secretion → tumour apoptosis.

> **Key points to remember:**
> - Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
> - CIS CR 53% at 3 months, 24% at 12 months.
> - Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

## SAQ 7 Von Bruns Nest

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 10 Paper 2, Q3e

*Read as: von Brunn nests of the urothelium.*

### Definition

- **Von Brunn nests** are solid nests of benign urothelial cells lying in the **lamina propria**. They form when the basal layer of the surface urothelium buds downward and the buds pinch off.
- A common normal or reactive finding in most adult bladders, especially at the **trigone and bladder neck**; also in the ureter and renal pelvis.

### Pathogenesis: the proliferative cystitis spectrum

`Chronic irritation (infection, stones, catheter, outlet obstruction) → urothelial buds → von Brunn nests (solid) → central degeneration and fluid → cystitis cystica (cysts) → glandular (columnar / goblet cell) metaplasia → cystitis glandularis`

| Lesion | Histology | Malignant potential |
|---|---|---|
| **Von Brunn nests** | Solid urothelial nests, lamina propria | None |
| **Cystitis cystica** | Nests with central cyst, urothelial lining | None |
| **Cystitis glandularis** (typical) | Cuboidal or columnar lining | Low |
| Cystitis glandularis, **intestinal type** | Goblet cells, mucin | Possible link to **adenocarcinoma** |

### Clinical features

- Usually incidental. When florid: irritative symptoms, haematuria, infection; cystoscopy shows submucosal nodules or cysts that can mimic tumour.
- **Associations:** chronic inflammation, outlet obstruction, **pelvic lipomatosis** (florid cystitis glandularis), bladder exstrophy.

### Differential diagnosis (pathologist's trap)

- **Nested variant of urothelial carcinoma:** deceptively bland but aggressive (male:female 6:1). Clues: deep infiltrative nests reaching muscle, nuclear atypia.
- **Inverted papilloma:** endophytic nests with trabeculae, usually at the trigone.
- Nephrogenic adenoma; urothelial carcinoma with glandular differentiation.

### Management

- None for isolated nests; treat the cause (infection, stone, catheter, obstruction).
- **TURBT (transurethral resection) or biopsy** if a mass is present, to exclude carcinoma.
- **Cystitis glandularis:** remove the irritant and arrange **endoscopic surveillance**, because rare change to adenocarcinoma is reported (especially the intestinal type and in exstrophy).
- Upper tract: benign, but coexisting upper-tract urothelial cancer cannot be excluded, so follow up.

**Diagram to draw:** urothelium budding into lamina propria, forming a solid nest, then a cyst, then glandular metaplasia.

> **Key points to remember:**
> - Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
> - Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
> - The nested variant of urothelial carcinoma mimics von Brunn nests.

## SAQ 8 Mitomycin

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 11 Paper 2, Q3b

### Pharmacology

- MMC (mitomycin C) is an antitumour antibiotic from *Streptomyces caespitosus*.
- **Mechanism:** after reductive activation in the cell it acts as an **alkylating agent**, cross-linking DNA and stopping DNA synthesis; cell-cycle non-specific.
- Molecular weight 334 Da: **little systemic absorption** from the bladder.

### Uses in NMIBC (non-muscle-invasive bladder cancer)

1. **Single immediate post-TURBT (transurethral resection of bladder tumour) instillation:** 40 mg. Given **within 6 hours** it reduces recurrence; at 24 hours it does not, because it works by preventing implantation of floating cells. Best for a solitary low-grade papillary tumour. Avoid if perforation or extensive resection is suspected.
2. **Adjuvant induction:** 40 mg weekly for 6 weeks (± maintenance) in low- and intermediate-risk disease. Reduces recurrence but **not progression**; BCG (bacille Calmette–Guérin) with maintenance is superior in high risk.
3. **Optimised delivery (Au, 2001):** empty bladder, overnight fluid restriction, sodium bicarbonate (less drug breakdown), 40 mg in 20 mL; retain 1–2 hours.
4. **Device-assisted:** EMDA (electromotive drug administration) increases tissue uptake; chemohyperthermia at 42 °C (RITE [radiofrequency-induced thermochemotherapeutic effect], HIVEC [hyperthermic intravesical chemotherapy]).
5. **Sequential gemcitabine then MMC** for BCG-exposed disease.

### Other urological uses

- **Upper tract:** single bladder dose after nephroureterectomy (ODMIT-C: bladder recurrence 28% to 16% at 1 year). **UGN-101 (mitomycin gel)**, FDA (Food and Drug Administration) approved 2020 for low-grade UTUC (upper tract urothelial carcinoma): complete response 59%, ureteric stricture about 44%. UGN-102 for low-grade intermediate-risk bladder tumours.
- **Bladder-sparing chemoradiation:** 5-FU (5-fluorouracil) + MMC (BC2001: better locoregional control than radiation alone).
- **Antifibrotic:** injected after incision of recurrent bladder neck contracture or urethral stricture.

### Side effects

- Chemical cystitis (30–40%); **skin rash, palmar or genital dermatitis** (signature toxicity; wash after voiding); contracted bladder (about 5%); encrustation cystitis; myelosuppression rare.
- Extravasation through a perforation can cause pelvic necrosis.

**Diagram to draw:** timeline: TURBT, single MMC within 6 hours, then weekly induction ×6.

> **Key points to remember:**
> - Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
> - Reduces recurrence, not progression.
> - Signature toxicity: skin rash; contracted bladder about 5%.

## SAQ 9 Mullerionosis of UB

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 2, Q3b

*"Mullerionosis of UB" is read as müllerianosis of the urinary bladder.*

### Definition

- Benign lesions of **müllerian-type epithelium** in the bladder wall, in three forms:

| Lesion | Tissue |
|---|---|
| **Endometriosis** (commonest) | Endometrial glands **and stroma** |
| **Endocervicosis** | Endocervical mucinous glands |
| **Endosalpingiosis** | Ciliated tubal epithelium |

- **Müllerianosis** strictly means two or more of these together.
- Women of reproductive age, often after **caesarean section**. Usually **posterior wall or dome** (vesicouterine pouch).
- Bladder is the commonest urinary tract site of endometriosis, ahead of the ureter.

### Pathogenesis

- **Implantation:** retrograde menstruation (Sampson) into the vesicouterine pouch.
- **Iatrogenic implantation** at caesarean or pelvic surgery (primary vs iatrogenic forms).
- **Metaplasia** of coelomic or subperitoneal müllerian remnants (the "secondary müllerian system").
- Lymphatic or vascular spread; direct extension of anterior uterine adenomyosis.

### Clinical features

- **Cyclical haematuria** or irritative LUTS (lower urinary tract symptoms) with menses; suprapubic pain, dysmenorrhoea, dyspareunia.
- Endocervicosis: non-cyclical mass or LUTS.
- **Also know:** menouria differential includes vesicouterine fistula (Youssef syndrome).

### Evaluation

- **Ultrasound** (transabdominal plus transvaginal): size and depth of a hypoechoic nodule.

- **MRI (magnetic resonance imaging):** extent; haemorrhagic foci **bright on T1**, fibrotic thickening dark on T2.
- **Cystoscopy:** bluish submucosal nodule with neovascularity, best seen during menses. Of **limited value**, because disease is mainly **extravesical** (from serosa inwards).
- **Biopsy:** excludes adenocarcinoma, especially with endocervicosis. Immunohistochemistry: ER/PR (oestrogen/progesterone receptor) positive; CD10 marks endometrial stroma.
- Ultrasound of upper tracts for ureteric involvement.
- **Differential:** adenocarcinoma, cystitis glandularis, nephrogenic adenoma, urachal remnant.

### Management

- **Hormonal suppression** (combined pill, progestins, GnRH [gonadotropin-releasing hormone] agonists): symptom relief; recurrence after stopping.
- **TUR (transurethral resection):** incomplete for transmural disease; perforation risk; recurs.
- **Partial cystectomy** (laparoscopic or robotic) with a margin: **primary treatment**. Ureteric stent or reimplantation if the orifice is involved.
- Excise other pelvic endometriosis with the gynaecologist.
- Malignant change (endometrioid or clear cell adenocarcinoma) is rare.

**Diagram to draw:** sagittal pelvis with a dome nodule growing from the vesicouterine pouch into the detrusor.

> **Key points to remember:**
> - Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
> - Cyclical haematuria; MRI bright on T1.
> - Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

## SAQ 10 Methods of preoperative bowel preparation.

P3

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 3, Q3d

### Rationale

- Bowel used in urinary diversion is opened into a sterile urinary tract. Faecal spillage causes wound infection, abscess, leak and sepsis.
- Flora: aerobes (*E. coli*, *Enterococcus faecalis*); anaerobes (*Bacteroides*, *Clostridium*).

| Segment | Organisms per gram |
|---|---|
| Jejunum | $10–10^5$ |
| Distal ileum | $10^5–10^7$ |
| Ascending colon | $10^6–10^8$ |
| Descending colon | $10^{10}–10^{12}$ |

### Methods

**1. Dietary:** clear liquids the day before; clear drinks until 2 hours before surgery.

**2. Mechanical bowel preparation (MBP)** reduces faecal **amount**, so total bacteria, not concentration.

- **PEG (polyethylene glycol)–electrolyte** lavage: non-absorbable; preferred in elderly, renal or heart failure, cirrhosis.
- Sodium phosphate, magnesium citrate; enemas.
- Historical: 3-day purgation, whole-gut irrigation by nasogastric tube (9–12 L saline), mannitol (feeds bacteria).
- Harms: dehydration, electrolyte loss; liquid stool may spill more.

**3. Oral antibiotic preparation** reduces bacterial **concentration**.

- **Neomycin + erythromycin base** (Nichols–Condon, 1977: septic complications 43% with MBP alone vs 9% with added antibiotics); neomycin + metronidazole; kanamycin.
- Harms: diarrhoea, *C. difficile* colitis, candidiasis; useless in obstruction.

**4. Intravenous prophylaxis (most important):** anaerobic plus aerobic cover (e.g. cephalosporin + metronidazole) within 1–2 hours before incision; it must be in tissue before contamination.

**5. ERAS (enhanced recovery after surgery): no routine MBP**; prehabilitation, multimodal opioid-sparing analgesia, chewing gum, **alvimopan** (peripheral μ-opioid antagonist, 12 mg before surgery), early feeding and mobilisation.

### Evidence

- Colorectal meta-analyses and Cochrane: MBP does not reduce leak or abscess; may raise SSI (surgical site infection).
- Oral antibiotics added to MBP reduce SSI and leaks in large colectomy series.
- Small cystectomy trials: no difference without preparation.

### Who still needs full preparation

- Salvage cystectomy after radiation, or cystectomy for benign disease (radiation cystitis, fistula, neurogenic bladder); immunosuppression.

**Also know:** wound protectors reduce SSI; enema before transrectal biopsy.

**Diagram to draw:** bacterial load along the gut.

> **Key points to remember:**
> - Mechanical lowers amount; antibiotics lower concentration.
> - Nichols–Condon: neomycin + erythromycin.
> - Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

## SAQ 11 Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

P3

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 3, Q3e

### Definition

- **CIS (carcinoma in situ, Tis):** flat, non-invasive, **high-grade** urothelial carcinoma. It is not "premalignant".
- Histology: severe atypia, loss of polarity, non-cohesive pleomorphic cells, **loss of umbrella cells** (dysplasia keeps them).
- **Types:** primary (isolated, about 10%), secondary or concurrent (with papillary tumour, about 90%).

### Pathogenesis

- **Flat pathway:** dysplasia → CIS → high-grade invasive cancer.
- Driven by **TP53, RB and PTEN** loss; **no FGFR3** mutation (unlike low-grade papillary tumours).
- **Field change:** multifocal; spreads to prostatic urethra, distal ureters and upper tract.

### Clinical importance

- Untreated, **40–83% progress** to muscle invasion. Primary CIS progresses in >50%.
- "CIS alone" hides invasion at cystectomy in up to 20%.
- **Second strongest prognostic factor after grade**; any CIS makes NMIBC (non-muscle-invasive bladder cancer) **high risk**.
- About 80% have **irritative symptoms**; it mimics cystitis.

### Diagnosis

- **Urine cytology:** most useful in CIS, because the cells are poorly cohesive. Positive cytology with normal cystoscopy needs mapping.
- Cystoscopy: red velvety patch or normal-looking mucosa; **blue-light cystoscopy** (hexaminolevulinate) detects about one-third more CIS; NBI (narrow-band imaging).
- **Biopsies:** cold-cup from suspicious and random sites; prostatic urethra; upper tract imaging.

### Management

- **BCG (bacillus Calmette–Guérin) is first line:**
- Induction: 6 weekly instillations.
- **Maintenance** (SWOG [Southwest Oncology Group] 2000: 3 weekly doses at 3 and 6 months, then 6-monthly for 3 years): median recurrence-free survival 76.8 vs 35.7 months.
- Complete response up to 84%; durable in about half.
- **Persistent CIS at 3 months:** second BCG course; 30–50% more respond.
- **BCG-unresponsive** (persistent or recurrent CIS within 12 months of adequate BCG): **radical cystectomy**.
- Unfit or unwilling: **pembrolizumab** (KEYNOTE-057: complete response 41%), nadofaragene firadenovec, gemcitabine–docetaxel.
- Prostatic urethral CIS: TURP plus BCG; ductal involvement needs cystoprostatectomy.

### Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly; upper-tract imaging.

**Diagram to draw:** flat versus papillary pathways of bladder cancer.

> **Key points to remember:**
>
> - Flat, high-grade; TP53/RB; no FGFR3.
> - Untreated 40–83% progress; always high risk.
> - BCG with maintenance; cystectomy if BCG-unresponsive.

## SAQ 12 Re-look (second-look) TUR bladder tumour

P4

**Asked as:**

- Re-look TUR bladder tumour. — *Summer 2017 Paper IV Q3d (SAQ)*
- Re-look TUR bladder tumour. — *Practice Set 5 Paper IV Q3d (earlier version) (SAQ)*

**Type:** SAQ (10 marks)

*Read as second-look (repeat or restaging) TURBT.*

### Definition

- A **planned second (restaging) TURBT (transurethral resection of bladder tumour)** of the primary site, including detrusor muscle, a few weeks after the first. It completes resection and corrects staging. It is a quality indicator of NMIBC (non-muscle-invasive bladder cancer) care.

### Rationale

- **TURBT is a staging operation first.** Completeness and muscle in the specimen are the quality markers.
- **Incomplete resection** — residual tumour is common:
- At the resection site in **≥40%** when re-resected within weeks; tumour at first check cystoscopy in up to 45%.
- **HG (high-grade) Ta:** residual up to 50%, upstaging up to 15%.
- **Understaging:** in **HG T1**, re-resection shows worse disease or upstaging to muscle invasion in **25–30%**. This rises to about half if the first specimen had **no muscle**.
- Understaging is about twice as likely without muscle.

### Indications (AUA [American Urological Association] / EAU [European Association of Urology])

| Recommended | Consider | Not needed |
|---|---|---|
| Incomplete first (initial) resection | HG Ta, large or multifocal (AUA) (residual up to 50%, upstaging up to 15%) | Low-grade Ta |
| **All T1 tumours** | Outside resection with no slides for review | Primary CIS (carcinoma in situ) |
| **No detrusor muscle** in specimen (except low-grade Ta and primary CIS) | | Clear muscle invasion |
| | | Cystectomy already planned |

### Timing

- **2–6 weeks** after the first TURBT (EAU); **within 6 weeks** (AUA), before BCG (bacille Calmette–Guérin). This allows healing yet avoids delaying intravesical BCG.

### Technique

- Resect the **primary-site scar and base deep into detrusor**, plus any residual tumour; send the base / specimens separately. Use blue light or NBI (narrow-band imaging) if available; it helps.

### Benefits / evidence

- Corrects stage and **changes treatment in about one-third**.
- **Divrik RCT (randomised controlled trial, 2010)** in T1: re-TUR improved recurrence-free and progression-free survival (lower recurrence and progression).
- Better BCG response with minimal residual tumour / when BCG follows re-TUR.

### Result-based plan (decision after re-TUR)

- **Residual T1 at re-resection** means (predicts) progression in up to **80%**: consider early cystectomy. `No tumour (T0) / Ta → BCG induction + maintenance Residual HG T1 → early cystectomy (or BCG with close review) Muscle invasion (≥T2) → treat as muscle-invasive: neoadjuvant chemotherapy + cystectomy`

### Recent advances

- **En bloc resection** gives muscle (more detrusor) more often, less residual tumour, and may reduce re-TUR. It is not yet a proven substitute.
- Expert pathology review changes management in about one-third.
- **Also know:** maximal TURBT before trimodality therapy.

**Diagram to draw:** flowchart: T1 or no muscle → re-TUR 2–6 weeks → T2: cystectomy pathway / residual T1: consider cystectomy / T0–Ta: BCG; and bladder wall showing first resection scar and deep second resection into detrusor.

> **Key points to remember:**
> - Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
> - Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
> - Residual T1 → progression (progresses) in up to 80%.

## SAQ 13 VI-RADS

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 4, Q3b

### Definition

- **VI-RADS (Vesical Imaging–Reporting and Data System):** a 5-point score for the likelihood of **muscle (detrusor) invasion** by bladder cancer on mpMRI (multiparametric magnetic resonance imaging). Panebianco et al., Eur Urol 2018.

- Standardises bladder MRI reporting, as PI-RADS (Prostate Imaging Reporting and Data System) does for the prostate.

### Technique

- Multiplanar **T2W (T2-weighted)**, **DWI (diffusion-weighted imaging)** with ADC (apparent diffusion coefficient) map, and **DCE (dynamic contrast-enhanced)** T1.
- Done **before TURBT (transurethral resection of bladder tumour)**: resection oedema mimics invasion. Distend the bladder adequately.

### Principle

- On T2W the detrusor is a **low-signal (dark) line**; urine, mucosa and submucosa are bright. The question is whether tumour **interrupts that dark line**.

### Scoring

| Score | Invasion | T2W | DCE | DWI |
|---|---|---|---|---|
| 1 | Highly unlikely | <1 cm, ± stalk; dark line intact | No early muscle enhancement | Muscle continuous |
| 2 | Unlikely | >1 cm with stalk and thick inner layer, or sessile; line intact | Inner layer enhances only | As 1 |
| 3 | Equivocal | No category 2 features; line intact | No muscle interruption | As 1 |
| 4 | Likely | **Line interrupted** | Enhancement extends into muscle | High signal into muscle |
| 5 | Highly likely | **Into perivesical fat** | Through wall into fat | High signal through wall and fat |

- **Cut-off ≥3 = suspicious for MIBC (muscle-invasive bladder cancer)**; the "equivocal" category already counts as positive.
- Sensitivity and specificity roughly 80–90%; DWI is the main functional sequence.

### Clinical uses

- Local staging before TURBT; earlier referral for neoadjuvant chemotherapy and cystectomy.
- Patients who cannot have iodinated contrast (renal impairment, allergy).
- **MRI-first pathway** (UK BladderPath trial) shortened time to correct treatment for MIBC.
- Response after neoadjuvant chemotherapy or bladder preservation: under study.

### Limitations

- Needs an experienced radiologist; post-TURBT or BCG (bacille Calmette–Guérin) inflammation; small, bladder-neck or diverticular tumours; poor for CIS (carcinoma in situ) and nodes; does not replace histology.

**Diagram to draw:** bladder wall with tumour breaching the dark detrusor line (VI-RADS 4).

> **Key points to remember:**
> - Five points; tests the dark muscle line on T2W.
> - ≥3 suspicious for MIBC; 4 line broken, 5 fat.
> - Scan before TURBT.
> - Panebianco 2018; accuracy about 80–90%.

## SAQ 14 Virtual cystoscopy

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3a

### Definition and principle

- **Virtual cystoscopy (VC)** is a computer-generated, three-dimensional (3D) endoluminal view of the bladder. It is reconstructed from thin-section **CT (computed tomography)** or **MRI (magnetic resonance imaging)** data of a distended bladder.
- Software "flies through" the lumen using **surface or volume rendering**; multiplanar images also show wall thickness and extravesical spread.

### Technique

| Step | CT virtual cystoscopy | MR virtual cystoscopy |
|---|---|---|
| Distension | Air or CO2 through a catheter (negative contrast), or excretory-phase contrast (positive contrast, no catheter) | Urine-filled bladder; urine bright on T2 |
| Acquisition | Thin slices, supine and prone (moves dependent urine off the posterior wall) | 3D T2 or T1 contrast sequences |
| Drawback | Radiation; catheter for gas | Cost, time, motion artefact |

### Advantages

- Non-invasive; no anaesthesia.
- Sees where a scope cannot: through a **tight urethral stricture**, inside **diverticula**, beyond a large median lobe.
- With **CTU (CT urography)**, upper tracts and staging are imaged in one study (Chapter 5: high accuracy for bladder tumours if well distended).

### Limitations

- Detects polypoid tumours ≥5 mm well; misses **small and flat lesions, including CIS (carcinoma in situ)**.
- No colour or mucosal detail, **no biopsy and no treatment**.
- False positives: trabeculation, clot, debris, folds of an underdistended bladder, prostatic impression.

### Indications (selective)

- Unfit for, or refusing, cystoscopy; urethral stricture; diverticulum; bladder evaluation during CTU for haematuria.
- **Guideline position:** EAU (European Association of Urology) and AUA (American Urological Association) do not recommend VC in place of cystoscopy. Flexible cystoscopy remains the standard.

### Related advances

- **Also know:** MRI **VI-RADS (Vesical Imaging–Reporting and Data System)** scores muscle invasion (Chapter 6); blue-light and narrow-band cystoscopy improve CIS detection (Chapter 8); virtual ureteroscopy uses the same principle.

**Diagram to draw:** thin-slice CT of a gas-distended bladder, then a 3D endoluminal fly-through showing a polypoid tumour.

> **Key points to remember:**
> - 3D endoluminal reconstruction from thin-section CT or MRI.
> - Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
> - No biopsy; does not replace cystoscopy.
> - Useful beyond strictures and inside diverticula.

## SAQ 15 Bladder transplantation

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3f

### Rationale

- After cystectomy, or when the bladder cannot be reconstructed, the usual substitute is **bowel**. Bowel in the urinary tract causes mucus, metabolic acidosis, stones, infection and malignancy risk.
- Chapter 135: two frontiers aim to **remove bowel from urinary reconstruction**: tissue engineering and bladder transplantation.

### Tissue engineering (why it has not succeeded)

- **Atala et al. (2006):** first seeded biodegradable construct for neurogenic augmentation. A later multi-institution trial showed **no significant gain in compliance or capacity** (Chapter 48).
- Engineered conduit trial: **half developed stenosis or stricture**, because of stiff material, poor cell coverage and **no blood supply** (Chapter 135).
- Engineered bladder: storage only (Chapter 111). See 2022-Summer Paper 4, Q2.

### Bladder transplantation: development (Chapter 135)

- A **VCA (vascularised composite allograft)**: the donor bladder is transplanted with its own blood vessels.
- Stepwise preclinical work at USC (University of Southern California): pigs, perfused cadavers, then brain-dead human donors.

### Technique (robotic)

1. **Robotic retrieval** of the bladder on its vesical vascular pedicles from the donor.
2. **Back-table vascular reconstruction**.
3. **Robotic implantation**: vessels to recipient iliac vessels; ureters joined to the graft.
4. **ICG (indocyanine green) fluorescence** confirms perfusion. Vascularity and mucosal integrity held for over 12 hours.

### First clinical case

- Reported in **2025** from Los Angeles: a combined kidney and bladder transplant from a deceased donor, in a dialysis patient who had lost most of his bladder.

### Problems

- **Lifelong immunosuppression**: infection, malignancy, drug nephrotoxicity. This is the major limitation (Chapter 135), especially after cancer.
- **Denervated graft**: stores urine; voiding may need straining or CIC (clean intermittent catheterisation).
- Rejection monitoring, donor supply, unknown long-term function.

### Likely candidates

- Already immunosuppressed patients (kidney transplant recipients) with an absent or non-functioning bladder.

**Diagram to draw:** donor bladder on its vascular pedicles anastomosed to recipient iliac vessels, ureters entering the graft.

> **Key points to remember:**
> - Goal: avoid bowel in the urinary tract.
> - Tissue-engineered bladders failed on blood supply and function.
> - Robotic VCA; first human case 2025.
> - Limits: immunosuppression and denervation.

## SAQ 16 Artificial urinary bladder

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3d

*Read as a man-made bladder substitute (prosthetic, engineered or transplanted). Bowel neobladder: see Practice Set 4 Paper 2, Q3e.*

### Need

- Bowel substitutes absorb urine and secrete mucus, causing acidosis, mucus, stones, infection, perforation and cancer. The aim is a substitute that **is bladder-like, not bowel**.

### Requirements of an ideal substitute

- Low-pressure storage of adequate volume; continence.
- Complete, voluntary emptying; no reflux or upper-tract damage.
- Biocompatible: no encrustation, infection, rejection or cancer risk; durable.

### Approaches

| Approach | Principle | Status |
|---|---|---|
| **Alloplastic (prosthetic) bladder** | Synthetic reservoir (silicone, polymers) joined to ureters and urethra | Failed: encrustation, infection, leak at the tissue–prosthesis junction, migration, fibrosis |
| **Tissue-engineered bladder** | Autologous urothelial and muscle cells seeded on a biodegradable scaffold, omental wrap | Atala (2006): 7 patients improved; phase II trial showed no significant gain in capacity or compliance; experimental |
| **Engineered conduit** | Seeded scaffold as urinary conduit | Neo-Urinary Conduit trial: half developed stenosis or stricture; stiff, no blood supply |

| Approach | Principle | Status |
|---|---|---|
| **Bladder transplantation** | Robotic retrieval and transplant of a vascularised composite bladder allograft | Feasible in models and brain-dead donors; first human case reported in 2025; needs immunosuppression |

### Why progress is slow

- **Vascularity:** thick constructs die centrally.
- **Innervation and contraction:** an engineered bladder stores urine but cannot yet empty voluntarily.
- **Urine–material interface:** foreign material encrusts and becomes infected.
- Fibrosis and contracture.

### Alternatives in clinical use

- Augmentation (bowel, ureterocystoplasty, autoaugmentation); continent or orthotopic diversion; ileal conduit.
- **Also know:** neuroprosthetic emptying (sacral anterior root stimulator), artificial urinary sphincter, intraurethral valve-pump devices for underactive bladders.

### Future

- Bioprinted vascularised constructs, stem cells (urine-derived, iPSCs, induced pluripotent stem cells), smart scaffolds.

**Diagram to draw:** biopsy → cell culture → seeding on bladder-shaped scaffold → omental wrap → anastomosis to bladder base.

> **Key points to remember:**
> - Prosthetic bladders failed (encrustation, infection, leak).
> - Atala 2006 promising; later trial no gain; still experimental.
> - Barriers: vascularity, innervation, urine–material interface.

# Rapid Revision Sheet · Section 12

### Intestinal segments in urology and orthotopic neobladder (LAQ)

- Ileum and colon: hyperchloraemic acidosis ($NH_4Cl$ absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

### Bladder-preserving treatment of TCC (LAQ)

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.

### Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic (LAQ)

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet; 500–600 mL by 1 year.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR (estimated glomerular filtration rate) >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

**Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy (LAQ)**

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab (adds durvalumab) improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

**Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation (LAQ)**

- Single instillation within 6 h (at most 24 h); never BCG immediately (SWOG S0337 gemcitabine).
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Withhold if perforation; best for low-risk solitary tumours.
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

**CA bladder, etiology, newer investigations and treatment. (LAQ)**

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

**Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman (LAQ)**

- Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

**Haemorrhagic cystitis: diagnosis and management (SAQ)**

- Acrolein (prevent with mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin only after cystogram.
- Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

**BCG therapy and BCG failure (SAQ)**

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

**Malignancy in augmented bladder and after urinary diversion (SAQ)**

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
- Excise defunctioned anastomoses.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

**Optical coherence tomography in carcinoma bladder (SAQ)**

- Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
- Loss of layering suggests tumour; boundary loss suggests T1.
- Role: Ta vs T1; cannot assess deep muscle.
- High sensitivity, moderate specificity, false positives from inflammation; investigational.

**Bladder diverticulum. (SAQ)**

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

**Intravesical gene therapy. (SAQ)**

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

**Von Bruns Nest (SAQ)**

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

**Mitomycin (SAQ)**

- Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
- Reduces recurrence, not progression.
- Signature toxicity: skin rash; contracted bladder about 5%.

**Mullerionosis of UB (SAQ)**

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Methods of preoperative bowel preparation. (SAQ)**

- Mechanical lowers amount; antibiotics lower concentration.
- Nichols–Condon: neomycin + erythromycin.
- Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

**Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder. (SAQ)**

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Re-look (second-look) TUR bladder tumour (SAQ)**

- Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
- Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
- Residual T1 → progression (progresses) in up to 80%.

**VI-RADS (SAQ)**

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

**Virtual cystoscopy (SAQ)**

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

**Bladder transplantation (SAQ)**

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

**Artificial urinary bladder (SAQ)**

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 13 · Kidney & UTUC

## Long answer questions (LAQ)

### LAQ 1 Nephron-sparing surgery for renal masses: indications, approaches and recent advances

P1 · P2 · P3 · P4 **merged from 4**

*Asked as:* P1 LAQ "Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy" · P2 LAQ "Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches" · P3 LAQ "Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy" · P4 SAQ "Nephron Saving Surgery (NSS)"

*"Nephron sparing nephrectomy" and "nephron saving surgery" are read as nephron-sparing surgery (NSS): partial nephrectomy and ablation. "Nepron" is read as nephron.*

#### Embryology of the kidney

- Three successive kidneys form from **intermediate mesoderm**, in cranial-to-caudal order:

| Kidney | Time and site | Fate |
|---|---|---|
| Pronephros | Week 3–4, cervical | Non-functional; regresses |
| Mesonephros | Week 4, thoracolumbar | Functions briefly; tubules become efferent ductules; **mesonephric (Wolffian) duct** persists |
| **Metanephros** | Week 5, sacral | **Definitive kidney** |

- **Reciprocal induction:** the **ureteric bud (UB)** grows from the caudal Wolffian duct (about week 5) into the **metanephric blastema**.
- UB branches into ureter, pelvis, calyces and collecting ducts.
- Blastema, induced by UB tips, forms nephrons (glomerulus to distal tubule).
- **Molecular control:** GDNF (glial cell line-derived neurotrophic factor) from the blastema acts on RET on the UB; SLIT2/ROBO2 restricts GDNF to one bud. Also WT1, PAX2.
- Urine production is established by 14 weeks; nephrogenesis ends before birth.
- **Ascent and rotation (weeks 6–9):** the kidney rises from the pelvis to the lumbar region and rotates 90° so the hilum faces medially. It takes successively higher arteries; persistent lower ones become accessory arteries.
- **Clinical correlation:** no UB gives agenesis; abnormal induction gives MCDK (multicystic dysplastic kidney); failed ascent gives ectopia; fused lower poles trapped by the IMA (inferior mesenteric artery) give horseshoe kidney; also malrotation, duplication.

#### Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3, lies on the cone-shaped psoas, so the upper pole is more medial and posterior, and the kidney is angled 30–50° behind the coronal plane. The right kidney is 1–2 cm lower (liver).
- **Relations:**

| Aspect | Right | Left |
|---|---|---|
| Superior | Liver, adrenal, 12th rib | Spleen, stomach, adrenal, 11th–12th ribs |

| Aspect | Right | Left |
|---|---|---|
| Medial/anterior | Duodenum (2nd part), hepatic flexure | Pancreatic tail with splenic vessels, splenic flexure, jejunum |
| Posterior (both) | Diaphragm, psoas, quadratus lumborum, subcostal nerve | Same |

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess (collections), yet fluid tracks into the pelvis.
- **Hilum (front to back):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Vascular anatomy (the basis of partial nephrectomy)**
- **Arteries:** renal artery from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava).
- **Five segmental end arteries**: posterior (first branch, behind the pelvis) plus four from the anterior division (apical, upper, middle, lower). With no anastomoses, occlusion causes segmental infarction, but each segment can be clamped or resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries.
- **Accessory arteries** in 25–28%, end arteries too.
- **Brödel line:** an avascular plane between anterior (two-thirds) and posterior (one-third) territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
- **Veins anastomose freely**, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma / internal structure:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio).
- **Also know:** lymph drains to para-aortic (left) or paracaval/interaortocaval nodes (right); sympathetic T8–L1.

## Small renal mass (SRM)

- **Definition:** an enhancing renal mass **≤4 cm (cT1a)**, mostly incidental.
- **Biology:** 20–30% of masses ≤4 cm and about 40% ≤2 cm are benign (oncocytoma, fat-poor angiomyolipoma) (P2: about **20%** of enhancing cT1 masses are benign); most cancers are indolent (<2% of the cancers metastasise). About 20% benign, 60% indolent, 20% potentially aggressive.

## Definition and rationale of NSS

- NSS (nephron-sparing surgery) is complete local removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why spare nephrons:** RN (radical nephrectomy) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN gives equal cancer control for small masses (cT1a) with better function and keeps about **90%** of global function (RN 65–70%).
- **CKD from surgery alone** is stable, with near-normal survival if the new GFR (glomerular filtration rate) stays >45; **CKD from medical disease** (diabetes, hypertension) progresses.

So NSS matters most when medical renal disease coexists.

- **Concept:** function after PN depends mainly on the **parenchyma preserved** (preserved parenchymal volume), less on warm ischaemia time.

### Indications for nephron-sparing surgery

| Category | Situations |
|---|---|
| **Imperative (absolute)** | Solitary (functioning) kidney; bilateral tumours (bilateral RCC, renal cell carcinoma); RN would mean dialysis |
| **Relative** | Contralateral kidney threatened: stones, renal artery stenosis, hydronephrosis, diabetes, (severe) hypertension, proteinuria; hereditary RCC |
| **Elective** | Localised tumour with a normal contralateral kidney; **cT1a (≤4 cm) with normal contralateral kidney: standard of care**; cT1b–T2 when feasible |
| **Benign** | Symptomatic AML (angiomyolipoma); focal atrophy in duplex kidney; infected calyceal diverticulum |

- Now extended (increasingly used) for selected cT1b–T2, complex or hilar tumours when function matters (better function).
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; collecting-system invasion; **venous tumour thrombus**; cT4; cN1 (nodes).

### Role of NSS by clinical setting

| Setting | Role of NSS |
|---|---|
| **cT1a (≤4 cm)** | PN is the **standard of care** when intervention is needed; RN is overtreatment |
| cT1b–T2 | PN when feasible and function matters; otherwise debatable |
| Hereditary RCC | Observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision) |
| Bilateral tumours, solitary kidney | Imperative PN, staged if bilateral; avoids dialysis |
| Frail, comorbid, cT1a <3 cm | Thermal ablation or active surveillance |
| Locally advanced, thrombus, cN1, metastatic | Generally RN or cytoreductive surgery |

### Choosing the option for a small renal mass (AUA, American Urological Association, 2021)

| Option | Role |
|---|---|
| **Active surveillance** | <2 cm or predominantly cystic; elderly, comorbid (frail patients), life expectancy <5 years. Imaging every 3–6 months. Intervene if >3 cm or growth >5 mm a year (DISSRM registry data: almost no metastases) |
| **Thermal ablation** (cryoablation, radiofrequency) | cT1a **<3 cm**, percutaneous, biopsy first. Higher local recurrence than PN, salvaged by repeat ablation |
| **Partial nephrectomy** | **Prioritise PN for cT1a** needing treatment. Imperative: solitary kidney, bilateral tumours, CKD or proteinuria; also familial RCC, young patients |

| Option | Role |
|---|---|
| **Radical nephrectomy** | Preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new (post-RN) eGFR likely **>45**. Otherwise PN |

- **PN principles:** negative margin (width does not matter); maximise preserved parenchyma; avoid long warm ischaemia; enucleation for familial or multifocal tumours.

## Preoperative evaluation and planning

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging) with 3D (three-dimensional) reconstruction; chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria (CKD stage).
- **Renal mass biopsy (RMB)** when it may change management, and before (preferred) or at ablation (non-diagnostic about 14%). Genetic counselling if age ≤46, multifocal or bilateral.
- **Nephrometry scores** predict complexity, complications, ischaemia time and conversion:

| R.E.N.A.L. component | Scored as |
|---|---|
| **R**adius | ≤4, 4–7, ≥7 cm |
| **E**xophytic/endophytic | ≥50% exophytic, <50%, entirely endophytic |
| **N**earness to collecting system | ≥7, 4–7, ≤4 mm |
| **A**nterior/posterior | Descriptor |
| **L**ocation to polar lines | Polar, crossing, central |
| - Total: **low 4–6, moderate 7–9, high 10–12** complexity. **PADUA** (Preoperative Aspects and Dimensions Used for an Anatomical) and C-index are alternatives. 3D models help (shorten ischaemia in a randomised trial). | |

## Methods of NSS

| Group | Methods |
|---|---|
| Surgical PN | Open, laparoscopic, robot-assisted (RAPN) |
| Excision technique | Standard (wedge) resection with a rim; enucleation; polar resection or heminephrectomy (≥30% of parenchyma; large polar tumours) |
| Thermal ablation | Cryoablation, RFA (radiofrequency ablation), MWA (microwave ablation) |
| Developmental | SBRT (stereotactic body radiotherapy), HIFU (high-intensity focused ultrasound), irreversible electroporation |

## Surgical approaches for PN

| Approach | Features |
|---|---|
| **Open, extraperitoneal flank** (supracostal 11th/12th rib, 8 cm mini-flank) | Posterior exposure; solitary kidney, complex or central tumours, cold ischaemia |
| **Open transperitoneal** (subcostal, chevron, midline) | Bilateral or large tumours, vascular reconstruction |
| **Laparoscopic transperitoneal** | Large space (more space), easier suturing; anterior tumours |

| Approach | Features |
|---|---|
| **Laparoscopic retroperitoneal** | No bowel mobilisation, direct hilum; posterior tumours |
| **Robot-assisted (RAPN)** | Easier suturing, shorter warm ischaemia, fewer conversions; transperitoneal for anterior and retroperitoneal for posterior tumours; equal oncology |

- Open PN suits complex tumours in a solitary kidney.
- First laparoscopic PN 1993 (Winfield).
- **Robotic vs laparoscopic:** shorter warm ischaemia, fewer conversions. **Robotic vs open:** less blood loss, shorter stay, fewer complications; same margins and eGFR.
- Haemostasis: sliding-clip renorrhaphy, barbed sutures, sealants.

## Operative steps of PN

1. Flank extraperitoneal or transperitoneal (open), laparoscopic or robotic approach.
2. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
3. Expose and defat the kidney; **intraoperative ultrasound** maps the tumour borders and depth.
4. Vascular control: clamp the artery (± vein); selective or off-clamp for small tumours. Use **ice slush** (cold ischaemia) if ischaemia is expected to be **≥25–30 minutes** (solitary kidney, CKD).
5. Excise with a thin rim of normal parenchyma, or **enucleate** along the pseudocapsule (hereditary, familial, multifocal). **A negative margin is the priority; its width does not matter.**
6. **Two-layer renorrhaphy** (sliding clip): inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.
7. Unclamp; check haemostasis; close Gerota fascia; drain.

## Enucleation vs standard resection

| | Standard (wedge) resection | Enucleation |
|---|---|---|
| Plane | Through normal parenchyma, thin rim | Along the tumour pseudocapsule |
| Advantage | Fewer positive margins | Less bleeding, maximal parenchyma, eases zero ischaemia |
| Use | Default | Hereditary RCC (e.g. VHL [von Hippel–Lindau]), familial, multifocal, solitary kidney, severe CKD |

## Vascular control and ischaemia

| Strategy | Verdict |
|---|---|
| En bloc or artery-only clamp | Equivalent; traditional warm ischaemia <25–30 minutes; ice slush if longer is expected |
| Early unclamping (after inner layer) | May reduce pseudoaneurysm |
| Selective or super-selective segmental clamping | Avoids global ischaemia; no functional gain in a randomised trial |
| Off-clamp (zero ischaemia) | Small exophytic tumours; more blood loss; no long-term eGFR benefit in randomised trials |

| Strategy | Verdict |
|---|---|
| Parenchymal compression | Peripheral tumours |
| Hypothermia, mannitol | No benefit with a normal contralateral kidney |

- Function depends mainly on the **quantity of parenchyma preserved**; ischaemia is secondary within accepted limits.
- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.

### Thermal ablation (ablative NSS)

- **Cryoablation** (double freeze–thaw), **RFA**, **MWA**; mostly **percutaneous** (CT or ultrasound guided).
- AUA 2021: alternative for **cT1a <3 cm**, especially elderly, comorbid, solitary kidney or VHL; biopsy before (preferred) or at ablation.
- **Cryoablation:** cell death at ≤−40 °C; iceball 5–10 mm beyond the tumour. **RFA/MWA:** heat causes coagulative necrosis.
- Success = loss of enhancement on CT/MRI. Local recurrence is higher than PN but treatable by repeat ablation (equal after repeat ablation).
- **Also know:** SBRT for non-surgical candidates; active surveillance for frail patients.

### Special situations

- **Hereditary RCC (VHL):** observe until the largest tumour is **3 cm**, then enucleate (except HLRCC: wide excision).
- **Bilateral tumours:** bilateral PN, usually staged; if one side needs RN, do the PN first.
- **Solitary kidney:** temporary dialysis about 3.5%; local recurrence 3–5%.
- **Also know:** bilateral Wilms tumour; kidney-sparing upper-tract urothelial cancer surgery.

### Role of neoadjuvant therapy

- **Aim:** downsize a tumour so that PN becomes possible, mainly an **imperative indication** (solitary kidney, bilateral tumours) where radical nephrectomy means dialysis.
- **TKIs (tyrosine kinase inhibitors):** pazopanib, sunitinib, axitinib. Tumour volume falls **10–30%**; resection becomes possible in **40–45%**, including PN candidates. Best in clear cell RCC.
- **Toxicity:** serious adverse events about 20%, mainly **wound healing**. Stop TKIs several half-lives before and after surgery.
- **Checkpoint inhibitors:** nivolumab is safe but shrinks little; trials ongoing.
- **Status:** not standard (trials only); **upfront surgery whenever feasible**. Adjuvant pembrolizumab (KEYNOTE-564) suits high-risk resected disease.

### Complications of PN

- **Urine leak** (drain, then stent); **haemorrhage, pseudoaneurysm or AV (arteriovenous) fistula** (**delayed haematuria**: angioembolisation); AKI (acute kidney injury), renal insufficiency; temporary dialysis in a solitary kidney.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** when >50% of a solitary kidney is lost: proteinuria, FSGS (focal segmental glomerulosclerosis); give an ACE (angiotensin-converting enzyme) inhibitor.

## Follow-up

- AUA risk-based imaging; eGFR and urine protein at every visit.

## Outcomes and evidence

- Elective PN: **local recurrence 1–2%**, cancer-free survival >90%; imperative 3–5%. Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN better function, similar oncology (cancer control); overall survival slightly favoured RN (10-year overall survival 81% RN vs 76% PN), but the trial is flawed.
- **DISSRM (Delayed Intervention and Surveillance for Small Renal Masses) registry:** surveillance almost never led to metastasis; about 24% had delayed intervention.

## Recent advances

| Area | Advance and evidence |
|---|---|
| **Planning** | R.E.N.A.L. and PADUA scores; **3D models**: randomised trial showed shorter operation and ischaemia, less blood loss |
| **Platform** | **RAPN:** shorter warm ischaemia than laparoscopic; less blood loss than open. **Single-port robot:** shorter stay, less opioid |
| **Vascular control** | Off-clamp, segmental, selective or super-selective clamping (guided by **NIRF (near-infrared fluorescence)** with ICG, indocyanine green), early unclamping. **Randomised trials: no long-term eGFR gain** |
| **Excision** | **Enucleation** along the pseudocapsule: less blood loss; hereditary, multifocal, solitary kidney |
| **Reconstruction** | Sliding-clip and **barbed suture** renorrhaphy; single-layer gives better function, fewer pseudoaneurysms; sutureless with sealants in selected |
| **Neoadjuvant** | TKI downsizing in imperative cases (see above) |
| Outcome reporting | **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes |

**Diagrams to draw:** segmental arteries with the Brödel line on a transverse kidney section; clamped renal artery, tumour excised with a rim, two-layer renorrhaphy; RAPN with selective segmental clamp and renorrhaphy.

**Key points to remember:**

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm (after biopsy), surveillance <2 cm or frail.
- Imperative, relative, elective; elective PN is standard for cT1a.
- AUA: RN only if complex + no CKD + new GFR likely >45.
- R.E.N.A.L. 4–6 / 7–9 / 10–12.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Enucleation for hereditary/multifocal.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials; 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

## LAQ 2 Metastatic RCC: assessment and management

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Metastatic RCC (mRCC): assessment and management, including recent concepts" · P3 SAQ "Discuss the management of Metastatic Renal Cell Carcinoma."

### Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis; 20–40% of localised tumours metastasise later.
- Five-year survival is about 12%, but median survival now approaches 3–5 years.
- Treatment moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

### Assessment (before treatment)

- **Clinical:** symptoms of metastases (cough, bone pain, neurological deficit), weight loss; paraneoplastic syndromes (hypercalcaemia, polycythaemia, hypertension, Stauffer syndrome); performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]); comorbidity.
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), ALP (alkaline phosphatase), creatinine, liver tests.
- **Imaging:** contrast CT (computed tomography) of chest, abdomen and pelvis (or MRI, also for IVC [inferior vena cava] thrombus). Bone scan only with bone pain, raised ALP or poor performance status; brain MRI (magnetic resonance imaging) with neurological symptoms; that is, bone scan or brain MRI only when symptoms or tests suggest spread; PET not routine.
- **Histology:** nephrectomy specimen or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change, which alters drug choice.

### Prognostic risk groups (decide first-line therapy)

| Adverse factor | MSKCC (Memorial Sloan Kettering), cytokine era | IMDC (International Metastatic RCC Database Consortium), TKI era |
|---|---|---|
| Karnofsky <80% | Yes | Yes |
| Haemoglobin below lower limit of normal | Yes | Yes |
| Corrected calcium raised | >10 mg/dL | > upper limit of normal (ULN) |
| LDH >1.5 × ULN | Yes | No |
| Time factor | No prior nephrectomy | <1 year from diagnosis to systemic therapy |
| Neutrophils, platelets > ULN | No | Yes (two factors) |

- IMDC 6 factors; MSKCC model: 5 factors including LDH.
- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

### Management: principles

- Multidisciplinary care: systemic therapy is the backbone; surgery and radiotherapy are added selectively.

### 1. First-line systemic therapy, clear cell RCC (NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

| IMDC risk | Preferred | Alternatives (if IO, immuno-oncology, not possible) |
|---|---|---|
| Favourable | Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib | Sunitinib or pazopanib; active surveillance for slow, low-volume (indolent) disease |
| Intermediate/poor | The same three, **or nivolumab + ipilimumab** | Cabozantinib; sunitinib; pazopanib |

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (the TKI target). ICIs block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing T-cell brakes and giving durable responses.
- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab: OS [overall survival] HR [hazard ratio] 0.63; overall survival benefit in intermediate/poor risk); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib: PFS [progression-free survival] 23.9 vs 9.2 months). Benefit is independent of PD-L1 (programmed death ligand-1) status, so PD-L1 does not select patients.

### 2. Later lines: switch mechanism

- After an IO combination: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor). IO rechallenge is not advised (CONTACT-03 negative).
- After both IO and TKI: **belzutifan**; LITESPARK-005 showed better PFS than everolimus.
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with OS gain.

### 3. Cytoreductive nephrectomy (CN): from routine to selective

- Cytokine era: SWOG 8949 and EORTC (European Organisation for Research and Treatment of Cancer) trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (median OS 18.4 vs 13.9 months).
- **SURTIME:** deferred CN after sunitinib was safe; OS favoured deferral.
- **Now:** immediate (upfront) CN only for fit patients with low metastatic burden who do not need immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk, poor performance status or brain metastases.
- Palliative nephrectomy or angioembolisation for haematuria (bleeding) or pain.

### 4. Metastasectomy and local therapy (radiotherapy)

- Complete resection of solitary or few metastases gives 5-year survival of 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year, metachronous disease.
- **SBRT (stereotactic body radiotherapy)** or ablation for oligometastases; radiosurgery for brain metastases; palliative radiotherapy or surgery for cord compression and weight-

bearing bones; palliative radiotherapy for bone pain.

### 5. Adjuvant therapy after complete resection

- **KEYNOTE-564:** pembrolizumab for 1 year after surgery in high-risk clear cell RCC, including **M1 NED** (no evidence of disease after complete metastasectomy): disease-free survival HR 0.63; FDA (Food and Drug Administration) approval 2021; a later report showed OS benefit.

### 6. Supportive care and special groups (non-clear cell RCC)

- Zoledronic acid or denosumab for bone metastases. Immune-related adverse events (colitis, hepatitis, hypophysitis, thyroiditis) need steroids and drug interruption.
- Non-clear cell RCC: clinical trials preferred; cabozantinib or sunitinib; gemcitabine + carboplatin (platinum) for collecting duct carcinoma.
- **Also know:** high-dose IL-2 (interleukin-2) gives durable complete response in 7–9% (another source: in a few percent); temsirolimus for poor risk; chemotherapy and hormones are ineffective.

### Follow-up

- CT every 8–12 weeks on therapy (response by RECIST [Response Evaluation Criteria in Solid Tumours]); monitor thyroid, liver, blood pressure and cortisol.

### Recent advances

- Belzutifan, also approved for VHL-associated tumours; neoadjuvant ICI trials; biomarkers to choose IO/IO or IO/TKI.

### Algorithm

```
Metastatic RCC -> biopsy + IMDC risk -> IO/TKI (all risks) or nivo + ipi (intermediate/poor) -> deferred CN / metastasectomy / SBRT in responders -> progression: switch class (TKI, belzutifan)
```

**Diagram to draw:** the flowchart above (metastatic RCC, biopsy and IMDC risk, IO combination ± deferred CN in responders, metastasectomy or SBRT, switch drug class on progression).

> **Key points to remember:**
>
> - IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
> - IO/TKI combinations for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
> - CARMENA and SURTIME: no routine upfront CN.
> - Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
> - Belzutifan after IO and TKI.

## LAQ 3 Upper tract urothelial carcinoma

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management" · P2 LAQ "Etiopathogenesis and management of Upper tract TCC"

*TCC (transitional cell carcinoma), including TCC of the ureter, is taken as UTUC (upper tract urothelial carcinoma).*

### Introduction and epidemiology

- **UTUC** is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Incidence is rising, **driven by ureteric tumours**.
- Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Concurrent bladder cancer in 17%.
- Bladder recurrence follows in 22–47%, mostly from downstream seeding.
- **Site (ureter): distal ureter 70%, mid 25%, proximal 5%**, explained by downstream implantation.
- Standard is **RNU (radical nephroureterectomy) with bladder cuff**; kidney-sparing versus radical surgery is chosen by **risk**.
- URS (ureteroscopy) biopsy is small and shallow, so it **cannot stage**. Stage is inferred from grade, endoscopic appearance, imaging and cytology together.

### Aetiology (etiopathogenesis)

| Factor | Key facts and mechanism |
|---|---|
| **Tobacco** | Commonest and most important modifiable factor; risk 2.5–7-fold. Aromatic amines are activated to mutagenic N-hydroxylamines |
| **Aristolochic acid** | Balkan endemic nephropathy (UTUC 40% of renal tumours there) and Chinese herb nephropathy. Aristolactam–DNA adducts cause A:T to T:A transversions in TP53. Familial but **not inherited**; often bilateral |
| **Lynch syndrome** (HNPCC, hereditary nonpolyposis colorectal cancer) | Germline MMR (mismatch repair) mutation: MLH1, MSH2, MSH6, PMS2. Risk ~14-fold, highest with MSH2; found in 5–9% of screened UTUC |
| **Occupation** | "Amino tumours": aniline dyes, benzidine, β-naphthylamine (dyes, rubber, petrochemicals); latency up to 20 years |
| **Analgesic abuse** | Phenacetin: papillary necrosis; capillarosclerosis is pathognomonic; history in 11% of ureteric tumours |
| **Arsenic** | Well water in south-west Taiwan (blackfoot disease) |

- **Also know:** cyclophosphamide (acrolein); chronic infection and stones (squamous carcinoma); alcohol.
- **Lynch screening:** suspect if young (age <60), or a personal or family history of Lynch-spectrum cancer (colon, endometrium). Test the tumour for MSI (microsatellite instability) or MMR immunohistochemistry, then germline testing.

## Pathogenesis

- Two theories both operate. **Intraluminal seeding** (one clone spreading downstream) explains distal predominance and frequent bladder recurrence (22–47%). **Field effect / field change** (carcinogen in urine, independent clones) explains multifocality.
- Sequence: hyperplasia, dysplasia, CIS (carcinoma in situ). Variant histology in ~25%. Molecular subtypes include FGFR3-mutant (low grade), TP53/MDM2-mutant and hypermutated (Lynch).

## Pathology and spread

- 85% papillary, 15% sessile; WHO (World Health Organization) 2004/2016 low or high grade. Ureteric tumours are low grade and low stage in 55–75%.
- **Why ureteric tumours do worse:** the renal parenchyma is a barrier for pelvic tumours, but periureteric fat holds a rich lymphovascular plexus. At equal stage, ureteric tumours have more local and distant failure.
- Spread: direct, lymphatic (**mid and distal ureter to pelvic nodes**; upper ureter to hilar and retroperitoneal), blood (lung, liver, bone).

## Clinical features

- **Haematuria** (56–98%), **flank pain** (20–32%, dull from hydronephrosis/obstruction or colic from clots); lumbar mass about 10%; incidental about 15%.
- Weight loss, bone pain or a mass suggest advanced disease.

## Diagnosis (investigations)

**1. Cystoscopy:** always, to exclude coexisting bladder cancer (concurrent in 17%). Positive cytology with a normal bladder and urethra means the upper tracts must be examined.

### 2. Imaging

| Test | Role and key points |
|---|---|
| **CTU (computed tomography urography)** | First line; best test; highest accuracy (sensitivity ~92%, specificity ~95%). Radiolucent **filling defect**, hydronephrosis, ureteric wall thickening, non-visualisation. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU. Misses flat lesions (CIS) |
| MRU (magnetic resonance urography) | If contrast or radiation is contraindicated; sensitivity ~75% for tumours <2 cm |
| Retrograde pyelography | Poorly filled calyces; accuracy ~75%; done with ureteroscopy. **Goblet (champagne-glass) sign**, the ureter dilated just below the tumour; **Bergman's sign**, the catheter coiling below it |
| FDG-PET/CT (fluorodeoxyglucose positron emission tomography) | Nodes and metastases when contrast is contraindicated |
| Renogram | Split function; contralateral kidney must be assessed |
| - **Hydronephrosis** predicts invasive disease (invasive ureteric tumour in up to 80%). | |

### 3. Urine cytology

- Sensitivity ~65%, grade-dependent (grade 1 ~20%, grade 3 ~75%). **Selective** ureteric washings are more accurate than voided urine. Take it before contrast. FISH (fluorescence in

situ hybridisation) is optional for surveillance.

**4. Flexible URS with biopsy: the key test**

- Adds accuracy (75–85% to ~90%). Biopsy gives **grade correctly in ~90%**, but **stage cannot be assessed** (shallow samples; a Ta biopsy is often T3 at surgery).
- Indications: diagnostic doubt, planned kidney-sparing treatment, solitary kidney.
- Diagnostic URS before RNU raises bladder recurrence, without changing survival.

## Staging (TNM [tumour, node, metastasis] 8th edition, 2017)

| T | Renal pelvis | Ureter |
|---|---|---|
| Ta / Tis | Non-invasive papillary / carcinoma in situ | Papillary non-invasive / carcinoma in situ |
| T1 | Subepithelial connective tissue | Subepithelial connective tissue |
| T2 | Muscularis | Muscularis |
| T3 | **Peripelvic fat or renal parenchyma** | Periureteric fat |
| T4 | Adjacent organs, or **through the kidney into perinephric fat** | Adjacent organs |
| - **N1** single node ≤2 cm; **N2** node >2 cm or multiple. **M1** distant metastasis. | | |
| - **AJCC (American Joint Committee on Cancer) grouping:** stage I = T1, II = T2, III = T3; **any N+, T4 or M1 = stage IV**. | | |
| - **Grade:** WHO 2004/2016: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. | | |
| - Regional nodes: pelvis and upper ureter to hilar and retroperitoneal; lower ureter to pelvic. | | |

## Risk stratification

### EAU (European Association of Urology)

| Low risk (all needed) | High risk (any one) |
|---|---|
| Unifocal | Multifocal |
| Under 2 cm (<2 cm) | 2 cm or more (P2: >2 cm) |
| Low-grade cytology and biopsy | High-grade cytology or biopsy |
| No invasion on CTU | Hydronephrosis; invasion on imaging |
| — | Previous radical cystectomy; variant histology |

- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023** splits each group into favourable and unfavourable (obstruction, multifocality, lower-tract involvement). Ablation is **preferred** in low-risk favourable, **offered** in low-risk unfavourable, **rare** in high-risk favourable, **palliative** in high-risk unfavourable. Endoscopic therapy suits tumours **under 1.5 cm**.

- **Prognostic factors:** stage (strongest), grade, LVI (lymphovascular invasion), nodes, sessile architecture, necrosis >10%, ureteric site, surgical delay >12 weeks.

## Management A: kidney-sparing (minimally invasive and endoscopic) treatment

### Indications

- **Elective**: low-risk disease, whatever the other kidney.
- **Imperative**: solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit for RNU, after shared decision.
- The patient must accept frequent recurrence and **lifelong endoscopic surveillance**.

### Ureteroscopic (retrograde) management

- Closed system, lowest morbidity; for small ureteric and renal tumours.
- **Steps**: cystoscopy, then selective cytology and saline barbotage, then no-touch semirigid then flexible URS, map all lesions, biopsy (3 Fr forceps or basket) into fixative, ablate the base with laser (holmium or thulium, after biopsy), stent optional.

| Laser | Penetration | Note |
|---|---|---|
| **Ho:YAG (holmium:yttrium-aluminium-garnet)** | Under 0.5 mm | Safe in thin ureter |
| **Nd:YAG (neodymium:YAG)** | Up to 5–6 mm | Coagulative necrosis; transmural injury risk |
| **Thulium** | Under 0.2 mm | Strong haemostasis |

- Avoid circumferential fulguration or ablation (stricture); stage large tumours.
- Results: upper-tract recurrence 23–90% (P2: about half), bladder recurrence 15–45%, progression 0–20%. **Grade is the strongest predictor.**

### Percutaneous (antegrade) management

- Percutaneous resection for large pelvic tumours, lower-pole or inaccessible calyces and urinary diversions.
- **Steps**: retrograde catheter, puncture a calyx away from the tumour (upper or middle calyx for pelvic tumours), dilate to 30 Fr, cold-cup, bipolar resection or laser with the base sent separately, nephrostomy left, **second-look nephroscopy**.
- Removes more tumour and can **stage**; the tract allows topical therapy. But it is an **open system**: tract seeding about 0.3%, bleeding.

### Segmental ureterectomy

- **Distal ureterectomy with bladder cuff and reimplantation** (psoas hitch or Boari flap) for distal tumours, including high-grade ones when the kidney must be saved. It gives a staging specimen; add pelvic lymphadenectomy if high grade.
- First examine the whole tract above for multifocality. Lower-ureteric tumours with unfavourable features need resection, not endoscopy.

| Site | Operation |
|---|---|
| **Distal ureter** | Distal ureterectomy with bladder cuff and reimplantation; psoas hitch (~5 cm extra) or Boari flap (10–15 cm) |

| Site | Operation |
|---|---|
| Mid or proximal | Excision with 1–2 cm margins (frozen section) and ureteroureterostomy (rare) |
| Whole ureter | Ileal ureter or autotransplantation |
| - Refluxing reimplant eases surveillance; non-refluxing limits seeding (debated). | |
| - Laparoscopic or robotic routes are feasible; outcomes equal RNU in selected patients. | |
| - Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); UGN-101 mitomycin gel for low grade (see topical therapy). Lifelong endoscopic surveillance. | |

### Management B: RNU with bladder cuff (standard for high risk)

- **Principles**: kidney outside Gerota's fascia; clip the ureter early; never open the collecting system; remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction intact; watertight bladder closure; template lymphadenectomy for high risk.
- **Laparoscopic and robotic RNU steps**: modified flank, 3–4 ports; reflect colon; clip ureter; artery then vein; kidney outside Gerota's fascia (adrenal spared); ureter traced under the superior vesical artery; cuff excised (extravesical or open); two-layer closure.
- Abandoned (higher recurrence): transurethral (endoscopic) "pluck", stripping, laparoscopic stapling.
- Open, laparoscopic and robotic routes are oncologically equivalent: laparoscopic **equivalent to open** (AUA/SUO), with less blood loss and shorter stay; less favoured in bulky T3/T4. Robotic yields more nodes.
- **Template lymphadenectomy** for high risk: hilar plus paracaval (right) or para-aortic (left); pelvic nodes for lower ureteric tumours.

### Management C: adjuvant topical therapy

- **Single postoperative intravesical chemotherapy** (AUA/SUO strong): **ODMIT-C (2011)** MMC 40 mg cut bladder recurrence 28% to 16% at 1 year; **Ito (2013)** pirarubicin, 42% to 17% at 2 years.
- **Upper-tract instillation** via nephrostomy (low pressure, 25 cm H2O), ureteric catheter or reflux up a double-J. **BCG** (or MMC): 6-week course for upper-tract CIS, and after complete ablation of high-risk favourable Ta/T1. Risks: sepsis, stricture.
- **UGN-101 (Jelmyto)**: mitomycin reverse-thermal gel, **FDA (Food and Drug Administration) 2020** for low-grade UTUC not manageable endoscopically. **OLYMPUS**: complete response 59%, 56% disease-free at 12 months; ureteric stricture about 44%.

### Management D: perioperative (systemic) chemotherapy

- **POUT (2020)**: adjuvant platinum after RNU for pT2–T4 or N+: 3-year DFS (disease-free survival) **71% vs 46%**. Gemcitabine–cisplatin; carboplatin if creatinine clearance 30–49.
- **Neoadjuvant** platinum is preferred when renal function is borderline, i.e. when post-RNU renal function would rule out cisplatin (RNU removes cisplatin eligibility).
- **Also know:** adjuvant nivolumab (CheckMate 274) if platinum-ineligible, but no clear benefit in the UTUC subgroup.

### Management E: metastatic disease

- First line: **enfortumab vedotin + pembrolizumab** (EV-302, 2023): overall survival ~31 vs ~16 months versus platinum chemotherapy. Alternative: gemcitabine–platinum, then avelumab maintenance.
- Later: pembrolizumab (KEYNOTE-045), erdafitinib for FGFR3 alterations. Node-positive: chemotherapy first, then consolidative surgery.

### Prognosis

- 5-year disease-specific survival: localised ~89%, regional ~63%, distant ~17%. Stage is the strongest factor.

### Surveillance and follow-up (AUA/SUO)

- **Kidney-sparing**: cystoscopy and URS within 3 months, URS again at 6 and 12 months; imaging 3–6-monthly (high risk) or 6–9-monthly (low risk), then yearly to 5 years; URS and CTU at closer intervals, for life.
- **After RNU**: cystoscopy and cytology from 3 months; CTU yearly (more often if pT2 or more).

### Recent advances

- Lynch syndrome screening; thulium fibre laser; single-use digital ureteroscopes.

**Diagrams to draw:** ureteroscopic laser ablation of a pelvic tumour beside RNU with bladder cuff; RNU specimen with bladder cuff beside lymph node templates; TNM levels of a renal pelvic tumour (T3 parenchyma versus T4 perinephric fat), beside the diagnostic flowchart (CTU, cystoscopy, URS biopsy, risk group); distal ureterectomy with psoas-hitch reimplantation, beside the goblet sign.

> **Key points to remember:**
>
> - Tobacco, aristolochic acid (A:T to T:A in TP53) and Lynch (MSH2) are the key causes.
> - CTU is the best test; URS biopsy gives grade, not stage.
> - Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
> - Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
> - EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
> - Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
> - Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
> - Distal ureterectomy for selected distal tumours.
> - RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
> - POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

## LAQ 4 Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.

P3

**Type:** LAQ (25 marks)
**Source:** 2019-Summer Paper 3, Q1

### Definition and biology

- **SRM (small renal mass):** an enhancing solid or complex cystic renal mass **≤4 cm (cT1a)**, usually found incidentally.
- **Benign share falls with size:** 20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign. At ≥7 cm few are benign and about half are high grade.
- Male sex triples the risk of malignancy. Under 2% of small cancers metastasise.

### Differential diagnosis

| Group | Entities | Pointers |
|---|---|---|
| RCC (renal cell carcinoma) | **Clear cell** (commonest), papillary (10–15%), chromophobe (3–5%), cystic RCC, rare molecular subtypes | Clear cell enhances strongly; papillary enhances weakly |
| Other malignant | Urothelial carcinoma, lymphoma, metastasis, sarcoma, epithelioid AML (angiomyolipoma) | Infiltrative growth |
| Benign solid | Oncocytoma, fat-poor AML, papillary adenoma, metanephric adenoma | See below |
| Cystic | Haemorrhagic or complex cyst, cystic nephroma | Bosniak category |
| Inflammatory, vascular | Abscess, focal pyelonephritis, tuberculosis; aneurysm, AVM (arteriovenous malformation), infarct, pseudotumour | Clinical picture |

- **Oncocytoma:** commonest benign enhancing SRM (up to half of benign SRMs). Central stellate scar and segmental enhancement inversion suggest it but are not specific. A core cannot separate it from chromophobe RCC, which has oncocytic areas. Sestamibi SPECT (single-photon emission computed tomography) is under evaluation.
- **Fat-poor AML:** no visible fat (fat below −20 HU [Hounsfield units] diagnoses classic AML); homogeneous, T2-dark, mimics RCC; biopsy HMB-45 and Melan-A positive.
- **Also know:** metanephric adenoma (hypovascular, polycythaemia, WT1 positive); papillary adenoma (≤15 mm).

### Narrowing the differential

- **Multiphase CT (computed tomography):** enhancement over 15–20 HU means solid tissue. MRI (magnetic resonance imaging) helps below 2 cm.
- **RMB (renal mass biopsy)** if it will change management (AUA [American Urological Association]); seeding almost unreported; non-diagnostic about 14%.
- Then surveillance, ablation or **PN (partial nephrectomy)**, preferred for cT1a.

## Haemostasis in LPN (laparoscopic partial nephrectomy)

**Principle:** a clear field for excision, then secure closure of the bed. Plan with R.E.N.A.L. score and 3D (three-dimensional) CT; intraoperative ultrasound.

**1. Vascular control**

| Technique | Method | Comment |
|---|---|---|
| Main artery ± vein clamp | Bulldog clamps, or laparoscopic Satinsky en bloc | Artery-only equals artery–vein. Aim warm ischaemia ≤25 min (trifecta) |
| Selective or super-selective | Clamp the tumour's segmental branch; 3D CT, NIRF (near-infrared fluorescence) | Avoids global ischaemia; no functional gain in an RCT (randomised controlled trial) |
| Off-clamp ("zero ischaemia") | No clamping; small exophytic tumours | More blood loss; same long-term eGFR (estimated glomerular filtration rate) |
| Early unclamping | Release after the inner suture layer | Shorter ischaemia; possibly fewer pseudoaneurysms |
| Cold ischaemia | Ice slush, cold saline via ureteric catheter, cold arterial perfusion | Solitary kidney, CKD (chronic kidney disease), long repair; benefit unproven |

- Function depends mainly on parenchyma preserved. Mannitol: no benefit in an RCT. **Also know:** parenchymal clamp for peripheral tumours.

**2. Excision:** score the capsule, cut with cold scissors, clip visible vessels. Enucleation along the pseudocapsule bleeds less.

**3. Renorrhaphy (most reliable step)**

- **Inner layer:** running 3-0 or 2-0 absorbable suture closes cut vessels and collecting system (opened in about 60%).
- **Outer layer, sliding-clip technique:** mattress sutures anchored by a locking clip; clips slid down to the capsule give tension without knots. Pledgets prevent capsular tears; oxidised cellulose bolster fills the defect.
- **Barbed suture:** holds tension without knots, so ischaemia is shorter.
- Single layer gave better function and fewer pseudoaneurysms than two layers.

**4. Energy:** argon beam or bipolar coagulation of the bed. Argon can destroy collecting-system sutures and adds gas, so vent the ports.

**5. Haemostatic agents (adjuncts, no proven benefit in RCTs):** gelatin–thrombin matrix; fibrin sealant; oxidised cellulose; PEG (polyethylene glycol) hydrogel; cyanoacrylate. Sutureless closure only for selected small tumours.

**6. Final check:** unclamp, inspect, **lower pneumoperitoneum** to unmask venous bleeding; drain.

## Postoperative bleeding

- Early (hypotension, falling haemoglobin): resuscitate, CT, angioembolisation.
- **Delayed haematuria (about day 12):** pseudoaneurysm or arteriovenous fistula; selective angioembolisation preserves function. Endophytic and posterior tumours carry higher risk.

**Diagram to draw:** sliding-clip renorrhaphy over a bolster, with bulldog clamp on the artery.

**Key points to remember:**

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

## LAQ 5 Radiology for solid renal masses.

P4

**Type:** LAQ (25 marks)
**Source:** 2018-Summer Paper 4, Q1

### Introduction

- Most renal masses are now found incidentally. Imaging is the **strongest predictor of malignancy**; size and **enhancement** are the two key variables.
- About 20% of solid enhancing cT1 masses are benign (oncocytoma, fat-poor AML (angiomyolipoma)).
- **Goals:** solid or cystic? enhancing? fat? subtype? stage and surgical anatomy.

### A. CT (computed tomography): the standard test

**Renal mass protocol** (dedicated, thin slices):

| Phase | Timing | Use |
|---|---|---|
| Non-contrast | – | Baseline HU (Hounsfield units), fat, calcification, haemorrhage |
| Corticomedullary | 30–70 s | Hypervascular masses; arteries; clear cell vs other subtypes |
| Nephrographic | 90–120 s | **Most sensitive** for detection (tumour enhances less than parenchyma); renal vein and IVC (inferior vena cava) thrombus |
| Excretory | 7–10 min | Collecting system relation |

**Enhancement thresholds** (rise from unenhanced value):

| Rise | Meaning |
|---|---|
| ≥20 HU | Enhancing: solid tumour |
| 10–20 HU | Equivocal (MRI, magnetic resonance imaging, or CEUS) |
| <10 HU | Non-enhancing: cyst |

- **Pitfall:** pseudoenhancement of small (<10–15 mm) cysts; dual-energy CT iodine maps help.
- **Macroscopic fat = AML:** fat ≤ −10 to −20 HU on unenhanced CT is diagnostic. Fat **with calcification** suggests RCC (renal cell carcinoma).
- **Fat-poor AML:** hyperdense on unenhanced CT, homogeneous, avid early enhancement and washout.
- **Subtype clues:** clear cell RCC enhances early, avidly and heterogeneously (necrosis); papillary RCC is homogeneous and **hypoenhancing**; chromophobe and oncocytoma intermediate, with a central scar. Overlap is large.

### B. MRI

- **Indications:** contrast allergy, CKD (chronic kidney disease), equivocal CT enhancement, young patients, thrombus. Enhancement = signal rise ≥15–20%. Macrocyclic gadolinium carries very low NSF (nephrogenic systemic fibrosis) risk.

| Feature | Clear cell | Papillary | Chromophobe / oncocytoma | AML |
|---|---|---|---|---|
| T2 signal | High, heterogeneous | **Low**, homogeneous (haemosiderin) | Central T2-bright scar | Fat signal |
| Enhancement | Avid (>200%) | Weak | Intermediate / avid | Avid |
| Diffusion (ADC, apparent diffusion coefficient) | High | Lowest | Intermediate | – |
| Chemical shift | **Signal drop** (intracellular lipid, ~59%) | No drop | – | **India-ink** artefact; loss on fat saturation |

- **ccLS (clear cell likelihood score, 1–5):** structured MRI algorithm for solid cT1a masses using T2 signal, corticomedullary enhancement and microscopic fat, plus diffusion and **segmental enhancement inversion** (favours oncocytoma). Score 4–5 suggests clear cell RCC; 1–2 a non-clear-cell tumour. It guides biopsy versus surveillance.

### C. Ultrasound and CEUS (contrast-enhanced ultrasound)

- Grey-scale: cyst versus solid; poor below 2 cm. AML is hyperechoic, but so are some small RCCs.
- **CEUS** (microbubbles, not nephrotoxic): shows enhancement; used in CKD or allergy, indeterminate lesions, ablation and surveillance.

### D. Cystic masses: Bosniak 2019 (CT and MRI)

| Class | Key feature | Malignancy / action |
|---|---|---|
| I | Simple, thin wall ≤2 mm, no septa | Benign; none |
| II | 1–3 thin septa; homogeneous hyperdense (≥70 HU unenhanced) | Benign; none |
| IIF | Wall or septa 3 mm, or ≥4 thin septa | <10%; follow 6, 12 months, then yearly for 5 years |
| III | Wall or septa ≥4 mm, or irregular | ~50%; surgery or surveillance |
| IV | **Enhancing nodule** | ~90%; treat |

### E. Functional and molecular imaging

- **99mTc-sestamibi SPECT/CT** (single-photon emission CT): mitochondria-rich **oncocytoma** and hybrid oncocytic or chromophobe tumours take it up; clear cell RCC does not.
- **Girentuximab PET (positron emission tomography)** targets CA-IX (carbonic anhydrase IX) on clear cell RCC. **ZIRCON** ($^{89}$Zr-girentuximab, indeterminate cT1 masses): about 85% sensitivity and 87% specificity for clear cell RCC.
- **FDG-PET** (fluorodeoxyglucose PET): not recommended (low sensitivity). PSMA (prostate-specific membrane antigen) PET: experimental.

### F. Renal mass biopsy

- **Indications:** result may change management; suspected lymphoma, metastasis or infection; before ablation or surveillance; metastatic disease. Not needed if surgery is planned regardless.

- Cores: PPV (positive predictive value) 98.8%; subtype accuracy ≥90%, grade 60–80%; non-diagnostic ~14%. Seeding almost nil. Oncocytoma cannot be confirmed on cores.

### G. Staging and surgical planning

- **Venous thrombus:** CT sensitivity 78% (renal vein), 96% (IVC). MRI or multiplanar CT shows the cephalad tip; enhancing thrombus is tumour, non-enhancing is bland.

| Mayo level | Extent | T stage |
|---|---|---|
| 0 | Renal vein | T3a |
| I | IVC ≤2 cm above renal vein | T3b |
| II | IVC below hepatic veins | T3b |
| III | Intrahepatic IVC, below diaphragm | T3b |
| IV | Above diaphragm or atrium | T3c |

- Nodes ≥2 cm usually malignant. Chest CT; bone scan only if symptomatic.
- **Nephrometry:** R.E.N.A.L. (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location), PADUA, C-index: predict complexity and complications; guide partial versus radical nephrectomy.
- **After ablation:** persistent or new enhancement = residual or recurrence.
- **Recent:** radiomics and AI (artificial intelligence).

**Diagram to draw:** algorithm: incidental mass → renal-protocol CT/MRI → fat (AML) / cyst (Bosniak) / enhancing solid → biopsy or treat.

> **Key points to remember:**
> - Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
> - MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
> - Sestamibi for oncocytoma; $^{89}$Zr-girentuximab PET (ZIRCON) for clear cell.
> - Biopsy PPV ~99%, non-diagnostic ~14%.

## Short answer questions (SAQ)

### SAQ 1 Immunotherapy / immunomodulation in RCC

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Immunotherapy in (metastatic) RCC" (2024-Winter Paper II Q3e; Practice Set 4 Paper II Q3f) · P4 SAQ "Immuno modulation in Renal, cell carcinoma" (2025-Winter Paper 4 Q3a)

*Assumed meaning: immunomodulation (immunotherapy) in renal cell carcinoma. Overall management of metastatic RCC: see 2024-Winter Paper 3, Q3b.*

#### Rationale

- RCC (renal cell carcinoma) is immunogenic and chemoresistant: rare (<1%) spontaneous regression after nephrectomy, and tumour-infiltrating T cells.
- **Checkpoints:** PD-1 (programmed death-1) on CD8 T cells binds tumour PD-L1 and switches the T cell off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) on regulatory T cells damps

priming. Blockade **releases the brake** on existing anti-tumour immunity.
- VEGF (vascular endothelial growth factor) is immunosuppressive, so TKIs (tyrosine kinase inhibitors) synergise with checkpoint inhibitors.

### Cytokine era (non-specific stimulation)

| | IFN-α (interferon-alpha) | High-dose IL-2 (interleukin-2) |
|---|---|---|
| Response | 10–15% | 15–20% |
| Durable complete response | <2% | **7–9%** |
| Toxicity | Flu-like | **Capillary leak**; needs intensive care |
| Role | Obsolete | Selected fit patients only |

- High-dose IL-2 dose: 600,000–720,000 IU/kg intravenously every 8 hours, maximum 15 doses; over 60% of complete responders stay disease-free.

### Checkpoint inhibitors: trials (all clear cell)

| Trial | Regimen vs comparator | Result |
|---|---|---|
| CheckMate 025 | Nivolumab vs everolimus, second line after TKI | OS (overall survival) 25.0 vs 19.6 months |
| **CheckMate 214** | Nivolumab + ipilimumab vs sunitinib | Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9% |
| KEYNOTE-426 | Pembrolizumab + axitinib vs sunitinib | Better PFS (progression-free survival) and OS |
| CheckMate 9ER | Nivolumab + cabozantinib vs sunitinib | Better PFS and OS; PFS 16.6 vs 8.3 months |
| **CLEAR** | Pembrolizumab + lenvatinib vs sunitinib | PFS 23.9 vs 9.2 months; response 71% |
| JAVELIN | Avelumab + axitinib | No OS benefit |

### Treatment sequencing

- **First line:** an IO/TKI (immuno-oncology plus TKI) combination for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF-targeted TKI. IO rechallenge is not recommended: it failed in CONTACT-03 and TiNivo-2. Belzutifan after IO and TKI.

### Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases (M1 NED, no evidence of disease). DFS (disease-free survival) HR 0.68 (P4 and another source: HR 0.63); later OS benefit (HR 0.62).
- Atezolizumab and nivolumab-based adjuvant trials were negative.

### Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

| Grade | Action |
|---|---|
| 1 | Continue, monitor |

| Grade | Action |
|---|---|
| 2 | Hold; prednisolone 0.5–1 mg/kg |
| 3–4 | Stop; high-dose steroids (methylprednisolone 1–2 mg/kg); infliximab or mycophenolate if refractory |

- Endocrinopathies: hormone replacement.

### Biomarkers and also know

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well. Gene signatures are investigational; antibiotics may reduce benefit.
- Non-clear cell data are scant.
- Vaccines and allogeneic transplant failed.

### Diagram to draw

- T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

> **Key points to remember:**
> - IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
> - High-dose IL-2: durable complete response 7–9%.
> - CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
> - KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
> - irAE grade ≥2: hold drug and give steroids.

## SAQ 2 Renal angiomyolipoma and its management

P2 · P3 **merged from 2**

*Asked as:* P2 SAQ "Renal aangiomyolipoma" (Practice Set 7 Paper 2 Q3f) · P3 SAQ "Management of AML." (2021-Summer Paper 3 Q3d)

*Assumed meaning: AML = renal angiomyolipoma (not acute myeloid leukaemia).*

### Introduction and pathogenesis

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; AML in 55–90%: younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34) leads to unchecked **mTOR (mammalian target of rapamycin)** signalling, which drives growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis); treat as RCC (renal cell carcinoma).

### Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock).
- CKD (chronic kidney disease) in TSC.

### Diagnosis

- **CT (computed tomography): macroscopic fat (below about −15 to −20 HU [Hounsfield units]) on non-contrast images is diagnostic.**
- US (ultrasound): hyperechoic (so are some RCCs).
- MRI (magnetic resonance imaging): signal loss on fat suppression, India-ink artefact.
- Fat-poor lesion: MRI ± biopsy (HMB-45 [human melanoma black-45], Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

### Management algorithm (Restrepo 2022)

| Situation | Management |
|---|---|
| Acute haemorrhage | Resuscitate, then **SAE (selective arterial embolisation)**; nephrectomy if SAE fails or is unavailable |
| Symptomatic (pain, haematuria) or poor access to care | SAE, NSS (nephron-sparing surgery), ablation in selected cases |
| Asymptomatic sporadic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy** | Active treatment |
| None of these | **Surveillance**: imaging (US or CT) at 6 and 12 months, then yearly |
| TSC, asymptomatic, **>3 cm** | **Everolimus**; MRI every 1–3 years |

- The old **4 cm** cut-off overtreats and should not be a general trigger; EAU (European Association of Urology) 2023 favours surveillance for most.

### Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis) is common; abscess, infarction.
- **Partial nephrectomy:** preferred surgery; essential in TSC (multifocal, recurrent).
- **Thermal ablation** (radiofrequency, cryo): limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%** (placebo); side effects stomatitis, acne, hyperlipidaemia.

### Diagram to draw

- Management algorithm by bleeding, symptoms, size and TSC.

> **Key points to remember:**
> - Macroscopic fat on non-contrast CT is diagnostic.
> - Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
> - SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

## SAQ 3 Role of renal nephrometry in management of RCC.

P1

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 1, Q3f

## Definition

- **Nephrometry scores** grade the anatomical complexity of a renal mass on CT (computed tomography) or MRI (magnetic resonance imaging), to standardise RCC (renal cell carcinoma) reporting and plan treatment.

## R.E.N.A.L. score (Kutikov and Uzzo, 2009)

| Component | 1 point | 2 points | 3 points |
|---|---|---|---|
| **R**adius (maximum diameter) | ≤4 cm | >4 to <7 cm | ≥7 cm |
| **E**xophytic/endophytic | ≥50% exophytic | <50% exophytic | Entirely endophytic |
| **N**earness to collecting system or sinus | ≥7 mm | >4 to <7 mm | ≤4 mm |
| **A**nterior/posterior | Suffix a, p or x (no points) | | |
| **L**ocation to polar lines | Entirely above or below | Crosses a polar line | >50% across, crosses midline, or entirely between |
| - Suffix "h": touches main renal vessels. **Total 4–12:** low 4–6, moderate 7–9, high 10–12. | | | |

## PADUA (Preoperative Aspects and Dimensions Used for an Anatomical classification) score (Ficarra, 2009)

- Polar location, exophytic rate, renal rim (lateral or medial), renal sinus, collecting system and size, plus anterior/posterior face. **Total 6–14:** 6–7 low, 8–9 intermediate, ≥10 high risk.

## C-index (centrality index)

- Distance from tumour centre to kidney centre ÷ tumour radius. **Lower value = more central, more complex.**

## Role in management

| Decision | Use |
|---|---|
| **Partial vs radical nephrectomy** | Feasibility and risk of PN (partial nephrectomy) vs RN (radical nephrectomy) |
| **Approach** | Open vs minimally invasive; hilar clamping |
| **Predict outcomes** | Warm ischaemia, blood loss, complications, conversion, collecting-system entry; urine leak (R and N components) |
| **Function** | Parenchymal loss, eGFR (estimated glomerular filtration rate) decline |
| **Ablation** | Cryoablation complications rise with complexity |
| **Pathology** | Higher scores: malignancy, higher grade |

- **AUA (American Urological Association) 2021:** with high oncological potential, prefer RN only if PN would be challenging even for experts, there is no CKD (chronic kidney disease) or proteinuria, and new baseline eGFR will likely exceed 45.

## Limitations

- Interobserver variation; ignores patient factors and surgeon experience.

**Also know:** ABC (arterial based complexity), DAP (diameter–axial–polar) scores; **MAP (Mayo Adhesive Probability) score** for adherent perinephric fat.

**Diagram to draw:** kidney with polar lines and the five R.E.N.A.L. components.

> **Key points to remember:**
> - RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
> - PADUA 6–14; C-index lower = more central.
> - Predicts PN feasibility, ischaemia time and complications.
> - MAP score: adherent perinephric fat.

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## SAQ 4 Acquired renal cystic disease

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3b

### Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

### Epidemiology

| Duration of renal failure | Prevalence |
|---|---|
| Before dialysis | About 10% |
| 3 years of dialysis | 44% |
| 5 years | 60% |
| ≥10 years | >90% |
| - Male:female 3:1; independent of the cause of renal failure. | |

### Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

### Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate crystals, good prognosis); papillary RCC is also common.

### Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

| Feature | ARCD | ADPKD (autosomal dominant polycystic kidney disease) |
|---|---|---|
| Kidney size | Small (<300 g) | Large (>800 g) |
| Cysts | Small | Large |
| Extrarenal | None | Liver cysts, aneurysms |

### Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.
- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

**Diagram to draw:** small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

## SAQ 5 ADPKD (Adult polycystic kidney disease)

P2

**Type:** SAQ (10 marks)
**Source:** 2017-Winter Paper 2, Q3d

### Genetics

- ADPKD (autosomal dominant polycystic kidney disease): commonest inherited cystic kidney disease (about 1 in 400–1000); 7–15% of patients on haemodialysis.

| Gene | Chromosome | Share | ESRD (end-stage renal disease) |
|---|---|---|---|
| **PKD1** (polycystin-1) | 16 | ~85% | 50s |
| **PKD2** (polycystin-2) | 4 | ~15% | 70s |

- Polycystins (primary cilium) restrain proliferation.

### Clinical features (30–50 years)

- **Pain** (commonest): mass, cyst haemorrhage, infection, stones. **Haematuria** about 50%; **stones** 20–30%.

- **Hypertension:** about 50% at 20–35 years with normal function; stretched vessels release renin.
- **Renal failure:** tracks kidney volume; worse with PKD1, male sex.
- **Extrarenal:** liver cysts (commonest; up to 90% by 50); **intracranial aneurysm 10–30%**; seminal vesicle (40%), pancreatic, arachnoid cysts; mitral valve prolapse; diverticulosis.
- RCC (renal cell carcinoma) is not commoner, but is more often bilateral and sarcomatoid.

### Diagnosis (ultrasound in at-risk relatives)

| Age | Ravine (PKD1 families) | Pei unified (2009) |
|---|---|---|
| 15–29 | ≥2 cysts, one or both kidneys | ≥3 cysts in total (15–39) |
| 30–59 | ≥2 cysts in each kidney | ≥2 in each kidney (40–59) |
| ≥60 | ≥4 cysts in each kidney | ≥4 in each kidney |

- CT or MRI shows TKV (total kidney volume), which predicts progression. Genetic testing if equivocal or for a living donor.

### Management

- **BP <130/80 mmHg** with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker); HALT-PKD: tighter control slowed TKV growth.
- **Tolvaptan** (V2-receptor antagonist; lowers cyclic adenosine monophosphate): **TEMPO 3:4 (2012)** slowed TKV growth and function loss; **REPRISE (2017)** slowed eGFR (estimated glomerular filtration rate) decline in later disease. Causes polyuria and liver toxicity.
- **Pain:** exclude infection, stone, tumour; avoid NSAIDs (non-steroidal anti-inflammatory drugs); aspiration, then laparoscopic unroofing.
- **Infected cyst:** lipid-soluble antibiotics (trimethoprim–sulfamethoxazole, fluoroquinolones); drain if no response.
- **Haemorrhage:** usually settles with rest. **Aneurysm screening** (MR angiography) if family history.
- **ESRD:** dialysis or transplant (no recurrence). **Native nephrectomy** before or at transplant for recurrent infection, bleeding, pain, suspected tumour or no room for the graft.

**Diagram to draw:** large cystic kidneys, liver cysts and berry aneurysm.

> **Key points to remember:**
> - PKD1: chromosome 16, 85%, earlier ESRD.
> - Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
> - Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

## SAQ 6 Biomarkers for patients with advanced RCC

P2

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 2, Q3d

### Introduction

- A **prognostic** biomarker predicts outcome regardless of treatment; a **predictive** biomarker predicts benefit from a specific drug.

- In advanced RCC (renal cell carcinoma), only clinical risk models are used routinely. No molecular biomarker is yet validated to select treatment (EAU, European Association of Urology; NCCN, National Comprehensive Cancer Network).

### Clinical and laboratory: risk models

| Factor | MSKCC (Memorial Sloan-Kettering Cancer Center) | IMDC (International Metastatic RCC Database Consortium) |
|---|---|---|
| Karnofsky performance status <80% | Yes | Yes |
| Haemoglobin < lower limit of normal | Yes | Yes |
| Corrected calcium high | >10 mg/dL | > upper limit of normal |
| LDH (lactate dehydrogenase) >1.5 × normal | Yes | No |
| Time factor | No prior nephrectomy | <1 year from diagnosis to treatment |
| Neutrophils, platelets > normal | No | Yes |

- IMDC: favourable 0 factors (median survival 43 months), intermediate 1–2 (22.5 months), poor ≥3 (7.8 months).
- IMDC group guides first-line choice (nivolumab–ipilimumab not for favourable risk).
- **Also know:** neutrophil-to-lymphocyte ratio, CRP (C-reactive protein), liver or bone metastases (poor prognosis).

### Tissue biomarkers

- **PD-L1 (programmed death-ligand 1):** benefit of checkpoint combinations is **independent of PD-L1**, so it is not used to select patients.
- **Sarcomatoid features:** respond well to checkpoint inhibitors.
- **Gene expression signatures:** IMmotion150 showed high T-effector signature tumours do better with atezolizumab–bevacizumab; high angiogenic signature tumours do well with sunitinib. Investigational.
- **Genomic:** BAP1 loss and CDKN2A loss indicate poor prognosis; PBRM1 data conflicting. MET alterations in papillary RCC (cabozantinib); FH (fumarate hydratase) deficiency. TMB (tumour mutational burden) is low and not predictive.

### Circulating and other biomarkers

- **KIM-1 (kidney injury molecule-1):** high serum levels predict recurrence and benefit from adjuvant immunotherapy (investigational).
- **ctDNA (circulating tumour DNA):** RCC sheds little DNA; methylation assays are promising.
- **Also know:** antibiotics during checkpoint therapy reduced survival (microbiome); sunitinib-induced hypertension is linked to better outcome.

**Diagram to draw:** IMDC risk groups with survival curves.

> **Key points to remember:**
>
> - IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
> - PD-L1 does not select patients in RCC.
> - Sarcomatoid tumours respond well to immunotherapy.

## SAQ 7 Tivozanib

P2

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 2, Q3b

### Introduction

- Oral **VEGFR TKI (vascular endothelial growth factor receptor tyrosine kinase inhibitor)** used in advanced RCC (renal cell carcinoma).
- Targets **VEGFR-1, -2 and -3** (also c-KIT and PDGFR-β, platelet-derived growth factor receptor-β); highly potent and selective.

### Mechanism and rationale

- In clear cell RCC, **VHL (von Hippel–Lindau) loss** lets HIF (hypoxia-inducible factor) accumulate, driving VEGF and angiogenesis.
- Tivozanib blocks VEGFR signalling, starving the tumour of new vessels.
- High selectivity means fewer off-target effects (less hand-foot skin reaction and diarrhoea than sorafenib).

### Dose and pharmacology

- **1.34 mg orally once daily, 21 days of a 28-day cycle.**
- Long half-life (about 4–5 days); dose reduction for toxicity or hepatic impairment.

### Key trials

| Trial | Setting | Result |
|---|---|---|
| **TIVO-1** (phase III) | 0–1 prior therapy, versus sorafenib | PFS (progression-free survival) 11.9 vs 9.1 months; response 33% vs 23%; OS (overall survival) not better (28.8 vs 29.3 months) |
| **TIVO-3** (phase III) | After 2–3 prior systemic therapies including a VEGFR TKI, versus sorafenib | PFS 5.6 vs 3.9 months (HR, hazard ratio, 0.73); response 18% vs 8%; OS not different (HR 0.97); benefit also after prior checkpoint inhibitor |
| **TiNivo-2** (phase III) | After checkpoint inhibitor; tivozanib ± nivolumab | **Negative:** adding nivolumab did not improve PFS (no immunotherapy re-challenge) |

### Approval and place

- FDA (Food and Drug Administration) declined in 2013 because of the OS concern in TIVO-1; EMA (European Medicines Agency) approved first-line use in 2017.
- **FDA March 2021:** relapsed or refractory advanced RCC **after ≥2 prior systemic therapies**.
- NCCN (National Comprehensive Cancer Network): later-line option, mainly after immunotherapy combinations.

### Adverse effects

- **Hypertension** (commonest; control before starting), dysphonia, fatigue, diarrhoea, proteinuria, hypothyroidism.
- Bleeding, arterial thrombosis, cardiac failure, **impaired wound healing** (stop before surgery), RPLS (reversible posterior leukoencephalopathy syndrome).
- **Monitoring:** blood pressure, thyroid tests, urine protein, liver tests.

**Diagram to draw:** VHL loss → HIF → VEGF → VEGFR, blocked by tivozanib.

> **Key points to remember:**
> - Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
> - TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
> - FDA 2021: after ≥2 prior systemic therapies.
> - TiNivo-2 negative; main toxicity hypertension.

## SAQ 8 VHL gene and von Hippel–Lindau disease

P2

**Asked as:**

- VHL gene. — *Practice Set 2 Paper II Q3f (SAQ)*
- Von Hippel Lindau disease — *Practice Set 8 Paper II Q3c (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 8 Paper II Q3c: VHL gene also answered in Practice Set 2 Paper 2, Q3f)

### Introduction

- **VHL (von Hippel–Lindau)** is a **tumour suppressor gene on chromosome 3p25**, identified in 1993; three exons, 213-amino-acid protein.
- Germline mutation causes **VHL disease**: an **autosomal dominant** familial cancer syndrome (about 1 in 36,000), penetrance >95% by 65 years.
- It is the familial form of **clear cell RCC** (renal cell carcinoma). Somatic VHL loss occurs in most sporadic clear cell RCC; chromosome 3p is altered in over 90%.

### Function / genetics and pathogenesis: the oxygen-sensing pathway

- VHL protein binds elongin B, elongin C, cullin-2 and RBX1 to form (VHL protein forms) an **E3 ubiquitin ligase** that degrades **HIF (hypoxia-inducible factor)** in normoxia.
- **Normoxia:** hydroxylated **HIF -1α and -2α** are tagged for proteasomal destruction.
- **VHL loss (or hypoxia):** loss of VHL stabilises HIF, which (chiefly HIF-2α) accumulates, joins ARNT (aryl hydrocarbon receptor nuclear translocator) and switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α (transforming growth factor-α), GLUT1, erythropoietin and CA-IX (carbonic anhydrase IX).
- Hence the **hypervascularity** (**highly vascular** tumours) of clear cell RCC.
- **Knudson two-hit:** one germline plus one somatic hit, so VHL tumours are **early, bilateral and multifocal**.

## VHL disease manifestations

| Lesion | Frequency |
|---|---|
| Craniospinal haemangioblastoma (cerebellum commonest) | 60–80% |
| Retinal angioma (earliest) | 25–60% |
| **Clear cell RCC** (mean age 39; commonest cause of death) | 25–75% |
| Renal cysts (commonest renal lesion) | Up to 76% |
| **Phaeochromocytoma** | 10–25% |
| Pancreatic cysts / neuroendocrine tumours | 45–75% / 11–17% |
| Epididymal cystadenoma | 25–60% of men |
| - Also endolymphatic sac tumour. | |
| - **RCC is now the commonest cause of death.** | |

## Genotype–phenotype

| Type | Mutation | Phaeochromocytoma | RCC |
|---|---|---|---|
| 1 | Deletions, truncations | Low | High |
| 2A | Missense | High | Low |
| 2B | Missense | High | High |
| 2C | Missense | Phaeochromocytoma only | Absent |

## Diagnosis

- Family history plus one lesion; otherwise ≥2 haemangioblastomas, or 1 plus a visceral lesion. Genetic testing of relatives.

## Surveillance (VHL Alliance 2020)

- Eye check (examination) yearly from infancy; metanephrines yearly from 5 years; CNS (brain and spine) MRI every 2 years from 11; **abdominal MRI every 2 years from 15**.

## Clinical implications / urological management

- **3 cm rule:** observe solid renal tumours <3 cm; intervene when largest reaches 3 cm; then nephron-sparing surgery (enucleation) or ablation, after excluding phaeochromocytoma.
- **Nephron-sparing surgery** (enucleation) is standard; not curative: local recurrence 75–85%, yet 10-year disease-specific survival 81–94%. Ablation for repeated small tumours.
- Exclude phaeochromocytoma before any surgery (normetanephrine raised; α-blockade). **Partial (cortical-sparing) adrenalectomy.**
- **Targeted therapy:** VEGF TKIs (tyrosine kinase inhibitors), bevacizumab, mTOR (mammalian target of rapamycin) inhibitors.
- **Belzutifan** (HIF-2α inhibitor): FDA (US Food and Drug Administration) approved 2021 for VHL disease; renal response 30–50% (another source: responses in about half of renal tumours); anaemia.
- **Also know:** PBRM1, BAP1, SETD2 on 3p.

**Diagram to draw:** VHL–HIF pathway in normoxia versus (vs) VHL loss, with drug targets.

**Key points to remember:**

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

## SAQ 9 Zero-ischaemia partial nephrectomy (laparoscopic and robotic)

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 4, Q3e (clamping strategies and laparoscopic partial nephrectomy haemostasis also in 2019-Summer Paper 3, Q1)

### Definition and rationale

- **Zero-ischaemia PN (partial nephrectomy):** excision and renorrhaphy **without clamping the main renal artery**, so the kidney stays perfused (Gill and colleagues, 2011).
- **Why:** warm ischaemia injures nephrons (historical limit 30 minutes; trifecta ≤25). Most relevant in a **solitary kidney or CKD (chronic kidney disease)**.

### Techniques

| Technique | Method |
|---|---|
| **Off-clamp** | Hilum looped for rescue but not clamped; cold-scissor excision with strong suction |
| **Anatomical (super-selective)** | Microdissect the higher-order arterial branches entering the tumour; occlude only these with small bulldogs |
| **Selective segmental clamping** | Only the tumour's segment is ischaemic ("minimal ischaemia") |
| **Parenchymal compression** | Parenchymal clamp across a polar tumour |
| **Enucleation** | Pseudocapsule plane bleeds less, so suits off-clamp work |

### Aids

- **3D (three-dimensional) CT (computed tomography) reconstruction** maps segmental arteries; intraoperative Doppler ultrasound.
- **NIRF (near-infrared fluorescence) with ICG (indocyanine green)** on robotic consoles: after selective clamping, perfused kidney glows and the ischaemic tumour zone stays dark.
- **Robot:** wristed instruments and fast suturing ease off-clamp work.
- **Haemostasis:** clip or suture bed vessels as met; sliding-clip or barbed renorrhaphy; clamp if bleeding hides the margin.

### Selection

- **Ideal:** small (cT1a), exophytic, peripheral tumour; experienced surgeon.
- **Caution:** endophytic, hilar or central tumours.

### Evidence

- Off- versus on-clamp: **better early eGFR (estimated glomerular filtration rate), more blood loss**; **no difference beyond the early period**.
- **Robotic RCTs (randomised controlled trials, e.g. CLOCK):** no difference in eGFR or split function up to 2 years.
- RCT of super-selective clamping with NIRF: no 6-month gain; stopped for futility.
- **Concept:** parenchymal volume preserved, not ischaemia time within accepted limits, governs function.
- **Also know:** early unclamping; hypothermia for long ischaemia; mannitol no benefit.

**Diagram to draw:** hilum with a micro-bulldog on a tertiary branch; tumour zone dark on NIRF.

> **Key points to remember:**
>
> - Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
> - NIRF with ICG confirms the right branch.
> - RCTs: no long-term eGFR gain; more blood loss.
> - Parenchyma preserved matters more than clamp time.

# Rapid Revision Sheet · Section 13

### Nephron-sparing surgery for renal masses: indications, approaches and recent advances (LAQ)

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm, surveillance <2 cm.
- Imperative, relative, elective; elective PN is standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- AUA: RN only if complex + no CKD + new eGFR/GFR likely >45.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function.
- Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

### Metastatic RCC: assessment and management (LAQ)

- IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI (IO combinations) for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
- Belzutifan after IO and TKI.

### Upper tract urothelial carcinoma (LAQ)

- Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

**Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy. (LAQ)**

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

**Radiology for solid renal masses. (LAQ)**

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; $^{89}$Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

**Immunotherapy / immunomodulation in RCC (SAQ)**

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

**Renal angiomyolipoma and its management (SAQ)**

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

**Role of renal nephrometry in management of RCC. (SAQ)**

- RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
- PADUA 6–14; C-index lower = more central.
- Predicts PN feasibility, ischaemia time and complications.
- MAP score: adherent perinephric fat. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Acquired renal cystic disease (SAQ)**

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

**ADPKD (Adult polycystic kidney disease) (SAQ)**

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

**Biomarkers for patients with advanced RCC (SAQ)**

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

**Tivozanib (SAQ)**

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

**VHL gene and von Hippel–Lindau disease (SAQ)**

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

**Zero-ischaemia partial nephrectomy (laparoscopic and robotic) (SAQ)**

- Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
- NIRF with ICG confirms the right branch.
- RCTs: no long-term eGFR gain; more blood loss.
- Parenchyma preserved matters more than clamp time. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*